

book_to_slide_BY_sections_V13_

July 19, 2025

1 Set up Paths

```
[26]: # Cell 1: Setup and Configuration
import os
import re
import logging
import warnings
from docx import Document
import pdfplumber
import ollama
from tenacity import retry, stop_after_attempt, wait_exponential, RetryError
import json
import logging

# Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %
↳ %(message)s')
logger = logging.getLogger(__name__)

# Extract Unit ID from the filename
# --- Helper Functions ---
def print_header(text: str, char: str = "="):
    """Prints a centered header to the console."""
    print("\n" + char * 80)
    print(text.center(80))
    print(char * 80)

def extract_uo_id_from_filename(filename: str) -> str:
    match = re.match(r'^[A-Z]+\d+', os.path.basename(filename))
    if match:
        return match.group(0)
    raise ValueError(f"Could not extract a valid Unit ID from filename:
↳ '{filename}'")

# To define outputs folder
TEST_MODE = True
```

```

# --- 1. CORE SETTINGS ---
# Set this to True for EPUB, False for PDF. This controls the entire notebook's
↳flow.
PROCESS_EPUB = True # for EPUB
# PROCESS_EPUB = False # for PDF

# Control process for testing
EXTRACT_UO = False
CREATE_RAG_BOOK = False

#=====INPUTS=====

# --- 2. INPUT FILE NAMES ---
# The name of the Unit Outline file (e.g., DOCX, PDF)
# UNIT_OUTLINE_FILENAME = "ICT312 Digital Forensic_Final.docx" # epub
UNIT_OUTLINE_FILENAME = "ICT311 Applied Cryptography.docx" # pdf

# The names of the book files
#EPUB_BOOK_FILENAME = "Bill Nelson, Amelia Phillips, Christopher Steuart -
↳Guide to Computer Forensics and Investigations_ Processing Digital
↳Evidence-Cengage Learning (2018).epub"
EPUB_BOOK_FILENAME = "/home/sebas_dev_linux/projects/course_generator/data/
↳books/Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell,
↳J.epub"
PDF_BOOK_FILENAME = "(Chapman & Hall_CRC Cryptography and Network Security
↳Series) Jonathan Katz, Yehuda Lindell - Introduction to Modern
↳Cryptography-CRC Press (2020).pdf"

# --- 3. DIRECTORY STRUCTURE ---
# Define the base path to your project to avoid hardcoding long paths everywhere
PROJECT_BASE_DIR = "/home/sebas_dev_linux/projects/course_generator"

CONFIG_DIR = os.path.join(PROJECT_BASE_DIR, "configs")
SETTINGS_DECK_PATH = os.path.join(CONFIG_DIR, "settings_deck.json")
#TEACHING_FLOWS_PATH = os.path.join(CONFIG_DIR, "teaching_flows.json") #
↳future add

# to check the layouts
LAYOUT_MAPPING_PATH = os.path.join(CONFIG_DIR, "layout_mapping_test_Mod.json")
↳# using the modified one

# New output path for the processed settings

```

```

PROCESSED_SETTINGS_PATH = os.path.join(CONFIG_DIR, "processed_settings.json")

# Define subdirectories relative to the base path
DATA_DIR_INPUT = os.path.join(PROJECT_BASE_DIR, "data")
INPUT_UO_DIR = os.path.join(DATA_DIR_INPUT, "UO")
INPUT_BOOKS_DIR = os.path.join(DATA_DIR_INPUT, "books")

#=====OUTPUTS=====

try:
    UNIT_ID = extract_uo_id_from_filename(UNIT_OUTLINE_FILENAME)
except ValueError as e:
    print(f"Error: {e}")
    UNIT_ID = "UNKNOWN_ID"

# Define outputs

if TEST_MODE:
    DATA_DIR_OUTPUT = os.path.join(PROJECT_BASE_DIR, "data_output/test_Output")
else:
    DATA_DIR_OUTPUT = os.path.join(PROJECT_BASE_DIR, "data_output")

os.makedirs(DATA_DIR_OUTPUT, exist_ok=True)
PARSE_DATA_DIR = os.path.join(PROJECT_BASE_DIR, DATA_DIR_OUTPUT, "1.
↳Parse_data")
os.makedirs(PARSE_DATA_DIR, exist_ok=True)

# Construct full paths for clarity

OUTPUT_PARSED_UO_DIR = os.path.join(PARSE_DATA_DIR, f"Parse_UO/{UNIT_ID}")
os.makedirs(OUTPUT_PARSED_UO_DIR, exist_ok=True)
OUTPUT_PARSED_TOC_DIR = os.path.join(PARSE_DATA_DIR, f"Parse_TOC_books/
↳{UNIT_ID}")
os.makedirs(OUTPUT_PARSED_TOC_DIR, exist_ok=True)
OUTPUT_DB_DIR = os.path.join(DATA_DIR_OUTPUT, f"0. DataBase_Chroma/{UNIT_ID}")
os.makedirs(OUTPUT_DB_DIR, exist_ok=True)

# to Save the individual FINAL plan to a file
PLAN_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"2. generated_plans/{UNIT_ID}")
os.makedirs(PLAN_OUTPUT_DIR, exist_ok=True)

#to Save the individual FINAL Content to a file

```

```

CONTENT_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"3. generated_content/
↳{UNIT_ID}")
os.makedirs(CONTENT_OUTPUT_DIR, exist_ok=True)

CONTENT_LLM_OUTPUT_DIR = os.path.join(DATA_DIR_OUTPUT, f"4.
↳generated_content_llm/{UNIT_ID}")
os.makedirs(CONTENT_LLM_OUTPUT_DIR, exist_ok=True)

SLIDE_TEMPLATE_PATH = "/home/sebas_dev_linux/projects/course_generator/data/
↳slide_style/slide_style_test_2.pptx"
FINAL_PRESENTATION_DIR = os.path.join(DATA_DIR_OUTPUT, f"5. final_presentations/
↳{UNIT_ID}")
os.makedirs(FINAL_PRESENTATION_DIR, exist_ok=True)

# --- 4. LLM & EMBEDDING CONFIGURATION ---
LLM_PROVIDER = "ollama" # Can be "ollama", "openai", "gemini"
OLLAMA_HOST = "http://localhost:11434"
OLLAMA_MODEL = "qwen3:8b" # "qwen3:8b", #"mistral:latest"
EMBEDDING_MODEL_OLLAMA = "nomic-embed-text"
CHUNK_SIZE = 800
CHUNK_OVERLAP = 100

# --- 5. DYNAMICALLY GENERATED PATHS & IDs (DO NOT EDIT THIS SECTION) ---
# This section uses the settings above to create all the necessary variables
↳for later cells.

# Full path to the unit outline file
FULL_PATH_UNIT_OUTLINE = os.path.join(INPUT_UO_DIR, UNIT_OUTLINE_FILENAME)

# Determine which book and output paths to use based on the PROCESS_EPUB flag
if PROCESS_EPUB:
    BOOK_PATH = os.path.join(INPUT_BOOKS_DIR, EPUB_BOOK_FILENAME)
    PRE_EXTRACTED_TOC_JSON_PATH = os.path.join(OUTPUT_PARSED_TOC_DIR,
↳f"{UNIT_ID}_epub_table_of_contents.json")
else:
    BOOK_PATH = os.path.join(INPUT_BOOKS_DIR, PDF_BOOK_FILENAME)
    PRE_EXTRACTED_TOC_JSON_PATH = os.path.join(OUTPUT_PARSED_TOC_DIR,
↳f"{UNIT_ID}_pdf_table_of_contents.json")

# Define paths for the vector database
file_type_suffix = 'epub' if PROCESS_EPUB else 'pdf'
CHROMA_PERSIST_DIR = os.path.join(OUTPUT_DB_DIR,
↳f"{UNIT_ID}_chroma_db_toc_chunks_{file_type_suffix}")

```

```

CHROMA_COLLECTION_NAME = f"book_toc_chunks_{file_type_suffix}"

# Define path for the parsed unit outline
PARSED_UO_JSON_PATH = os.path.join(OUTPUT_PARSED_UO_DIR, f"{os.path.
↳splitext(UNIT_OUTLINE_FILENAME)[0]}_parsed.json")

# --- Sanity Check Printout ---
print("--- CONFIGURATION SUMMARY ---")
print(f"Processing Mode: {' EPUB' if PROCESS_EPUB else ' PDF'}")
print(f" Unit ID: {UNIT_ID}")
print(f" Unit Outline Path: {FULL_PATH_UNIT_OUTLINE}")
print(f" Book Path: {BOOK_PATH}")
print(f" Parsed UO Output Path: {PARSED_UO_JSON_PATH}")
print(f" Parsed ToC Output Path: {PRE_EXTRACTED_TOC_JSON_PATH}")
print(f" Vector DB Path: {CHROMA_PERSIST_DIR}")
print(f" Vector DB Collection: {CHROMA_COLLECTION_NAME}")
print(f" Plan Develop output: {PLAN_OUTPUT_DIR}")
print(f" Plan Fetched output: {CONTENT_OUTPUT_DIR}")
print(f" Plan Generated LLM output: {CONTENT_LLM_OUTPUT_DIR}")
print(f" Final slides: {FINAL_PRESENTATION_DIR}")
print("--- SETUP COMPLETE ---")

```

--- CONFIGURATION SUMMARY ---

Processing Mode: EPUB

Unit ID: ICT311

Unit Outline Path:

/home/sebas_dev_linux/projects/course_generator/data/UO/ICT311 Applied

Cryptography.docx

Book Path:

/home/sebas_dev_linux/projects/course_generator/data/books/Introduction to
Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub

Parsed UO Output Path:

/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_UO/ICT311/ICT311 Applied Cryptography_parsed.json

Parsed ToC Output Path:

/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_TOC_books/ICT311/ICT311_epub_table_of_contents.json

Vector DB Path:

/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/0.
DataBase_Chroma/ICT311/ICT311_chroma_db_toc_chunks_epub

Vector DB Collection: book_toc_chunks_epub

Plan Develop output:

/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311

Plan Fetched output:

/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.

```

generated_content/ICT311
  Plan Generated LLM output:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
  Final slides:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311
--- SETUP COMPLETE ---

```

2 System Prompt

```

[27]: UNIT_OUTLINE_SYSTEM_PROMPT_TEMPLATE = """
You are an expert academic assistant tasked with parsing a university unit
  ↳outline document and extracting key information into a structured JSON
  ↳format.

The input will be the raw text content of a unit outline. Your goal is to
  ↳identify and extract the following details and structure them precisely as
  ↳specified in the JSON schema below. Note: do not change any key name

**JSON Output Schema:**

```json
{{
 "unitInformation": {{
 "unitCode": "string | null",
 "unitName": "string | null",
 "creditPoints": "integer | null",
 "unitRationale": "string | null",
 "prerequisites": "string | null"
 }},
 "learningOutcomes": [
 "string"
],
 "assessments": [
 {{
 "taskName": "string",
 "description": "string",
 "dueWeek": "string | null",
 "weightingPercent": "integer | null",
 "learningOutcomesAssessed": "string | null"
 }}
],
 "weeklySchedule": [
 {{
 "week": "string",

```

```

 "contentTopic": "string",
 "requiredReading": "string | null"
 }}
],
"requiredReadings": [
 "string"
],
"recommendedReadings": [
 "string"
]
}}

```

Instructions for Extraction:

Unit Information: Locate Unit Code, Unit Name, Credit Points. Capture 'Unit Overview / Rationale' as unitRationale. Identify prerequisites.

Learning Outcomes: Extract each learning outcome statement.

Assessments: Each task as an object. Capture full task name, description, Due Week, Weighting % (number), and Learning Outcomes Assessed.

weeklySchedule: Each week as an object. Capture Week, contentTopic, and requiredReading.

Required and Recommended Readings: List full text for each.

**\*\*Important Considerations for the LLM\*\*:**

Pay close attention to headings and table structures.

If information is missing, use null for string/integer fields, or an empty list [] for array fields.

Do not change keys in the template given

Ensure the output is ONLY the JSON object, starting with {{{ and ending with }}}. No explanations or conversational text before or after the JSON.

Now, parse the following unit outline text:

```

--- UNIT_OUTLINE_TEXT_START ---
{outline_text}
--- UNIT_OUTLINE_TEXT_END ---
"""

```

[28]: *# Place this in a new cell after your imports, or within Cell 3 before the functions.*

*# This code is based on the schema from your screenshot on page 4.*

```

from pydantic import BaseModel, Field, ValidationError
from typing import List, Optional
import time

```

*# Define Pydantic models that match your JSON schema*

```

class UnitInformation(BaseModel):
 unitCode: Optional[str] = None
 unitName: Optional[str] = None
 creditPoints: Optional[int] = None

```

```

 unitRationale: Optional[str] = None
 prerequisites: Optional[str] = None

class Assessment(BaseModel):
 taskName: str
 description: str
 dueWeek: Optional[str] = None
 weightingPercent: Optional[int] = None
 learningOutcomesAssessed: Optional[str] = None

class WeeklyScheduleItem(BaseModel):
 week: str
 contentTopic: str
 requiredReading: Optional[str] = None

class ParsedUnitOutline(BaseModel):
 unitInformation: UnitInformation
 learningOutcomes: List[str]
 assessments: List[Assessment]
 weeklySchedule: List[WeeklyScheduleItem]
 requiredReadings: List[str]
 recommendedReadings: List[str]

```

### 3 Helper functions

```

[29]: # Cell 10: Configuration and Scoping for Content Generation (Corrected)

import os
import json
import logging

Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - _
↳ %(message)s')
logger = logging.getLogger(__name__)

--- Global Test Overrides (for easy testing) ---
TEST_OVERRIDE_WEEKS = None
TEST_OVERRIDE_FLOW_ID = None
TEST_OVERRIDE_SESSIONS_PER_WEEK = None
TEST_OVERRIDE_DISTRIBUTION_STRATEGY = None

def process_and_load_configurations():

```



```

"""
 PHASE 1: Loads configurations, calculates a PRELIMINARY time-based slide_
↳budget,
 and saves the result as 'processed_settings.json' for the Planning Agent.
"""

print_header("Phase 1: Configuration and Scoping Process", char="-")

--- Load all input files ---
logger.info("Loading all necessary configuration and data files...")
try:
 os.makedirs(CONFIG_DIR, exist_ok=True)
 with open(PARSED_UO_JSON_PATH, 'r', encoding='utf-8') as f:
↳unit_outline = json.load(f)
 with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f:
↳book_toc = json.load(f)
 with open(SETTINGS_DECK_PATH, 'r', encoding='utf-8') as f:
↳settings_deck = json.load(f)
 # with open(TEACHING_FLOWS_PATH, 'r', encoding='utf-8') as f:
↳teaching_flows = json.load(f)
 logger.info("All files loaded successfully.")
except FileNotFoundError as e:
 logger.error(f"FATAL: A required configuration file was not found: {e}")
 return None

--- Pre-process and Refine Settings ---
logger.info("Pre-processing settings_deck for definitive plan...")
processed_settings = json.loads(json.dumps(settings_deck))

unit_info = unit_outline.get("unitInformation", {})
processed_settings['course_id'] = unit_info.get("unitCode",
↳"UNKNOWN_COURSE")
processed_settings['unit_name'] = unit_info.get("unitName", "Unknown Unit_
↳Name")

--- Apply test overrides IF they are not None ---
logger.info("Applying overrides if specified...")
This block now correctly sets the teaching_flow_id based on the_
↳interactive flag.
if TEST_OVERRIDE_FLOW_ID is not None:
 processed_settings['teaching_flow_id'] = TEST_OVERRIDE_FLOW_ID
 logger.info(f"OVERRIDE: teaching_flow_id set to_
↳'{TEST_OVERRIDE_FLOW_ID}'")
else:
 # If no override, use the 'interactive' boolean from the file as the_
↳source of truth.
 is_interactive = processed_settings.get('interactive', False)

```

```

 if is_interactive:
 processed_settings['teaching_flow_id'] = 'apply_topic_interactive'
 else:
 processed_settings['teaching_flow_id'] = 'standard_lecture'
 logger.info(f"Loaded from settings: 'interactive' is {is_interactive}.\n
↳Set teaching_flow_id to '{processed_settings['teaching_flow_id']}'".)

 # The 'interactive' flag is now always consistent with the teaching_flow_id.
 processed_settings['interactive'] = "interactive" in\
↳processed_settings['teaching_flow_id'].lower()

 if TEST_OVERRIDE_SESSIONS_PER_WEEK is not None:
 processed_settings['week_session_setup']['sessions_per_week'] =\
↳TEST_OVERRIDE_SESSIONS_PER_WEEK
 logger.info(f"OVERRIDE: sessions_per_week set to\
↳{TEST_OVERRIDE_SESSIONS_PER_WEEK}")

 if TEST_OVERRIDE_DISTRIBUTION_STRATEGY is not None:
 processed_settings['week_session_setup']['distribution_strategy'] =\
↳TEST_OVERRIDE_DISTRIBUTION_STRATEGY
 logger.info(f"OVERRIDE: distribution_strategy set to\
↳'{TEST_OVERRIDE_DISTRIBUTION_STRATEGY}'")

 if TEST_OVERRIDE_WEEKS is not None:
 processed_settings['generation_scope']['weeks'] = TEST_OVERRIDE_WEEKS
 logger.info(f"OVERRIDE: generation_scope weeks set to\
↳{TEST_OVERRIDE_WEEKS}")

 # --- DYNAMIC SLIDE BUDGET CALCULATION (Phase 1) ---
 logger.info("Calculating preliminary slide budget based on session time...")

 params = processed_settings.get('parameters_slides', {})
 SLIDES_PER_HOUR = params.get('slides_per_hour', 18)

 duration_hours = processed_settings['week_session_setup'].
↳get('session_time_duration_in_hour', 1.0)
 sessions_per_week = processed_settings['week_session_setup'].
↳get('sessions_per_week', 1)

 logger.info(f"Duration Hours {duration_hours}.")
 logger.info(f"Sessions per week {sessions_per_week}.")

 slides_content_per_session = int(duration_hours * SLIDES_PER_HOUR)

```

```

 target_total_slides = slides_content_per_session * sessions_per_week

 processed_settings['slide_count_strategy']['target_total_slides'] = ␣
↪target_total_slides
 processed_settings['slide_count_strategy']['slides_content_per_session'] = ␣
↪slides_content_per_session
 logger.info(f"Preliminary weekly content slide target calculated:␣
↪#{target_total_slides} slides.")

 # --- Resolve Generation Scope if not overridden ---
 if TEST_OVERRIDE_WEEKS is None and processed_settings.
↪get('generation_scope', {}).get('weeks') == "all":
 num_weeks = len(unit_outline.get('weeklySchedule', []))
 processed_settings['generation_scope']['weeks'] = list(range(1,␣
↪num_weeks + 1))

 # --- Save the processed settings to disk ---
 logger.info(f"Saving preliminary processed configuration to:␣
↪{PROCESSED_SETTINGS_PATH}")
 with open(PROCESSED_SETTINGS_PATH, 'w', encoding='utf-8') as f:
 json.dump(processed_settings, f, indent=2)
 logger.info("File saved successfully.")

 # --- Assemble master config for optional preview ---
 master_config = {
 "processed_settings": processed_settings,
 "unit_outline": unit_outline,
 "book_toc": book_toc,
 # "teaching_flows": teaching_flows
 }

 print_header("Phase 1 Configuration Complete", char="-")
 logger.info("Master configuration object is ready for the Planning Agent.")
 return master_config

```

### 3.0.1 Component: Definitive PowerPoint Layout Inspector

Component: Definitive PowerPoint Layout Inspector Primary Purpose The inspect\_and\_generate\_layout\_config function serves as a critical pre-processing utility for the automated presentation generation system. Its primary purpose is to bridge the gap between a visual PowerPoint template and the programmatic logic of the content generation agents. It achieves this by performing a deep inspection of a given PowerPoint (.pptx) template file and auto-generating a detailed, structured, and human-readable JSON configuration file (layout\_mapping.json). This configuration file acts as the “API documentation” for the presentation template, allowing both human users and a Large Language Model (LLM) to understand and utilize the available slide layouts effectively. Key Functions and GoOf course. Here is a formal description of the purpose and functionality of the “Definitive PowerPoint Layout Inspector”

script. This description is suitable for project documentation, a README file, or for explaining its role to other developers.

**Primary Purpose** The `inspect_and_generate_layout_config` function serves as a critical pre-processing utility for the automated presentation generation system. Its primary purpose is to **bridge the gap between a visual PowerPoint template and the programmatic logic of the content generation agents.**

It achieves this by performing a deep inspection of a given PowerPoint (.pptx) template file and auto-generating a detailed, structured, and human-readable JSON configuration file (`layout_mapping.json`). This configuration file acts as the “API documentation” for the presentation template, allowing both human users and a Large Language Model (LLM) to understand and utilize the available slide layouts effectively.

## Key Functions and Goals

### 1. Comprehensive Layout Discovery:

- The script guarantees that **every single slide layout** present in the PowerPoint template’s Slide Master is detected and analyzed. This prevents a common problem where unused or unconventionally named layouts might be missed by simpler scripts.

### 2. Detailed Placeholder Analysis:

- For each layout, the script extracts an exhaustive list of all its placeholders. For every placeholder, it records crucial metadata:
  - **type**: The functional role of the placeholder (e.g., `TITLE`, `BODY`, `OBJECT`, `TABLE`, `PICTURE`).
  - **name**: The unique name given to the placeholder in the PowerPoint interface (e.g., “Title 1”, “Content Placeholder 2”).
  - **idx**: The internal identification number of the placeholder.
  - **position and size**: The physical coordinates (`left`, `top`) and dimensions (`width`, `height`), converted to an intuitive unit (inches) for easy comprehension of the layout’s visual structure.

### 3. Intelligent Capability Summarization:

- The script’s core innovation is its ability to generate a **machine-readable capabilities summary** for each layout. Instead of just listing raw data, it synthesizes the placeholder information into a concise description of what the layout is designed for. For example:
  - `{"title_support": "standard_title", "body_layout": "2_column"}`
  - `{"title_support": "centered_title_with_subtitle", "body_layout": "no_body"}`
  - `{"specific_content_types": ["TABLE", "CHART"]}`
- This summary is specifically designed to be passed to an LLM as part of a prompt, enabling the LLM to make an informed, logical choice about the best layout for presenting a given piece of information.

### 4. User-Friendly Configuration:

- While providing a detailed summary for the LLM, the script also generates a simplified `user_selections` section. This allows a human operator to easily map the system’s semantic slide types (e.g., “Agenda”, “Summary”) to a specific layout index, providing a robust fallback and manual override capability.

## How It Solves Critical Problems

- **Eliminates Ambiguity:** By capturing the name, index, and position of every placeholder, it solves the problem of layouts with multiple placeholders of the same type (e.g., two content boxes). The system can now programmatically target the “left column” vs. the “right column”.
- **Decouples Logic from Design:** The presentation generation agent no longer needs hard-coded assumptions about the template’s design. All the logic for choosing and populating layouts is driven by the generated JSON file. This means the visual template can be updated or completely replaced without requiring changes to the core Python code.
- **Empowers the LLM:** It transforms a visual, unstructured design asset (the .pptx file) into a structured, well-defined set of “tools” (the layouts) that an LLM can reason about. This is the key to enabling more advanced tasks where the LLM doesn’t just fill in content, but also makes decisions about the *visual structure* of the presentation.

In summary, the Definitive PowerPoint Layout Inspector is the foundational component that makes the entire presentation generation process intelligent, configurable, and robust. It translates the abstract design of a template into concrete, actionable data.

```
[30]: # this process the layouts from the slides provide to use them to process the
 ↪ plan (This require be change to the course of the system)
 # https://aistudio.google.com/prompts/18YaU5pG96eFMbM1l6yeG1v9j63PkLc5H

import logging
import os
import json
import random
from pptx import Presentation
from pptx.util import Inches
from pptx.enum.shapes import MSO_SHAPE_TYPE
from pptx.enum.shapes import PP_PLACEHOLDER

Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -
 ↪ %(message)s')
logger = logging.getLogger(__name__)

def print_header(text: str, char: str = "="):
 """Prints a centered header to the console."""
 print("\n" + char * 80)
 print(text.center(80))
 print(char * 80)

def generate_layout_capabilities(layout_name: str, placeholders: list) -> dict:

 essential_placeholders = [p for p in placeholders if p['type'] !=
 ↪ 'SLIDE_NUMBER']
 capabilities = {"title_support": "none", "body_layout": "none",
 ↪ "specific_content_types": []}
```

```

 has_center_title = any(p['type'] == 'CENTER_TITLE' for p in_
↪essential_placeholders)
 has_standard_title = any(p['type'] == 'TITLE' for p in_
↪essential_placeholders)
 has_subtitle = any(p['type'] == 'SUBTITLE' for p in essential_placeholders)

 if has_center_title: capabilities["title_support"] = "centered_title"
 elif has_standard_title: capabilities["title_support"] = "standard_title"
 if has_subtitle: capabilities["title_support"] += "_with_subtitle"

 body_placeholders = [p for p in essential_placeholders if p['type'] not in_
↪('TITLE', 'CENTER_TITLE', 'SUBTITLE')]

 if len(body_placeholders) == 0: capabilities["body_layout"] = "no_body"
 elif len(body_placeholders) == 1: capabilities["body_layout"] =_
↪"single_column"
 elif len(body_placeholders) > 1:
 body_placeholders.sort(key=lambda p: p['left'])
 if len(body_placeholders) > 1 and body_placeholders[1]['left'] >_
↪(body_placeholders[0]['left'] + body_placeholders[0]['width'] * 0.5):
 capabilities["body_layout"] = f"{len(body_placeholders)}_column"
 else: capabilities["body_layout"] = "stacked_sections"

 specific_types = {p['type'] for p in body_placeholders if p['type'] not in_
↪('BODY', 'OBJECT')}
 if specific_types: capabilities["specific_content_types"] =_
↪sorted(list(specific_types))

 return capabilities

def inspect_and_generate_layout_config(template_path: str, output_path: str):
 """
 Inspects a template, generates a machine-readable capabilities summary, and_
↪creates
 a complete and definitive JSON configuration file for user and LLM use.
 """
 try:
 prs = Presentation(template_path)
 except Exception as e:
 logger.error(f"Could not open or parse the presentation template at_
↪'{template_path}'. Error: {e}")
 return

 # --- 1. Analyze all Layouts ---
 available_layouts = []

```

```

 for i, layout in enumerate(prs.slide_layouts):
 placeholder_details = []
 for p in layout.placeholders:
 placeholder_details.append({
 "idx": p.placeholder_format.idx, "type": p.placeholder_format.
↳ type.name,
 "name": p.name, "left": round(p.left.inches, 2), "top": round(p.
↳ top.inches, 2),
 "width": round(p.width.inches, 2), "height": round(p.height.
↳ inches, 2)
 })
 capabilities = generate_layout_capabilities(layout.name,
↳ placeholder_details)
 layout_info = {"layout_index": i, "layout_name": layout.name,
↳ "capabilities": capabilities, "placeholders": placeholder_details}
 available_layouts.append(layout_info)

 # --- 2. Create the Complete User-Facing Configuration Structure ---
 def find_default_by_capability(layouts, capability_key, capability_value,
↳ fallback_index=1):
 for layout in layouts:
 if layout['capabilities'].get(capability_key) == capability_value:
 return layout['layout_index']
 # If no perfect match is found, return a safe fallback index
 return fallback_index if len(layouts) > fallback_index else 0

 # REFACTORED: This map is now updated to match the desired JSON output
↳ exactly.
 user_selection_map = {
 "Title": {
 "description": "Main title slide of the deck.",
 "selected_layout_index":
↳ find_default_by_capability(available_layouts, "title_support",
↳ "centered_title_with_subtitle")
 },
 "Agenda": {
 "description": "Agenda/Table of Contents slide.",
 "selected_layout_index":
↳ find_default_by_capability(available_layouts, "body_layout", "single_column")
 },
 "Content": {
 "description": "Default slide for a standard topic with bullet
↳ points.",
 "selected_layout_index":
↳ find_default_by_capability(available_layouts, "body_layout", "single_column")
 },
 }

```

```

 "Content_Two_Column": {
 "description": "Slide for side-by-side content, like comparisons.",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "2_column")
 },
 "Content_child": {
 "description": "Default slide for a standard topic with bullet
 ↪points.",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "single_column")
 },
 "Content_Two_Column_child": {
 "description": "Slide for side-by-side content, like comparisons.",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "2_column")
 },
 "Application": {
 "description": "Slide for interactive questions ('Let's Apply This!
 ↪').",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "single_column")
 },
 # REFACTORED: Removed trailing space from the key
 ↪"Application_Two_Column"
 "Application_Two_Column": {
 "description": "Slide for side-by-side content ('Let's Apply This!
 ↪').",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "2_column")
 },
 "Summary": {
 "description": "The final summary slide of the deck.",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "body_layout", "single_column")
 },
 "End": {
 "description": "The final 'Thank You / Questions?' slide.",
 "selected_layout_index": 0
 ↪find_default_by_capability(available_layouts, "title_support",
 ↪"centered_title", fallback_index=7)
 },
 "Divider": {
 "description": "Slide that divide sections general use base on the
 ↪agenda.",

```



```

 "selected_layout_index":
↪find_default_by_capability(available_layouts, "title_support",
↪"centered_title", fallback_index=7)
 }
}

config_to_save = {
 "-//_INSTRUCTIONS": "This file describes the available slide layouts.
↪The 'available_layouts' section is for an LLM to read. The 'user_selections'
↪section is for you to edit. Please verify the 'selected_layout_index' for
↪each slide type.",
 "template_file": os.path.basename(template_path),
 "user_selections": user_selection_map,
 "available_layouts": available_layouts
}

--- 3. Save the Configuration File ---
try:
 with open(output_path, 'w', encoding='utf-8') as f:
 json.dump(config_to_save, f, indent=4)
 print_header("Configuration Generated Successfully", char="=")
 print(f"A new, human-readable configuration file has been saved to:
↪\n{output_path}")
 print("\nPlease open this file to review the classifications and edit
↪your layout selections.")
 except Exception as e:
 logger.error(f"Failed to write the layout configuration file to
↪'{output_path}'. Error: {e}")

--- Execution ---
SLIDE_TEMPLATE_PATH = "/home/sebas_dev_linux/projects/course_generator/data/
↪slide_style/slide_style_test_2.pptx"
LAYOUT_MAPPING_PATH = "/home/sebas_dev_linux/projects/course_generator/
↪configs/layout_mapping_test.json"
You run this function to generate the config file.
Make sure SLIDE_TEMPLATE_PATH and LAYOUT_MAPPING_PATH are defined.
inspect_and_generate_layout_config(SLIDE_TEMPLATE_PATH, LAYOUT_MAPPING_PATH)

```

#### 4 Extrac Unit outline details to process following steps - output raw json with UO details

[31]: # Cell 3: Parse Unit Outline

```
--- Helper Functions for Parsing ---
```

```

def extract_text_from_file(filepath: str) -> str:
 _, ext = os.path.splitext(filepath.lower())
 if ext == '.docx':
 doc = Document(filepath)
 full_text = [p.text for p in doc.paragraphs]
 for table in doc.tables:
 for row in table.rows:
 full_text.append(" | ".join(cell.text for cell in row.cells))
 return '\n'.join(full_text)
 elif ext == '.pdf':
 with pdfplumber.open(filepath) as pdf:
 return "\n".join(page.extract_text() for page in pdf.pages if page.
↪extract_text())
 else:
 raise TypeError(f"Unsupported file type: {ext}")

def parse_llm_json_output(content: str) -> dict:
 try:
 match = re.search(r'\{.*\}', content, re.DOTALL)
 if not match: return None
 return json.loads(match.group(0))
 except (json.JSONDecodeError, TypeError):
 return None

@retry(stop=stop_after_attempt(3), wait=wait_exponential(min=2, max=10))
def call_ollama_with_retry(client, prompt):
 logger.info(f"Calling Ollama model '{OLLAMA_MODEL}'...")
 response = client.chat(
 model=OLLAMA_MODEL,
 messages=[{"role": "user", "content": prompt}],
 format="json",
 options={"temperature": 0.0}
)
 if not response or 'message' not in response or not response['message'].
↪get('content'):
 raise ValueError("Ollama returned an empty or invalid response.")
 return response['message']['content']

--- Main Orchestration Function for this Cell ---
def parse_and_save_outline_robust(
 input_filepath: str,
 output_filepath: str,
 prompt_template: str,
 max_retries: int = 3
):
 logger.info(f"Starting to robustly process Unit Outline: {input_filepath}")

```

```

if not os.path.exists(input_filepath):
 logger.error(f"Input file not found: {input_filepath}")
 return

try:
 outline_text = extract_text_from_file(input_filepath)
 if not outline_text.strip():
 logger.error("Extracted text is empty. Aborting.")
 return
except Exception as e:
 logger.error(f"Failed to extract text from file: {e}", exc_info=True)
 return

client = ollama.Client(host=OLLAMA_HOST)
current_prompt = prompt_template.format(outline_text=outline_text)

for attempt in range(max_retries):
 logger.info(f"Attempt {attempt + 1}/{max_retries} to parse outline.")

 try:
 # Call the LLM
 llm_output_str = call_ollama_with_retry(client, current_prompt)

 # Find the JSON blob in the response
 json_blob = parse_llm_json_output(llm_output_str) # Your existing_
↪helper

 if not json_blob:
 raise ValueError("LLM did not return a parsable JSON object.")

 # *** THE KEY VALIDATION STEP ***
 # Try to parse the dictionary into your Pydantic model.
 # This will raise a `ValidationError` if keys are wrong, types are_
↪wrong, or fields are missing.
 parsed_data = ParsedUnitOutline.model_validate(json_blob)

 # If successful, save the validated data and exit the loop
 logger.info("Successfully validated JSON structure against Pydantic_
↪model.")

 os.makedirs(os.path.dirname(output_filepath), exist_ok=True)
 with open(output_filepath, 'w', encoding='utf-8') as f:
 # Use .model_dump_json() for clean, validated output
 f.write(parsed_data.model_dump_json(indent=2))

 logger.info(f"Successfully parsed and saved Unit Outline to:
↪{output_filepath}")
 return # Exit function on success

```

```

 except ValidationError as e:
 logger.warning(f"Validation failed on attempt {attempt + 1}. Error:␣
↪{e}")

 # Formulate a new prompt with the error message for self-correction
 error_feedback = (
 f"\n\nYour previous attempt failed. You MUST correct the␣
↪following errors:\n"
 f"{e}\n\n"
 f"Please regenerate the entire JSON object, ensuring it␣
↪strictly adheres to the schema "
 f"and corrects these specific errors. Do not change any key␣
↪names."
)
 current_prompt = current_prompt + error_feedback # Append the error␣
↪to the prompt

 except Exception as e:
 # Catch other errors like network issues from call_ollama_with_retry
 logger.error(f"An unexpected error occurred on attempt {attempt +␣
↪1}: {e}", exc_info=True)
 # You might want to wait before retrying for non-validation errors
 time.sleep(5)

 logger.error(f"Failed to get valid structured data from the LLM after␣
↪{max_retries} attempts.")

--- In your execution block, call the new function ---
parse_and_save_outline(...) becomes:

if EXTRACT_UO:
 parse_and_save_outline_robust(
 input_filepath=FULL_PATH_UNIT_OUTLINE,
 output_filepath=PARSED_UO_JSON_PATH,
 prompt_template=UNIT_OUTLINE_SYSTEM_PROMPT_TEMPLATE
)

```

## 5 Extract TOC from epub or PDF

```

[32]: # Cell 4: Extract Book Table of Contents (ToC) with Pre-assigned IDs & Links in␣
↪Order

from ebooklib import epub, ITEM_NAVIGATION
from bs4 import BeautifulSoup
import fitz # PyMuPDF
import json

```

```

import os
from typing import List, Dict
import urllib.parse # Needed to clean up links

=====
1. HELPER FUNCTIONS (MODIFIED TO INCLUDE ID ASSIGNMENT AND LINK EXTRACTION)
=====

def clean_epub_href(href: str) -> str:
 """Removes URL fragments and decodes URL-encoded characters."""
 if not href: return ""
 # Remove fragment identifier (e.g., '#section1')
 cleaned_href = href.split('#')[0]
 # Decode any URL-encoded characters (e.g., %20 -> space)
 return urllib.parse.unquote(cleaned_href)

--- EPUB Extraction Logic ---
def parse_navpoint(navpoint: BeautifulSoup, counter: List[int], level: int = 0) -> Dict:
 """Recursively parses EPUB 2 navPoints and assigns a toc_id and link_filename."""
 title = navpoint.navLabel.text.strip()
 if not title: return None

 # --- MODIFICATION: Extract the linked filename ---
 content_tag = navpoint.find('content', recursive=False)
 link_filename = clean_epub_href(content_tag['src']) if content_tag else ""

 node = {
 "level": level,
 "toc_id": counter[0],
 "title": title,
 "link_filename": link_filename, # Add the cleaned link
 "children": []
 }
 counter[0] += 1

 for child_navpoint in navpoint.find_all('navPoint', recursive=False):
 child_node = parse_navpoint(child_navpoint, counter, level + 1)
 if child_node: node["children"].append(child_node)

 return node

def parse_li(li_element: BeautifulSoup, counter: List[int], level: int = 0) -> Dict:
 """Recursively parses EPUB 3 elements and assigns a toc_id and link_filename."""

```

```

a_tag = li_element.find('a', recursive=False)
if a_tag:
 title = a_tag.get_text(strip=True)
 if not title: return None

 # --- MODIFICATION: Extract the linked filename ---
 link_filename = clean_epub_href(a_tag.get('href'))

 node = {
 "level": level,
 "toc_id": counter[0],
 "title": title,
 "link_filename": link_filename, # Add the cleaned link
 "children": []
 }
 counter[0] += 1

 nested_ol = li_element.find('ol', recursive=False)
 if nested_ol:
 for sub_li in nested_ol.find_all('li', recursive=False):
 child_node = parse_li(sub_li, counter, level + 1)
 if child_node: node["children"].append(child_node)
 return node
 return None

def extract_epub_toc(epub_path, output_json_path):
 print(f"Processing EPUB ToC for: {epub_path}")
 toc_data = []
 book = epub.read_epub(epub_path)
 id_counter = [0]

 for nav_item in book.get_items_of_type(ITEM_NAVIGATION):
 soup = BeautifulSoup(nav_item.get_content(), 'xml')
 # Logic to handle both EPUB 2 (NCX) and EPUB 3 (XHTML)
 if nav_item.get_name().endswith('.ncx'):
 print("INFO: Found EPUB 2 (NCX) Table of Contents. Parsing...")
 navmap = soup.find('navMap')
 if navmap:
 for navpoint in navmap.find_all('navPoint', recursive=False):
 node = parse_navpoint(navpoint, id_counter, level=0)
 if node: toc_data.append(node)
 else: # Assumes EPUB 3
 print("INFO: Found EPUB 3 (XHTML) Table of Contents. Parsing...")
 toc_nav = soup.select_one('nav[epub:type="toc"]')
 if toc_nav:
 top_ol = toc_nav.find('ol', recursive=False)
 if top_ol:

```

```

 for li in top_ol.find_all('li', recursive=False):
 node = parse_li(li, id_counter, level=0)
 if node: toc_data.append(node)
 if toc_data: break

 if toc_data:
 os.makedirs(os.path.dirname(output_json_path), exist_ok=True)
 with open(output_json_path, 'w', encoding='utf-8') as f:
 json.dump(toc_data, f, indent=2, ensure_ascii=False)
 print(f" Successfully wrote EPUB ToC with IDs and links to:␣
↪{output_json_path}")
 else:
 print(" WARNING: No ToC data extracted from EPUB.")

--- PDF Extraction Logic (Unchanged) ---
def build_pdf_hierarchy_with_ids(toc_list: List) -> List[Dict]:
 root = []
 parent_stack = {-1: {"children": root}}
 id_counter = [0]
 for level, title, page in toc_list:
 normalized_level = level - 1
 node = {"level": normalized_level, "toc_id": id_counter[0], "title":␣
↪title.strip(), "page": page, "children": []}
 id_counter[0] += 1
 parent_node = parent_stack.get(normalized_level - 1)
 if parent_node: parent_node["children"].append(node)
 parent_stack[normalized_level] = node
 return root

def extract_pdf_toc(pdf_path, output_json_path):
 print(f"Processing PDF ToC for: {pdf_path}")
 try:
 doc = fitz.open(pdf_path)
 toc = doc.get_toc()
 hierarchical_toc = []
 if not toc: print(" WARNING: This PDF has no embedded bookmarks (ToC).
↪")
 else:
 print(f"INFO: Found {len(toc)} bookmark entries. Building hierarchy␣
↪and assigning IDs...")
 hierarchical_toc = build_pdf_hierarchy_with_ids(toc)
 os.makedirs(os.path.dirname(output_json_path), exist_ok=True)
 with open(output_json_path, 'w', encoding='utf-8') as f:
 json.dump(hierarchical_toc, f, indent=2, ensure_ascii=False)
 print(f" Successfully wrote PDF ToC with assigned IDs to:␣
↪{output_json_path}")

```

```

 except Exception as e: print(f"An error occurred during PDF ToC extraction:␣
↪{e}")

=====
2. EXECUTION BLOCK
=====

if PROCESS_EPUB:
 extract_epub_toc(BOOK_PATH, PRE_EXTRACTED_TOC_JSON_PATH)
else:
 extract_pdf_toc(BOOK_PATH, PRE_EXTRACTED_TOC_JSON_PATH)

```

Processing EPUB ToC for:

```

/home/sebas_dev_linux/projects/course_generator/data/books/Introduction to
Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub
INFO: Found EPUB 2 (NCX) Table of Contents. Parsing...
 Successfully wrote EPUB ToC with IDs and links to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/1.
Parse_data/Parse_TOC_books/ICT311/ICT311_epub_table_of_contents.json

```

## 6 Hirachical DB base on TOC

### 6.1 Process Book

```

[33]: # Cell 5: Create Hierarchical Vector Database (with Sequential ToC ID and Chunk␣
↪ID)
This cell processes the book, enriches it with hierarchical and sequential␣
↪metadata,
chunks it, and creates the final vector database.

import os
import json
import shutil
import logging
from typing import List, Dict, Any, Tuple
from langchain_core.documents import Document
from langchain_community.document_loaders import PyPDFLoader,␣
↪UnstructuredEPubLoader
from langchain_ollama.embeddings import OllamaEmbeddings
from langchain_chroma import Chroma
from langchain.text_splitter import RecursiveCharacterTextSplitter

Setup Logger for this cell
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
logger = logging.getLogger(__name__)

--- Helper: Clean metadata values for ChromaDB ---

```



```

def clean_metadata_for_chroma(value: Any) -> Any:
 """Sanitizes metadata values to be compatible with ChromaDB."""
 if isinstance(value, list): return ", ".join(map(str, value))
 if isinstance(value, dict): return json.dumps(value)
 if isinstance(value, (str, int, float, bool)) or value is None: return value
 return str(value)

--- Core Function to Process Book with Pre-extracted ToC ---
def process_book_with_extracted_toc(
 book_path: str,
 extracted_toc_json_path: str,
 chunk_size: int,
 chunk_overlap: int
) -> Tuple[List[Document], List[Dict[str, Any]]]:

 logger.info(f"Processing book '{os.path.basename(book_path)}' using ToC_
 ↪from '{os.path.basename(extracted_toc_json_path)}'".)

 # 1. Load the pre-extracted hierarchical ToC
 try:
 with open(extracted_toc_json_path, 'r', encoding='utf-8') as f:
 hierarchical_toc = json.load(f)
 if not hierarchical_toc:
 logger.error(f"Pre-extracted ToC at '{extracted_toc_json_path}' is_
 ↪empty or invalid.")
 return [], []
 logger.info(f"Successfully loaded pre-extracted ToC with_
 ↪{len(hierarchical_toc)} top-level entries.")
 except Exception as e:
 logger.error(f"Error loading pre-extracted ToC JSON: {e}",_
 ↪exc_info=True)
 return [], []

 # 2. Load all text elements/pages from the book
 all_raw_book_docs: List[Document] = []
 _, file_extension = os.path.splitext(book_path.lower())

 if file_extension == ".epub":
 loader = UnstructuredEPubLoader(book_path, mode="elements",_
 ↪strategy="fast")
 try:
 all_raw_book_docs = loader.load()
 logger.info(f"Loaded {len(all_raw_book_docs)} text elements from_
 ↪EPUB.")
 except Exception as e:
 logger.error(f"Error loading EPUB content: {e}", exc_info=True)

```

```

 return [], hierarchical_toc
 elif file_extension == ".pdf":
 loader = PyPDFLoader(book_path)
 try:
 all_raw_book_docs = loader.load()
 logger.info(f"Loaded {len(all_raw_book_docs)} pages from PDF.")
 except Exception as e:
 logger.error(f"Error loading PDF content: {e}", exc_info=True)
 return [], hierarchical_toc
 else:
 logger.error(f"Unsupported book file format: {file_extension}")
 return [], hierarchical_toc

 if not all_raw_book_docs:
 logger.error("No text elements/pages loaded from the book.")
 return [], hierarchical_toc

 # 3. Create enriched LangChain Documents by matching ToC to content
 final_documents_with_metadata: List[Document] = []

 # Flatten the ToC, AND add a unique sequential ID for sorting and
 ↪validation.
 flat_toc_entries: List[Dict[str, Any]] = []

 def _add_ids_and_flatten_recursive(nodes: List[Dict[str, Any]], ↪
 ↪current_titles_path: List[str], counter: List[int]):
 """
 Recursively traverses ToC nodes to flatten them and assign a unique, ↪
 ↪sequential toc_id.
 """
 for node in nodes:
 toc_id = counter[0]
 counter[0] += 1
 title = node.get("title", "").strip()
 if not title: continue
 new_titles_path = current_titles_path + [title]
 entry = {
 "titles_path": new_titles_path,
 "level": node.get("level"),
 "full_title_for_matching": title,
 "toc_id": toc_id
 }
 if "page" in node: entry["page"] = node["page"]
 flat_toc_entries.append(entry)
 if node.get("children"):
 _add_ids_and_flatten_recursive(node.get("children", []), ↪
 ↪new_titles_path, counter)

```

```

 toc_id_counter = [0]
 _add_ids_and_flatten_recursive(hierarchical_toc, [], toc_id_counter)
 logger.info(f"Flattened ToC and assigned sequential IDs to
↳{len(flat_toc_entries)} entries.")

 # Logic for PDF metadata assignment
 if file_extension == ".pdf" and any("page" in entry for entry in
↳flat_toc_entries):
 logger.info("Assigning metadata to PDF pages based on ToC page numbers..
↳.")
 flat_toc_entries.sort(key=lambda x: x.get("page", -1) if x.get("page")
↳is not None else -1)
 for page_doc in all_raw_book_docs:
 page_num_0_indexed = page_doc.metadata.get("page", -1)
 page_num_1_indexed = page_num_0_indexed + 1
 assigned_metadata = {"source": os.path.basename(book_path),
↳"page_number": page_num_1_indexed}
 best_match_toc_entry = None
 for toc_entry in flat_toc_entries:
 toc_page = toc_entry.get("page")
 if toc_page is not None and toc_page <= page_num_1_indexed:
 if best_match_toc_entry is None or toc_page >
↳best_match_toc_entry.get("page", -1):
 best_match_toc_entry = toc_entry
 elif toc_page is not None and toc_page > page_num_1_indexed:
 break
 if best_match_toc_entry:
 for i, title_in_path in
↳enumerate(best_match_toc_entry["titles_path"]):
 assigned_metadata[f"level_{i+1}_title"] = title_in_path
 assigned_metadata['toc_id'] = best_match_toc_entry.get('toc_id')
 else:
 assigned_metadata["level_1_title"] = "Uncategorized PDF Page"
 cleaned_meta = {k: clean_metadata_for_chroma(v) for k, v in
↳assigned_metadata.items()}
 final_documents_with_metadata.append(Document(page_content=page_doc.
↳page_content, metadata=cleaned_meta))

 # Logic for EPUB metadata assignment
 elif file_extension == ".epub":
 logger.info("Assigning metadata to EPUB elements by matching ToC titles
↳in text...")
 toc_titles_for_search = [entry for entry in flat_toc_entries if entry.
↳get("full_title_for_matching")]
 current_hierarchy_metadata = {}

```

```

 for element_doc in all_raw_book_docs:
 element_text = element_doc.page_content.strip() if element_doc.
↪page_content else ""
 if not element_text: continue
 for toc_entry in toc_titles_for_search:
 if element_text == toc_entry["full_title_for_matching"]:
 current_hierarchy_metadata = {"source": os.path.
↪basename(book_path)}
 for i, title_in_path in enumerate(toc_entry["titles_path"]):
 current_hierarchy_metadata[f"level_{i+1}_title"] =
↪title_in_path
 current_hierarchy_metadata['toc_id'] = toc_entry.
↪get('toc_id')
 if "page" in toc_entry:
↪current_hierarchy_metadata["epub_toc_page"] = toc_entry["page"]
 break
 if not current_hierarchy_metadata:
 doc_metadata_to_assign = {"source": os.path.
↪basename(book_path), "level_1_title": "EPUB Preamble", "toc_id": -1}
 else:
 doc_metadata_to_assign = current_hierarchy_metadata.copy()
 cleaned_meta = {k: clean_metadata_for_chroma(v) for k, v in
↪doc_metadata_to_assign.items()}
 final_documents_with_metadata.
↪append(Document(page_content=element_text, metadata=cleaned_meta))

 else: # Fallback
 final_documents_with_metadata = all_raw_book_docs

 if not final_documents_with_metadata:
 logger.error("No documents were processed or enriched with hierarchical_
↪metadata.")
 return [], hierarchical_toc

 logger.info(f"Total documents prepared for chunking:
↪{len(final_documents_with_metadata)}")

 text_splitter = RecursiveCharacterTextSplitter(
 chunk_size=chunk_size,
 chunk_overlap=chunk_overlap,
 length_function=len
)
 final_chunks = text_splitter.split_documents(final_documents_with_metadata)
 logger.info(f"Split into {len(final_chunks)} final chunks, inheriting_
↪hierarchical metadata.")

```

```

--- MODIFICATION START: Add a unique, sequential chunk_id to each chunk
↪---
logger.info("Assigning sequential chunk_id to all final chunks...")
for i, chunk in enumerate(final_chunks):
 chunk.metadata['chunk_id'] = i
logger.info(f"Assigned chunk_ids from 0 to {len(final_chunks) - 1}.")
--- MODIFICATION END ---

return final_chunks, hierarchical_toc

--- Main Execution Block for this Cell ---
if CREATE_RAG_BOOK:
 if not os.path.exists(PRE_EXTRACTED_TOC_JSON_PATH):
 logger.error(f"CRITICAL: Pre-extracted ToC file not found at
↪'{PRE_EXTRACTED_TOC_JSON_PATH}'.")
 logger.error("Please run the 'Extract Book Table of Contents (ToC)'
↪cell (Cell 4) first.")
 else:
 final_chunks_for_db, toc_reloaded = process_book_with_extracted_toc(
 book_path=BOOK_PATH,
 extracted_toc_json_path=PRE_EXTRACTED_TOC_JSON_PATH,
 chunk_size=CHUNK_SIZE,
 chunk_overlap=CHUNK_OVERLAP
)

 if final_chunks_for_db:
 if os.path.exists(CHROMA_PERSIST_DIR):
 logger.warning(f"Deleting existing ChromaDB directory:
↪{CHROMA_PERSIST_DIR}")
 shutil.rmtree(CHROMA_PERSIST_DIR)

 logger.info(f"Initializing embedding model
↪'{EMBEDDING_MODEL_OLLAMA}' and creating new vector database...")
 embedding_model = OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA)

 vector_db = Chroma.from_documents(
 documents=final_chunks_for_db,
 embedding=embedding_model,
 persist_directory=CHROMA_PERSIST_DIR,
 collection_name=CHROMA_COLLECTION_NAME
)

 reloaded_db = Chroma(persist_directory=CHROMA_PERSIST_DIR,
↪embedding_function=embedding_model, collection_name=CHROMA_COLLECTION_NAME)
 count = reloaded_db._collection.count()

```

```

 print("-" * 50)
 logger.info(f" Vector DB created successfully at:␣
↪{CHROMA_PERSIST_DIR}")
 logger.info(f" Collection '{CHROMA_COLLECTION_NAME}' contains␣
↪{count} documents.")
 print("-" * 50)
 else:
 logger.error(" Failed to generate chunks. Vector DB not created.")

```

[34]: *# Cell 5a: Inspecting EPUB Documents and Metadata BEFORE Chunking*

```

import json
import os
import logging
from langchain_community.document_loaders import UnstructuredEPubLoader
from langchain_core.documents import Document

--- Setup Logger for this inspection cell ---
logger = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')

def inspect_epub_preprocessing():
 """
 This function replicates the pre-chunking logic from Cell 5 for EPUB files
 to show the list of large documents with their assigned ToC metadata.
 """
 if not PROCESS_EPUB:
 print("This inspection cell is for EPUB processing. Please set␣
↪PROCESS_EPUB = True in Cell 1.")
 return

 print_header("EPUB Pre-Processing Inspection", char="~")

 # --- 1. Load the necessary data (replicating start of Cell 5) ---
 logger.info("Loading pre-extracted ToC and raw EPUB elements...")
 try:
 with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f:
 hierarchical_toc = json.load(f)

 loader = UnstructuredEPubLoader(BOOK_PATH, mode="elements",␣
↪strategy="fast")
 all_raw_book_docs = loader.load()
 logger.info(f"Successfully loaded {len(all_raw_book_docs)} raw text␣
↪elements from the EPUB.")
 except Exception as e:
 logger.error(f"Failed to load necessary files: {e}")

```

```

 return

--- 2. Flatten the ToC (replicating logic from Cell 5) ---
logger.info("Flattening the hierarchical ToC for matching...")
flat_toc_entries = []
def _add_ids_and_flatten_recursive(nodes, current_titles_path, counter):
 for node in nodes:
 toc_id = counter[0]
 counter[0] += 1
 title = node.get("title", "").strip()
 if not title: continue
 new_titles_path = current_titles_path + [title]
 entry = {
 "titles_path": new_titles_path,
 "level": node.get("level"),
 "full_title_for_matching": title,
 "toc_id": toc_id
 }
 flat_toc_entries.append(entry)
 if node.get("children"):
 _add_ids_and_flatten_recursive(node.get("children", []),
↪new_titles_path, counter)

_add_ids_and_flatten_recursive(hierarchical_toc, [], [0])
logger.info(f"Flattened ToC into {len(flat_toc_entries)} entries.")

--- 3. The Core Matching Logic for EPUB (the part you want to see) ---
logger.info("Assigning metadata to EPUB elements by matching ToC titles...")

final_documents_with_metadata = []
toc_titles_for_search = [entry for entry in flat_toc_entries if entry.
↪get("full_title_for_matching")]
current_hierarchy_metadata = {}

for element_doc in all_raw_book_docs:
 element_text = element_doc.page_content.strip() if element_doc.
↪page_content else ""
 if not element_text:
 continue

 # Check if this element is a heading that matches a ToC entry
 is_heading = False
 for toc_entry in toc_titles_for_search:
 if element_text == toc_entry["full_title_for_matching"]:
 # It's a heading! Update the current context.
 current_hierarchy_metadata = {"source": os.path.
↪basename(BOOK_PATH)}

```

```

 for i, title_in_path in enumerate(toc_entry["titles_path"]):
 current_hierarchy_metadata[f"level_{i+1}_title"] =
↪title_in_path
 current_hierarchy_metadata['toc_id'] = toc_entry.get('toc_id')
 is_heading = True
 break # Found the match, no need to search further

 # Assign metadata
 if not current_hierarchy_metadata:
 # Content before the first ToC entry (e.g., cover, title page)
 doc_metadata_to_assign = {"source": os.path.basename(BOOK_PATH),
↪"level_1_title": "EPUB Preamble", "toc_id": -1}
 else:
 doc_metadata_to_assign = current_hierarchy_metadata.copy()

 final_documents_with_metadata.
↪append(Document(page_content=element_text, metadata=doc_metadata_to_assign))

 logger.info(f"Processing complete. Generated
↪{len(final_documents_with_metadata)} documents with assigned metadata.")

 # --- 4. Print the result for inspection ---
 print_header("INSPECTION RESULTS: Documents Before Chunking", char="=")
 print(f"Total documents created: {len(final_documents_with_metadata)}\n")

 for i, doc in enumerate(final_documents_with_metadata[:100]): # Print first
↪30 to avoid flooding the output
 print(f"--- Document [{i+1}] ---")
 print(f" Assigned Metadata: {doc.metadata}")
 print(f" Content (Un-chunked Element):")
 print(f" >> '{doc.page_content}'")
 print(f" -" * 25 + "\n")

 # --- Execute the inspection ---
 inspect_epub_preprocessing()

```

2025-07-19 20:24:26,324 - INFO - Loading pre-extracted ToC and raw EPUB elements...

~~~~~  
EPUB Pre-Processing Inspection  
~~~~~

[WARNING] Could not load translations for en-US  
data file translations/en.yaml not found

2025-07-19 20:24:30,978 - WARNING - Could not load translations for en-US









```
\[= \frac{1}{2} \cdot \text{\Pr}\left\lbrack\right.
```

$$\left.\text{\PrivK}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \mid b = 0\right\rbrace + \frac{1}{2} \cdot \text{\Pr}\left\lbrack\right.$$
$$\left.\text{\PrivK}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \mid b = 1\right\rbrace \\$$
$$=\frac{1}{2} \cdot \text{\Pr}\lbrack\mathcal{A}\text{\outputs}0 \mid b = 0\rbrace + \frac{1}{2} \cdot \text{\Pr}\lbrack\mathcal{A}\text{\outputs}1 \mid b = 1\rbrace,$$
$$\end{array}, \text{rendering as TeX}$$

2025-07-19 20:24:30,990 - WARNING - Could not convert TeX math \begin{array}{l}

$$\text{\Pr}\left\lbrack\right. {\mathtt{PrivK}}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \left.\right\rbrace \\$$
$${=} \frac{1}{2} \cdot \text{\Pr}\left\lbrack\right.$$
$${\mathtt{PrivK}}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \mid b = 0 \left.\right\rbrace + \frac{1}{2} \cdot \text{\Pr}\left\lbrack\right.$$
$${\mathtt{PrivK}}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \mid b = 1 \left.\right\rbrace \\$$
$${=} \frac{1}{2} \cdot \text{\Pr}\lbrack\mathcal{A}\text{\outputs}0 \mid b = 0\rbrace + \frac{1}{2} \cdot \text{\Pr}\lbrack\mathcal{A}\text{\outputs}1 \mid b = 1\rbrace,$$
$$\end{array}, \text{rendering as TeX}$$

[WARNING] Could not convert TeX math \frac{1}{2}, rendering as TeX

2025-07-19 20:24:30,991 - WARNING - Could not convert TeX math \frac{1}{2}, rendering as TeX

[WARNING] Could not convert TeX math \frac{1}{2} \cdot \frac{1}{26}, rendering as TeX

2025-07-19 20:24:30,992 - WARNING - Could not convert TeX math \frac{1}{2} \cdot \frac{1}{26}, rendering as TeX

[WARNING] Could not convert TeX math

$$\text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 0 \mid b = 0\rbrace = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} \approx 0.52., \text{rendering as TeX}$$

2025-07-19 20:24:30,992 - WARNING - Could not convert TeX math

$$\text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 0 \mid b = 0\rbrace = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} \approx 0.52., \text{rendering as TeX}$$

[WARNING] Could not convert TeX math \frac{1}{2} \cdot \frac{1}{26}, rendering as TeX

2025-07-19 20:24:30,993 - WARNING - Could not convert TeX math \frac{1}{2} \cdot \frac{1}{26}, rendering as TeX

[WARNING] Could not convert TeX math

$$\text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 1 \mid b = 1\rbrace = 1 - \text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 0 \mid b = 1\rbrace = 1 - \frac{1}{2} \cdot \frac{1}{26} \approx 0.98., \text{rendering as TeX}$$

2025-07-19 20:24:30,994 - WARNING - Could not convert TeX math

$$\text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 1 \mid b = 1\rbrace = 1 - \text{\Pr}\lbrack\mathcal{A}\backslash,\text{\outputs}\backslash, 0 \mid b = 1\rbrace = 1 - \frac{1}{2} \cdot \frac{1}{26} \approx 0.98., \text{rendering as TeX}$$

[WARNING] Could not convert TeX math \text{\Pr}\left\lbrack\right.

$${\mathtt{PrivK}}_{\mathcal{A},\text{\Pi}}^{\mathtt{eav}} = 1 \left.\right\rbrace \text{\right\lbrack} =$$

$$\frac{1}{2} \cdot \frac{1}{26} \left( \frac{1}{2} + \frac{1}{26} + 1 - \frac{1}{2} \cdot \frac{1}{26} \right) = 0.75 > \frac{1}{2},$$
 rendering as TeX

2025-07-19 20:24:30,995 - WARNING - Could not convert TeX math

$$\text{Pr} \left[ \text{PrivK}_{\mathcal{A}}(\Pi)^{\text{eav}} = 1 \right] = \frac{1}{2} \cdot \left( \frac{1}{2} + \frac{1}{26} + 1 - \frac{1}{2} \cdot \frac{1}{26} \right) = 0.75 > \frac{1}{2},$$
 rendering as TeX

[WARNING] Could not convert TeX math

$$\text{Pr} \left[ C = c \mid M = m \right] = \text{Pr} \left[ C = c \oplus m \mid M = m \right]$$

$$\{ \text{Pr} \left[ C = m \oplus c \mid M = m \right] = 2^{-\ell} \}$$
 rendering as TeX

2025-07-19 20:24:30,996 - WARNING - Could not convert TeX math

$$\text{Pr} \left[ C = c \mid M = m \right] = \text{Pr} \left[ C = c \oplus m \mid M = m \right]$$

$$\{ \text{Pr} \left[ C = m \oplus c \mid M = m \right] = 2^{-\ell} \}$$
 rendering as TeX

[WARNING] Could not convert TeX math

$$\text{Pr} \left[ C = c \right] = \sum_{M \in \mathcal{M}} \text{Pr} \left[ C = c \mid M = m \right] \cdot \text{Pr} \left[ M = m \right]$$

$$\{ 2^{-\ell} \cdot \sum_{M \in \mathcal{M}} \text{Pr} \left[ M = m \right] = 2^{-\ell} \}$$
 rendering as TeX

2025-07-19 20:24:30,997 - WARNING - Could not convert TeX math

$$\text{Pr} \left[ C = c \right] = \sum_{M \in \mathcal{M}} \text{Pr} \left[ C = c \mid M = m \right] \cdot \text{Pr} \left[ M = m \right]$$

$$\{ 2^{-\ell} \cdot \sum_{M \in \mathcal{M}} \text{Pr} \left[ M = m \right] = 2^{-\ell} \}$$
 rendering as TeX

[WARNING] Could not convert TeX math

$$\text{Pr} \left[ M = m \mid C = c \right] = \frac{\text{Pr} \left[ C = c \mid M = m \right] \cdot \text{Pr} \left[ C = c \right]}{\text{Pr} \left[ C = c \right]}$$

$$\{ \text{Pr} \left[ M = m \mid C = c \right] = \frac{2^{-\ell} \cdot 2^{-\ell}}{2^{-\ell}} = 2^{-\ell} \}$$
 rendering as TeX





$\frac{1}{2} + \text{negl}(n)$ , rendering as TeX  
 2025-07-19 20:24:31,012 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \text{PrivK}_{\mathcal{A}}(\Pi)^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k(m) \right) = m^i \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX  
 2025-07-19 20:24:31,013 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k(m) \right) = m^i \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{array}{l} \text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k(m) \right) = m^i \right] \\ = \frac{1}{2} \cdot \text{Pr}_{m_0} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_0 \right) \right) = 0 \right] + \frac{1}{2} \cdot \text{Pr}_{m_1} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_1 \right) \right) = 1 \right] \end{array}$   
 $\end{array}$ , rendering as TeX  
 2025-07-19 20:24:31,014 - WARNING - Could not convert TeX math  $\begin{array}{l} \text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k(m) \right) = m^i \right] \\ = \frac{1}{2} \cdot \text{Pr}_{m_0} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_0 \right) \right) = 0 \right] + \frac{1}{2} \cdot \text{Pr}_{m_1} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_1 \right) \right) = 1 \right] \end{array}$   
 $\end{array}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{array}{l} \text{PrivK}_{\mathcal{A}}^{\prime}(\Pi)^{\text{av}}(n) = 1 \\ \text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_b \right) \right) = b \right] \\ = \frac{1}{2} \cdot \text{Pr}_{m_0} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_0 \right) \right) = 0 \right] + \frac{1}{2} \cdot \text{Pr}_{m_1} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k \left( m_1 \right) \right) = 1 \right] \end{array}$   
 $\end{array}$ , rendering as TeX  
 2025-07-19 20:24:31,015 - WARNING - Could not convert TeX math  $\begin{array}{l} \text{PrivK}_{\mathcal{A}}^{\prime}(\Pi)^{\text{av}}(n) = 1 \\ \text{Pr} \left[ \mathcal{A} \left( 1^n, \text{Enc}_k(m) \right) = m^i \right] \end{array}$



```

{\text{Pr}\left\lbrack
{\mathtt{PrivK}}_{\mathcal{A}}^{\prime},\text{\Pi}\}^{\text{av}}(n) = 1}
\right\rbrack} \\\
{\text{Pr}\left\lbrack \mathcal{A}\left(1^n,\mathtt{Enc}_k\left(
m_{\text{b}} \right)\right) \right) = b} \right\rbrack}
\end{array} \\\
\begin{array}{l}
{\text{Pr}\left\lbrack \mathcal{A}\left(1^n,\mathtt{Enc}_k\left(m_0 \right)\right) \right) = 0}
\right\rbrack + \frac{1}{2} \cdot \text{Pr}\left\lbrack \mathcal{A}\left(1^n,\mathtt{Enc}_k\left(m_1 \right)\right) \right) = 1}
\right\rbrack} \\\
{\text{Pr}\left\lbrack \mathcal{A}\left(1^n,\mathtt{Enc}_k(m) \right) \right) = m^i} \right\rbrack.}
\end{array}
\end{array}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack
{\mathtt{PrivK}}_{\mathcal{A}}^{\prime},\Pi\}^{\text{eav}}(n) = 1} \right\rbrack
\leq \frac{1}{2} + \text{negl}(n), \text{rendering as TeX}
2025-07-19 20:24:31,017 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}^{\prime},\Pi\}^{\text{eav}}(n)
= 1} \right\rbrack \leq \frac{1}{2} + \text{negl}(n), \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack \mathcal{A}\left(
1^n,\mathtt{Enc}_k(m) \right) \right) = m^i} \right\rbrack \leq \frac{1}{2} +
\mathtt{negl}(n)., \text{rendering as TeX}
2025-07-19 20:24:31,018 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack \mathcal{A}\left(1^n,\mathtt{Enc}_k(m) \right) \right) =
m^i} \right\rbrack \leq \frac{1}{2} + \mathtt{negl}(n)., \text{rendering as TeX}
[WARNING] Could not convert TeX math \frac{1}{2}, \text{rendering as TeX}
2025-07-19 20:24:31,018 - WARNING - Could not convert TeX math \frac{1}{2},
\text{rendering as TeX}
[WARNING] Could not convert TeX math \left| \frac{1}{2} - 1 \right|, \text{rendering
as TeX}
2025-07-19 20:24:31,019 - WARNING - Could not convert TeX math \left|
\frac{1}{2} - 1 \right|, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack
{\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{eav}}(n) = 1} \right\rbrack \leq
\frac{1}{2} + \mathtt{negl}(n)., \text{rendering as TeX}
2025-07-19 20:24:31,019 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{eav}}(n) =
1} \right\rbrack \leq \frac{1}{2} + \mathtt{negl}(n)., \text{rendering as TeX}
[WARNING] Could not convert TeX math \widetilde{\Pi} = \left(
\{\widetilde{\mathtt{Gen}}\},\widetilde{\mathtt{Enc}}\},\widetilde{\mathtt{Dec}}\}
\right), \text{rendering as TeX}
2025-07-19 20:24:31,020 - WARNING - Could not convert TeX math \widetilde{\Pi} =
\left(
\{\widetilde{\mathtt{Gen}}\},\widetilde{\mathtt{Enc}}\},\widetilde{\mathtt{Dec}}\}
\right), \text{rendering as TeX}

```

[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Gen}}\left(1^n\right)$ , rendering as TeX

2025-07-19 20:24:31,021 - WARNING - Could not convert TeX math  $\widetilde{\mathtt{Gen}}\left(1^n\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\widetilde{\Pi}}^{\text{eav}}(n) = 1\right] = \frac{1}{2}$ , rendering as TeX

2025-07-19 20:24:31,022 - WARNING - Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\widetilde{\Pi}}^{\text{eav}}(n) = 1\right] = \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathtt{PrivK}_{\mathcal{A},\overline{\Pi}}^{\text{eav}}(n)$ , rendering as TeX

2025-07-19 20:24:31,022 - WARNING - Could not convert TeX math  $\mathtt{PrivK}_{\mathcal{A},\overline{\Pi}}^{\text{eav}}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[w\leftarrow\{0,1\}^{\ell(n)}\right]D(w) = 1$ , rendering as TeX

2025-07-19 20:24:31,023 - WARNING - Could not convert TeX math  $\text{Pr}\left[w\leftarrow\{0,1\}^{\ell(n)}\right]D(w) = 1$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathtt{PrivK}_{\mathcal{A},\overline{\Pi}}^{\text{eav}}(n)$ , rendering as TeX

2025-07-19 20:24:31,024 - WARNING - Could not convert TeX math  $\mathtt{PrivK}_{\mathcal{A},\overline{\Pi}}^{\text{eav}}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left|\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1\right] - \frac{1}{2}\right| \leq \mathtt{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,025 - WARNING - Could not convert TeX math  $\left|\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1\right] - \frac{1}{2}\right| \leq \mathtt{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,027 - WARNING - Could not convert TeX math  $\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,028 - WARNING - Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overrightarrow{M}_0$ , rendering as TeX

as TeX

2025-07-19 20:24:31,029 - WARNING - Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_0$ , rendering as TeX

[WARNING] Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_1$ , rendering as TeX

2025-07-19 20:24:31,030 - WARNING - Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_1$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overset{\rightarrow}{C}$ , rendering as TeX

2025-07-19 20:24:31,030 - WARNING - Could not convert TeX math  $\overset{\rightarrow}{C}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,031 - WARNING - Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_0$ , rendering as TeX

2025-07-19 20:24:31,032 - WARNING - Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_0$ , rendering as TeX

[WARNING] Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_1$ , rendering as TeX

2025-07-19 20:24:31,033 - WARNING - Could not convert TeX math  $\{\overset{\rightarrow}{M}\}_1$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overset{\rightarrow}{C}$ , rendering as TeX

2025-07-19 20:24:31,034 - WARNING - Could not convert TeX math  $\overset{\rightarrow}{C}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{ }}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,035 - WARNING - Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{ }}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{LR} - \text{cpa}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,036 - WARNING - Could not convert TeX math  $\text{Pr}\left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{LR} - \text{cpa}}(n) = 1\right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $F_{\mathbf{k}}(x) \setminus, \underset{-}{\underset{-}{\text{def}}}, F(\mathbf{k}, x)$ , rendering as TeX

2025-07-19 20:24:31,037 - WARNING - Could not convert TeX math  $F_{\mathbf{k}}(x) \setminus, \underset{-}{\underset{-}{\text{def}}}, F(\mathbf{k}, x)$ , rendering as TeX

[WARNING] Could not convert TeX math  $G(s) \setminus, \underset{-}{\underset{-}{\text{def}}}, F_s(1) \left| F_s(2) \right.$

$\right|\ldots\left|F_s(\ell)\right|$ , rendering as TeX  
2025-07-19 20:24:31,037 - WARNING - Could not convert TeX math  
 $G(s)\underset{\text{def}}{F_s(1)}\left|F_s(2)\right|\ldots\left|F_s(\ell)\right|$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\Pi} = \left(\widetilde{\text{Gen}}, \widetilde{\text{Enc}}, \widetilde{\text{Dec}}\right)$ , rendering as TeX  
2025-07-19 20:24:31,039 - WARNING - Could not convert TeX math  $\widetilde{\Pi} = \left(\widetilde{\text{Gen}}, \widetilde{\text{Enc}}, \widetilde{\text{Dec}}\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\text{Gen}}\left(1^n\right)$ , rendering as TeX  
2025-07-19 20:24:31,040 - WARNING - Could not convert TeX math  $\widetilde{\text{Gen}}\left(1^n\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left|\text{Pr}\right|\left[\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n) = 1\right] - 1\left[\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n) = 1\right] \leq \text{negl}(n)$ , rendering as TeX  
2025-07-19 20:24:31,040 - WARNING - Could not convert TeX math  $\left|\text{Pr}\right|\left[\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n) = 1\right] - 1\left[\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n) = 1\right] \leq \text{negl}(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n)$ , rendering as TeX  
2025-07-19 20:24:31,041 - WARNING - Could not convert TeX math  $\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}_{f\leftarrow\text{Func}_n}\left[D^f(\cdot)\left(1^n\right) = 1\right] \left[\text{Priv}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1\right] \leq \frac{1}{2} + \frac{q(n)}{2^n}$ , rendering as TeX  
2025-07-19 20:24:31,042 - WARNING - Could not convert TeX math  $\text{Pr}_{f\leftarrow\text{Func}_n}\left[D^f(\cdot)\left(1^n\right) = 1\right] \left[\text{Priv}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1\right] \leq \frac{1}{2} + \frac{q(n)}{2^n}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}_{f\leftarrow\text{Func}_n}\left[D^f(\cdot)\left(1^n\right) = 1\right] \left[\text{Priv}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1\right] \leq \frac{1}{2} + \frac{q(n)}{2^n}$ , rendering as TeX  
2025-07-19 20:24:31,043 - WARNING - Could not convert TeX math  $\text{Priv}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n)$ , rendering as TeX

$\mathtt{PrivK}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\mathtt{PrivK}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}$ , rendering as TeX  
 2025-07-19 20:24:31,044 - WARNING - Could not convert TeX math  
 $\mathtt{PrivK}_{\mathcal{A}, \overline{\Pi}}^{\text{cpa}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{\text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1$   
 $\right\rbrack \setminus \setminus$   
 $\{\setminus, \setminus, \setminus, \setminus, \begin{array}{l}$   
 $\{= \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge$   
 $\mathtt{repeat}\} \right\rbrack + \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge$   
 $\overline{\mathtt{repeat}}\} \right\rbrack \setminus \setminus$   
 $\{\leq \text{Pr}\}_{\text{left}} \lbrack \mathtt{repeat}\} \right\rbrack + \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \mid$   
 $\overline{\mathtt{repeat}}\} \right\rbrack \leq \frac{q(n)}{2^n} +$   
 $\frac{1}{2}.$   
 $\end{array}$   
 $\end{array}$ , rendering as TeX  
 2025-07-19 20:24:31,045 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{\text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1$   
 $\right\rbrack \setminus \setminus$   
 $\{\setminus, \setminus, \setminus, \setminus, \begin{array}{l}$   
 $\{= \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge$   
 $\mathtt{repeat}\} \right\rbrack + \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge$   
 $\overline{\mathtt{repeat}}\} \right\rbrack \setminus \setminus$   
 $\{\leq \text{Pr}\}_{\text{left}} \lbrack \mathtt{repeat}\} \right\rbrack + \text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \mid$   
 $\overline{\mathtt{repeat}}\} \right\rbrack \leq \frac{q(n)}{2^n} +$   
 $\frac{1}{2}.$   
 $\end{array}$   
 $\end{array}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}\}_{\text{left}} \lbrack$   
 $\{\mathtt{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1\} \right\rbrack \leq$   
 $\frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$ , rendering as TeX  
 2025-07-19 20:24:31,045 - WARNING - Could not convert TeX math  
 $\text{Pr}\}_{\text{left}} \lbrack \{\mathtt{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1$   
 $\right\rbrack \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$ , rendering  
 as TeX  
 [WARNING] Could not convert TeX math  $\frac{q(n)}{2^n}$ , rendering as TeX  
 2025-07-19 20:24:31,046 - WARNING - Could not convert TeX math  
 $\frac{q(n)}{2^n}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}\}_{\text{left}} \lbrack$

$\{\mathtt{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1\} \right\rbrace \leq \frac{1}{2} \text{negl}'(n)$ , rendering as TeX  
2025-07-19 20:24:31,047 - WARNING - Could not convert TeX math  
 $\text{Pr}\left\{\mathtt{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1\right\} \leq \frac{1}{2} \text{negl}'(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\mathtt{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1\right\} \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$ , rendering as TeX  
2025-07-19 20:24:31,048 - WARNING - Could not convert TeX math  
 $\text{Pr}\left\{\mathtt{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1\right\} \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{q(n)}{2^n}$ , rendering as TeX  
2025-07-19 20:24:31,048 - WARNING - Could not convert TeX math  
 $\frac{q(n)}{2^n}$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $G^{\text{ell}}(s) \overset{\text{def}}{=} \mathtt{GetBits}_1\left(\mathtt{Init}(s), 1^{\text{ell}}\right)$ , rendering as TeX  
2025-07-19 20:24:31,049 - WARNING - Could not convert TeX math  
 $G^{\text{ell}}(s) \overset{\text{def}}{=} \mathtt{GetBits}_1\left(\mathtt{Init}(s), 1^{\text{ell}}\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_s^{\text{ell}}(\text{IV}) \overset{\text{def}}{=} \mathtt{GetBits}_1\left(\mathtt{Init}(s, \text{IV}), 1^{\text{ell}}\right)$ , rendering as TeX  
2025-07-19 20:24:31,050 - WARNING - Could not convert TeX math  
 $F_s^{\text{ell}}(\text{IV}) \overset{\text{def}}{=} \mathtt{GetBits}_1\left(\mathtt{Init}(s, \text{IV}), 1^{\text{ell}}\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_k(x) \setminus, \underset{\text{def}}{\setminus}, F\left(\{k, x\}\right)$ , rendering as TeX  
2025-07-19 20:24:31,051 - WARNING - Could not convert TeX math  
 $F_k(x) \setminus, \underset{\text{def}}{\setminus}, F\left(\{k, x\}\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\Pi} = \left(\widetilde{\mathtt{Gen}} \text{,} \widetilde{\mathtt{Enc}} \text{,} \widetilde{\mathtt{Dec}}\right)$ , rendering as TeX  
2025-07-19 20:24:31,052 - WARNING - Could not convert TeX math  $\widetilde{\Pi} = \left(\widetilde{\mathtt{Gen}} \text{,} \widetilde{\mathtt{Enc}} \text{,} \widetilde{\mathtt{Dec}}\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Gen}} \left(1^n\right)$ , rendering as TeX  
2025-07-19 20:24:31,053 - WARNING - Could not convert TeX math  
 $\widetilde{\mathtt{Gen}} \left(1^n\right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Enc}}$ , rendering as TeX  
2025-07-19 20:24:31,053 - WARNING - Could not convert TeX math  
 $\widetilde{\mathtt{Enc}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Gen}}$ , rendering as TeX  
2025-07-19 20:24:31,054 - WARNING - Could not convert TeX math

$\widetilde{\mathtt{Gen}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\widetilde{\mathtt{Enc}}$ , rendering as TeX  
 2025-07-19 20:24:31,055 - WARNING - Could not convert TeX math  
 $\widetilde{\mathtt{Enc}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left| \{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \text{\Pi}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil - \{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \widetilde{\text{\Pi}}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \mathtt{negl}(n).$ , rendering as TeX  
 2025-07-19 20:24:31,056 - WARNING - Could not convert TeX math  $\left| \{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \text{\Pi}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil - \{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \widetilde{\text{\Pi}}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \mathtt{negl}(n).$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \widetilde{\text{\Pi}}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}}.$ , rendering as TeX  
 2025-07-19 20:24:31,057 - WARNING - Could not convert TeX math  
 $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \widetilde{\text{\Pi}}\}^{\{\text{LR} - \text{cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}}.$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \text{\Pi}\}^{\{\text{LR} - \text{Cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}} + \mathtt{negl}(n).$ , rendering as TeX  
 2025-07-19 20:24:31,057 - WARNING - Could not convert TeX math  
 $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{PrivK}_{\mathcal{A}}, \text{\Pi}\}^{\{\text{LR} - \text{Cpa}\}(n)} = 1\}$   
 $\right\rceil \leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}} + \mathtt{negl}(n).$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX  
 2025-07-19 20:24:31,059 - WARNING - Could not convert TeX math  
 $\frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX  
 2025-07-19 20:24:31,059 - WARNING - Could not convert TeX math  
 $\frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\mathtt{PrivK}_{\mathcal{A}}, \overline{\Pi}^{\{\text{LR-cpa}\}}$ , rendering as TeX  
 2025-07-19 20:24:31,060 - WARNING - Could not convert TeX math  
 $\mathtt{PrivK}_{\mathcal{A}}, \overline{\Pi}^{\{\text{LR-cpa}\}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{repeat}$   
 $\right\rceil \leq \frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX  
 2025-07-19 20:24:31,061 - WARNING - Could not convert TeX math  
 $\{\text{Pr}\}\left\lceil \right\rceil \right\{\mathtt{repeat}$   
 $\right\rceil \leq \frac{q(n)^2}{2^{3n/4 + 1}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$

$$\{\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{pa}}(n) = 1\}$$

$\begin{array}{l}$

$$\{\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \text{repeat}\}$$

$\leq \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \mid \overline{\text{repeat}} \wedge \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \text{repeat}\}$

$\leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}},$

$\end{array}$ , rendering as TeX

2025-07-19 20:24:31,062 - WARNING - Could not convert TeX math  $\begin{array}{l}$

$$\{\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{pa}}(n) = 1\}$$

$\begin{array}{l}$

$$\{\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \text{repeat}\}$$

$\leq \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \mid \overline{\text{repeat}} \wedge \text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{cpa}}(n) = 1 \wedge \text{repeat}\}$

$\leq \frac{1}{2} + \frac{q(n)^2}{2^{3n/4 + 1}},$

$\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{ncpa}}(n) = 1$

$$\leq \frac{1}{2} + \text{negl}(n),,$$

2025-07-19 20:24:31,063 - WARNING - Could not convert TeX math  $\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{LR} - \text{ncpa}}(n) = 1$

$$\leq \frac{1}{2} + \text{negl}(n),,$$

rendering as TeX

[WARNING] Could not convert TeX math  $\left.$

$$G(s), \underset{\text{def}}{\text{middle}}, \text{middle}, s \text{right},,$$

rendering as TeX

2025-07-19 20:24:31,063 - WARNING - Could not convert TeX math  $\left.$

$$G(s), \underset{\text{def}}{\text{middle}}, \text{middle}, s \text{right},,$$

rendering as TeX

[WARNING] Could not convert TeX math  $G'(s), \underset{\text{def}}{\text{left}}(\overline{s} \text{right}),$

rendering as TeX



2025-07-19 20:24:31,064 - WARNING - Could not convert TeX math  
 $G'(\underset{\text{def}}{\underset{\text{def}}{\overline{s}}})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{s}$ , rendering as TeX  
2025-07-19 20:24:31,065 - WARNING - Could not convert TeX math  $\overline{s}$ ,  
rendering as TeX  
[WARNING] Could not convert TeX math  
 $G'(\underset{\text{def}}{\underset{\text{def}}{\overline{G(s)}}})$ , rendering  
as TeX  
2025-07-19 20:24:31,065 - WARNING - Could not convert TeX math  
 $G'(\underset{\text{def}}{\underset{\text{def}}{\overline{G(s)}}})$ , rendering  
as TeX  
[WARNING] Could not convert TeX math  
 $G'(\underset{\text{def}}{\underset{\text{def}}{0^{|s|} \text{ middle } s}})$ , rendering as TeX  
2025-07-19 20:24:31,066 - WARNING - Could not convert TeX math  
 $G'(\underset{\text{def}}{\underset{\text{def}}{0^{|s|} \text{ middle } s}})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left.$   
 $G'(\underset{\text{def}}{\underset{\text{def}}{G(s) \text{ middle } \text{middle } \left( s + 1 \right) \text{ right.}}})$ , rendering as TeX  
2025-07-19 20:24:31,067 - WARNING - Could not convert TeX math  $\left.$   
 $G'(\underset{\text{def}}{\underset{\text{def}}{G(s) \text{ middle } \text{middle } \left( s + 1 \right) \text{ right.}}})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( 0 \text{ middle } \text{middle } x \text{ right} \right) \text{ right.}}})$ , rendering as TeX  
2025-07-19 20:24:31,068 - WARNING - Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( 0 \text{ middle } \text{middle } x \text{ right} \right) \text{ right.}}})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( 1 \text{ middle } \text{middle } x \text{ right} \right) \text{ right.}}})$ , rendering as TeX  
2025-07-19 20:24:31,069 - WARNING - Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( 1 \text{ middle } \text{middle } x \text{ right} \right) \text{ right.}}})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( x \text{ middle } \text{middle } 0 \text{ right} \right) \text{ right.}}})$ , rendering as TeX  
2025-07-19 20:24:31,070 - WARNING - Could not convert TeX math  $\left.$   
 $F_k'(\underset{\text{def}}{\underset{\text{def}}{\text{middle } \text{middle } x \text{ right } \text{middle } \text{middle } F_k \left( x \text{ middle } \text{middle } 0 \text{ right} \right) \text{ right.}}})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left$ .  
 $F_k^{\prime}(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{\left(0\right)}}, F_k\left(0\right)\right)\,,$  rendering as TeX  
2025-07-19 20:24:31,071 - WARNING - Could not convert TeX math  $\left$ .  
 $F_k^{\prime}(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{\left(0\right)}}, F_k\left(0\right)\right)\,,$  rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_k(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{k\&x}},$  rendering as TeX  
2025-07-19 20:24:31,072 - WARNING - Could not convert TeX math  
 $F_k(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{k\&x}},$  rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_{A,b}(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{Ax+b}},$  rendering as TeX  
2025-07-19 20:24:31,073 - WARNING - Could not convert TeX math  
 $F_{A,b}(x)\,,\,\underset{\text{def}}{\underset{\text{def}}{Ax+b}},$  rendering as TeX  
[WARNING] Could not convert TeX math  
 $G(s)\,,\,\underset{\text{def}}{\underset{\text{def}}{F_s\left(\left\langle 1\right\rangle \right)\left\langle F_s\left(\left\langle 2\right\rangle \right)\right\rangle \dots\left\langle F_s\left(\left\langle \ell\right\rangle \right)\right\rangle \right)}\right),$  rendering as TeX  
2025-07-19 20:24:31,073 - WARNING - Could not convert TeX math  
 $G(s)\,,\,\underset{\text{def}}{\underset{\text{def}}{F_s\left(\left\langle 1\right\rangle \right)\left\langle F_s\left(\left\langle 2\right\rangle \right)\right\rangle \dots\left\langle F_s\left(\left\langle \ell\right\rangle \right)\right\rangle \right)}\right),$  rendering as TeX  
[WARNING] Could not convert TeX math  $F_k(x)\overset{\text{def}}{=}\left\{\begin{array}{l} F_k^{\prime}(x)\,,\,\text{if}\,x,\,\text{is even} \\ F_k^{\prime}(x+1)\text{if}\,x,\,\text{is odd} \end{array}\right\}$   
 $\left\{ F_k^{\prime}(x)\,,\,\text{if}\,x,\,\text{is even} \right\}$   
 $\left\{ F_k^{\prime}(x+1)\text{if}\,x,\,\text{is odd} \right\}$   
 $\end{array}\right\}$  rendering as TeX  
2025-07-19 20:24:31,075 - WARNING - Could not convert TeX math  
 $F_k(x)\overset{\text{def}}{=}\left\{\begin{array}{l} F_k^{\prime}(x)\,,\,\text{if}\,x,\,\text{is even} \\ F_k^{\prime}(x+1)\text{if}\,x,\,\text{is odd} \end{array}\right\}$   
 $\left\{ F_k^{\prime}(x)\,,\,\text{if}\,x,\,\text{is even} \right\}$   
 $\left\{ F_k^{\prime}(x+1)\text{if}\,x,\,\text{is odd} \right\}$   
 $\end{array}\right\}$  rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\Pi} = \left( \widetilde{\mathtt{Gen}}\text{,}\widetilde{\mathtt{Mac}},\widetilde{\mathtt{Vrfy}} \right),$   
rendering as TeX  
2025-07-19 20:24:31,077 - WARNING - Could not convert TeX math  $\widetilde{\Pi} = \left( \widetilde{\mathtt{Gen}}\text{,}\widetilde{\mathtt{Mac}},\widetilde{\mathtt{Vrfy}} \right),$  rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Gen}}\left(1^n\right),$  rendering as TeX  
2025-07-19 20:24:31,077 - WARNING - Could not convert TeX math  
 $\widetilde{\mathtt{Gen}}\left(1^n\right),$  rendering as TeX  
[WARNING] Could not convert TeX math  $\widetilde{\mathtt{Mac}},$  rendering as TeX  
2025-07-19 20:24:31,078 - WARNING - Could not convert TeX math

$\widetilde{\mathtt{Mac}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $t\,,\underset{-}{\underset{-}{\{?\}}}\backslash,$   
 $F_{\{k\}}(m)$ , rendering as TeX  
2025-07-19 20:24:31,079 - WARNING - Could not convert TeX math  
 $t\,,\underset{-}{\underset{-}{\{?\}}}\backslash,$   $F_{\{k\}}(m)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mid \text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1\} \right\rbrack - \text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n) = 1\} \right\rbrack$   
 $\mid \leq \mathtt{negl}(n).$ , rendering as TeX  
2025-07-19 20:24:31,079 - WARNING - Could not convert TeX math  $\mid$   
 $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1\}$   
 $\right\rbrack - \text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n) = 1\} \right\rbrack \mid \leq$   
 $\mathtt{negl}(n).$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathtt{Mac}$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\overline{\text{\texttt{Pi}}}\}}(n)$ , rendering as TeX  
2025-07-19 20:24:31,080 - WARNING - Could not convert TeX math  $\mathtt{Mac}$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\overline{\text{\texttt{Pi}}}\}}(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathtt{Mac}$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\overline{\text{\texttt{Pi}}}\}}(n) = 1$ , rendering as TeX  
2025-07-19 20:24:31,081 - WARNING - Could not convert TeX math  $\mathtt{Mac}$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\overline{\text{\texttt{Pi}}}\}}(n) = 1$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\texttt{Pr}}\left\lbrack D^{\{f(\cdot)\}}\left(1^n\right) = 1\} \right\rbrack = \text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n) = 1\} \right\rbrack,$ , rendering as  
TeX  
2025-07-19 20:24:31,082 - WARNING - Could not convert TeX math  
 $\text{\texttt{Pr}}\left\lbrack D^{\{f(\cdot)\}}\left(1^n\right) = 1\} \right\rbrack =$   
 $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n)$   
 $= 1\} \right\rbrack,$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n) = 1\} \right\rbrack \leq 2^{-n},$   
rendering as TeX  
2025-07-19 20:24:31,083 - WARNING - Could not convert TeX math  
 $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\widetilde{\text{\texttt{Pi}}}\}}(n)$   
 $= 1\} \right\rbrack \leq 2^{-n},$  rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1\}$   
 $\right\rbrack = \text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$   
 $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1 \wedge \mathtt{repeat}\} \right\rbrack \backslash$   
 $\begin{matrix}$   
 $\{+$   $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1$   
 $\wedge \overline{\mathtt{repeat}}\} \wedge \mathtt{NewBlock}\} \right\rbrack \backslash$   
 $\begin{matrix}$   
 $\{+$   $\text{\texttt{Pr}}\left\lbrack \{\mathtt{Mac}\right.$  $\text{\texttt{forge}}_{\{\mathcal{A},\text{\texttt{Pi}}\}}(n) = 1$   
 $\wedge \overline{\mathtt{repeat}}\} \wedge \overline{\mathtt{NewBlock}}\} \right\rbrack$   
 $\right\rbrack \backslash$   
 $\{\leq \text{\texttt{Pr}}\left\lbrack \mathtt{repeat}\right\rbrack\}$   $\backslash$

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\begin{array}{l}
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \mathtt{NewBlock}} \right\rbrack} \\
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \overline{\mathtt{repeat}} \land \overline{\mathtt{NewBlock}}} \\
\right\rbrack.} \\
\end{array} \\
\end{array} \\
\end{array} \\
\end{matrix}, \text{rendering as TeX}
2025-07-19 20:24:31,083 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\right\rbrack = \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \land \mathtt{repeat} \right\rbrack} \\
\begin{array}{l}
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \overline{\mathtt{repeat}} \land \mathtt{NewBlock}} \right\rbrack} \\
\begin{array}{l}
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \overline{\mathtt{repeat}} \land \overline{\mathtt{NewBlock}}} \\
\right\rbrack} \\
\leq \text{Pr}\left\lbrack \mathtt{repeat} \right\rbrack} \\
\begin{array}{l}
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \mathtt{NewBlock}} \right\rbrack} \\
{+ \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \right\rbrack} \\
\land \overline{\mathtt{repeat}} \land \overline{\mathtt{NewBlock}}} \\
\right\rbrack.} \\
\end{array} \\
\end{array} \\
\end{array} \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \land \overline{\mathtt{repeat}} \land \overline{\mathtt{NewBlock}} \right\rbrack = 0., \text{rendering as TeX}
2025-07-19 20:24:31,084 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\text{\Pi}\}(n) = 1 \land \overline{\mathtt{repeat}} \land \overline{\mathtt{NewBlock}} \right\rbrack = 0., \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\Pi\}(n) = 1\hat{\mathtt{NewBlock}} \right\rbrack, \text{rendering as TeX}
2025-07-19 20:24:31,085 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A}},\Pi\}(n) = 1\hat{\mathtt{NewBlock}} \right\rbrack, \text{rendering as TeX}
[WARNING] Could not convert TeX math
t\,,\underset{-}{\underset{-}{\{?\}}}\,,\mathtt{Mac}_{\mathcal{k}}(m), \text{rendering as TeX}
2025-07-19 20:24:31,086 - WARNING - Could not convert TeX math

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t\,\underset{-}{\texttt{Mac}}\_{k}(m), rendering as TeX  
[WARNING] Could not convert TeX math \texttt{CBC}\_{k}\left(\right.\overset{\text{def}}{=}F\_k)\left(F\_k\left(\cdots F\_k\left(F\_k\left(x\_1\right) \oplus x\_2\right) \oplus \cdots\right) \oplus x\_\ell\right),, rendering as TeX  
2025-07-19 20:24:31,086 - WARNING - Could not convert TeX math \texttt{CBC}\_{k}\left(\right.\overset{\text{def}}{=}F\_k)\left(F\_k\left(\cdots F\_k\left(F\_k\left(x\_1\right) \oplus x\_2\right) \oplus \cdots\right) \oplus x\_\ell\right),, rendering as TeX  
[WARNING] Could not convert TeX math \mathtt{CBC}\_g\left(\right.\overset{\text{def}}{=}g\left(g\left(\cdots g\left(g\left(x\_1\right) \oplus x\_2\right) \oplus \cdots\right) \oplus x\_\ell\right), rendering as TeX  
2025-07-19 20:24:31,087 - WARNING - Could not convert TeX math \mathtt{CBC}\_g\left(\right.\overset{\text{def}}{=}g\left(g\left(\cdots g\left(g\left(x\_1\right) \oplus x\_2\right) \oplus \cdots\right) \oplus x\_\ell\right), rendering as TeX  
[WARNING] Could not convert TeX math \left| \text{Pr} \left\lceil D^{\mathtt{CBC}\_g}(\cdot) \left(1^n\right)=1\right\rceil-\text{Pr} \left\lceil D^{f(\cdot)}\left(1^n\right)=1\right\rceil \leq \frac{q^{-2} \ell^{2}}{2^n},, rendering as TeX  
2025-07-19 20:24:31,089 - WARNING - Could not convert TeX math \left| \text{Pr} \left\lceil D^{\mathtt{CBC}\_g}(\cdot) \left(1^n\right)=1\right\rceil-\text{Pr} \left\lceil D^{f(\cdot)}\left(1^n\right)=1\right\rceil \leq \frac{q^{-2} \ell^{2}}{2^n},, rendering as TeX  
[WARNING] Could not convert TeX math \mathrm{C}\_g(X)\overset{\text{def}}{=}\left(x\_1,\mathtt{CBC}\_g\left(x\_1\right) \oplus x\_2,\ldots,\mathtt{CBC}\_g\left(x\_1,\ldots,x\_{m-1}\right) \oplus x\_m\right)., rendering as TeX  
2025-07-19 20:24:31,089 - WARNING - Could not convert TeX math \mathrm{C}\_g(X)\overset{\text{def}}{=}\left(x\_1,\mathtt{CBC}\_g\left(x\_1\right) \oplus x\_2,\ldots,\mathtt{CBC}\_g\left(x\_1,\ldots,x\_{m-1}\right) \oplus x\_m\right)., rendering as TeX  
[WARNING] Could not convert TeX math \text{Pr} \left\lceil \forall i:\mathtt{CBC}\_g\left(X\_i\right)=t\_i \mid \overline{\mathtt{Coll}}\right\rceil = 2^{-n}., rendering as TeX  
2025-07-19 20:24:31,090 - WARNING - Could not convert TeX math \text{Pr} \left\lceil \forall i:\mathtt{CBC}\_g\left(X\_i\right)=t\_i \mid \overline{\mathtt{Coll}}\right\rceil = 2^{-n}., rendering as TeX  
[WARNING] Could not convert TeX math \overline{\mathtt{Coll}}, rendering as TeX  
2025-07-19 20:24:31,091 - WARNING - Could not convert TeX math \overline{\mathtt{Coll}}, rendering as TeX  
[WARNING] Could not convert TeX math \text{Pr} \left\lceil \sum \limits\_{i,j:i < j} \text{Pr} \left\lceil \mathtt{Coll} \right\rceil\_{i,j} \right\rceil \leq \frac{q^{-2}}{2} \cdot \max \limits\_{i < j} \left\{ \text{Pr} \left\lceil \mathtt{Coll} \right\rceil\_{i,j} \right\}., rendering as TeX

2025-07-19 20:24:31,091 - WARNING - Could not convert TeX math  

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \leq \sum_{i,j} \frac{q^2}{\max_{i < j} \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\}} \right\}$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \right\} = \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \left\{ \text{V}_k \right\} \right\} \leq \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (1) \right\} + \sum_{k=2}^{\ell-t-2} \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\}$$
, rendering as TeX  
2025-07-19 20:24:31,092 - WARNING - Could not convert TeX math  $\begin{matrix}$   

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \right\} = \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \left\{ \text{V}_k \right\} \right\} \leq \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (1) \right\} + \sum_{k=2}^{\ell-t-2} \left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\}$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\} \right\} = k/2^n$ , rendering as TeX  
2025-07-19 20:24:31,093 - WARNING - Could not convert TeX math  

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\} \right\} = k/2^n$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (t) \mid \overline{\left\{ \text{Coll} \right\} (t-1)} \right\} \right\} \leq 2t/2^n$ , rendering as TeX  
2025-07-19 20:24:31,093 - WARNING - Could not convert TeX math  

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (t) \mid \overline{\left\{ \text{Coll} \right\} (t-1)} \right\} \right\} \leq 2t/2^n$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\} \right\} = \left\{ \left\{ k + 1 \right\} \right\} / 2^n$ , rendering as TeX  
2025-07-19 20:24:31,095 - WARNING - Could not convert TeX math  

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} (k) \mid \overline{\left\{ \text{Coll} \right\} (k-1)} \right\} \right\} = \left\{ \left\{ k + 1 \right\} \right\} / 2^n$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \right\} \leq 2^{-n} \cdot \left\{ \sum_{k=1}^{t-1} k + 2t + \sum_{k=t+1}^{\ell-t-2} (k+1) \right\} \right\}$$
  

$$= 2^{-n} \cdot \sum_{k=2}^{\ell-t-1} k < 2^{\ell-2} \cdot 2^{-n}$$
  

$$\end{matrix}$$
, rendering as TeX  
2025-07-19 20:24:31,095 - WARNING - Could not convert TeX math  $\begin{matrix}$   

$$\left\{ \left\{ \text{Pr} \right\} \left\{ \text{Coll} \right\} \right\} \right\} \leq 2^{-n} \cdot \left\{ \sum_{k=1}^{t-1} k + 2t + \sum_{k=t+1}^{\ell-t-2} (k+1) \right\} \right\}$$
  

$$= 2^{-n} \cdot \sum_{k=2}^{\ell-t-1} k < 2^{\ell-2} \cdot 2^{-n}$$
  

$$\end{matrix}$$

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\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i} \right\rbrack \geq \text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i \mid \overline{\text{Coll}}} \right\rbrack \cdot \text{Pr}\left\lbrack \overline{\text{Coll}} \right\rbrack} \\
{= 2^{-n} \cdot \text{Pr}\left\lbrack \overline{\text{Coll}} \right\rbrack, \geq (1 - \delta) \cdot 2^{-n}},
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,096 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i} \right\rbrack \geq \text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i \mid \overline{\text{Coll}}} \right\rbrack \cdot \text{Pr}\left\lbrack \overline{\text{Coll}} \right\rbrack} \\
{= 2^{-n} \cdot \text{Pr}\left\lbrack \overline{\text{Coll}} \right\rbrack, \geq (1 - \delta) \cdot 2^{-n}},
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \overline{X} = \left(\{X_1, \ldots, X_q\} \right), rendering as TeX
2025-07-19 20:24:31,097 - WARNING - Could not convert TeX math \overline{X} = \left(\{X_1, \ldots, X_q\} \right), rendering as TeX
[WARNING] Could not convert TeX math \overline{t} = \left(\{t_1, \ldots, t_q\} \right), rendering as TeX
2025-07-19 20:24:31,098 - WARNING - Could not convert TeX math \overline{t} = \left(\{t_1, \ldots, t_q\} \right), rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {D^{\text{CBC}_g}(\cdot) \left(1^n \right) = 1} \right\rbrack = \sum\limits_{\overline{X}} \text{Pr}\left\lbrack {\overline{X} \xrightarrow{\alpha(\overline{X}, \overline{t})} \cdot \text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i} \right\rbrack} \right\rbrack} \\
{\geq \sum\limits_{\overline{X}} \left(1 - \delta \right) \cdot \text{Pr}\left\lbrack {D^f(\cdot) \left(1^n \right) = 1} \right\rbrack}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,099 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {D^{\text{CBC}_g}(\cdot) \left(1^n \right) = 1} \right\rbrack = \sum\limits_{\overline{X}} \text{Pr}\left\lbrack {\overline{X} \xrightarrow{\alpha(\overline{X}, \overline{t})} \cdot \text{Pr}\left\lbrack {\forall i:\text{CBC}_g\left(X_i \right) = t_i} \right\rbrack} \right\rbrack} \\
{\geq \sum\limits_{\overline{X}} \left(1 - \delta \right) \cdot \text{Pr}\left\lbrack {D^f(\cdot) \left(1^n \right) = 1} \right\rbrack}
\end{matrix}, rendering as TeX

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free;}\vec{t}}\alpha(\overline{X},\overline{t}) \cdot (1 -
\delta) \cdot \text{Pr}\left\{\text{forall } i:f\left(X_{i} \right) = t_{i}\right\}
\right\}\right\} \\\
\quad = (1 - \delta) \cdot \text{Pr}\left\{D^{f(\cdot)}\left(1^{n} \right) = 1\right\} \right\}
\end{array}
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math s\,,\underset{\sim}{\underset{\sim}{\{?\}}},
h_{k_h}(m) + F_{k_F}(r), \text{rendering as TeX}
2025-07-19 20:24:31,099 - WARNING - Could not convert TeX math
s\,,\underset{\sim}{\underset{\sim}{\{?\}}}, h_{k_h}(m) + F_{k_F}(r), \text{rendering as TeX}
[WARNING] Could not convert TeX math \mid \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1\right\} \right\} -
\text{Pr}\left\{\text{Mathtt{Mac-sforge}}_{\text{mathcal{A}}},\text{\Pi}\right\}(n) = 1\right\}
\right\} \mid \leq \text{Mathtt{negl}}(n)., \text{rendering as TeX}
2025-07-19 20:24:31,100 - WARNING - Could not convert TeX math \mid
\text{Pr}\left\{\text{Mathtt{Mac-sforge}}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n)
= 1\right\} \right\} - \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\text{\Pi}\right\}(n) = 1\right\} \right\} \mid \leq
\text{Mathtt{negl}}(n)., \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\overline{\Pi}(n), \text{rendering as TeX}
2025-07-19 20:24:31,101 - WARNING - Could not convert TeX math \text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\overline{\Pi}(n), \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{matrix}
\left\{\text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1\right\} \right\} \leq
\text{Pr}\left\{\text{repeat}\right\} \right\} + \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1 \wedge \text{Mathtt{new-r}}\right\}
\right\} \\\
\quad + \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1 \wedge
\overline{\text{Mathtt{repeat}}} \wedge \overline{\text{Mathtt{new-r}}}\right\} \right\}
\end{matrix}, \text{rendering as TeX}
2025-07-19 20:24:31,102 - WARNING - Could not convert TeX math \begin{matrix}
\left\{\text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1\right\} \right\} \leq
\text{Pr}\left\{\text{repeat}\right\} \right\} + \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1 \wedge \text{Mathtt{new-r}}\right\}
\right\} \\\
\quad + \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\widetilde{\text{\Pi}}\right\}(n) = 1 \wedge
\overline{\text{Mathtt{repeat}}} \wedge \overline{\text{Mathtt{new-r}}}\right\} \right\}
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\{\text{Mathtt{Mac-}
sforge}_{\text{mathcal{A}}},\overline{\Pi}(n) = 1\right\} \hat{\text{Mathtt{new-r}}} \right\}
\leq \text{varepsilon}(n), \text{rendering as TeX}
2025-07-19 20:24:31,103 - WARNING - Could not convert TeX math

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$\text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \overline{\Pi}}(n) = 1 \wedge \text{new-r} \right\} \leq \varepsilon(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \overline{\text{new-r}} \right\} \leq \varepsilon(n)$ ., rendering as TeX  
 2025-07-19 20:24:31,103 - WARNING - Could not convert TeX math  
 $\text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \overline{\text{new-r}} \right\} \leq \varepsilon(n)$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Mac-sforge}_{\mathcal{A}, \overline{\Pi}}(n)$ , rendering as TeX  
 2025-07-19 20:24:31,104 - WARNING - Could not convert TeX math  $\text{Mac-sforge}_{\mathcal{A}, \overline{\Pi}}(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \overline{\text{new-r}} \right\} \leq \text{Pr}_{k \leftarrow \mathcal{K}_n} \left\{ h_{k_h}(m) - h_{k_h} \left( m_i \right) = \Delta \right\} \leq \varepsilon(n) \end{matrix}$   
 $\end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,105 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \wedge \overline{\text{repeat}} \wedge \overline{\text{new-r}} \right\} \leq \text{Pr}_{k \leftarrow \mathcal{K}_n} \left\{ h_{k_h}(m) - h_{k_h} \left( m_i \right) = \Delta \right\} \leq \varepsilon(n) \end{matrix}$   
 $\end{matrix}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \right\} \leq 2 \cdot \varepsilon(n) + \frac{q^2}{2^{n+1}}$ ., rendering as TeX  
 2025-07-19 20:24:31,106 - WARNING - Could not convert TeX math  
 $\text{Pr} \left\{ \text{Mac-sforge}_{\mathcal{A}, \widetilde{\Pi}}(n) = 1 \right\} \leq 2 \cdot \varepsilon(n) + \frac{q^2}{2^{n+1}}$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  $m(X) \overset{\text{def}}{=} m_1 \cdot X^{\ell'_{\text{prime}}} + m_2 \cdot X^{\ell'_{\text{prime}} - 1} + \dots + m_{\ell'_{\text{prime}}} \cdot X$ ., rendering as TeX  
 2025-07-19 20:24:31,107 - WARNING - Could not convert TeX math  
 $m(X) \overset{\text{def}}{=} m_1 \cdot X^{\ell'_{\text{prime}}} + m_2 \cdot X^{\ell'_{\text{prime}} - 1} + \dots + m_{\ell'_{\text{prime}}} \cdot X$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  $P(X) \overset{\text{def}}{=} m(X) - m^{\ell'_{\text{prime}}}(X) - \Delta$ ., rendering as TeX  
 2025-07-19 20:24:31,108 - WARNING - Could not convert TeX math  
 $P(X) \overset{\text{def}}{=} m(X) - m^{\ell'_{\text{prime}}}(X) - \Delta$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  $\left. \underset{\cdot}{\cdot} \right|_k \in$

$\mathbb{F}\}\{\text{Pr}\}\left\lbrack h_{\{k\}}(m) - h_{\{k\}}\left( m^{\{\prime\}} \right) = \right.\text{\Delta}\}\right\rbrack = \underset{\{k \in \mathbb{F}\}}{\text{Pr}}\left\lbrack P(k) = 0\right\rbrack \leq \ell / \middle| \mathbb{F} \middle| , \right.,$  rendering as TeX

2025-07-19 20:24:31,109 - WARNING - Could not convert TeX math \left.

$\underset{\{k \in \mathbb{F}\}}{\text{Pr}}\left\lbrack h_{\{k\}}(m) - h_{\{k\}}\left( m^{\{\prime\}} \right) = \right.\text{\Delta}\}\right\rbrack = \underset{\{k \in \mathbb{F}\}}{\text{Pr}}\left\lbrack P(k) = 0\right\rbrack \leq \ell / \middle| \mathbb{F} \middle| , \right.,$  rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left\lbrack h_{\{k\}}(m) = t \wedge h_{\{k\}}\left( m^{\{\prime\}} \right) = t^{\{\prime\}} \right\rbrack = \frac{1}{\left| \mathcal{T} \right|^2},,$  rendering as TeX

2025-07-19 20:24:31,110 - WARNING - Could not convert TeX math

$\text{Pr}\left\lbrack h_{\{k\}}(m) = t \wedge h_{\{k\}}\left( m^{\{\prime\}} \right) = t^{\{\prime\}} \right\rbrack = \frac{1}{\left| \mathcal{T} \right|^2},,$  rendering as TeX

[WARNING] Could not convert TeX math  $t\,,\underset{-}{\underset{-}{\{?\}}}\backslash, h_{\{k\}}(m),$  rendering as TeX

2025-07-19 20:24:31,110 - WARNING - Could not convert TeX math

$t\,,\underset{-}{\underset{-}{\{?\}}}\backslash, h_{\{k\}}(m),$  rendering as TeX

[WARNING] Could not convert TeX math \begin{matrix}

$\left\{ \text{Pr} \left\lbrack \mathtt{Mac-forge}_{\mathcal{A}}, \text{\Pi} \right\rbrack^{\{1 - \text{time}\}} = 1 \right\rbrack = \sum \limits_{t^{\{\prime\}} \in \mathcal{T}} \left\{ \text{Pr} \left\lbrack \mathtt{Mac-forge}_{\mathcal{A}}, \text{\Pi} \right\rbrack^{\{1 - \text{time}\}} = 1 \wedge h_{\{k\}}\left( m^{\{\prime\}} \right) = t^{\{\prime\}} \right\rbrack \right\}$

$$\begin{array}{l} \begin{array}{l} \{= \sum \limits_{\begin{array}{l} t^{\{\prime\}} \in \mathcal{T} \end{array}} \{ (m, t) \mathcal{A} (t^{\{\prime\}}) \} \end{array} \\ \text{Pr} \left\lbrack h_{\{k\}}(m) = t \wedge h_{\{k\}}\left( m^{\{\prime\}} \right) = t^{\{\prime\}} \right\rbrack \\ \{= \sum \limits_{\begin{array}{l} t^{\{\prime\}} \in \mathcal{T} \end{array}} \{ (m, t) \mathcal{A} (t^{\{\prime\}}) \} \} \\ \frac{1}{\left| \mathcal{T} \right|^2} = \frac{1}{\left| \mathcal{T} \right|} \end{array}$$

$$\end{array}$$

$$\end{matrix},$$
 rendering as TeX

2025-07-19 20:24:31,111 - WARNING - Could not convert TeX math \begin{matrix}

$\left\{ \text{Pr} \left\lbrack \mathtt{Mac-forge}_{\mathcal{A}}, \text{\Pi} \right\rbrack^{\{1 - \text{time}\}} = 1 \right\rbrack = \sum \limits_{t^{\{\prime\}} \in \mathcal{T}} \left\{ \text{Pr} \left\lbrack \mathtt{Mac-forge}_{\mathcal{A}}, \text{\Pi} \right\rbrack^{\{1 - \text{time}\}} = 1 \wedge h_{\{k\}}\left( m^{\{\prime\}} \right) = t^{\{\prime\}} \right\rbrack \right\}$

$$\begin{array}{l}$$

$$\{= \sum \limits_{\begin{array}{l}$$

```

{t^{\prime} \in \mathcal{T}} \\\
{(m,t) \in \mathcal{A} \{(t^{\prime})\}}\}
\end{array}}\text{\text{Pr}}\left\lbrack h_{\{k\}}(m) = t \ \text{\text{and}} \ h_{\{k\}}\left(m^{\prime} \right) = t^{\prime} \right\rbrack \\\
{= \sum\limits_{\begin{array}{l} {t^{\prime} \in \mathcal{T}} \\\
{(m,t) \in \mathcal{A} \{(t^{\prime})\}}\} \end{array}} \\\
\end{array}}\frac{1}{\left| \mathcal{T} \right|^2}} = \frac{1}{\left| \mathcal{T} \right|} \} \} \\\
\end{array}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math
{\mathbb{Z}}_p\,,\underset{-}{\underset{-}{\text{def}}}\,,\left\{ \{0,\ldots,p - 1\} \right\}, rendering as TeX
2025-07-19 20:24:31,111 - WARNING - Could not convert TeX math
{\mathbb{Z}}_p\,,\underset{-}{\underset{-}{\text{def}}}\,,\left\{ \{0,\ldots,p - 1\} \right\}, rendering as TeX
[WARNING] Could not convert TeX math h_{\{a,b\}}(m)\overset{\text{def}}{=}\lbrack a \cdot m + b\text{mod}p\rbrack,, rendering as TeX
2025-07-19 20:24:31,113 - WARNING - Could not convert TeX math
h_{\{a,b\}}(m)\overset{\text{def}}{=}\lbrack a \cdot m + b\text{mod}p\rbrack,, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
\text{\text{Pr}}\left\lbrack h_{\{k\}}(m) - h_{\{k\}}\left(m^{\prime} \right) = \text{\text{A}} \right\rbrack = \text{\text{Pr}}\left\lbrack k \cdot \left(m - m^{\prime} \right) = \text{\text{A}}\text{mod}p \right\rbrack \\\
{= \text{\text{Pr}}\left\lbrack k = \text{\text{A}} \cdot \left(m - m^{\prime} \right)^{-1}\text{mod}p \right\rbrack = 1/p,}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,113 - WARNING - Could not convert TeX math \begin{matrix}
\text{\text{Pr}}\left\lbrack h_{\{k\}}(m) - h_{\{k\}}\left(m^{\prime} \right) = \text{\text{A}} \right\rbrack = \text{\text{Pr}}\left\lbrack k \cdot \left(m - m^{\prime} \right) = \text{\text{A}}\text{mod}p \right\rbrack \\\
{= \text{\text{Pr}}\left\lbrack k = \text{\text{A}} \cdot \left(m - m^{\prime} \right)^{-1}\text{mod}p \right\rbrack = 1/p,}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math t\,,\underset{-}{\underset{-}{\text{def}}}\,, h_{\{k\}}(m) + r, rendering as TeX
2025-07-19 20:24:31,114 - WARNING - Could not convert TeX math
t\,,\underset{-}{\underset{-}{\text{def}}}\,, h_{\{k\}}(m) + r, rendering as TeX
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack \text{\text{Macforge}}_{\mathcal{A},\text{\text{Π}}}^{\{1 - \text{\text{time}}\}} = 1 \right\rbrack = \sum\limits_{\substack{t^{\prime} \in \mathcal{T} \\\
\mathcal{A} \{(t^{\prime})\}}} \text{\text{Pr}}\left\lbrack h_{\{k\}}(m) + r = t \ \text{\text{and}} \ h_{\{k\}}\left(m^{\prime} \right) + r = t^{\prime} \right\rbrack., rendering as TeX
2025-07-19 20:24:31,115 - WARNING - Could not convert TeX math

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$$\left\{ \text{Pr} \left\{ \text{Mac-forge} \right\}_{\mathcal{A}}, \text{Pr} \right\}^{1 - \text{time}} = 1 \right\} = \sum \limits_{\substack{t \in \mathcal{T} \\ t = \mathcal{A}(t^{\text{prime}})}} \left\{ \text{Pr} \left\{ h_k(m) + r = t \wedge h_k \left( m^{\text{prime}} \right) + r = t^{\text{prime}} \right\} \right\} \right\}$$
, rendering as TeX

[WARNING] Could not convert TeX math  $h_k(m) - h_k \left( m^{\text{prime}} \right) = t - t^{\text{prime}}, \underset{\text{def}}{\Delta}$ , rendering as TeX

2025-07-19 20:24:31,116 - WARNING - Could not convert TeX math  $h_k(m) - h_k \left( m^{\text{prime}} \right) = t - t^{\text{prime}}, \underset{\text{def}}{\Delta}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \left\{ \text{Pr} \left\{ h_k(m) + r = t \wedge h_k \left( m^{\text{prime}} \right) + r = t^{\text{prime}} \right\} \right\} \\ = \left\{ \text{Pr} \left\{ h_k(m) + r = t \wedge h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \\ \begin{array}{l} = \left\{ \text{Pr} \left\{ r = t - h_k(m) \mid h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \cdot \left\{ \text{Pr} \left\{ h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \\ = \left( \frac{1}{\left| \mathcal{T} \right|} \right) \cdot \text{varepsilon}, \end{array} \end{array}$ , rendering as TeX

2025-07-19 20:24:31,116 - WARNING - Could not convert TeX math  $\begin{array}{l} \left\{ \text{Pr} \left\{ h_k(m) + r = t \wedge h_k \left( m^{\text{prime}} \right) + r = t^{\text{prime}} \right\} \right\} \\ = \left\{ \text{Pr} \left\{ h_k(m) + r = t \wedge h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \\ \begin{array}{l} = \left\{ \text{Pr} \left\{ r = t - h_k(m) \mid h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \cdot \left\{ \text{Pr} \left\{ h_k(m) - h_k \left( m^{\text{prime}} \right) = \Delta \right\} \right\} \\ = \left( \frac{1}{\left| \mathcal{T} \right|} \right) \cdot \text{varepsilon}, \end{array} \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{K} \left( t_0 \right) \overset{\text{def}}{=} \left\{ k \mid \text{Vrfy}_k \left( m_0, t_0 \right) = 1 \right\}$ , rendering as TeX

2025-07-19 20:24:31,117 - WARNING - Could not convert TeX math  $\mathcal{K} \left( t_0 \right) \overset{\text{def}}{=} \left\{ k \mid \text{Vrfy}_k \left( m_0, t_0 \right) = 1 \right\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} \sum \limits_{t_0} \left\{ \text{Pr} \left\{ \text{Mac} \right\}_k \left( m_0 \right) \right\} \right\} \cdot \frac{1}{\left| \mathcal{K} \right|} \left( t_0 \right) \\ \geq \sum \limits_{t_0} \left\{ \text{Pr} \left\{ \text{Mac} \right\}_k \left( m_0 \right) \right\} \right\} \cdot \frac{1}{\left. \text{varepsilon} \cdot \text{middle} \right| \mathcal{K} \right\}} \\ = \frac{1}{\left. \text{varepsilon} \cdot \text{middle} \right| \mathcal{K} \right\}} \end{matrix}$ , rendering as TeX



$\cdots$  &  $\text{K}_{\ell}$   
 $\end{bmatrix}$ ., rendering as TeX  
2025-07-19 20:24:31,125 - WARNING - Could not convert TeX math  $\begin{bmatrix}$   
 $\text{K}_n$  &  $\text{K}_{n-1}$  &  $\text{K}_{n-2}$  &  $\cdots$  &  $\text{K}_1$  \\
 $\text{K}_{n+1}$  &  $\text{K}_n$  &  $\text{K}_{n-1}$  &  $\cdots$  &  $\text{K}_2$  \\
 $\text{K}_{n+2}$  &  $\text{K}_{n+1}$  &  $\text{K}_n$  &  $\cdots$  &  $\text{K}_3$  \\
 $\vdots$  &  $\vdots$  &  $\vdots$  &  $\vdots$  &  $\vdots$  \\
 $\text{K}_{n+\ell-1}$  &  $\text{K}_{n+\ell-2}$  &  $\text{K}_{n+\ell-3}$  &  $\cdots$  &  $\text{K}_{\ell}$   
 $\end{bmatrix}$ ., rendering as TeX  
[WARNING] Could not convert TeX math  
 $\underset{b}{\text{times}}$  $\underset{\text{b}}{\text{dots}}$ , rendering as TeX  
2025-07-19 20:24:31,126 - WARNING - Could not convert TeX math  
 $\underset{b}{\text{times}}$  $\underset{\text{b}}{\text{dots}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\Delta_i$   $\overset{\text{def}}{=}$   $0 \text{ times}$   
 $00 \text{ times}$   $\overset{b}{\text{times}}$   $\overset{\text{b}}{\text{dots}}$   
 $0 \text{ times}$   $\end{matrix}$  \\
 $\text{times}$   $00 \text{ times}$   
 $00 \text{ times}$   $\overset{b}{\text{times}}$   $\overset{\text{b}}{\text{dots}}$   $0 \text{ times}$   $b$ ;  
 $\end{matrix}$ ., rendering as TeX  
2025-07-19 20:24:31,128 - WARNING - Could not convert TeX math  $\begin{matrix}$   
 $\Delta_i$   $\overset{\text{def}}{=}$   $0 \text{ times}$   
 $00 \text{ times}$   $\overset{b}{\text{times}}$   $\overset{\text{b}}{\text{dots}}$   
 $0 \text{ times}$   $\end{matrix}$  \\
 $\text{times}$   $00 \text{ times}$   
 $00 \text{ times}$   $\overset{b}{\text{times}}$   $\overset{\text{b}}{\text{dots}}$   $0 \text{ times}$   $b$ ;  
 $\end{matrix}$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr} \left[ \text{PrivK}_{\mathcal{A}, \Pi}^{\text{cca}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ ., rendering as TeX  
2025-07-19 20:24:31,129 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \text{PrivK}_{\mathcal{A}, \Pi}^{\text{cca}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr} \left[ \text{PrivK}_{\mathcal{A}, \Pi}^{\text{ae}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ ., rendering as TeX  
2025-07-19 20:24:31,130 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \text{PrivK}_{\mathcal{A}, \Pi}^{\text{ae}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Vrfy}_{k_M}(\{c, t\})$   
 $\right)$ ,  $\underset{\text{b}}{\text{times}}$  $\underset{\text{b}}{\text{dots}}$ , 1, rendering as TeX  
2025-07-19 20:24:31,131 - WARNING - Could not convert TeX math  
 $\text{Vrfy}_{k_M}(\{c, t\})$ ,  $\underset{\text{b}}{\text{times}}$  $\underset{\text{b}}{\text{dots}}$ , 1, rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l}$

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{\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cCa}}}(n)
= 1\right\rbrack} \\\
{\,,\leq \text{\text{Pr}}\lbrack\mathtt{ValidQuery}\rbrack + \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cca}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack.}
\end{array}, rendering as TeX
2025-07-19 20:24:31,132 - WARNING - Could not convert TeX math \begin{array}{l}
{\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cCa}}}(n)
= 1\right\rbrack} \\\
{\,,\leq \text{\text{Pr}}\lbrack\mathtt{ValidQuery}\rbrack + \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cca}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack.}
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{CCa}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack \leq \frac{1}{2} +
\mathtt{negl}(n)., rendering as TeX
2025-07-19 20:24:31,132 - WARNING - Could not convert TeX math
\text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{CCa}}}(n) = 1
\land \overline{\mathtt{ValidQuery}}\right\rbrack \leq \frac{1}{2} +
\mathtt{negl}(n)., rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n) = 1}
\right\rbrack \geq \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack} \\\
{= \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cca}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack.}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,133 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n) = 1}
\right\rbrack \geq \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack} \\\
{= \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}},\text{\Pi}\}^{\text{\text{cca}}}(n) = 1\land
\overline{\mathtt{ValidQuery}}\right\rbrack.}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n) = 1\right\rbrack
\leq \frac{1}{2} + \text{\text{negl}}(n), rendering as TeX
2025-07-19 20:24:31,134 - WARNING - Could not convert TeX math
\text{\text{Pr}}\left\lbrack {\mathtt{PrivK}}_{\mathcal{A}}_{\text{E}},\text{\Pi}_{\text{E}}\}^{\text{\text{cpa}}}(n)
= 1\right\rbrack \leq \frac{1}{2} + \text{\text{negl}}(n), rendering as TeX
[WARNING] Could not convert TeX math

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$H^s(x)$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $H\left( \{s,x\} \right)$ ,  
 rendering as TeX  
 2025-07-19 20:24:31,135 - WARNING - Could not convert TeX math  
 $H^s(x)$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $H\left( \{s,x\} \right)$ ,  
 rendering as TeX  
 [WARNING] Could not convert TeX math  $\left.$   
 $I_i$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $z_{i-1} \mid \mid x_i \right.$ , rendering as TeX  
 2025-07-19 20:24:31,136 - WARNING - Could not convert TeX math  $\left.$   
 $I_i$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $z_{i-1} \mid \mid x_i \right.$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $I_B +$   
 $1$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $z_B$ , rendering as TeX  
 2025-07-19 20:24:31,137 - WARNING - Could not convert TeX math  $I_B +$   
 $1$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $z_B$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\mathtt{Vrfy}_k\left( H^s(m),t \right)$ ,  
 $\underset{-}{\underset{-}{\text{def}}}$ ,  $1$ , rendering as TeX  
 2025-07-19 20:24:31,138 - WARNING - Could not convert TeX math  
 $\mathtt{Vrfy}_k\left( H^s(m),t \right)$ ,  $\underset{-}{\underset{-}{\text{def}}}$ ,  $1$ ,  
 rendering as TeX  
 [WARNING] Could not convert TeX math  $H^s\left( \mathcal{Q} \right)$ ,  
 $\underset{-}{\underset{-}{\text{def}}}$ ,  $\left\{ H^s(m) \mid m \in \mathcal{Q} \right\}$ , rendering as TeX  
 2025-07-19 20:24:31,139 - WARNING - Could not convert TeX math  $H^s\left( \mathcal{Q} \right)$ ,  
 $\underset{-}{\underset{-}{\text{def}}}$ ,  $\left\{ H^s(m) \mid m \in \mathcal{Q} \right\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil$   $\backslash\backslash$   
 $= \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and } \mathtt{coll}$   
 $\right\rceil + \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and }$   
 $\overline{\mathtt{coll}}$   $\right\rceil$   $\backslash\backslash$   
 $\leq \text{Pr}\left\lceil \mathtt{coll} \right\rceil + \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and }$   
 $\overline{\mathtt{coll}}$   $\right\rceil$ .  
 $\end{array}$ , rendering as TeX  
 2025-07-19 20:24:31,140 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil$   $\backslash\backslash$   
 $= \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and } \mathtt{coll}$   
 $\right\rceil + \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and }$   
 $\overline{\mathtt{coll}}$   $\right\rceil$   $\backslash\backslash$   
 $\leq \text{Pr}\left\lceil \mathtt{coll} \right\rceil + \text{Pr}\left\lceil \mathtt{Mac-}$   
 $\text{forge}_{\mathcal{A}^{\prime}}(\Pi^{\prime})(n) = 1 \right\rceil \text{ and }$



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\overline{\mathtt{coll}}}} \right\rbrace.}
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack \{\mathtt{Mac-
forge}_{\mathcal{A},\text{\Pi}}\}(n) = 1\right\rbrace = \text{Pr}\left\lbrack
\{\mathtt{Mac-forge}_{\mathcal{A}^{\prime},\text{\Pi}^{\prime}}\}(n) = 1 \land
\overline{\mathtt{coll}}}} \right\rbrace,, rendering as TeX
2025-07-19 20:24:31,141 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack \{\mathtt{Mac-forge}_{\mathcal{A},\text{\Pi}}\}(n) = 1\right\rbrace = \text{Pr}\left\lbrack \{\mathtt{Mac-
forge}_{\mathcal{A}^{\prime},\text{\Pi}^{\prime}}\}(n) = 1 \land
\overline{\mathtt{coll}}}} \right\rbrace,, rendering as TeX
[WARNING] Could not convert TeX math \left. t, \underset{-}{\underset{-}{\{?\}}},
H^s((k \oplus \text{opad})) \middle| \middle| H^s((k \oplus \text{ipad}))
\middle| \middle| m) \right., rendering as TeX
2025-07-19 20:24:31,142 - WARNING - Could not convert TeX math \left.
t, \underset{-}{\underset{-}{\{?\}}}, H^s((k \oplus \text{opad})) \middle|
\middle| H^s((k \oplus \text{ipad})) \middle| \middle| m) \right., rendering
as TeX
[WARNING] Could not convert TeX math
k_{\text{out}} \backslash, \underset{-}{\underset{-}{\{\text{def}\}}}, h^s \left(IV \middle|
\middle| \left(\{k \oplus \text{opad}\} \right) \right) \right), rendering as TeX
2025-07-19 20:24:31,142 - WARNING - Could not convert TeX math
k_{\text{out}} \backslash, \underset{-}{\underset{-}{\{\text{def}\}}}, h^s \left(IV \middle|
\middle| \left(\{k \oplus \text{opad}\} \right) \right) \right), rendering as TeX
[WARNING] Could not convert TeX math \{\widetilde{\mathtt{Gen}}\}^s = \left(\{\widetilde{\mathtt{Mac}}\}^s, \{\widetilde{\mathtt{Vrfy}}\}^s \right), rendering as TeX
2025-07-19 20:24:31,143 - WARNING - Could not convert TeX math
\{\widetilde{\Pi}\}^s = \left(\{\widetilde{\mathtt{Gen}}\}^s, \{\widetilde{\mathtt{Mac}}\}^s, \{\widetilde{\mathtt{Vrfy}}\}^s \right), rendering as TeX
[WARNING] Could not convert TeX math
\{\widetilde{\mathtt{Mac}}\}_{k_{\text{out}}}^s \left(\{\hat{m}\}^* \right) =
h^s \left(k_{\text{out}} \middle| \middle| \{\hat{m}\}^* \right), rendering as TeX
2025-07-19 20:24:31,144 - WARNING - Could not convert TeX math
\{\widetilde{\mathtt{Mac}}\}_{k_{\text{out}}}^s \left(\{\hat{m}\}^* \right) =
h^s \left(k_{\text{out}} \middle| \middle| \{\hat{m}\}^* \right), rendering as TeX
[WARNING] Could not convert TeX math
k_{\text{in}} \backslash, \underset{-}{\underset{-}{\{\text{def}\}}}, h^s \left(IV \middle|
\middle| \left(\{k \oplus \text{ipad}\} \right) \right) \right), rendering as TeX
2025-07-19 20:24:31,144 - WARNING - Could not convert TeX math
k_{\text{in}} \backslash, \underset{-}{\underset{-}{\{\text{def}\}}}, h^s \left(IV \middle|
\middle| \left(\{k \oplus \text{ipad}\} \right) \right) \right), rendering as TeX
[WARNING] Could not convert TeX math G^s(k) \overset{\text{def}}{=} h^s(IV \parallel (k \oplus \text{ipad})) \parallel h^s(IV \parallel (k \oplus \text{opad})) = k_{\text{in}} \parallel k_{\text{out}}., rendering as TeX
2025-07-19 20:24:31,145 - WARNING - Could not convert TeX math
G^s(k) \overset{\text{def}}{=} h^s(IV \parallel (k \oplus \text{ipad}))
\parallel h^s(IV \parallel (k \oplus \text{opad})) = k_{\text{in}} \parallel

```

$k_{\text{out}}$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\Delta_{\underset{\sim}{\underset{\sim}{\text{def}}}}, J - I$ , rendering as TeX  
 2025-07-19 20:24:31,145 - WARNING - Could not convert TeX math  
 $\Delta_{\underset{\sim}{\underset{\sim}{\text{def}}}}, J - I$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $f(x)_{\underset{\sim}{\underset{\sim}{\text{def}}}}, H\left( \{g(x)\} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,146 - WARNING - Could not convert TeX math  
 $f(x)_{\underset{\sim}{\underset{\sim}{\text{def}}}}, H\left( \{g(x)\} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $x_i_{\underset{\sim}{\underset{\sim}{\text{def}}}}, H^{\{i\}}\left( x_0 \right)$ , rendering as TeX  
 2025-07-19 20:24:31,146 - WARNING - Could not convert TeX math  
 $x_i_{\underset{\sim}{\underset{\sim}{\text{def}}}}, H^{\{i\}}\left( x_0 \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,148 - WARNING - Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,148 - WARNING - Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left( \{x_i \cdot \sqrt{N}\}, x_{\{i+1\}} \cdot \sqrt{N} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,149 - WARNING - Could not convert TeX math  $\left( \{x_i \cdot \sqrt{N}\}, x_{\{i+1\}} \cdot \sqrt{N} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sqrt{N} - 1$ , rendering as TeX  
 2025-07-19 20:24:31,149 - WARNING - Could not convert TeX math  $\sqrt{N} - 1$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\mathcal{O}\left( \sqrt{N} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,151 - WARNING - Could not convert TeX math  $\mathcal{O}\left( \sqrt{N} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,152 - WARNING - Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $x = x_{\{i+1\}} \cdot \sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,153 - WARNING - Could not convert TeX math  $x = x_{\{i+1\}} \cdot \sqrt{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $x' = x_i \cdot \sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,153 - WARNING - Could not convert TeX math  $x' = x_i \cdot \sqrt{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
 2025-07-19 20:24:31,154 - WARNING - Could not convert TeX math  $\sqrt{N}$ , rendering as TeX



$0^{t-1} \frac{1}{2} = \frac{\text{st}}{2N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\frac{\text{st}}{2N} = \frac{1}{4t}$ , rendering as TeX  
 2025-07-19 20:24:31,159 - WARNING - Could not convert TeX math  $\frac{\text{st}}{2N} = \frac{1}{4t}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $1 - \text{\text{Pr}}\lbracket\text{\text{no table contains}}\rbracket, x\rbracket = 1 - \left(1 - \frac{1}{4t}\right)^{4t} \approx 1 - e^{-1} = 0.63$ , rendering as TeX  
 2025-07-19 20:24:31,160 - WARNING - Could not convert TeX math  $1 - \text{\text{Pr}}\lbracket\text{\text{no table contains}}\rbracket, x\rbracket = 1 - \left(1 - \frac{1}{4t}\right)^{4t} \approx 1 - e^{-1} = 0.63$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $H_i, \underset{\text{\text{def}}}{\text{\text{Pr}}\lbracket\text{\text{no table contains}}\rbracket, f_i \circ H$ , rendering as TeX  
 2025-07-19 20:24:31,160 - WARNING - Could not convert TeX math  
 $H_i, \underset{\text{\text{def}}}{\text{\text{Pr}}\lbracket\text{\text{no table contains}}\rbracket, f_i \circ H$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $H(k), \underset{\text{\text{def}}}{F_k(m)}$ , rendering as TeX  
 2025-07-19 20:24:31,161 - WARNING - Could not convert TeX math  
 $H(k), \underset{\text{\text{def}}}{F_k(m)}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $F_k(x) \overset{\text{\text{def}}}{=} H(k \parallel x)$ , rendering as TeX  
 2025-07-19 20:24:31,162 - WARNING - Could not convert TeX math  
 $F_k(x) \overset{\text{\text{def}}}{=} H(k \parallel x)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $h_{1 \ldots 8}^{\prime}$ , rendering as TeX  
 2025-07-19 20:24:31,163 - WARNING - Could not convert TeX math  $h_{1 \ldots 8}^{\prime}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $H \left( \text{pw} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,164 - WARNING - Could not convert TeX math  $H \left( \text{pw} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{\text{Com}} \left( \text{\text{params}}, m; r \right)$ , rendering as TeX  
 2025-07-19 20:24:31,165 - WARNING - Could not convert TeX math  
 $\text{\text{Com}} \left( \text{\text{params}}, m; r \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{\text{Pr}} \left( \text{\text{Hiding}}_{\text{\text{A}}}, \text{\text{Com}}(n) = 1 \right) \leq \frac{1}{2} + \text{\text{negl}}(n)$ , rendering as TeX  
 2025-07-19 20:24:31,166 - WARNING - Could not convert TeX math  
 $\text{\text{Pr}} \left( \text{\text{Hiding}}_{\text{\text{A}}}, \text{\text{Com}}(n) = 1 \right) \leq \frac{1}{2} + \text{\text{negl}}(n)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\hat{H}_s(x), \underset{\text{\text{def}}}{H^s \left( \hat{H}_s(x) \right)}$ , rendering as TeX  
 2025-07-19 20:24:31,167 - WARNING - Could not convert TeX math  
 $\hat{H}_s(x), \underset{\text{\text{def}}}{H^s \left( \hat{H}_s(x) \right)}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{s_i^{(t+1)} := s_{i+1}^{(t)}, \backslash, i = 0, \ldots, n-2\} \backslash \backslash$   
 $\{s_{n-1}^{(t+1)} := \underset{i=0}{\overset{n-1}{\oplus}} c_i s_i^{(t)}\}.$   
 $\end{array}$ , rendering as TeX

2025-07-19 20:24:31,168 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{s_i^{(t+1)} := s_{i+1}^{(t)}, \backslash, i = 0, \ldots, n-2\} \backslash \backslash$   
 $\{s_{n-1}^{(t+1)} := \underset{i=0}{\overset{n-1}{\oplus}} c_i s_i^{(t)}\}.$   
 $\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{y_i = s_i^{(0)} \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, i = 0, \ldots, n-1\} \backslash \backslash$   
 $\{y_i = \underset{j=0}{\overset{n-1}{\oplus}} c_j y_{i-n+j} \backslash, i > n-1\}.$   
 $\end{array}$ , rendering as TeX

2025-07-19 20:24:31,169 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{y_i = s_i^{(0)} \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, i = 0, \ldots, n-1\} \backslash \backslash$   
 $\{y_i = \underset{j=0}{\overset{n-1}{\oplus}} c_j y_{i-n+j} \backslash, i > n-1\}.$   
 $\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{(0,0,1,1)\} \backslash \backslash$   
 $\{(1,0,0,1)\} \backslash \backslash$   
 $\{(1,1,0,0)\} \backslash \backslash$   
 $\{(1,1,1,0)\} \backslash \backslash$   
 $\{(1,1,1,1)\}$   
 $\end{array}$ , rendering as TeX

2025-07-19 20:24:31,170 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{(0,0,1,1)\} \backslash \backslash$   
 $\{(1,0,0,1)\} \backslash \backslash$   
 $\{(1,1,0,0)\} \backslash \backslash$   
 $\{(1,1,1,0)\} \backslash \backslash$   
 $\{(1,1,1,1)\}$   
 $\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $y_n = c_{n-1} y_{n-1} \oplus \cdots \oplus c_0 y_0 \backslash \backslash$   
 $\vdots \backslash \backslash$   
 $y_{2n-1} = c_{n-1} y_{2n-2} \oplus \cdots \oplus c_0 y_{n-1}.$   
 $\end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,170 - WARNING - Could not convert TeX math  $\begin{matrix}$   
 $y_n = c_{n-1} y_{n-1} \oplus \cdots \oplus c_0 y_0 \backslash \backslash$   
 $\vdots \backslash \backslash$   
 $y_{2n-1} = c_{n-1} y_{2n-2} \oplus \cdots \oplus c_0 y_{n-1}.$   
 $\end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{s_i^{(t+1)} := s_{i+1}^{(t)}, \backslash, i = 0, \ldots, n-2\} \backslash \backslash$   
 $\{s_{n-1}^{(t+1)} := g\left( \{s_{n-1}^{(t)}, \ldots, s_0^{(t)}\} \right) \}$   
 $\end{array}$ , rendering as TeX

2025-07-19 20:24:31,171 - WARNING - Could not convert TeX math  $\begin{array}{l} \{s_i\}^{(t+1)} = s_{i+1}^{(t)}, \backslash, i = 0, \ldots, n-2 \backslash \backslash \\ \{s_{n-1}\}^{(t+1)} = g\left(\{s_{n-1}\}^{(t)}, \ldots, s_0^{(t)}\right) \backslash \right) \backslash \\ \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $X \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $S_0 \backslash \lbrack 1 \rbrack \neq 2$ , rendering as TeX

2025-07-19 20:24:31,172 - WARNING - Could not convert TeX math  $X \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $S_0 \backslash \lbrack 1 \rbrack \neq 2$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} \{\text{Pr}\left\lbrack S_0 \right\rbrack 2 \right\rbrack = 0 \text{and} S_0 \backslash \lbrack 1 \rbrack \neq 2 \right\rbrack + \frac{1}{256} \cdot \{\text{Pr}\left\lbrack S_0 \right\rbrack 2 \right\rbrack \neq 0 \right\rbrack = \frac{1}{256} + \frac{1}{256} \cdot \left(1 - \frac{1}{256}\right) \backslash \backslash \\ \{\approx \frac{2}{256}\}, \\ \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,172 - WARNING - Could not convert TeX math  $\begin{matrix} \{\text{Pr}\left\lbrack S_0 \right\rbrack 2 \right\rbrack = 0 \text{and} S_0 \backslash \lbrack 1 \rbrack \neq 2 \right\rbrack + \frac{1}{256} \cdot \{\text{Pr}\left\lbrack S_0 \right\rbrack 2 \right\rbrack \neq 0 \right\rbrack = \frac{1}{256} + \frac{1}{256} \cdot \left(1 - \frac{1}{256}\right) \backslash \backslash \\ \{\approx \frac{2}{256}\}, \\ \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $F_k(x) \overset{\text{def}}{=} P(\text{const} \parallel k \parallel x) \text{const} \parallel k \parallel x$ , rendering as TeX

2025-07-19 20:24:31,173 - WARNING - Could not convert TeX math  $F_k(x) \overset{\text{def}}{=} P(\text{const} \parallel k \parallel x) \text{const} \parallel k \parallel x$ , rendering as TeX

[WARNING] Could not convert TeX math  $F_k(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $F\left(\{k, x\}\right)$ , rendering as TeX

2025-07-19 20:24:31,173 - WARNING - Could not convert TeX math  $F_k(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $F\left(\{k, x\}\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $f_i(R) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $\{\hat{f}\}_i \left(\{k_i, R\}\right)$ , rendering as TeX

2025-07-19 20:24:31,174 - WARNING - Could not convert TeX math  $f_i(R) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}}$ ,  $\{\hat{f}\}_i \left(\{k_i, R\}\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left(\{L_2, R_2\}\right) \overset{\text{def}}{=} F_k \left(0^{\ell/2}, 0^{\ell/2}\right) \right) \text{and} \left(\{L_2'\}, R_2'\right) \overset{\text{def}}{=} F_k \left(1^{\ell/2}, 0^{\ell/2}\right)$ , rendering as TeX

2025-07-19 20:24:31,175 - WARNING - Could not convert TeX math  $\left(\{L_2, R_2\}\right) \overset{\text{def}}{=} F_k \left(0^{\ell/2}, 0^{\ell/2}\right)$

$\right)\text{and}\left( L_2^{\prime}, R_2^{\prime} \right)$   
 $\right)\overset{\text{def}}{=} F_k \left( 1^{\ell/2}, 0^{\ell/2} \right),,$   
 rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{L_1 = R_0\} \\\$   
 $\{R_1 = L_0 \oplus f_1 \left( R_0 \right)\} \\\$   
 $\{L_2 = R_1 = L_0 \oplus f_1 \left( R_0 \right)\} \\\$   
 $\{R_2 = L_1 \oplus f_2 \left( R_1 \right)\}.$   
 $\end{array},$  rendering as TeX  
 2025-07-19 20:24:31,176 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{L_1 = R_0\} \\\$   
 $\{R_1 = L_0 \oplus f_1 \left( R_0 \right)\} \\\$   
 $\{L_2 = R_1 = L_0 \oplus f_1 \left( R_0 \right)\} \\\$   
 $\{R_2 = L_1 \oplus f_2 \left( R_1 \right)\}.$   
 $\end{array},$  rendering as TeX  
 [WARNING] Could not convert TeX math  
 $F_{k_1, k_2}^{\prime}(x) \overset{\text{def}}{=} F_{k_2} \left( F_{k_1}(x) \right)$   
 $\right),,$  rendering as TeX  
 2025-07-19 20:24:31,177 - WARNING - Could not convert TeX math  
 $F_{k_1, k_2}^{\prime}(x) \overset{\text{def}}{=} F_{k_2} \left( F_{k_1}(x) \right)$   
 $\right),,$  rendering as TeX  
 [WARNING] Could not convert TeX math  
 $F_{k_1, k_2, k_3}^{\prime\prime}(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}},$   
 $F_{k_3} \left( F_{k_2}^{\sim-1} \left( F_{k_1}(x) \right) \right) \right),$   
 rendering as TeX  
 2025-07-19 20:24:31,178 - WARNING - Could not convert TeX math  
 $F_{k_1, k_2, k_3}^{\prime\prime}(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}},$   
 $F_{k_3} \left( F_{k_2}^{\sim-1} \left( F_{k_1}(x) \right) \right) \right),$   
 rendering as TeX  
 [WARNING] Could not convert TeX math  
 $F_{k_1, k_2}^{\prime\prime}(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}},$   
 $F_{k_1} \left( F_{k_2}^{\sim-1} \left( F_{k_2}^{\sim-1}(x) \right) \right) \right)$   
 $\right),$  rendering as TeX  
 2025-07-19 20:24:31,179 - WARNING - Could not convert TeX math  
 $F_{k_1, k_2}^{\prime\prime}(x) \backslash, \underset{\sim}{\underset{\sim}{\text{def}}},$   
 $F_{k_1} \left( F_{k_2}^{\sim-1} \left( F_{k_2}^{\sim-1}(x) \right) \right) \right)$   
 $\right),$  rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\{\overline{y}\}_1^i, \{\overline{y}\}_2^i,$  rendering as TeX  
 2025-07-19 20:24:31,179 - WARNING - Could not convert TeX math  
 $\{\overline{y}\}_1^i, \{\overline{y}\}_2^i,$  rendering as TeX  
 [WARNING] Could not convert TeX math  $\frac{1}{4} \cdot \frac{1}{4} \cdot$   
 $\frac{1}{4} = \frac{1}{64},$  rendering as TeX  
 2025-07-19 20:24:31,180 - WARNING - Could not convert TeX math  $\frac{1}{4} \cdot$   
 $\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{64},$  rendering as TeX  
 [WARNING] Could not convert TeX math  $y \backslash, \underset{\sim}{\underset{\sim}{\text{def}}},$   
 $F_k(x),$  rendering as TeX  
 2025-07-19 20:24:31,180 - WARNING - Could not convert TeX math

$y, \underset{-}{\underset{-}{\text{def}}}, F_k(x)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left| \text{Pr} \left[ \left\{ x_I \oplus y_0 \oplus k_{\text{K}} = 0 \right\} \right] - \frac{1}{2} \right| = \varepsilon$ , rendering as TeX  
 2025-07-19 20:24:31,181 - WARNING - Could not convert TeX math  $\left| \text{Pr} \left[ \left\{ x_I \oplus y_0 \oplus k_{\text{K}} = 0 \right\} \right] - \frac{1}{2} \right| = \varepsilon$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left. h: \left\{ 0, 1 \right\}^n \rightarrow \left\{ 0, 1 \right\}^{\ell} \text{by} \right|, h \left( \{k, x\} \right), \underset{-}{\underset{-}{\text{def}}}, F_k(x) \oplus x$ , rendering as TeX  
 2025-07-19 20:24:31,182 - WARNING - Could not convert TeX math  $\left. h: \left\{ 0, 1 \right\}^n \rightarrow \left\{ 0, 1 \right\}^{\ell} \text{by} \right|, h \left( \{k, x\} \right), \underset{-}{\underset{-}{\text{def}}}, F_k(x) \oplus x$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $h_i, \underset{-}{\underset{-}{\text{def}}}, h \left( \{k_i, x_i\} \right) = F \left( \{k_i, x_i\} \right) \oplus x_i$ , rendering as TeX  
 2025-07-19 20:24:31,183 - WARNING - Could not convert TeX math  $h_i, \underset{-}{\underset{-}{\text{def}}}, h \left( \{k_i, x_i\} \right) = F \left( \{k_i, x_i\} \right) \oplus x_i$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $h_i, \underset{-}{\underset{-}{\text{def}}}, h \left( \{k_i, x_i\} \right) = y_i \oplus F^{-1} \left( \{k_i, y_i\} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,184 - WARNING - Could not convert TeX math  $h_i, \underset{-}{\underset{-}{\text{def}}}, h \left( \{k_i, x_i\} \right) = y_i \oplus F^{-1} \left( \{k_i, y_i\} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left( \begin{array}{l} q \\ 2 \end{array} \right) < q^2/2$ , rendering as TeX  
 2025-07-19 20:24:31,185 - WARNING - Could not convert TeX math  $\left( \begin{array}{l} q \\ 2 \end{array} \right) < q^2/2$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $P(x), \underset{-}{\underset{-}{\text{def}}}, F \left( \{0^n, x\} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,185 - WARNING - Could not convert TeX math  $P(x), \underset{-}{\underset{-}{\text{def}}}, F \left( \{0^n, x\} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\frac{q^2}{2^v} + \frac{q \cdot \left( \{q + 1\} \right)}{2^c}$ , rendering as TeX  
 2025-07-19 20:24:31,186 - WARNING - Could not convert TeX math  $\frac{q^2}{2^v} + \frac{q \cdot \left( \{q + 1\} \right)}{2^c}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr} \left[ \mathbf{E1} \vee \right]$



$\mathbf{E2} \vee \mathbf{E3}$  \right\rbrack \leq \frac{q^2}{2^v} + \frac{q}{\cdot \left( q + 1 \right)}^{2^c}, rendering as TeX  
2025-07-19 20:24:31,187 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \mathbf{E1} \vee \mathbf{E2} \vee \mathbf{E3} \right] \leq \frac{q^2}{2^v} + \frac{q}{\cdot \left( q + 1 \right)}^{2^c}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr} \left[ \mathbf{E1} \right] = \text{Pr} \left[ \underset{i < j}{V} \text{Coll}_{i,j} \right] \leq \sum \limits_{i < j} \text{Pr} \left[ \text{Coll}_{i,j} \right] \leq \left( \begin{array}{l} q \\ 2 \end{array} \right) \cdot 2 \cdot 2^{-v} \leq q^2/2^v$ ., rendering as TeX  
2025-07-19 20:24:31,187 - WARNING - Could not convert TeX math  
 $\text{Pr} \left[ \mathbf{E1} \right] = \text{Pr} \left[ \underset{i < j}{V} \text{Coll}_{i,j} \right] \leq \sum \limits_{i < j} \text{Pr} \left[ \text{Coll}_{i,j} \right] \leq \left( \begin{array}{l} q \\ 2 \end{array} \right) \cdot 2 \cdot 2^{-v} \leq q^2/2^v$ ., rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_k(x) \cdot \underset{-}{\underset{-}{\text{def}}} \cdot P \left( \text{const} \left( k \right) \right)$ , rendering as TeX  
2025-07-19 20:24:31,189 - WARNING - Could not convert TeX math  
 $F_k(x) \cdot \underset{-}{\underset{-}{\text{def}}} \cdot P \left( \text{const} \left( k \right) \right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{\text{DES}_k(x)} = \overline{\overline{\text{DES}_k(x)}}$ , rendering as TeX  
2025-07-19 20:24:31,189 - WARNING - Could not convert TeX math  $\overline{\text{DES}_k(x)} = \overline{\overline{\text{DES}_k(x)}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{z}$ , rendering as TeX  
2025-07-19 20:24:31,190 - WARNING - Could not convert TeX math  $\overline{z}$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $F_{k_1, k_1'} \cdot \underset{-}{\underset{-}{\text{def}}} \cdot F_{k_1} \left( F_{k_2} \left( F_{k_1}(x) \right) \right)$ , rendering as TeX  
2025-07-19 20:24:31,191 - WARNING - Could not convert TeX math  
 $F_{k_1, k_1'} \cdot \underset{-}{\underset{-}{\text{def}}} \cdot F_{k_1} \left( F_{k_2} \left( F_{k_1}(x) \right) \right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left. h: \left\{ 0,1 \right\}^{112} \rightarrow \left\{ 0,1 \right\}^{64} \right|, \text{as} \left| \left\{ x_1, x_2 \right\} \right| \cdot \underset{-}{\underset{-}{\text{def}}} \cdot \text{DES}_{x_1} \left( \text{DES}_{x_2} \left( 0^{64} \right) \right) \right|$ ., rendering as TeX  
2025-07-19 20:24:31,192 - WARNING - Could not convert TeX math  $\left. h: \left\{ 0,1 \right\}^{112} \rightarrow \left\{ 0,1 \right\}^{64} \right|, \text{as} \left| \left\{ x_1, x_2 \right\} \right| \cdot \underset{-}{\underset{-}{\text{def}}} \cdot \text{DES}_{x_1} \left( \text{DES}_{x_2} \left( 0^{64} \right) \right) \right|$ ., rendering as TeX



G(s)\overset{\text{def}}{=}f(s) \parallel \text{hc}(s), rendering as TeX  
[WARNING] Could not convert TeX math  $g(x,r)\overset{\text{def}}{=}(f(x),r)$ , rendering as TeX

2025-07-19 20:24:31,204 - WARNING - Could not convert TeX math  $g(x,r)\overset{\text{def}}{=}(f(x),r)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{gl}(x,r)\overset{\text{def}}{=}\oplus_{i=1}^n x_i \cdot r_i$ , rendering as TeX

2025-07-19 20:24:31,205 - WARNING - Could not convert TeX math  $\text{gl}(x,r)\overset{\text{def}}{=}\oplus_{i=1}^n x_i \cdot r_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x_i = \mathcal{A}\left(f(\hat{x}), e^i\right) = \text{gl}\left(\hat{x}, e^i\right) = \underset{j=1}{\overset{n}{\oplus}} \hat{x}_j \cdot e_j^i = \hat{x}_i$ , rendering as TeX

2025-07-19 20:24:31,207 - WARNING - Could not convert TeX math  $x_i = \mathcal{A}\left(f(\hat{x}), e^i\right) = \text{gl}\left(\hat{x}, e^i\right) = \underset{j=1}{\overset{n}{\oplus}} \hat{x}_j \cdot e_j^i = \hat{x}_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}_{x,r}\leftarrow\{0,1\}^n\lbrack\mathcal{A}(f(x),r) = \text{gl}(x,r)\rbrack \geq \frac{3}{4} + \frac{1}{p(n)}$ , rendering as TeX

2025-07-19 20:24:31,208 - WARNING - Could not convert TeX math  $\text{Pr}_{x,r}\leftarrow\{0,1\}^n\lbrack\mathcal{A}(f(x),r) = \text{gl}(x,r)\rbrack \geq \frac{3}{4} + \frac{1}{p(n)}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}_x\leftarrow\{0,1\}^n\left\lbrack\mathcal{A}^{\prime}\left(1^n,f(x)\right)\text{in } f^{-1}(f(x))\right\rbrack \geq \frac{1}{4} \cdot p(n)$ , rendering as TeX

2025-07-19 20:24:31,209 - WARNING - Could not convert TeX math  $\text{Pr}_x\leftarrow\{0,1\}^n\left\lbrack\mathcal{A}^{\prime}\left(1^n,f(x)\right)\text{in } f^{-1}(f(x))\right\rbrack \geq \frac{1}{4} \cdot p(n)$ , rendering as TeX  
[WARNING] Could not convert TeX math 
$$\begin{array}{l} \text{gl}(x,r) \oplus \text{gl}\left(\{x,r \oplus e^i\}\right) \\ \backslash, \backslash, = \left(\underset{j=1}{\overset{n}{\oplus}} x_j \cdot r_j\right) \oplus \left(\underset{j=1}{\overset{n}{\oplus}} x_j \cdot \left(r_j \oplus e_j^i\right)\right) = x_i \cdot r_i \oplus \left(x_i \cdot \overline{r}_i\right) = x_i, \end{array}$$
, rendering as TeX

2025-07-19 20:24:31,210 - WARNING - Could not convert TeX math 
$$\begin{array}{l} \text{gl}(x,r) \oplus \text{gl}\left(\{x,r \oplus e^i\}\right) \\ \backslash, \backslash, = \left(\underset{j=1}{\overset{n}{\oplus}} x_j \cdot r_j\right) \oplus \left(\underset{j=1}{\overset{n}{\oplus}} x_j \cdot \left(r_j \oplus e_j^i\right)\right) = x_i \cdot r_i \oplus \left(x_i \cdot \overline{r}_i\right) = x_i, \end{array}$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{r}_i$ , rendering as TeX

2025-07-19 20:24:31,210 - WARNING - Could not convert TeX math  $\overline{r}_i$ , rendering as TeX



[illegible]

$\frac{\text{varepsilon}(n)}{2}$ ., rendering as TeX

2025-07-19 20:24:31,223 - WARNING - Could not convert TeX math  $\text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}(f(x), r) \neq g(x, r) \right] \leq \frac{1}{4} - \frac{\text{varepsilon}(n)}{2}$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{\{0,1\}^n} \left[ \mathcal{A} \left( \left( f(x), r \oplus e^i \right) \neq g \left( \left( f(x), r \oplus e^i \right) \right) \right) \right] \leq \frac{1}{4} - \frac{\text{varepsilon}(n)}{2}$ ., rendering as TeX

2025-07-19 20:24:31,224 - WARNING - Could not convert TeX math  $\text{Pr}_{\{0,1\}^n} \left[ \mathcal{A} \left( \left( f(x), r \oplus e^i \right) \neq g \left( \left( f(x), r \oplus e^i \right) \right) \right) \right] \leq \frac{1}{4} - \frac{\text{varepsilon}(n)}{2}$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\left( \frac{1}{4} - \frac{\text{varepsilon}(n)}{2} \right) + \left( \frac{1}{4} - \frac{\text{varepsilon}(n)}{2} \right) = \frac{1}{2} - \text{varepsilon}(n)$ ., rendering as TeX

2025-07-19 20:24:31,225 - WARNING - Could not convert TeX math  $\left( \frac{1}{4} - \frac{\text{varepsilon}(n)}{2} \right) + \left( \frac{1}{4} - \frac{\text{varepsilon}(n)}{2} \right) = \frac{1}{2} - \text{varepsilon}(n)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}(f(x), r) = g(x, r) \right] \geq \frac{3}{4} + \text{varepsilon}(n)$ ., rendering as TeX

2025-07-19 20:24:31,226 - WARNING - Could not convert TeX math  $\text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}(f(x), r) = g(x, r) \right] \geq \frac{3}{4} + \text{varepsilon}(n)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \right] \\ \geq \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \mid x \in S_n \right] \\ \geq \frac{1}{2} \cdot \frac{\text{varepsilon}(n)}{2} = \frac{1}{4p(n)}. \end{matrix}$

2025-07-19 20:24:31,227 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \right] \\ \geq \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \mid x \in S_n \right] \\ \geq \frac{1}{2} \cdot \frac{\text{varepsilon}(n)}{2} = \frac{1}{4p(n)}. \end{matrix}$

2025-07-19 20:24:31,227 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \right] \\ \geq \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \mid x \in S_n \right] \\ \geq \frac{1}{2} \cdot \frac{\text{varepsilon}(n)}{2} = \frac{1}{4p(n)}. \end{matrix}$

2025-07-19 20:24:31,227 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \right] \\ \geq \text{Pr}_{\{0,1\}^n} \left[ \mathcal{A}^{\prime} \left( f^{(-1)}(f(x)) \right) \mid x \in S_n \right] \\ \geq \frac{1}{2} \cdot \frac{\text{varepsilon}(n)}{2} = \frac{1}{4p(n)}. \end{matrix}$

[WARNING] Could not convert TeX math  $\frac{1}{2} + \varepsilon(n)$ , rendering as TeX

2025-07-19 20:24:31,229 - WARNING - Could not convert TeX math  $\frac{1}{2} + \varepsilon(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $1 - \frac{1}{2n}$ , rendering as TeX

2025-07-19 20:24:31,230 - WARNING - Could not convert TeX math  $1 - \frac{1}{2n}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\frac{1}{2n}$ , rendering as TeX

2025-07-19 20:24:31,231 - WARNING - Could not convert TeX math  $\frac{1}{2n}$ , rendering as TeX

[WARNING] Could not convert TeX math  $n \cdot \frac{1}{2n} = \frac{1}{2}$ , rendering as TeX

2025-07-19 20:24:31,232 - WARNING - Could not convert TeX math  $n \cdot \frac{1}{2n} = \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $1 - \frac{1}{2} = \frac{1}{2}$ , rendering as TeX

2025-07-19 20:24:31,233 - WARNING - Could not convert TeX math  $1 - \frac{1}{2} = \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \frac{1}{p(n)}\}$ , rendering as TeX

2025-07-19 20:24:31,234 - WARNING - Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \frac{1}{p(n)}\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_x \leftarrow \{0,1\}^n \mid \text{cal}_A^{\prime} \left( \{1\}^n, f(x) \right) \in f^{\{-1\}}(f(x)) \mid \geq \frac{1}{p^{\prime}(n)}\}$ , rendering as TeX

2025-07-19 20:24:31,234 - WARNING - Could not convert TeX math  $\text{Pr}_x \leftarrow \{0,1\}^n \mid \text{cal}_A^{\prime} \left( \{1\}^n, f(x) \right) \in f^{\{-1\}}(f(x)) \mid \geq \frac{1}{p^{\prime}(n)}\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \frac{1}{p(n)}\}$ , rendering as TeX

2025-07-19 20:24:31,235 - WARNING - Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \frac{1}{p(n)}\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \varepsilon(n)\}$ , rendering as TeX

2025-07-19 20:24:31,237 - WARNING - Could not convert TeX math  $\text{Pr}_{x,r} \leftarrow \{0,1\}^n \mid \text{cal}_A(f(x),r) = \text{gl}(x,r) \mid \geq \frac{1}{2} + \varepsilon(n)\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\frac{\varepsilon(n)}{2} \cdot 2^n$ , rendering as TeX

2025-07-19 20:24:31,238 - WARNING - Could not convert TeX math  $\frac{\varepsilon(n)}{2} \cdot 2^n$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{\text{Pr}}_r \leftarrow \{0,1\}^n \left[ \mathcal{A}(f(x),r) = g(x,r) \right] \geq \frac{1}{2} + \frac{\varepsilon(n)}{2}$ ., rendering as TeX

2025-07-19 20:24:31,239 - WARNING - Could not convert TeX math  $\text{\text{Pr}}_r \leftarrow \{0,1\}^n \left[ \mathcal{A}(f(x),r) = g(x,r) \right] \geq \frac{1}{2} + \frac{\varepsilon(n)}{2}$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\frac{1}{2} + \frac{\varepsilon(n)}{2}$ , rendering as TeX

2025-07-19 20:24:31,240 - WARNING - Could not convert TeX math  $\frac{1}{2} + \frac{\varepsilon(n)}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{\text{Pr}} \left[ x_i^I = \hat{x}_i \right] \geq \frac{1}{2} + \frac{\varepsilon(n)}{2}$ , rendering as TeX

2025-07-19 20:24:31,242 - WARNING - Could not convert TeX math  $\text{\text{Pr}} \left[ x_i^I = \hat{x}_i \right] \geq \frac{1}{2} + \frac{\varepsilon(n)}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} \text{\text{Pr}} \left[ x_i \neq \hat{x}_i \right] \right. \\ \left. \frac{1}{4} \cdot \left( \frac{\varepsilon(n)}{2} \right)^2 \cdot \left( 2^{\ell} - 1 \right) \right] \\ \leq \frac{1}{4} \cdot \left( \frac{\varepsilon(n)}{2} \right)^2 \cdot \left( 2n / \varepsilon(n)^2 \right) \\ = \frac{1}{2n} \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,243 - WARNING - Could not convert TeX math  $\begin{matrix} \text{\text{Pr}} \left[ x_i \neq \hat{x}_i \right] \right. \\ \left. \frac{1}{4} \cdot \left( \frac{\varepsilon(n)}{2} \right)^2 \cdot \left( 2^{\ell} - 1 \right) \right] \\ \leq \frac{1}{4} \cdot \left( \frac{\varepsilon(n)}{2} \right)^2 \cdot \left( 2n / \varepsilon(n)^2 \right) \\ = \frac{1}{2n} \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\frac{1}{2^{\ell}} \geq \frac{1}{2} \cdot \left( 2n / \varepsilon(n)^2 + 1 \right) > \frac{\varepsilon(n)^2}{5n}$ , rendering as TeX

2025-07-19 20:24:31,245 - WARNING - Could not convert TeX math  $\frac{1}{2^{\ell}} \geq \frac{1}{2} \cdot \left( 2n / \varepsilon(n)^2 + 1 \right) > \frac{\varepsilon(n)^2}{5n}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\underset{r \leftarrow \{0,1\}^n}{1} \left[ D(r) = 1 \right] - \underset{s \leftarrow \{0,1\}^n}{\text{\text{Pr}}} \left[ D(G(s)) = 1 \right] \leq \text{\text{negl}}(n)$ ., rendering as TeX

2025-07-19 20:24:31,246 - WARNING - Could not convert TeX math  $\underset{r \leftarrow \{0,1\}^{n+1}}{\text{\text{Pr}}} \left[ D(r) = 1 \right] - \underset{s \leftarrow \{0,1\}^n}{\text{\text{Pr}}} \left[ D(G(s)) = 1 \right] \leq$





$\text{hc}(s) = 1$  \rbrack, rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{1}{2} \cdot \left( \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] - \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 1 \right] \right) \leq \text{negl}(n)$ ., rendering as TeX  
2025-07-19 20:24:31,250 - WARNING - Could not convert TeX math  $\frac{1}{2} \cdot \left( \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] - \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 1 \right] \right) \leq \text{negl}(n)$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{r}'$ , rendering as TeX  
2025-07-19 20:24:31,251 - WARNING - Could not convert TeX math  $\overline{r}'$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l} \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \right] \\ = \frac{1}{2} \cdot \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \mid r' = \text{hc}(s) \right] \\ + \frac{1}{2} \cdot \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \mid r' \neq \text{hc}(s) \right] \\ = \frac{1}{2} \cdot \left( \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 0 \right] + \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] \right) \\ = \frac{1}{2} \cdot \left( \left( 1 - \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 1 \right] \right) + \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] \right) \\ = \frac{1}{2} + \frac{1}{2} \cdot \left( \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] - \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 1 \right] \right) \end{array}$ , rendering as TeX  
2025-07-19 20:24:31,252 - WARNING - Could not convert TeX math  $\begin{array}{l} \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \right] \\ = \frac{1}{2} \cdot \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \mid r' = \text{hc}(s) \right] \\ + \frac{1}{2} \cdot \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ \mathcal{A}(f(s)) = \text{hc}(s) \mid r' \neq \text{hc}(s) \right] \\ = \frac{1}{2} \cdot \left( \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \text{hc}(s)) = 0 \right] + \underset{s \leftarrow \{0,1\}^n}{\text{Pr}} \left[ D(f(s) \parallel \overline{\text{hc}(s)} = 1) \right] \right) \end{array}$

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\overline{\text{hc}}(s)) = 1\rbrace \right)} \\\
{= \frac{1}{2} \cdot \left({\left({1 - \underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel \text{hc}(s)) = 1\rbrace \right)}
+ \underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel
\overline{\text{hc}}(s)) = 1\rbrace \right)} \\\
{= \frac{1}{2} + \frac{1}{2} \cdot \left({\underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel \overline{\text{hc}}(s)) =
1\rbrace - \underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s)
\parallel \text{hc}(s)) = 1\rbrace \right).}
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \frac{1}{2} \cdot \left(
{\underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel
\overline{\text{hc}}(s)) = 1\rbrace - \underset{s}{\leftarrow}{\{
0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel \text{hc}(s)) = 1\rbrace \right)
\leq \text{negl}(n), rendering as TeX
2025-07-19 20:24:31,253 - WARNING - Could not convert TeX math \frac{1}{2} \cdot
\left({\underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel
\overline{\text{hc}}(s)) = 1\rbrace - \underset{s}{\leftarrow}{\{
0,1\}}^n}{\text{Pr}}\right)\lbrack D(f(s) \parallel \text{hc}(s)) = 1\rbrace \right)
\leq \text{negl}(n), rendering as TeX
[WARNING] Could not convert TeX math \left| {\underset{t_2}{\leftarrow}
H_n^0}{\text{Pr}}\right)\left\lbrack D\left(t_2 \right) = 1 \right\rbrace -
\underset{t_2}{\leftarrow} H_n^1}{\text{Pr}}\right)\left\lbrack D\left(t_2
\right) = 1 \right\rbrace \right| \leq \text{negl}^{\prime}(n)., rendering as TeX
2025-07-19 20:24:31,254 - WARNING - Could not convert TeX math \left|
{\underset{t_2}{\leftarrow} H_n^0}{\text{Pr}}\right)\left\lbrack D\left(t_2
\right) = 1 \right\rbrace - \underset{t_2}{\leftarrow}
H_n^1}{\text{Pr}}\right)\left\lbrack D\left(t_2 \right) = 1 \right\rbrace
\right| \leq \text{negl}^{\prime}(n)., rendering as TeX
[WARNING] Could not convert TeX math \underset{t_1}{\leftarrow}{\{ 0,1\}}^{n +
1}}{\text{Pr}}\right)\left\lbrack D^{\prime}\left(t_1 \right) = 1 \right\rbrace =
\underset{t_2}{\leftarrow} H_n^1}{\text{Pr}}\right)\left\lbrack D\left(t_2
\right) = 1 \right\rbrace., rendering as TeX
2025-07-19 20:24:31,254 - WARNING - Could not convert TeX math
\underset{t_1}{\leftarrow}{\{ 0,1\}}^{n + 1}}{\text{Pr}}\right)\left\lbrack
D^{\prime}\left(t_1 \right) = 1 \right\rbrace = \underset{t_2}{\leftarrow}
H_n^1}{\text{Pr}}\right)\left\lbrack D\left(t_2 \right) = 1 \right\rbrace.,
rendering as TeX
[WARNING] Could not convert TeX math \underset{s}{\leftarrow}{\{
0,1\}}^n}{\text{Pr}}\right)\left\lbrack D^{\prime}(G(s)) = 1 \right\rbrace =
\underset{t_2}{\leftarrow} H_n^0}{\text{Pr}}\right)\left\lbrack D\left(t_2
\right) = 1 \right\rbrace., rendering as TeX
2025-07-19 20:24:31,255 - WARNING - Could not convert TeX math
\underset{s}{\leftarrow}{\{ 0,1\}}^n}{\text{Pr}}\right)\left\lbrack D^{\prime}(G(s)) =
1 \right\rbrace = \underset{t_2}{\leftarrow} H_n^0}{\text{Pr}}\right)\left\lbrack
D\left(t_2 \right) = 1 \right\rbrace., rendering as TeX
[WARNING] Could not convert TeX math \left. \left| {\underset{s}{\leftarrow}{\{
0,1\}}^n}{\text{Pr}}\right)\left\lbrack D^{\prime}\left(G(s) \right) = 1 \right\rbrace}

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$\underset{t}{\leftarrow} H_n^{j \ast} \{\text{Pr}\} \lbrack D(t) = 1 \rbrack.$ ,  
 rendering as TeX  
 2025-07-19 20:24:31,264 - WARNING - Could not convert TeX math  
 $\underset{w}{\leftarrow} \{0,1\}^{n+1} \{\text{Pr}\} \lbrack D^{\prime}(w) = 1 \mid j = j^{\ast} \rbrack = \underset{t}{\leftarrow} H_n^{j \ast} \{\text{Pr}\} \lbrack D(t) = 1 \rbrack.$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\{\text{Pr}\}_w \leftarrow \{0,1\}^{n+1} \lbrack D^{\prime}(w) = 1 \rbrack$   
 $\rbrack = \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} =$   
 $1^{p(n)} \underset{w}{\leftarrow} \{0,1\}^{n+1} \{\text{Pr}\} \lbrack D^{\prime}(w) = 1 \mid j = j^{\ast} \rbrack \setminus$   
 $\{= \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} =$   
 $1^{p(n)} \text{Pr}_t \leftarrow H_n^{j^{\ast}} \rbrack D(t) = 1 \rbrack.$   
 $\end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,264 - WARNING - Could not convert TeX math  $\begin{matrix}$   
 $\{\text{Pr}\}_w \leftarrow \{0,1\}^{n+1} \lbrack D^{\prime}(w) = 1 \rbrack$   
 $\rbrack = \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} =$   
 $1^{p(n)} \underset{w}{\leftarrow} \{0,1\}^{n+1} \{\text{Pr}\} \lbrack D^{\prime}(w) = 1 \mid j = j^{\ast} \rbrack \setminus$   
 $\{= \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} =$   
 $1^{p(n)} \text{Pr}_t \leftarrow H_n^{j^{\ast}} \rbrack D(t) = 1 \rbrack.$   
 $\end{matrix}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $t \overset{\text{def}}{=} t_{p(n)}$ , rendering  
 as TeX  
 2025-07-19 20:24:31,266 - WARNING - Could not convert TeX math  
 $t \overset{\text{def}}{=} t_{p(n)}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\underset{s}{\leftarrow} \{0,1\}^n \{\text{Pr}\} \lbrack D^{\prime}(G(s)) = 1 \mid j = j^{\ast} \rbrack$   
 $\rbrack = \underset{t}{\leftarrow} H_n^{j+1} \{\text{Pr}\} \lbrack D(t) = 1 \rbrack.$ , rendering as TeX  
 2025-07-19 20:24:31,267 - WARNING - Could not convert TeX math  
 $\underset{s}{\leftarrow} \{0,1\}^n \{\text{Pr}\} \lbrack D^{\prime}(G(s)) = 1 \mid j = j^{\ast} \rbrack \rbrack = \underset{t}{\leftarrow} H_n^{j+1} \{\text{Pr}\} \lbrack D(t) = 1 \rbrack.$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\{\underset{s}{\leftarrow} \{0,1\}^n \{\text{Pr}\} \lbrack D^{\prime}(\hat{G}(s)) = 1 \rbrack = \frac{1}{p(n)} \cdot$   
 $\sum \limits_{j^{\ast}} = 1^{p(n)} \underset{s}{\leftarrow} \{0,1\}^n \{\text{Pr}\} \lbrack D^{\prime}(G(s)) = 1 \mid j = j^{\ast} \rbrack \setminus$   
 $\begin{array}{l}$   
 $\{= \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} = 1^{p(n)} \underset{t}{\leftarrow} H_n^{j^{\ast}} - 1 \{\text{Pr}\} \lbrack D(t) = 1 \rbrack \setminus$   
 $\{= \frac{1}{p(n)} \cdot \sum \limits_{j^{\ast}} = 0\}^{p(n)} -$   
 $1 \underset{t}{\leftarrow} H_n^{j^{\ast}} \{\text{Pr}\} \lbrack D(t) = 1 \rbrack.$   
 $\end{array}$   
 $\end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,269 - WARNING - Could not convert TeX math  $\begin{matrix}$

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{\underset{s\leftarrow\{\{ 0,1\}\}^n\}}{\text{Pr}}\left\lbrack
{D^{\prime}(\hat{G}(s)) = 1} \right\rbrack = \frac{1}{p(n)} \cdot
\sum\limits_{j^{\ast} = 1}^{p(n)}\underset{s\leftarrow\{\{
0,1\}\}^n\}}{\text{Pr}}\left\lbrack {D^{\prime}(G(s)) = 1} \mid j = j^{\ast}\right\rbrack \\\
\begin{array}{l}
{=} \frac{1}{p(n)} \cdot \sum\limits_{j^{\ast} = 1}^{p(n)}\underset{t\leftarrow
H_n^{j^{\ast}} - 1}{\text{Pr}}\left\lbrack D(t) = 1\right\rbrack \\\
{=} \frac{1}{p(n)} \cdot \sum\limits_{j^{\ast} = 0}^{p(n)} -
1\underset{t\leftarrow H_n^{j^{\ast}}}{\text{Pr}}\left\lbrack D(t) = 1\right\rbrack.}
\end{array}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
\left| \{\underset{s\leftarrow\{\{ 0,1\}\}^n\}}{\text{Pr}}\left\lbrack
{D^{\prime}(G(s)) = 1} \right\rbrack - \underset{w\leftarrow\{\{ 0,1\}\}^{n +
1}}{\text{Pr}}\left\lbrack {D^{\prime}(w) = 1} \right\rbrack} \right| \\\
\begin{array}{l}
{=} \left| \frac{1}{p(n)} \cdot \mid \sum\limits_{j^{\ast} = 0}^{p(n)} -
1\text{Pr}_{t\leftarrow H_n^{j^{\ast}}}\left\lbrack D(t) = 1\right\rbrack -
\sum\limits_{j^{\ast} = 1}^{p(n)}\text{Pr}_{t\leftarrow H_n^{\ast}}\left\lbrack
D(t) = 1\right\rbrack} \right| \\\
{=} \frac{1}{p(n)} \cdot \left| \text{Pr}_{t\leftarrow H_n^0}\left\lbrack D(t)
= 1\right\rbrack - \text{Pr}_{t\leftarrow H_n^{p(n)}}\left\lbrack D(t) = 1\right\rbrack}
\right|,}
\end{array}
\end{array}, rendering as TeX
2025-07-19 20:24:31,269 - WARNING - Could not convert TeX math \begin{array}{l}
\left| \{\underset{s\leftarrow\{\{ 0,1\}\}^n\}}{\text{Pr}}\left\lbrack
{D^{\prime}(G(s)) = 1} \right\rbrack - \underset{w\leftarrow\{\{ 0,1\}\}^{n +
1}}{\text{Pr}}\left\lbrack {D^{\prime}(w) = 1} \right\rbrack} \right| \\\
\begin{array}{l}
{=} \left| \frac{1}{p(n)} \cdot \mid \sum\limits_{j^{\ast} = 0}^{p(n)} -
1\text{Pr}_{t\leftarrow H_n^{j^{\ast}}}\left\lbrack D(t) = 1\right\rbrack -
\sum\limits_{j^{\ast} = 1}^{p(n)}\text{Pr}_{t\leftarrow H_n^{\ast}}\left\lbrack
D(t) = 1\right\rbrack} \right| \\\
{=} \frac{1}{p(n)} \cdot \left| \text{Pr}_{t\leftarrow H_n^0}\left\lbrack D(t)
= 1\right\rbrack - \text{Pr}_{t\leftarrow H_n^{p(n)}}\left\lbrack D(t) = 1\right\rbrack}
\right|,}
\end{array}
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
{F_k(00) = G_0\left({G_0(k)} \right)} \\\
{F_k(01) = G_1\left({G_0(k)} \right)} \\\
\end{array},\,,\begin{array}{l}
{F_k(10) = G_0\left({G_1(k)} \right)} \\\
{F_k(11) = G_1\left({G_1(k)} \right).}
\end{array}
\end{array}, rendering as TeX
2025-07-19 20:24:31,270 - WARNING - Could not convert TeX math \begin{array}{l}

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{F_{k}}(00) = G_{0}\left({G_{0}}(k)} \right)} \\\
{F_{k}}(01) = G_{1}\left({G_{0}}(k)} \right)} \\
\end{array}\,,\,,\begin{array}{l}
{F_{k}}(10) = G_{0}\left({G_{1}}(k)} \right)} \\\
{F_{k}}(11) = G_{1}\left({G_{1}}(k)} \right)} \end{array}.) \\
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
\left| {\underset{y}{\leftarrow G_n^{\{t(n)\}}}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack \\
- \underset{y}{\leftarrow G_n^{\{0\}}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack \right| \\\
{=} \left| {\text{Pr}}\right\{\left\lbrack A\left({r_1} \parallel \cdots \parallel r_{t(n)} \right) \right\rbrack} \right\} \right| \right\rbrack - \text{Pr}\left\{\left\lbrack A\left({G\left(s_1 \right) \parallel \cdots \parallel G\left(s_{t(n)} \right)} \right) \right\rbrack} \right\} \right| \right\rbrack \\
\end{array}, rendering as TeX
2025-07-19 20:24:31,271 - WARNING - Could not convert TeX math \begin{array}{l}
\left| {\underset{y}{\leftarrow G_n^{\{t(n)\}}}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack \\
- \underset{y}{\leftarrow G_n^{\{0\}}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack \right| \\\
{=} \left| {\text{Pr}}\right\{\left\lbrack A\left({r_1} \parallel \cdots \parallel r_{t(n)} \right) \right\rbrack} \right\} \right| \right\rbrack - \text{Pr}\left\{\left\lbrack A\left({G\left(s_1 \right) \parallel \cdots \parallel G\left(s_{t(n)} \right)} \right) \right\rbrack} \right\} \right| \right\rbrack \\
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \underset{r}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{2n\}} \right\}\text{Pr}\right\}\left\lbrack A^{\{\prime\}}(r) = 1 \right\rbrack = \\
\frac{1}{t(n)} \cdot \sum\limits_{j = 1}^{\{t(n)\}}\text{Pr}_{y\leftarrow} G_n^{\{j\}}\lbrack A(y) = 1\rbrack, rendering as TeX
2025-07-19 20:24:31,272 - WARNING - Could not convert TeX math \\
\underset{r}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{2n\}} \right\}\text{Pr}\right\}\left\lbrack A^{\{\prime\}}(r) = 1 \right\rbrack = \frac{1}{t(n)} \cdot \sum\limits_{j = 1}^{\{t(n)\}}\text{Pr}_{y\leftarrow} G_n^{\{j\}}\lbrack A(y) = 1\rbrack, rendering as TeX
[WARNING] Could not convert TeX math \underset{s}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{n\}} \right\}\text{Pr}\right\}\left\lbrack A^{\{\prime\}}(G(s)) = 1 \right\rbrack = \frac{1}{t(n)} \cdot \sum\limits_{j = 0}^{\{t(n) - 1\}}\text{Pr}_{y\leftarrow} G_n^{\{j\}}\lbrack A(y) = 1\rbrack., rendering as TeX
2025-07-19 20:24:31,273 - WARNING - Could not convert TeX math \\
\underset{s}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{n\}} \right\}\text{Pr}\right\}\left\lbrack A^{\{\prime\}}(G(s)) = 1 \right\rbrack = \frac{1}{t(n)} \cdot \sum\limits_{j = 0}^{\{t(n) - 1\}}\text{Pr}_{y\leftarrow} G_n^{\{j\}}\lbrack A(y) = 1\rbrack., rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
\left| {\underset{r}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{2n\}} \right\}\text{Pr}}\right\}\left\lbrack A^{\{\prime\}}(r) = 1 \right\rbrack - \underset{s}{\leftarrow}\left\{ \left\{ 0,1 \right\}^{\{n\}} \right\}\text{Pr}}\right\}\left\lbrack A^{\{\prime\}}(G(s)) = 1 \right\rbrack \right| \\
\\
{=} \frac{1}{t(n)} \cdot \left| {\underset{y}{\leftarrow} G_n^{\{t(n)\}}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack - \underset{y}{\leftarrow} G_n^{\{0\}}\right\{\text{Pr}}\}\lbrack A(y) = 1\rbrack \right| .} \\
\end{array}, rendering as TeX

```







$$\{\mathcal{A}^{\prime}\}_{\text{outputs}0 \mid b = 0} \mid \mathcal{A}^{\prime}\}_{\text{outputs}1 \mid b = 1}$$

$$\geq \frac{1}{2} \cdot \{\text{Pr}\}_{\left[\text{Invert}_{\mathcal{A},f}(n) = 1\right]} + \frac{1}{2} \cdot \left(\{1 - 2^{-n}\}\right)$$

$$\frac{1}{2} + \frac{1}{2} \cdot \left(\{\text{Pr}\}_{\left[\text{Invert}_{\mathcal{A},f}(n) = 1\right]} - 2^{-n}\right)$$

\end{array}, rendering as TeX  
 [WARNING] Could not convert TeX math  

$$\{\text{Pr}\}_{\left[\text{PrivK}_{\mathcal{A}^{\prime},\text{prod}}^{\text{eav}}(n) = 1\right]} \leq \frac{1}{2} + \text{negl}(n)$$
, rendering as TeX  
 2025-07-19 20:24:31,282 - WARNING - Could not convert TeX math  

$$\{\text{Pr}\}_{\left[\text{PrivK}_{\mathcal{A}^{\prime},\text{prod}}^{\text{eav}}(n) = 1\right]} \leq \frac{1}{2} + \text{negl}(n)$$
, rendering as TeX  
 [WARNING] Could not convert TeX math  $\left| \{\underset{s}{\leftarrow} X\}_{\left[\text{Pr}\right]} D(s) = 1\right| - \left| \{\underset{s}{\leftarrow} Y\}_{\left[\text{Pr}\right]} D(s) = 1\right|$ , rendering as TeX  
 2025-07-19 20:24:31,283 - WARNING - Could not convert TeX math  $\left| \{\underset{s}{\leftarrow} X\}_{\left[\text{Pr}\right]} D(s) = 1\right| - \left| \{\underset{s}{\leftarrow} Y\}_{\left[\text{Pr}\right]} D(s) = 1\right|$ , rendering as TeX  
 [WARNING] Could not convert TeX math  

$$\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Y}$$
, rendering as TeX  
 2025-07-19 20:24:31,284 - WARNING - Could not convert TeX math  

$$\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Y}$$
, rendering as TeX  
 [WARNING] Could not convert TeX math  $\left| \{\underset{x}{\leftarrow} X_n\}_{\left[\text{Pr}\right]} \left[ D\left( \{1^n, x\} \right) = 1 \right] - \left| \{\underset{y}{\leftarrow} Y_n\}_{\left[\text{Pr}\right]} \left[ D\left( \{1^n, y\} \right) = 1 \right] \right| \leq \text{negl}(n)$ ., rendering as TeX  
 2025-07-19 20:24:31,285 - WARNING - Could not convert TeX math  $\left| \{\underset{x}{\leftarrow} X_n\}_{\left[\text{Pr}\right]} \left[ D\left( \{1^n, x\} \right) = 1 \right] - \left| \{\underset{y}{\leftarrow} Y_n\}_{\left[\text{Pr}\right]} \left[ D\left( \{1^n, y\} \right) = 1 \right] \right| \leq \text{negl}(n)$ ., rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{\mathcal{X}} = \left\{ (X_n^{(1)}, \dots, X_n^{(t(n))}) \mid n \in \mathbb{N} \right\}$ , rendering as TeX  
 2025-07-19 20:24:31,285 - WARNING - Could not convert TeX math  

$$\overline{\mathcal{X}} = \left\{ (X_n^{(1)}, \dots, X_n^{(t(n))}) \mid n \in \mathbb{N} \right\}$$
, rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{\mathcal{Y}} = \left\{ (Y_n^{(1)}, \dots, Y_n^{(t(n))}) \mid n \in \mathbb{N} \right\}$ , rendering as TeX  
 2025-07-19 20:24:31,286 - WARNING - Could not convert TeX math  

$$\overline{\mathcal{Y}} = \left\{ (Y_n^{(1)}, \dots, Y_n^{(t(n))}) \mid n \in \mathbb{N} \right\}$$
, rendering as TeX  
 [WARNING] Could not convert TeX math  $\{\underset{t(n)}{\leftarrow}\}_{\left[\text{Pr}\right]}$

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\left(\G\left(U_{\{n\}} \right), \ldots, G\left(U_{\{n\}} \right) \right) \right) \}_{n \in \{\mathbb{N}\}}, rendering as TeX
2025-07-19 20:24:31,286 - WARNING - Could not convert TeX math
\underiset{t{(n)}}{\underiset{}{\left\{ \left(\G\left(U_{\{n\}} \right), \ldots, G\left(U_{\{n\}} \right) \right) \right\} \right)}_n \in \{\mathbb{N}\}} \}, rendering as TeX
[WARNING] Could not convert TeX math
g(x_{\{1\}}, x_{\{2\}}) \overset{\text{def}}{=} (f(x_{\{1\}}), x_{\{2\}}), rendering as TeX
2025-07-19 20:24:31,287 - WARNING - Could not convert TeX math
g(x_{\{1\}}, x_{\{2\}}) \overset{\text{def}}{=} (f(x_{\{1\}}), x_{\{2\}}), rendering as TeX
[WARNING] Could not convert TeX math f' \overset{\text{def}}{=} f(x)
\parallel 10^{p(|x|) - |f(x)|}, rendering as TeX
2025-07-19 20:24:31,287 - WARNING - Could not convert TeX math
f' \overset{\text{def}}{=} f(x) \parallel 10^{p(|x|) - |f(x)|}, rendering as TeX
[WARNING] Could not convert TeX math g(x) \overset{\text{def}}{=} f(f(x)), rendering as TeX
2025-07-19 20:24:31,288 - WARNING - Could not convert TeX math
g(x) \overset{\text{def}}{=} f(f(x)), rendering as TeX
[WARNING] Could not convert TeX math g'(x) \overset{\text{def}}{=} f(x)
\parallel f(f(x)), rendering as TeX
2025-07-19 20:24:31,290 - WARNING - Could not convert TeX math
g'(x) \overset{\text{def}}{=} f(x) \parallel f(f(x)), rendering as TeX
[WARNING] Could not convert TeX math \underset{I}{\leftarrow}
Gen((1^n)), x \leftarrow Samp(I)) \{\text{Pr}\} \left[\mathcal{A} \left(\{I, f_I(x)\} \right) = hc(I, x) \right] \leq \frac{1}{2} + \negl(n)., rendering as TeX
2025-07-19 20:24:31,291 - WARNING - Could not convert TeX math
\underiset{I}{\leftarrow Gen((1^n)), x \leftarrow Samp(I)) \{\text{Pr}\} \left[\mathcal{A} \left(\{I, f_I(x)\} \right) = hc(I, x) \right] \leq \frac{1}{2} + \negl(n)., rendering as TeX
[WARNING] Could not convert TeX math \underset{x}{\leftarrow} \{ 0, 1 \}^n \} \{\text{Pr}\} \left[A_i(f(x)) = x_i \right] \geq \frac{1}{2} + \frac{1}{2n}. , rendering as TeX
2025-07-19 20:24:31,291 - WARNING - Could not convert TeX math
\underiset{x}{\leftarrow \{ 0, 1 \}^n \} \{\text{Pr}\} \left[A_i(f(x)) = x_i \right] \geq \frac{1}{2} + \frac{1}{2n}. , rendering as TeX
[WARNING] Could not convert TeX math
F_k^{-\ast}(x) \overset{\text{def}}{=} \text{Feistel}_{F_k, F_k, F_k}(x). , rendering as TeX
2025-07-19 20:24:31,292 - WARNING - Could not convert TeX math
F_k^{-\ast}(x) \overset{\text{def}}{=} \text{Feistel}_{F_k, F_k, F_k}(x). , rendering as TeX
[WARNING] Could not convert TeX math
\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Y}, rendering as TeX
2025-07-19 20:24:31,293 - WARNING - Could not convert TeX math

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$\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Y}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\mathcal{Y} \overset{\text{c}}{\equiv} \mathcal{Z}$ , rendering as TeX  
 2025-07-19 20:24:31,293 - WARNING - Could not convert TeX math  
 $\mathcal{Y} \overset{\text{c}}{\equiv} \mathcal{Z}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Z}$ , rendering as TeX  
 2025-07-19 20:24:31,294 - WARNING - Could not convert TeX math  
 $\mathcal{X} \overset{\text{c}}{\equiv} \mathcal{Z}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $I \overset{\text{def}}{=} \left\{ \hat{X}a + \hat{Y}b \middle| \hat{X}, \hat{Y} \in \mathbb{Z} \right\}$ , rendering as TeX  
 2025-07-19 20:24:31,294 - WARNING - Could not convert TeX math  
 $I \overset{\text{def}}{=} \left\{ \hat{X}a + \hat{Y}b \middle| \hat{X}, \hat{Y} \in \mathbb{Z} \right\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} \\ \end{matrix}$   

$$\begin{matrix} 1093028 \cdot 190301 = \lbrack 1093028 \bmod 100 \rbrack \cdot \lbrack 190301 \bmod 100 \rbrack \bmod 100 \\ \{ = 28 \cdot 1 = 28 \bmod 100 \} \end{matrix}$$
, rendering as TeX  
 2025-07-19 20:24:31,295 - WARNING - Could not convert TeX math  $\begin{matrix} \\ \end{matrix}$   

$$\begin{matrix} 1093028 \cdot 190301 = \lbrack 1093028 \bmod 100 \rbrack \cdot \lbrack 190301 \bmod 100 \rbrack \bmod 100 \\ \{ = 28 \cdot 1 = 28 \bmod 100 \} \end{matrix}$$
, rendering as TeX  
 [WARNING] Could not convert TeX math  $\lbrack$   
 $a/b \bmod N \rbrack \overset{\text{def}}{=} \lbrack ab^{-1} \bmod N \rbrack$ , rendering as TeX  
 2025-07-19 20:24:31,295 - WARNING - Could not convert TeX math  $\lbrack$   
 $a/b \bmod N \rbrack \overset{\text{def}}{=} \lbrack ab^{-1} \bmod N \rbrack$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $a + b \overset{\text{def}}{=} \lbrack a + b \bmod N \rbrack$ , rendering as TeX  
 2025-07-19 20:24:31,297 - WARNING - Could not convert TeX math  $a + b \overset{\text{def}}{=} \lbrack a + b \bmod N \rbrack$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $mg = m \cdot g \overset{\text{def}}{=} \underset{m}{\text{times}} \{ \underset{g}{g} + \cdots + g \}$ , rendering as TeX  
 2025-07-19 20:24:31,298 - WARNING - Could not convert TeX math  $mg = m \cdot g \overset{\text{def}}{=} \underset{m}{\text{times}} \{ \underset{g}{g} + \cdots + g \}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $g^m \overset{\text{def}}{=} \underset{m}{\text{times}} \{ \underset{g}{g \cdots g} \}$ , rendering as TeX  
 2025-07-19 20:24:31,299 - WARNING - Could not convert TeX math  
 $g^m \overset{\text{def}}{=} \underset{m}{\text{times}} \{ \underset{g}{g \cdots g} \}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $0 \cdot g \overset{\text{def}}{=} 0$ , rendering as TeX

2025-07-19 20:24:31,300 - WARNING - Could not convert TeX math  $0 \cdot \overset{\text{def}}{=}{0}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $(-m) \cdot \overset{\text{def}}{=}{m} \cdot (-g)$ , rendering as TeX  
2025-07-19 20:24:31,300 - WARNING - Could not convert TeX math  $(-m) \cdot \overset{\text{def}}{=}{m} \cdot (-g)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g^0 \overset{\text{def}}{=}{1}$ , rendering as TeX  
2025-07-19 20:24:31,301 - WARNING - Could not convert TeX math  $g^0 \overset{\text{def}}{=}{1}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g^{-m} \overset{\text{def}}{=}{(g^{-1})^m}$ , rendering as TeX  
2025-07-19 20:24:31,303 - WARNING - Could not convert TeX math  $g^{-m} \overset{\text{def}}{=}{(g^{-1})^m}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\{\mathbb{Z}\}_N^{\ast} \overset{\text{def}}{=}{\left\{ b \in \left\{ 1, \dots, N-1 \right\} \mid \gcd(b, N) = 1 \right\};}$ , rendering as TeX  
2025-07-19 20:24:31,303 - WARNING - Could not convert TeX math  $\{\mathbb{Z}\}_N^{\ast} \overset{\text{def}}{=}{\left\{ b \in \left\{ 1, \dots, N-1 \right\} \mid \gcd(b, N) = 1 \right\};}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $a \overset{\text{def}}{=}$ , rendering as TeX  
2025-07-19 20:24:31,304 - WARNING - Could not convert TeX math  $a \overset{\text{def}}{=}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left. \phi(N) \overset{\text{def}}{=} \middle| \{\mathbb{Z}\}_N^{\ast} \right|$ , rendering as TeX  
2025-07-19 20:24:31,305 - WARNING - Could not convert TeX math  $\left. \phi(N) \overset{\text{def}}{=} \middle| \{\mathbb{Z}\}_N^{\ast} \right|$ , rendering as TeX  
[WARNING] Could not convert TeX math  $(g, h) \circ \left( g^{\prime}, h^{\prime} \right) \overset{\text{def}}{=} \left( g \circ_{\mathbb{G}} g^{\prime}, h \circ_{\mathbb{H}} h^{\prime} \right)$ , rendering as TeX  
2025-07-19 20:24:31,306 - WARNING - Could not convert TeX math  $(g, h) \circ \left( g^{\prime}, h^{\prime} \right) \overset{\text{def}}{=} \left( g \circ_{\mathbb{G}} g^{\prime}, h \circ_{\mathbb{H}} h^{\prime} \right)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $f(x) \overset{\text{def}}{=} (\lfloor x \bmod p \rfloor, \lfloor x \bmod q \rfloor)$ , rendering as TeX  
2025-07-19 20:24:31,307 - WARNING - Could not convert TeX math  $f(x) \overset{\text{def}}{=} (\lfloor x \bmod p \rfloor, \lfloor x \bmod q \rfloor)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{ll} \{f(\left\{ a +_N b \right\} \right) \& \{= \left( \left\{ \left\{ \left\{ a +_N b \right\} \right\} \bmod p \right\} \right\} \left\{ \left\{ \left\{ a +_N b \right\} \right\} \bmod q \right\} \right\} \\ & \& \begin{array}{l} = (\lfloor a + b \rfloor \bmod p, \lfloor a + b \rfloor \bmod q) \\ = (\lfloor a \rfloor \bmod p, \lfloor a \rfloor \bmod q) (\lfloor b \rfloor \bmod p, \lfloor b \rfloor \bmod q) = f(a) f(b) \end{array} \end{array}$

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\end{array}
\end{array}, rendering as TeX
2025-07-19 20:24:31,307 - WARNING - Could not convert TeX math \begin{array}{ll}
{f\left({a + _N b} \right)} & {= \left({\left\{ \left\{ \left({a + _N b} \right. \right. \right.} \right.} \right.} \\
\left. \right\} \bmod p} \right\} \bmod q} \\
\left. \right\} \bmod p} \right\} \bmod q} = f(a) f(b).} \\
\end{array}
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math N\overset{\text{def}}{=}\{\prod_{i = 1}^{\ell} p_i\}, rendering as TeX
2025-07-19 20:24:31,308 - WARNING - Could not convert TeX math
N\overset{\text{def}}{=}\{\prod_{i = 1}^{\ell} p_i\}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
\left. 1 \rightleftrightharpoons (1,1) \right. & \left. 2 \rightleftrightharpoons (2,2) \right. & \left. 4 \rightleftrightharpoons (4,1) \right. & \left. 7 \rightleftrightharpoons (2,1) \right. \\
\left. 8 \rightleftrightharpoons (3,2) \right. & \left. 11 \rightleftrightharpoons (1,2) \right. & \left. 13 \rightleftrightharpoons (3,1) \right. & \left. 14 \rightleftrightharpoons (4,2) \right. \\
\end{matrix},, rendering as TeX
2025-07-19 20:24:31,309 - WARNING - Could not convert TeX math \begin{matrix}
\left. 1 \rightleftrightharpoons (1,1) \right. & \left. 2 \rightleftrightharpoons (2,2) \right. & \left. 4 \rightleftrightharpoons (4,1) \right. & \left. 7 \rightleftrightharpoons (2,1) \right. \\
\left. 8 \rightleftrightharpoons (3,2) \right. & \left. 11 \rightleftrightharpoons (1,2) \right. & \left. 13 \rightleftrightharpoons (3,1) \right. & \left. 14 \rightleftrightharpoons (4,2) \right. \\
\end{matrix},, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{(4,3) = 4 \cdot (1,0) + 3 \cdot (0,1)} \\
\begin{array}{l}
\left. \left\{ 4 \cdot 1_p + 3 \cdot 1_q \bmod 35 \right\} \right. \\
= \left\{ 4 \cdot 21 + 3 \cdot 15 \bmod 35 \right\} = 24.}
\end{array}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,309 - WARNING - Could not convert TeX math \begin{matrix}
{(4,3) = 4 \cdot (1,0) + 3 \cdot (0,1)} \\
\begin{array}{l}
\left. \left\{ 4 \cdot 1_p + 3 \cdot 1_q \bmod 35 \right\} \right. \\
= \left\{ 4 \cdot 21 + 3 \cdot 15 \bmod 35 \right\} = 24.}
\end{array}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \mathcal{O}(\sqrt{N} \cdot \text{polylog}(N)), rendering as TeX
2025-07-19 20:24:31,310 - WARNING - Could not convert TeX math
\mathcal{O}(\sqrt{N} \cdot \text{polylog}(N)), rendering as TeX

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[WARNING] Could not convert TeX math  $\left\lfloor\sqrt{N}\right\rfloor$ , rendering as TeX  
2025-07-19 20:24:31,310 - WARNING - Could not convert TeX math  $\left\lfloor\sqrt{N}\right\rfloor$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
2025-07-19 20:24:31,311 - WARNING - Could not convert TeX math  $\sqrt{N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left\lfloor\sqrt{N}\right\rfloor$ , rendering as TeX  
2025-07-19 20:24:31,311 - WARNING - Could not convert TeX math  $\left\lfloor\sqrt{N}\right\rfloor$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left(1 - \frac{1}{3n}\right)^t = \left(\left(1 - \frac{1}{3n}\right)^{3n}\right)^{\frac{t}{3n}} \leq \left(e^{-1}\right)^{\frac{t}{3n}} = e^{-\frac{t}{3n}}$ , rendering as TeX  
2025-07-19 20:24:31,312 - WARNING - Could not convert TeX math  $\left(1 - \frac{1}{3n}\right)^t = \left(\left(1 - \frac{1}{3n}\right)^{3n}\right)^{\frac{t}{3n}} \leq \left(e^{-1}\right)^{\frac{t}{3n}} = e^{-\frac{t}{3n}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $a^{N-1}\overset{?}{=}{1}$ , rendering as TeX  
2025-07-19 20:24:31,313 - WARNING - Could not convert TeX math  $a^{N-1}\overset{?}{=}{1}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{h}$ , rendering as TeX  
2025-07-19 20:24:31,313 - WARNING - Could not convert TeX math  $\overline{h}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{h}$ , rendering as TeX  
2025-07-19 20:24:31,314 - WARNING - Could not convert TeX math  $\overline{h}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{\mathbb{H}}\overset{\text{def}}{=}\left\{\overline{h}h\mid h\in\mathbb{H}\right\}$ , rendering as TeX  
2025-07-19 20:24:31,317 - WARNING - Could not convert TeX math  $\overline{\mathbb{H}}\overset{\text{def}}{=}\left\{\overline{h}h\mid h\in\mathbb{H}\right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{||}\overline{\mathbb{H}}\text{||} = \text{||}\mathbb{H}\text{||}$ , rendering as TeX  
2025-07-19 20:24:31,317 - WARNING - Could not convert TeX math  $\text{||}\overline{\mathbb{H}}\text{||} = \text{||}\mathbb{H}\text{||}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
2025-07-19 20:24:31,318 - WARNING - Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
2025-07-19 20:24:31,319 - WARNING - Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
2025-07-19 20:24:31,319 - WARNING - Could not convert TeX math  $\overline{\mathbb{H}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{||}\mathbb{G}\text{||} \geq \text{||}\mathbb{H}\text{||} + \text{||}\overline{\mathbb{H}}\text{||}$



$2\text{\texttt{\textbackslash text\{|\}\{\mathbb{H}\}\text{\texttt{\textbackslash text\{|}}}$ , rendering as TeX  
 2025-07-19 20:24:31,320 - WARNING - Could not convert TeX math  
 $\text{\texttt{\textbackslash text\{|\}\mathbb{G}\text{\texttt{\textbackslash text\{|}} \geq \text{\texttt{\textbackslash text\{|\}\{\mathbb{H}\}\text{\texttt{\textbackslash text\{|}} +$   
 $\text{\texttt{\textbackslash text\{|\}\overline{\mathbb{H}}\text{\texttt{\textbackslash text\{|}} = 2\text{\texttt{\textbackslash text\{|\}\{\mathbb{H}\}\text{\texttt{\textbackslash text\{|}}$ , rendering  
 as TeX  
 [WARNING] Could not convert TeX math  $\overline{h}h_1 = \overline{h}h_2$ ,  
 rendering as TeX  
 2025-07-19 20:24:31,321 - WARNING - Could not convert TeX math  $\overline{h}h_1$   
 $= \overline{h}h_2$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\{\overline{h}\}^{-1}$ , rendering as TeX  
 2025-07-19 20:24:31,322 - WARNING - Could not convert TeX math  $\{\overline{h}\}^{-1}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h}h \in \overline{\mathbb{H}}$ ,  
 rendering as TeX  
 2025-07-19 20:24:31,322 - WARNING - Could not convert TeX math  $\overline{h}h \in$   
 $\overline{\mathbb{H}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h}h \in \{\mathbb{H}\}$ , rendering  
 as TeX  
 2025-07-19 20:24:31,323 - WARNING - Could not convert TeX math  $\overline{h}h \in$   
 $\{\mathbb{H}\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h}h = h\prime$ , rendering as TeX  
 2025-07-19 20:24:31,324 - WARNING - Could not convert TeX math  $\overline{h}h =$   
 $h\prime$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h} = h\prime h^{-1}$ , rendering  
 as TeX  
 2025-07-19 20:24:31,324 - WARNING - Could not convert TeX math  $\overline{h} =$   
 $h\prime h^{-1}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h} = \{\mathbb{H}\}$ , rendering as  
 TeX  
 2025-07-19 20:24:31,325 - WARNING - Could not convert TeX math  $\overline{h} =$   
 $\{\mathbb{H}\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\overline{h}$ , rendering as TeX  
 2025-07-19 20:24:31,326 - WARNING - Could not convert TeX math  $\overline{h}$ ,  
 rendering as TeX  
 [WARNING] Could not convert TeX math  $\left( \{a^u, a^{2u}, \ldots, a^{2^r u}\} \right) = (\underset{i+1}{\text{terms}}\{\underset{\star}{\star}, \ldots, \star, -$   
 $1, \}1, \ldots, 1).$ , rendering as TeX  
 2025-07-19 20:24:31,328 - WARNING - Could not convert TeX math  $\left( \{a^u, a^{2u}, \ldots, a^{2^r u}\} \right) = (\underset{i+1}{\text{terms}}\{\underset{\star}{\star}, \ldots, \star, -$   
 $1, \}1, \ldots, 1).$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\text{\texttt{\textbackslash text\{Bad\}\prime\overset{\text{\texttt{\textbackslash text\{def\}}{=}}{\left\{ \{a \mid a^{2^i u} = \pm$   
 $1 \bmod N \right\}}}$ , rendering as TeX  
 2025-07-19 20:24:31,329 - WARNING - Could not convert TeX math  
 $\text{\texttt{\textbackslash text\{Bad\}\prime\overset{\text{\texttt{\textbackslash text\{def\}}{=}}{\left\{ \{a \mid a^{2^i u} = \pm$   
 $1 \bmod N \right\}}}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $y\overset{\text{\texttt{\textbackslash text\{def\}}{=}}{\lbrack$



$\right)\overset{\text{def}}{=}\left\{ (x,y) \mid x,y \in \mathbb{Z}_p \text{ and } y^2 = x^3 + Ax + B \pmod{p} \right\} \cup \left\{ \mathcal{O} \right\}$ , rendering as TeX  
2025-07-19 20:24:31,336 - WARNING - Could not convert TeX math  $E\left(\mathbb{Z}_p\right)\overset{\text{def}}{=}\left\{ (x,y) \mid x,y \in \mathbb{Z}_p \text{ and } y^2 = x^3 + Ax + B \pmod{p} \right\} \cup \left\{ \mathcal{O} \right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $f(x)\overset{\text{def}}{=}x^3 + 3x + 3$ , rendering as TeX  
2025-07-19 20:24:31,337 - WARNING - Could not convert TeX math  $f(x)\overset{\text{def}}{=}x^3 + 3x + 3$ , rendering as TeX  
[WARNING] Could not convert TeX math  $P_1 + P_2\overset{\text{def}}{=}(x, -y)$ , rendering as TeX  
2025-07-19 20:24:31,337 - WARNING - Could not convert TeX math  $P_1 + P_2\overset{\text{def}}{=}(x, -y)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $P_1 + P_2\overset{\text{def}}{=}\mathcal{O}$ , rendering as TeX  
2025-07-19 20:24:31,338 - WARNING - Could not convert TeX math  $P_1 + P_2\overset{\text{def}}{=}\mathcal{O}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $-P\overset{\text{def}}{=}(x, -y)$ , rendering as TeX  
2025-07-19 20:24:31,340 - WARNING - Could not convert TeX math  $-P\overset{\text{def}}{=}(x, -y)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $s\overset{\text{def}}{=}\left[\frac{y_2 - y_1}{x_2 - x_1} \pmod{p}\right]$ , rendering as TeX  
2025-07-19 20:24:31,342 - WARNING - Could not convert TeX math  $s\overset{\text{def}}{=}\left[\frac{y_2 - y_1}{x_2 - x_1} \pmod{p}\right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x_3\overset{\text{def}}{=}\left[s^2 - x_1 - x_2 \pmod{p}\right]$ , rendering as TeX  
2025-07-19 20:24:31,343 - WARNING - Could not convert TeX math  $x_3\overset{\text{def}}{=}\left[s^2 - x_1 - x_2 \pmod{p}\right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $s = \left[\frac{y_2 - y_1}{x_2 - x_1} \pmod{p}\right]$ , rendering as TeX  
2025-07-19 20:24:31,346 - WARNING - Could not convert TeX math  $s = \left[\frac{y_2 - y_1}{x_2 - x_1} \pmod{p}\right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l} x_3 = \left[s^2 - x_1 - x_2 \pmod{p}\right] \\ \right.\text{quad}\text{and}\text{quad} y_3 = \left[s \cdot \left(x_1 - x_3\right) - y_1 \pmod{p}\right] \\ \end{array}$ , rendering as TeX  
2025-07-19 20:24:31,347 - WARNING - Could not convert TeX math  $\begin{array}{l} x_3 = \left[s^2 - x_1 - x_2 \pmod{p}\right] \\ \right.\text{quad}\text{and}\text{quad} y_3 = \left[s \cdot \left(x_1 - x_3\right) - y_1 \pmod{p}\right] \\ \end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $s = \left[\frac{3x_1^2 +$

$A\{2y_{\{1\}}\{\text{mod}\}p\} \right\rbrace$ , rendering as TeX  
 2025-07-19 20:24:31,348 - WARNING - Could not convert TeX math  $s = \left\lbrack \frac{3x_{\{1\}}^2 + A\{2y_{\{1\}}\{\text{mod}\}p\} \right\rbrace$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $Q\overset{\{\text{def}\}}{=}P_{\{1\}} + P_{\{2\}} = (3,\lbrack 1 \cdot (1 - 3) - 0\{\text{mod}\}7\rbrace) = (3,5);$ , rendering as TeX  
 2025-07-19 20:24:31,349 - WARNING - Could not convert TeX math  
 $Q\overset{\{\text{def}\}}{=}P_{\{1\}} + P_{\{2\}} = (3,\lbrack 1 \cdot (1 - 3) - 0\{\text{mod}\}7\rbrace) = (3,5);$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $Q^{\{\prime\}}\overset{\{\text{def}\}}{=}P_{\{2\}} + P_{\{3\}} = (3,\lbrack 5 \cdot (4 - 3) - 3\{\text{mod}\}7\rbrace) = (3,2).$ , rendering as TeX  
 2025-07-19 20:24:31,349 - WARNING - Could not convert TeX math  
 $Q^{\{\prime\}}\overset{\{\text{def}\}}{=}P_{\{2\}} + P_{\{3\}} = (3,\lbrack 5 \cdot (4 - 3) - 3\{\text{mod}\}7\rbrace) = (3,2).$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\left( \{x_{\{3\}},y_{\{3\}} \right) = \left( \frac{x_{\{1\}}y_{\{2\}} + x_{\{2\}}y_{\{1\}}\{1 + dx_{\{1\}}x_{\{2\}}y_{\{1\}}y_{\{2\}}\}, \frac{y_{\{1\}}y_{\{2\}} - ax_{\{1\}}x_{\{2\}}\{1 - dx_{\{1\}}x_{\{2\}}y_{\{1\}}y_{\{2\}}\}} \right).$ , rendering as TeX  
 2025-07-19 20:24:31,350 - WARNING - Could not convert TeX math  $\left( \{x_{\{3\}},y_{\{3\}} \right) = \left( \frac{x_{\{1\}}y_{\{2\}} + x_{\{2\}}y_{\{1\}}\{1 + dx_{\{1\}}x_{\{2\}}y_{\{1\}}y_{\{2\}}\}, \frac{y_{\{1\}}y_{\{2\}} - ax_{\{1\}}x_{\{2\}}\{1 - dx_{\{1\}}x_{\{2\}}y_{\{1\}}y_{\{2\}}\}} \right).$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $p + 1 - 2\sqrt{p} \leq \text{\textit{E}}(\{\mathbb{Z}\}_{\{p\}}) \leq p + 1 + 2\sqrt{p}$ , rendering as TeX  
 2025-07-19 20:24:31,351 - WARNING - Could not convert TeX math  $p + 1 - 2\sqrt{p} \leq \text{\textit{E}}(\{\mathbb{Z}\}_{\{p\}}) \leq p + 1 + 2\sqrt{p}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{\textit{t}} \leq 2\sqrt{p}$ , rendering as TeX  
 2025-07-19 20:24:31,353 - WARNING - Could not convert TeX math  $\text{\textit{t}} \leq 2\sqrt{p}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\mathbb{G}\overset{\{\text{def}\}}{=}\left\{ \{rP \mid P \in E\left( \{\mathbb{Z}\}_{\{p\}} \right) \right\} \right\} \subset E\left( \{\mathbb{Z}\}_{\{p\}} \right)$ , rendering as TeX  
 2025-07-19 20:24:31,354 - WARNING - Could not convert TeX math  
 $\mathbb{G}\overset{\{\text{def}\}}{=}\left\{ \{rP \mid P \in E\left( \{\mathbb{Z}\}_{\{p\}} \right) \right\} \right\} \subset E\left( \{\mathbb{Z}\}_{\{p\}} \right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} P_{\{3\}}\overset{\{\text{def}\}}{=}P_{\{1\}} + P_{\{2\}} = (s^2 - X_{\{1\}}/Z_{\{1\}} - X_{\{2\}}/Z_{\{2\}}) \\ \{s \cdot \left( \{X_{\{1\}}/Z_{\{1\}} - s^2 + X_{\{1\}}/Z_{\{1\}} + X_{\{2\}}/Z_{\{2\}} \right) - Y_{\{1\}}/Z_{\{1\}}, 1\} \\ \end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,354 - WARNING - Could not convert TeX math  $\begin{matrix} P_{\{3\}}\overset{\{\text{def}\}}{=}P_{\{1\}} + P_{\{2\}} = (s^2 - X_{\{1\}}/Z_{\{1\}} - X_{\{2\}}/Z_{\{2\}}) \\ \{s \cdot \left( \{X_{\{1\}}/Z_{\{1\}} - s^2 + X_{\{1\}}/Z_{\{1\}} + X_{\{2\}}/Z_{\{2\}} \right) - Y_{\{1\}}/Z_{\{1\}}, 1\} \\ \end{matrix}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix}$

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{u = Y_{2}Z_{1} - Y_{1}Z_{2},\quad v = X_{2}Z_{1} - X_{1}Z_{2}} \\
{w = u^{2}Z_{1}Z_{2} - v^{3} - 2v^{2}X_{1}Z_{2}}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,355 - WARNING - Could not convert TeX math \begin{matrix}
{u = Y_{2}Z_{1} - Y_{1}Z_{2},\quad v = X_{2}Z_{1} - X_{1}Z_{2}} \\
{w = u^{2}Z_{1}Z_{2} - v^{3} - 2v^{2}X_{1}Z_{2}}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math
\varepsilon(n)\overset{\text{def}}{=}\text{Pr}\left\lbrack \text{Hash-}
coll\right\}_{\mathcal{A},\text{\Pi}}(n) = 1 \right\rbrack, rendering as TeX
2025-07-19 20:24:31,356 - WARNING - Could not convert TeX math
\varepsilon(n)\overset{\text{def}}{=}\text{Pr}\left\lbrack \text{Hash-}
coll\right\}_{\mathcal{A},\text{\Pi}}(n) = 1 \right\rbrack, rendering as TeX
[WARNING] Could not convert TeX math f_{0}^{s}(x)\overset{\text{def}}{=}\lbrack
x^{e}\bmod N\rbrack, rendering as TeX
2025-07-19 20:24:31,357 - WARNING - Could not convert TeX math
f_{0}^{s}(x)\overset{\text{def}}{=}\lbrack x^{e}\bmod N\rbrack, rendering as TeX
[WARNING] Could not convert TeX math f_{1}^{s}(x)\overset{\text{def}}{=}\lbrack
y \cdot x^{e}\bmod N\rbrack, rendering as TeX
2025-07-19 20:24:31,358 - WARNING - Could not convert TeX math
f_{1}^{s}(x)\overset{\text{def}}{=}\lbrack y \cdot x^{e}\bmod N\rbrack, rendering
as TeX
[WARNING] Could not convert TeX math
H^{s}(x)\overset{\text{def}}{=}f_{x_{1}}^{s}\left({f_{x_{2}}^{s}\left(
\cdots(1)\cdots \right)} \right), rendering as TeX
2025-07-19 20:24:31,358 - WARNING - Could not convert TeX math
H^{s}(x)\overset{\text{def}}{=}f_{x_{1}}^{s}\left({f_{x_{2}}^{s}\left(
\cdots(1)\cdots \right)} \right), rendering as TeX
[WARNING] Could not convert TeX math f(x)\overset{\text{def}}{=}\lbrack
x\bmod p\rbrack, rendering as TeX
2025-07-19 20:24:31,359 - WARNING - Could not convert TeX math
f(x)\overset{\text{def}}{=}\lbrack x\bmod p\rbrack, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
\left. y = \left\lbrack x^{B} - 1\bmod N \right\rbrack \leftrightsquigarrow \left(
\left\lbrack x_{p},x_{q} \right\rbrack \right)^{B} - (1,1) \right. \\
\left. \begin{array}{l}
\left. \left\lbrack x_{p}^{B} - 1\bmod p,x_{q}^{B} - 1\bmod q \right\rbrack \right. \\
\left. \left\lbrack \left(x_{p}^{p-1} \right)^{\gamma} - 1\bmod p,x_{q}^{B} -
1\bmod q \right\rbrack \right. \\
\left. \left\lbrack 0,\left\lbrack x_{q}^{B} - 1\bmod q \right\rbrack \right\rbrack \right) \end{array} \right.
\end{array}, rendering as TeX
2025-07-19 20:24:31,360 - WARNING - Could not convert TeX math \begin{array}{l}
\left. y = \left\lbrack x^{B} - 1\bmod N \right\rbrack \leftrightsquigarrow \left(
\left\lbrack x_{p},x_{q} \right\rbrack \right)^{B} - (1,1) \right. \\
\left. \begin{array}{l}
\left. \left\lbrack x_{p}^{B} - 1\bmod p,x_{q}^{B} - 1\bmod q \right\rbrack \right. \\
\left. \left\lbrack \left(x_{p}^{p-1} \right)^{\gamma} - 1\bmod p,x_{q}^{B} -
1\bmod q \right\rbrack \right. \\
\left. \left\lbrack 0,\left\lbrack x_{q}^{B} - 1\bmod q \right\rbrack \right\rbrack \right) \end{array} \right.
\end{array}, rendering as TeX

```

$1 \bmod q \text{ \right)} \text{ \right)} \text{ \right)}$   
 $\{ = \left( \{ 0, \left[ x_q^B - 1 \bmod q \right] \right) \}$   
 $\end{array}$   
 $\end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x_q \overset{\text{def}}{=} \left[ x \bmod q \right]$ , rendering as TeX  
2025-07-19 20:24:31,361 - WARNING - Could not convert TeX math  $x_q \overset{\text{def}}{=} \left[ x \bmod q \right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{\phi(q-1)}{q-1} =$ , rendering as TeX  
2025-07-19 20:24:31,362 - WARNING - Could not convert TeX math  $\frac{\phi(q-1)}{q-1} =$ , rendering as TeX  
[WARNING] Could not convert TeX math  $B = \{\prod_{i=1}^k \{p_i\}^{\lfloor n/\log p_i \rfloor}\}$ , rendering as TeX  
2025-07-19 20:24:31,363 - WARNING - Could not convert TeX math  $B = \{\prod_{i=1}^k \{p_i\}^{\lfloor n/\log p_i \rfloor}\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\prod_{i=1}^k p_i^{e_i}$ , rendering as TeX  
2025-07-19 20:24:31,364 - WARNING - Could not convert TeX math  $\prod_{i=1}^k p_i^{e_i}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $k = 2^{n/2} = \mathcal{O}(\sqrt{p})$ , rendering as TeX  
2025-07-19 20:24:31,364 - WARNING - Could not convert TeX math  $k = 2^{n/2} = \mathcal{O}(\sqrt{p})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x_p^{(i)} \overset{\text{def}}{=} \left[ x^{(i)} \bmod p \right]$ , rendering as TeX  
2025-07-19 20:24:31,366 - WARNING - Could not convert TeX math  $x_p^{(i)} \overset{\text{def}}{=} \left[ x^{(i)} \bmod p \right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $k = \mathcal{O}(\sqrt{p})$ , rendering as TeX  
2025-07-19 20:24:31,366 - WARNING - Could not convert TeX math  $k = \mathcal{O}(\sqrt{p})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{p}) = \mathcal{O}(N^{1/4})$ , rendering as TeX  
2025-07-19 20:24:31,368 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{p}) = \mathcal{O}(N^{1/4})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x = \sqrt{N} + 1$ , rendering as TeX  
2025-07-19 20:24:31,368 - WARNING - Could not convert TeX math  $x = \sqrt{N} + 1$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\sqrt{N} + 2$ , rendering as TeX  
2025-07-19 20:24:31,369 - WARNING - Could not convert TeX math  $\sqrt{N} + 2$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x > \sqrt{N}$ , rendering as TeX  
2025-07-19 20:24:31,370 - WARNING - Could not convert TeX math  $x > \sqrt{N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q \overset{\text{def}}{=} \left[ x^2 \bmod N \right] = x^2 - N$ , rendering as TeX  
2025-07-19 20:24:31,370 - WARNING - Could not convert TeX math

$$q \overset{\text{def}}{=} \left[ x^2 \bmod N \right] = x^2 - N$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l}$   

$$q_1 = \left[ x_1^2 \bmod N \right] \prod \limits_{i=1}^k p_i^{e_{1,i}}$$
  

$$\vdots$$
  

$$q_{\ell} = \left[ x_{\ell}^2 \bmod N \right] \prod \limits_{i=1}^k p_i^{e_{\ell,i}}$$
  

$$\end{array}$$
, rendering as TeX  
2025-07-19 20:24:31,371 - WARNING - Could not convert TeX math  $\begin{array}{l}$   

$$q_1 = \left[ x_1^2 \bmod N \right] \prod \limits_{i=1}^k p_i^{e_{1,i}}$$
  

$$\vdots$$
  

$$q_{\ell} = \left[ x_{\ell}^2 \bmod N \right] \prod \limits_{i=1}^k p_i^{e_{\ell,i}}$$
  

$$\end{array}$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $z = \prod \limits_{j \in S} q_j = \prod \limits_{i=1}^k p_i^{\sum \limits_{j \in S} e_{j,i}}$ , rendering as TeX  
2025-07-19 20:24:31,371 - WARNING - Could not convert TeX math  $z = \prod \limits_{j \in S} q_j = \prod \limits_{i=1}^k p_i^{\sum \limits_{j \in S} e_{j,i}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{pmatrix} \gamma_{1,1} & \gamma_{1,2} & \cdots & \gamma_{1,k} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{\ell,1} & \gamma_{\ell,2} & \cdots & \gamma_{\ell,k} \end{pmatrix} \overset{\text{def}}{=} \begin{pmatrix} \left[ e_{1,1} \bmod 2 \right] & \left[ e_{1,2} \bmod 2 \right] & \cdots & \left[ e_{1,k} \bmod 2 \right] \\ \vdots & \vdots & \ddots & \vdots \\ \left[ e_{\ell,1} \bmod 2 \right] & \left[ e_{\ell,2} \bmod 2 \right] & \cdots & \left[ e_{\ell,k} \bmod 2 \right] \end{pmatrix}$   

$$\end{pmatrix}$$
, rendering as TeX  
2025-07-19 20:24:31,372 - WARNING - Could not convert TeX math  $\begin{pmatrix} \gamma_{1,1} & \gamma_{1,2} & \cdots & \gamma_{1,k} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{\ell,1} & \gamma_{\ell,2} & \cdots & \gamma_{\ell,k} \end{pmatrix} \overset{\text{def}}{=} \begin{pmatrix} \left[ e_{1,1} \bmod 2 \right] & \left[ e_{1,2} \bmod 2 \right] & \cdots & \left[ e_{1,k} \bmod 2 \right] \\ \vdots & \vdots & \ddots & \vdots \\ \left[ e_{\ell,1} \bmod 2 \right] & \left[ e_{\ell,2} \bmod 2 \right] & \cdots & \left[ e_{\ell,k} \bmod 2 \right] \end{pmatrix}$   

$$\end{pmatrix}$$
, rendering as TeX  
[WARNING] Could not convert TeX math  $z \overset{\text{def}}{=} \prod \limits_{j \in S} q_j = \prod \limits_{i=1}^k p_i^{\sum \limits_{j \in S} e_{j,i}} = \left( \prod \limits_{i=1}^k p_i^{\frac{\sum \limits_{j \in S} e_{j,i}}{2}} \right)^2$ , rendering as TeX

2025-07-19 20:24:31,373 - WARNING - Could not convert TeX math  $z^{\overset{\text{def}}{=}} \prod_{j \in S} q_j = \prod_{i=1}^k p_i^{\sum_{j \in S} e_{j,i}} = \left( \prod_{i=1}^k p_i^{\left( \frac{\sum_{j \in S} e_{j,i}}{2} \right)} \right)^2$ , rendering as TeX

[WARNING] Could not convert TeX math  $z = \prod_{j \in S} q_j = \prod_{j \in S} x_j^2 = \left( \prod_{j \in S} x_j \right)^2 \pmod{N}$ , rendering as TeX

2025-07-19 20:24:31,373 - WARNING - Could not convert TeX math  $z = \prod_{j \in S} q_j = \prod_{j \in S} x_j^2 = \left( \prod_{j \in S} x_j \right)^2 \pmod{N}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \left[ 621^2 \pmod{N} \right] \right] = 2^4 \cdot 17 \cdot 29 \\ \left[ 645^2 \pmod{N} \right] \right] = 2^7 \cdot 13 \cdot 23 \\ \left[ 655^2 \pmod{N} \right] \right] = 2^3 \cdot 13 \cdot 17 \cdot 29. \end{array}$ , rendering as TeX

2025-07-19 20:24:31,374 - WARNING - Could not convert TeX math  $\begin{array}{l} \left[ 621^2 \pmod{N} \right] \right] = 2^4 \cdot 17 \cdot 29 \\ \left[ 645^2 \pmod{N} \right] \right] = 2^7 \cdot 13 \cdot 23 \\ \left[ 655^2 \pmod{N} \right] \right] = 2^3 \cdot 13 \cdot 17 \cdot 29. \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} 620^2 \cdot 621^2 \cdot 645^2 \cdot 655^2 = 2^{14} \cdot 13^2 \cdot 17^4 \cdot 23^2 \cdot 29^2 \pmod{N} \\ \left. \begin{array}{l} 620 \cdot 621 \cdot 645 \cdot 655 \pmod{N} \end{array} \right]^2 = \left[ 2^7 \cdot 13 \cdot 17^2 \cdot 23 \cdot 29 \pmod{N} \right] \pmod{N} \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,375 - WARNING - Could not convert TeX math  $\begin{matrix} 620^2 \cdot 621^2 \cdot 645^2 \cdot 655^2 = 2^{14} \cdot 13^2 \cdot 17^4 \cdot 23^2 \cdot 29^2 \pmod{N} \\ \left. \begin{array}{l} 620 \cdot 621 \cdot 645 \cdot 655 \pmod{N} \end{array} \right]^2 = \left[ 2^7 \cdot 13 \cdot 17^2 \cdot 23 \cdot 29 \pmod{N} \right] \pmod{N} \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $2^{\mathcal{O}(\sqrt{\log N \log \log N})}$ , rendering as TeX

2025-07-19 20:24:31,377 - WARNING - Could not convert TeX math  $2^{\mathcal{O}(\sqrt{\log N \log \log N})}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,378 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX



rendering as TeX  
2025-07-19 20:24:31,379 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,380 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
2025-07-19 20:24:31,380 - WARNING - Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
2025-07-19 20:24:31,381 - WARNING - Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
2025-07-19 20:24:31,381 - WARNING - Could not convert TeX math  $q = \prod_{i=1}^k q_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g_i \overset{\text{def}}{=} g^{q/q_i}$ , rendering as TeX  
2025-07-19 20:24:31,383 - WARNING - Could not convert TeX math  $g_i \overset{\text{def}}{=} g^{q/q_i}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $h_i \overset{\text{def}}{=} h^{q/q_i}$ , rendering as TeX  
2025-07-19 20:24:31,383 - WARNING - Could not convert TeX math  $h_i \overset{\text{def}}{=} h^{q/q_i}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix} x = x_1 \bmod q_1 \\ \vdots \\ x = x_k \bmod q_k \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,384 - WARNING - Could not convert TeX math  $\begin{matrix} x = x_1 \bmod q_1 \\ \vdots \\ x = x_k \bmod q_k \end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l} \left. \left( g^{30/5} \right)^x = h^{30/5} \right\} \rightarrow \left( 3^6 \right)^x = 26^6 \rightarrow 16^x = 1 \end{array}$   
 $\left. \left( g^{30/3} \right)^x = h^{30/3} \right\} \rightarrow \left( 3^{10} \right)^x = 26^{10} \rightarrow 25^x = 5$   
 $\left. \left( g^{30/2} \right)^x = h^{30/2} \right\} \rightarrow \left( 3^{15} \right)^x = 26^{15} \rightarrow 30^x = 30$   
rendering as TeX  
2025-07-19 20:24:31,385 - WARNING - Could not convert TeX math  $\begin{array}{l} \left. \left( g^{30/5} \right)^x = h^{30/5} \right\} \rightarrow \left( 3^6 \right)^x = 26^6 \rightarrow 16^x = 1 \end{array}$   
 $\left. \left( g^{30/5} \right)^x = h^{30/5} \right\} \rightarrow \left( 3^6 \right)^x = 26^6 \rightarrow 16^x = 1$

$\left. \left( g^{\{30/3\}} \right)^{\{x\}} = h^{\{30/3\}} \right\} \rightarrow \left( 3^{\{10\}} \right)^{\{x\}} = 26^{\{10\}} \rightarrow 25^{\{x\}} = 5 \right. \backslash \backslash$   
 $\left. \left( g^{\{30/2\}} \right)^{\{x\}} = h^{\{30/2\}} \right\} \rightarrow \left( 3^{\{15\}} \right)^{\{x\}} = 26^{\{15\}} \rightarrow 30^{\{x\}} = 30. \right.$   
 $\end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q = \prod_{i=1}^k p_i^{e_i}$ , rendering as TeX  
2025-07-19 20:24:31,386 - WARNING - Could not convert TeX math  $q = \prod_{i=1}^k p_i^{e_i}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,387 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $t \overset{\text{def}}{=} \left\lfloor \sqrt{q} \right\rfloor$ , rendering as TeX  
2025-07-19 20:24:31,388 - WARNING - Could not convert TeX math  $t \overset{\text{def}}{=} \left\lfloor \sqrt{q} \right\rfloor$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left\lfloor \sqrt{q} \right\rfloor + 1 = \mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,389 - WARNING - Could not convert TeX math  $\left\lfloor \sqrt{q} \right\rfloor + 1 = \mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $t := \left\lfloor \sqrt{q} \right\rfloor$ , rendering as TeX  
2025-07-19 20:24:31,389 - WARNING - Could not convert TeX math  $t := \left\lfloor \sqrt{q} \right\rfloor$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,390 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,391 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q} \cdot \log q)$ , rendering as TeX  
2025-07-19 20:24:31,392 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q} \cdot \log q)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q} \cdot \text{polylog}(q))$ , rendering as TeX  
2025-07-19 20:24:31,393 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q} \cdot \text{polylog}(q))$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
2025-07-19 20:24:31,393 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{\text{G}}) = \mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,394 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{\text{\texttt{\texttt{G}}}}) = \mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,395 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $H(k)\overset{\text{\texttt{def}}}{=}H_{g,h}(F(k))$ , rendering as TeX

2025-07-19 20:24:31,396 - WARNING - Could not convert TeX math  $H(k)\overset{\text{\texttt{def}}}{=}H_{g,h}(F(k))$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{\text{\texttt{\texttt{G}}}}) = \mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,397 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{\text{\texttt{\texttt{G}}}}) = \mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $g_i\overset{\text{\texttt{def}}}{=}\lbrack g^{x_i}\bmod p\rbrack$ , rendering as TeX

2025-07-19 20:24:31,397 - WARNING - Could not convert TeX math  $g_i\overset{\text{\texttt{def}}}{=}\lbrack g^{x_i}\bmod p\rbrack$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} g^{x_1} = \prod\limits_{i=1}^k p_i^{e_{1,i}}\bmod p \\ \vdots \\ g^{x_{\ell}} = \prod\limits_{i=1}^k p_i^{e_{\ell,i}}\bmod p \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,398 - WARNING - Could not convert TeX math  $\begin{matrix} g^{x_1} = \prod\limits_{i=1}^k p_i^{e_{1,i}}\bmod p \\ \vdots \\ g^{x_{\ell}} = \prod\limits_{i=1}^k p_i^{e_{\ell,i}}\bmod p \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} x_1 = \sum\limits_{i=1}^k e_{1,i} \log_g p_i \bmod (p-1) \\ \vdots \\ x_{\ell} = \sum\limits_{i=1}^k e_{\ell,i} \log_g p_i \bmod (p-1) \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,398 - WARNING - Could not convert TeX math  $\begin{matrix} x_1 = \sum\limits_{i=1}^k e_{1,i} \log_g p_i \bmod (p-1) \\ \vdots \\ x_{\ell} = \sum\limits_{i=1}^k e_{\ell,i} \log_g p_i \bmod (p-1) \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} g^x \cdot h = \prod\limits_{i=1}^k p_i^{e_i} \bmod p \\ \leftarrow x + \log_g h = \sum\limits_{i=1}^k e_i \log_g p_i \bmod (p-1) \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,399 - WARNING - Could not convert TeX math  $\begin{matrix} g^x \cdot h = \prod_{i=1}^k p_i^{e_i} \pmod{p} \\ \leftarrow x + \log_g h = \sum_{i=1}^k e_i \cdot \log_g p_i \pmod{p-1}, \right. \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \{10 = \log_3 5 + \log_3 13 \pmod{100} \\ \{12 = 4 \cdot \log_3 2 + \log_3 5 \pmod{100} \\ \{14 = \log_3 13 \pmod{100}. \end{array}$ , rendering as TeX

2025-07-19 20:24:31,401 - WARNING - Could not convert TeX math  $\begin{array}{l} \{10 = \log_3 5 + \log_3 13 \pmod{100} \\ \{12 = 4 \cdot \log_3 2 + \log_3 5 \pmod{100} \\ \{14 = \log_3 13 \pmod{100}. \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $2^{\mathcal{O}(\sqrt{\log p \log \log p})}$ , rendering as TeX

2025-07-19 20:24:31,401 - WARNING - Could not convert TeX math  $2^{\mathcal{O}(\sqrt{\log p \log \log p})}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,402 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{b-a})$ , rendering as TeX

2025-07-19 20:24:31,404 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{b-a})$ , rendering as TeX

[WARNING] Could not convert TeX math  $x^{(i)} \overset{\text{def}}{=} F^{(i)}(x)$ , rendering as TeX

2025-07-19 20:24:31,404 - WARNING - Could not convert TeX math  $x^{(i)} \overset{\text{def}}{=} F^{(i)}(x)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{p})$ , rendering as TeX

2025-07-19 20:24:31,405 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{p})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{p})$ , rendering as TeX

2025-07-19 20:24:31,406 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{p})$ , rendering as TeX

[WARNING] Could not convert TeX math  $q = \{\prod_{i=1}^k p_i^{e_i}\}$ , rendering as TeX

2025-07-19 20:24:31,406 - WARNING - Could not convert TeX math  $q = \{\prod_{i=1}^k p_i^{e_i}\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sum_{i=1}^k e_i \sqrt{p_i})$ , rendering as TeX

2025-07-19 20:24:31,407 - WARNING - Could not convert TeX math  $\mathcal{O}(\sum_{i=1}^k e_i \sqrt{p_i})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{b-a})$ , rendering as TeX

rendering as TeX

2025-07-19 20:24:31,407 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{b-a})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

2025-07-19 20:24:31,409 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{q})$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1) \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,410 - WARNING - Could not convert TeX math  $\text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1) \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}(n)$ , rendering as TeX

2025-07-19 20:24:31,411 - WARNING - Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}$ , rendering as TeX

2025-07-19 20:24:31,412 - WARNING - Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1) \leq \frac{1}{2} \cdot \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1 \mid b = 0) \\ \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1 \mid b = 1) \end{array}$

$\end{array}$ , rendering as TeX

2025-07-19 20:24:31,413 - WARNING - Could not convert TeX math  $\begin{array}{l} \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1) \leq \frac{1}{2} \cdot \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1 \mid b = 0) \\ \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1 \mid b = 1) \end{array}$

$\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}(n)$ , rendering as TeX

2025-07-19 20:24:31,414 - WARNING - Could not convert TeX math  $\hat{\text{KE}}_A(\Pi)^{\text{eav}}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1) \leq \frac{1}{2} \cdot \text{Pr}(\text{Pr}(\text{KE}_A, \Pi)^{\text{eav}}(n) = 1 \mid b = 0) \end{array}$

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\right\rbrace + \frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack
\{\hat{\text{KE}}\}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1 \mid b = 1\right\rbrace \\
= \frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,xy\right) = 0\right\rbrace + \frac{1}{2}
\cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace \\
= \frac{1}{2} \cdot \left(\{1 - \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,xy\right) = 1\right\rbrace\right) +
\frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace \right) \\
= \frac{1}{2} + \frac{1}{2} \cdot \left(\text{\text{Pr}}\left\lbrack
\mathcal{A}\left(\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace
- \text{\text{Pr}}\left\lbrack \mathcal{A}\left(\mathbb{G},q,g,x,y,xy\right)
= 1\right\rbrace \right) \\
\leq \frac{1}{2} + \frac{1}{2} \cdot \mid \text{\text{Pr}}\left\lbrack
\mathcal{A}\left(\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace
- \text{\text{Pr}}\left\lbrack \mathcal{A}\left(\mathbb{G},q,g,x,y,xy\right)
= 1\right\rbrace,
\end{array}, \text{rendering as TeX}
2025-07-19 20:24:31,415 - WARNING - Could not convert TeX math \begin{array}{l}
\text{\text{Pr}}\left\lbrack
\{\hat{\text{KE}}\}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1\right\rbrace \\
= \frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack
\{\hat{\text{KE}}\}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1 \mid b = 0\right\rbrace
+ \frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack
\{\hat{\text{KE}}\}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1 \mid b = 1\right\rbrace \\
= \frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,xy\right) = 0\right\rbrace + \frac{1}{2}
\cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace \\
= \frac{1}{2} \cdot \left(\{1 - \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,xy\right) = 1\right\rbrace\right) +
\frac{1}{2} \cdot \text{\text{Pr}}\left\lbrack \mathcal{A}\left(
\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace \right) \\
= \frac{1}{2} + \frac{1}{2} \cdot \left(\text{\text{Pr}}\left\lbrack
\mathcal{A}\left(\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace
- \text{\text{Pr}}\left\lbrack \mathcal{A}\left(\mathbb{G},q,g,x,y,xy\right)
= 1\right\rbrace \right) \\
\leq \frac{1}{2} + \frac{1}{2} \cdot \mid \text{\text{Pr}}\left\lbrack
\mathcal{A}\left(\mathbb{G},q,g,x,y,z\right) = 1\right\rbrace
- \text{\text{Pr}}\left\lbrack \mathcal{A}\left(\mathbb{G},q,g,x,y,xy\right)
= 1\right\rbrace,
\end{array}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack
\{\hat{\text{KE}}\}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1\right\rbrace
\leq \frac{1}{2} + \frac{1}{2} \cdot \text{negl}(n),, \text{rendering as TeX}

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2025-07-19 20:24:31,415 - WARNING - Could not convert TeX math  

$$\text{Pr} \left\{ \hat{\text{KE}}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2} + \frac{1}{2} \cdot \text{negl}(n),$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n),$  rendering as TeX

2025-07-19 20:24:31,416 - WARNING - Could not convert TeX math  

$$\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n),$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2},$  rendering as TeX

2025-07-19 20:24:31,417 - WARNING - Could not convert TeX math  

$$\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2},$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{LR} - \text{cpa}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n),$  rendering as TeX

2025-07-19 20:24:31,418 - WARNING - Could not convert TeX math  

$$\text{Pr} \left\{ \text{Pub} \text{KE}_{\mathcal{A}, \Pi}^{\text{LR} - \text{cpa}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n),$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\overrightarrow{C}_0,$  rendering as TeX

2025-07-19 20:24:31,418 - WARNING - Could not convert TeX math  

$$\overrightarrow{C}_0,$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\overrightarrow{C}_1,$  rendering as TeX

2025-07-19 20:24:31,419 - WARNING - Could not convert TeX math  

$$\overrightarrow{C}_1,$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\overrightarrow{C}_0,$  rendering as TeX

2025-07-19 20:24:31,419 - WARNING - Could not convert TeX math  

$$\overrightarrow{C}_0,$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\overrightarrow{C}_1,$  rendering as TeX

2025-07-19 20:24:31,420 - WARNING - Could not convert TeX math  

$$\overrightarrow{C}_1,$$
 rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \mid \text{Pr} \left\{ \mathcal{A} \left( \text{pk}, \text{Enc}_{\text{pk}} \left( m_{1,0} \right), \text{Enc}_{\text{pk}} \left( m_{2,0} \right) \right) \right\} \leq \frac{1}{2} \\ \text{Pr} \left\{ \mathcal{A} \left( \text{pk}, \text{Enc}_{\text{pk}} \left( m_{1,1} \right), \text{Enc}_{\text{pk}} \left( m_{2,1} \right) \right) \right\} \leq \text{negl}(n) \end{array}$  rendering as TeX

2025-07-19 20:24:31,420 - WARNING - Could not convert TeX math  $\begin{array}{l} \mid \text{Pr} \left\{ \mathcal{A} \left( \text{pk}, \text{Enc}_{\text{pk}} \left( m_{1,0} \right), \text{Enc}_{\text{pk}} \left( m_{2,0} \right) \right) \right\} \leq \frac{1}{2} \\ \text{Pr} \left\{ \mathcal{A} \left( \text{pk}, \text{Enc}_{\text{pk}} \left( m_{1,1} \right), \text{Enc}_{\text{pk}} \left( m_{2,1} \right) \right) \right\} \leq \text{negl}(n) \end{array}$





$\{\overset{\rightarrow}{C}\}_{0}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\{\overset{\rightarrow}{C}\}_{01}$ , rendering as TeX  
2025-07-19 20:24:31,431 - WARNING - Could not convert TeX math  
 $\{\overset{\rightarrow}{C}\}_{01}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{\mid \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,0}\right)\right)\right\}\right\}=1\right\}$  \\
 $\{\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,, - \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}\leq \text{negl}(n).$   
 $\end{array}$ , rendering as TeX  
2025-07-19 20:24:31,432 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{\mid \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,0}\right)\right)\right\}\right\}=1\right\}$  \\
 $\{\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,, - \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}\leq \text{negl}(n).$   
 $\end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l}$   
 $\{\mid \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}$  \\
 $\{\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,, - \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,1}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}\leq \text{negl}(n).$   
 $\end{array}$ , rendering as TeX  
2025-07-19 20:24:31,433 - WARNING - Could not convert TeX math  $\begin{array}{l}$   
 $\{\mid \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,0}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}$  \\
 $\{\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,,\,, - \text{Pr}\}\left\{\mathcal{A}\left(\{pk,\text{Enc}\}_{pk}\left(m_{1,1}\right),\text{Enc}_{pk}\left(m_{2,1}\right)\right)\right\}\right\}=1\right\}\leq \text{negl}(n).$   
 $\end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\mathcal{A},\text{Pr}\right\}^{\text{LR}} - \text{cpa}(n) = 1$   
 $\right\} = \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{LR}}_{pk}(\text{pk}) = 0\right\} + \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{LR}}_{pk}(0)(\text{pk}) = 1\right\}$ , rendering as TeX  
2025-07-19 20:24:31,434 - WARNING - Could not convert TeX math  
 $\text{Pr}\left\{\mathcal{A},\text{Pr}\right\}^{\text{LR}} - \text{cpa}(n) = 1$  \\
 $= \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{LR}}_{pk}(\text{pk}) = 0\right\} + \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{LR}}_{pk}(0)(\text{pk}) = 1\right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\text{Pr}\left\{\mathcal{A}^{\text{prime}}\right\}_1 \mid b = 0$   
 $\right\} = \sum \limits_{i^{\ast} = 1}^t \text{Pr}\left\{i =$

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i^{\ast}} \right\rbrack \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 0 \wedge i = i^{\ast}}
\right\rbrack \backslash
{= \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack.}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,434 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 0}
\right\rbrack = \sum\limits_{i^{\ast} = 1}^t \text{Pr}\left\lbrack {i =
i^{\ast}} \right\rbrack \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 0 \wedge i = i^{\ast}}
\right\rbrack} \backslash
{= \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack.}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 1}
\right\rbrack = \sum\limits_{i^{\ast} = 1}^t \text{Pr}\left\lbrack {i =
i^{\ast}} \right\rbrack \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 1 \wedge i = i^{\ast}}
\right\rbrack} \backslash
\begin{array}{l}
{= \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}} - 1}(pk) = 1} \right\rbrack} \backslash
{= \sum\limits_{i^{\ast} = 0}^{t-1} \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack.}
\end{array}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,436 - WARNING - Could not convert TeX math \begin{matrix}
{\text{Pr}\left\lbrack {\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 1}
\right\rbrack = \sum\limits_{i^{\ast} = 1}^t \text{Pr}\left\lbrack {i =
i^{\ast}} \right\rbrack \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{outputs}1 \mid b = 1 \wedge i = i^{\ast}}
\right\rbrack} \backslash
\begin{array}{l}
{= \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}} - 1}(pk) = 1} \right\rbrack} \backslash
{= \sum\limits_{i^{\ast} = 0}^{t-1} \frac{1}{t} \cdot \text{Pr}\left\lbrack
{\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack.}
\end{array}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\text{negl}(n) \geq \left| \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot \text{Pr}\left\lbrack {\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack - \sum\limits_{i^{\ast} = 0}^{t-1} \frac{1}{t} \cdot \text{Pr}\left\lbrack {\mathcal{A}}^{\prime}\text{LR}_{pk}^{i^{\ast}}(pk) = 1} \right\rbrack} \right|} \backslash
{= \frac{1}{t} \cdot \left| \text{Pr}\left\lbrack

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{\mathcal{A}}^{\{\text{L}R_{pk}^t\}}(pk) = 1} \right\rbrack - \text{\text{Pr}}\left\lbrack \right.
{\mathcal{A}}^{\{\text{LR}_{pk}^0\}}(pk) = 1} \right\rbrack \right|,}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,436 - WARNING - Could not convert TeX math \begin{matrix}
{\neg l}(n) \geq \left| \sum\limits_{i^{\ast} = 1}^t \frac{1}{t} \cdot
\text{\text{Pr}}\left\lbrack {\mathcal{A}}^{\{\text{LR}_{pk}^{\ast}\}}(pk) = 1} \right\rbrack - \sum\limits_{i^{\ast} = 0}^{t-1} \frac{1}{t} \cdot
\text{\text{Pr}}\left\lbrack {\mathcal{A}}^{\{\text{LR}_{pk}^{\ast}\}}(pk) = 1} \right\rbrack \right| \setminus
{=} \frac{1}{t} \cdot \left| \text{\text{Pr}}\left\lbrack
{\mathcal{A}}^{\{\text{L}R_{pk}^t\}}(pk) = 1} \right\rbrack - \text{\text{Pr}}\left\lbrack
{\mathcal{A}}^{\{\text{LR}_{pk}^0\}}(pk) = 1} \right\rbrack \right|,}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack
{\text{Pub}}\text{\text{K}}_{\{\mathcal{A},\text{\text{II}}\}^{\{\text{cca}\}}}(n) = 1} \right\rbrack \leq
\frac{1}{2} + {\neg l}(n)., rendering as TeX
2025-07-19 20:24:31,437 - WARNING - Could not convert TeX math
\text{\text{Pr}}\left\lbrack {\text{Pub}}\text{\text{K}}_{\{\mathcal{A},\text{\text{II}}\}^{\{\text{cca}\}}}(n) = 1} \right\rbrack \leq \frac{1}{2} + {\neg l}(n)., rendering as TeX
[WARNING] Could not convert TeX math
\gamma\overset{\{\text{\text{~def~}}\}}{=}\ell/\ell^{\prime}, rendering as TeX
2025-07-19 20:24:31,438 - WARNING - Could not convert TeX math
\gamma\overset{\{\text{\text{~def~}}\}}{=}\ell/\ell^{\prime}, rendering as TeX
[WARNING] Could not convert TeX math \frac{\left. \alpha + \beta \cdot \middle|
m \right|}{|m|} = \frac{\alpha}{|m|} + \beta,, rendering as TeX
2025-07-19 20:24:31,439 - WARNING - Could not convert TeX math \frac{\left.
\alpha + \beta \cdot \middle| m \right|}{|m|} = \frac{\alpha}{|m|} + \beta,,
rendering as TeX
[WARNING] Could not convert TeX math \alpha + 2^{20} \cdot \frac{\alpha}{2^{11}}
\approx 512 \cdot \alpha,, rendering as TeX
2025-07-19 20:24:31,440 - WARNING - Could not convert TeX math \alpha + 2^{20}
\cdot \frac{\alpha}{2^{11}} \approx 512 \cdot \alpha,, rendering as TeX
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack
{\text{\text{K}}EM}_{\{\mathcal{A},\text{\text{II}}\}^{\{\text{cpa}\}}}(n) = 1} \right\rbrack \leq
\frac{1}{2} + {\neg l}(n)., rendering as TeX
2025-07-19 20:24:31,442 - WARNING - Could not convert TeX math
\text{\text{Pr}}\left\lbrack {\text{\text{K}}EM}_{\{\mathcal{A},\text{\text{II}}\}^{\{\text{cpa}\}}}(n) = 1} \right\rbrack \leq \frac{1}{2} + {\neg l}(n)., rendering as TeX
[WARNING] Could not convert TeX math ``X\overset{\{\text{c}\}}{\equiv}Y", rendering
as TeX:
``X\overset{\{\text{c}\}}{\equiv}Y"
~
unexpected ``'
expecting "%", "\\label", "\\tag", "\\nonumber", whitespace, "\\bangle",
"\\brace", "\\brack", "\\choose", "\\displaystyle", "\\textstyle",
"\\scriptstyle", "\\scriptscriptstyle", "{", "\\operatorname", letter, digit,
".", "!", "'", '"', "'''", "''''", "'''''", "*", "+", ",", "-", ".", "/", ":", ":",
";", "<", "=", ">", "?", "@", "~", "_", "^", "\\left", "\\right", "\\hyperref" or end

```

of input

2025-07-19 20:24:31,442 - WARNING - Could not convert TeX math

`\overset{\text{c}}{\equiv}Y`, rendering as TeX:

`\overset{\text{c}}{\equiv}Y`

^

unexpected ``

expecting "%", "\\label", "\\tag", "\\nonumber", whitespace, "\\bangle",  
"\\brace", "\\brack", "\\choose", "\\displaystyle", "\\textstyle",  
"\\scriptstyle", "\\scriptscriptstyle", "{", "\\operatorname", letter, digit,  
".", "!", "'", "''", "'''", "''''", "'''''", "\*", "+", ",", "-", ".", "/", ":", ":",  
";", "<", "=", ">", "?", "@", "~", "\_", "^", "\\left", "\\right", "\\hyperref" or end  
of input

[WARNING] Could not convert TeX math `{Enc}_k^{\prime}\left(m_0\right)\overset{\text{c}}{\equiv}{Enc}_k^{\prime}\left(m_1\right)`,  
rendering as TeX

2025-07-19 20:24:31,443 - WARNING - Could not convert TeX math

`{Enc}_k^{\prime}\left(m_0\right)\overset{\text{c}}{\equiv}{Enc}_k^{\prime}\left(m_1\right)`,  
rendering as TeX

[WARNING] Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`,  
rendering as TeX

2025-07-19 20:24:31,444 - WARNING - Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`,  
rendering as TeX

[WARNING] Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`,  
rendering as TeX

2025-07-19 20:24:31,445 - WARNING - Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`,  
rendering as TeX

[WARNING] Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`,  
rendering as TeX

2025-07-19 20:24:31,446 - WARNING - Could not convert TeX math `\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)\overset{\text{c}}{\equiv}\left(\text{pk},\text{Encaps}_{\text{pk}}^{\left(1\right)}\left(1^n\right)\right)`



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\mid b = 0 \right\rbrace + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{0} \right) \right) \right\rbrace \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{0} \right) \right) \right\rbrace \\
\quad + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{0} \right) \right) \right\rbrace = 1 \\
\right\rbrace \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{matrix}
\frac{1}{2} + \text{negl}^{\text{\text{prime}}}(n) \geq \text{\text{Pr}} \left\lbrack \text{\text{Priv}}_{\text{\text{K}}} \left\{ \mathcal{A}^{\text{\text{prime}}}, \text{\text{Enc}}_{\text{\text{K}}}^{\text{\text{prime}}} \right\}^{\text{\text{eav}}}(n) = 1 \right\rbrace \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{prime}}} \left\{ \text{\text{Outputs}}_0 \mid b = 0 \right\} + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{prime}}} \left\{ \text{\text{Outputs}}_1 \mid b = 1 \right\} \right\rbrace \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{0} \right) \right) \right\rbrace = 0 \\
\quad + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{1} \right) \right) \right\rbrace = 1 \\
\right\rbrace. \\
\end{matrix}, \text{rendering as TeX}
2025-07-19 20:24:31,451 - WARNING - Could not convert TeX math \begin{matrix}
\frac{1}{2} + \text{negl}^{\text{\text{prime}}}(n) \geq \text{\text{Pr}} \left\lbrack \text{\text{Priv}}_{\text{\text{K}}} \left\{ \mathcal{A}^{\text{\text{prime}}}, \text{\text{Enc}}_{\text{\text{K}}}^{\text{\text{prime}}} \right\}^{\text{\text{eav}}}(n) = 1 \right\rbrace \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{prime}}} \left\{ \text{\text{Outputs}}_0 \mid b = 0 \right\} + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{prime}}} \left\{ \text{\text{Outputs}}_1 \mid b = 1 \right\} \right\rbrace \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{0} \right) \right) \right\rbrace = 0 \\
\quad + \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}^{\text{\text{hy}}} \left(\text{\text{Encaps}}_{\text{\text{pk}}}^{\text{\text{(1)}}} \left(1^n \right), \text{\text{Enc}}_{\text{\text{k}}}^{\text{\text{prime}}} \left(m_{1} \right) \right) \right\rbrace = 1 \\
\right\rbrace. \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{matrix}
\frac{1}{2} + \text{negl}_2(n) \geq \text{\text{Pr}} \left\lbrack \text{\text{EM}}_{\mathcal{A}_2}, \text{\text{Enc}}_{\text{\text{K}}}^{\text{\text{cpa}}} \right\rbrace = 1 \\
\quad = \frac{1}{2} \cdot \text{\text{Pr}} \left\lbrack \mathcal{A}_2 \left\{ \text{\text{Outputs}}_0 \right\} \right\rbrace
\end{matrix}

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\mid b = 0} \right\rbrace + \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}_2\right\} \mid b = 1} \right\rbrace \\
= \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk}^{(1)} \left(1^n \right), \text{Enc}_k^{\text{prime}} \left(m_1 \right) \right) = 1} \right\rbrace \\
+ \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk}^{(1)} \left(1^n \right), \text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_1 \right) \right) = 0} \right\rbrace \\
\end{matrix}, \text{rendering as TeX}
2025-07-19 20:24:31,451 - WARNING - Could not convert TeX math \begin{matrix}
\frac{1}{2} + \text{negl}_2(n) \geq \text{Pr}\left\{\mathcal{A}_2\right\} \mid b = 0} \right\rbrace + \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}_2\right\} \mid b = 1} \right\rbrace \\
= \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk}^{(1)} \left(1^n \right), \text{Enc}_k^{\text{prime}} \left(m_1 \right) \right) = 1} \right\rbrace \\
+ \frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk}^{(1)} \left(1^n \right), \text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_1 \right) \right) = 0} \right\rbrace \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{array}{l}
\frac{3}{2} + \text{negl}(n) \geq \\
\left(\frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_k^{\text{'}} \left(m_0 \right) \right) \right) = 0 \right\} + \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_0 \right) \right) \right) = 1 \right\} \right) \\
\left(\frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_0 \right) \right) \right) = 0 \right\} + \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_1 \right) \right) \right) = 1 \right\} \right) \\
\left(\frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_1 \right) \right) \right) = 1 \right\} + \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_1 \right) \right) \right) = 0 \right\} \right) \\
\end{array}, \text{rendering as TeX}
2025-07-19 20:24:31,452 - WARNING - Could not convert TeX math \begin{array}{l}
\frac{3}{2} + \text{negl}(n) \geq \\
\left(\frac{1}{2} \cdot \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_k^{\text{'}} \left(m_0 \right) \right) \right) = 0 \right\} + \text{Pr}\left\{\mathcal{A}^{\text{hy}}\right\} \left(\text{Enc}_{pk,c} \left(\text{Enc}_{k^{\text{prime}}}^{\text{prime}} \left(m_0 \right) \right) \right) = 1 \right\} \right)

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\right\rbrace \\
{\,\,,\,, + \text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_0 \right)\right) \right) = 0}
\right\rbrace + \text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_1 \right)\right) \right) = 1}
\right\rbrace \\
{\,\,,\,,\,, + \text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_1 \right)\right) \right) = 1\right\lbrack +
\text{Pr}\right\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_1 \right)\right) \right) = 0}
\right\rbrace),}
\end{array}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{array}{l}
{\frac{1}{2}} + \text{negl}(n) \\
{\,\,,\,,\,, \geq \frac{1}{2} \cdot \left({\text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_0 \right)\right) \right) = 0} \right\lbrack +
\text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_1 \right)\right) \right) = 1} \right\lbrack} \right) \\
{\,\,,\,,\,, = \text{Pr}\left\lbrack
\text{Pub}\text{K}_{\mathcal{A}^{\text{ny}}},\text{Pr}^{\text{hy}}\text{eav}(n) = 1}
\right\rbrace}
\end{array}, \text{rendering as TeX}
2025-07-19 20:24:31,453 - WARNING - Could not convert TeX math \begin{array}{l}
{\frac{1}{2}} + \text{negl}(n) \\
{\,\,,\,,\,, \geq \frac{1}{2} \cdot \left({\text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_0 \right)\right) \right) = 0} \right\lbrack +
\text{Pr}\left\lbrack {\mathcal{A}^{\text{hy}}}\left(
\text{pk},c,\text{Enc}_{k^{\prime}}^{\prime}\left(m_1 \right)\right) \right) = 1} \right\lbrack} \right) \\
{\,\,,\,,\,, = \text{Pr}\left\lbrack
\text{Pub}\text{K}_{\mathcal{A}^{\text{ny}}},\text{Pr}^{\text{hy}}\text{eav}(n) = 1}
\right\rbrace}
\end{array}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack
\text{K}_{\text{EM}_{\mathcal{A}}},\text{Pr}^{\text{cca}}(n) = 1} \right\rbrace \leq
\frac{1}{2} + \text{negl}(n)., \text{rendering as TeX}
2025-07-19 20:24:31,454 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack {\text{K}_{\text{EM}_{\mathcal{A}}},\text{Pr}^{\text{cca}}(n) = 1}
\right\rbrace \leq \frac{1}{2} + \text{negl}(n)., \text{rendering as TeX}
[WARNING] Could not convert TeX math \hat{m} = \frac{c_2}{c_1^x} =
\frac{h^y}{\left(g^y \right)^x} = \frac{\left(g^x \right)^y}{\left(g^{xy} \right)} = \frac{g^{xy}}{\left(g^{xy} \right)} = m., \text{rendering as TeX}
2025-07-19 20:24:31,454 - WARNING - Could not convert TeX math \hat{m} =
\frac{c_2}{c_1^x} = \frac{h^y}{\left(g^y \right)^x} =
\frac{\left(g^x \right)^y}{\left(g^{xy} \right)} = \frac{g^{xy}}{\left(g^{xy} \right)} = m., \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack

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{{Pub}}\text{K}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1} \right\rbrack \leq
\frac{1}{2} + \text{negl}(n)., rendering as TeX
2025-07-19 20:24:31,455 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack {{Pub}}\text{K}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1}
\right\rbrack \leq \frac{1}{2} + \text{negl}(n)., rendering as TeX
[WARNING] Could not convert TeX math \widetilde{\prod}, rendering as TeX
2025-07-19 20:24:31,456 - WARNING - Could not convert TeX math
\widetilde{\prod}, rendering as TeX
[WARNING] Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
2025-07-19 20:24:31,457 - WARNING - Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
[WARNING] Could not convert TeX math \widetilde{\prod}, rendering as TeX
2025-07-19 20:24:31,458 - WARNING - Could not convert TeX math
\widetilde{\prod}, rendering as TeX
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack
{{Pub}}\text{K}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n) = 1}
\right\rbrack = \frac{1}{2}., rendering as TeX
2025-07-19 20:24:31,459 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack
{{Pub}}\text{K}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n) = 1}
\right\rbrack = \frac{1}{2}., rendering as TeX
[WARNING] Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
2025-07-19 20:24:31,460 - WARNING - Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
[WARNING] Could not convert TeX math \text{Pr}\left\lbrack \mathbb{D}\left(
\{\mathbb{G},q,g,g^x,g^y,g^z\} \right) = 1} \right\rbrack =
\text{Pr}\left\lbrack {{Pub}}\text{K}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n)
= 1} \right\rbrack = \frac{1}{2}., rendering as TeX
2025-07-19 20:24:31,461 - WARNING - Could not convert TeX math
\text{Pr}\left\lbrack \mathbb{D}\left(\{\mathbb{G},q,g,g^x,g^y,g^z\} \right) = 1}
\right\rbrack = \text{Pr}\left\lbrack
{{Pub}}\text{K}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n) = 1} \right\rbrack =
\frac{1}{2}., rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
\{\text{negl}(n) \geq \left| \text{Pr}\left\lbrack \mathbb{D}\left(
\{\mathbb{G},q,g,g^x,g^y,g^z\} \right) = 1} \right\rbrack -
\text{Pr}\left\lbrack \mathbb{D}\left(\{\mathbb{G},q,g,g^x,g^y,g^{xy}\} \right) = 1}
\right\rbrack} \right|} \\
\{= \left| \frac{1}{2} - \text{Pr}\left\lbrack
{{Pub}}\text{K}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1} \right\rbrack
\right|} \\
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,462 - WARNING - Could not convert TeX math \begin{matrix}
\{\text{negl}(n) \geq \left| \text{Pr}\left\lbrack \mathbb{D}\left(
\{\mathbb{G},q,g,g^x,g^y,g^z\} \right) = 1} \right\rbrack -
\text{Pr}\left\lbrack \mathbb{D}\left(\{\mathbb{G},q,g,g^x,g^y,g^{xy}\} \right) = 1}
\right\rbrack} \right|} \\
\end{matrix}

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\right\rbrace \right|} \\\
{= \left| \left\{ \frac{1}{2} - \text{Pr} \left\{ \text{Pub} \right\}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \right\} \right|} \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr} \left\{ \text{Pub} \right\}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n), \text{rendering as TeX}
2025-07-19 20:24:31,463 - WARNING - Could not convert TeX math \text{Pr} \left\{ \text{Pub} \right\}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right\} \leq \frac{1}{2} + \text{negl}(n), \text{rendering as TeX}
[WARNING] Could not convert TeX math \frac{1}{2} \cdot \sum \limits_k \in \{0, 1\}^{\ell(n)} \left| \left\{ \text{Pr} \left\{ H(g) = k \right\} - 2^{-\ell(n)} \right\} \right| \leq \text{negl}(n),, \text{rendering as TeX}
2025-07-19 20:24:31,464 - WARNING - Could not convert TeX math \frac{1}{2} \cdot \sum \limits_k \in \{0, 1\}^{\ell(n)} \left| \left\{ \text{Pr} \left\{ H(g) = k \right\} - 2^{-\ell(n)} \right\} \right| \leq \text{negl}(n),, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right\} \leq \frac{1}{2} + \text{neg} \mid (n)., \text{rendering as TeX}
2025-07-19 20:24:31,465 - WARNING - Could not convert TeX math \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right\} \leq \frac{1}{2} + \text{neg} \mid (n)., \text{rendering as TeX}
[WARNING] Could not convert TeX math \begin{matrix} \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{ca}}(n) = 1 \right\} \right\} \\ \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \\ \overline{\left\{ \text{Query} \right\}} \right\} \end{matrix} \\
{+ \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \left\{ \text{Query} \right\} \right\} \\ \leq \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \overline{\left\{ \text{Query} \right\}} \right\} + \\ \left\{ \text{Pr} \left\{ \text{Query} \right\} \right\} } \\
\end{matrix}, \text{rendering as TeX}
2025-07-19 20:24:31,466 - WARNING - Could not convert TeX math \begin{matrix} \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{ca}}(n) = 1 \right\} \right\} \\ \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \\ \overline{\left\{ \text{Query} \right\}} \right\} \\
{+ \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \left\{ \text{Query} \right\} \right\} \\ \leq \left\{ \text{Pr} \left\{ \text{K} \right\}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \} \wedge \overline{\left\{ \text{Query} \right\}} \right\} + \\ \left\{ \text{Pr} \left\{ \text{Query} \right\} \right\} } \\
\end{matrix}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \left\{ \text{Pr} \left\{ \overline{\left\{ \text{Query} \right\}} \right\} \right\} = 0, \text{rendering as TeX}
2025-07-19 20:24:31,467 - WARNING - Could not convert TeX math

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$\text{Pr}\}\overline{\text{Query}}\} = 0$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}\right\}$   
 $\text{Pr}\}\right\} = 0$ , rendering as TeX  
 2025-07-19 20:24:31,468 - WARNING - Could not convert TeX math  
 $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}\right\}$   $\text{Pr}\}\right\} = 0$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \wedge \overset{\text{Query}}{\rightarrow}\right\} \right\} = \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \overset{\text{Query}}{\rightarrow}\right\} \right\} \cdot \\ \text{Pr}\}\overline{\text{Query}}\} \\ \leq \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \text{Query}\right\} \right\}. \end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,469 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \wedge \overset{\text{Query}}{\rightarrow}\right\} \right\} = \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \overset{\text{Query}}{\rightarrow}\right\} \right\} \cdot \\ \text{Pr}\}\overline{\text{Query}}\} \\ \leq \text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \text{Query}\right\} \right\}. \end{matrix}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $k\overset{\text{def}}{=}H\left(h^y\right)$ , rendering as TeX  
 2025-07-19 20:24:31,469 - WARNING - Could not convert TeX math  $k\overset{\text{def}}{=}H\left(h^y\right)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \overset{\text{Query}}{\rightarrow}\right\} \right\} = \frac{1}{2}.$ , rendering as TeX  
 2025-07-19 20:24:31,471 - WARNING - Could not convert TeX math  
 $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1 \mid \overset{\text{Query}}{\rightarrow}\right\} \right\} = \frac{1}{2}.$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1\right\} \right\} \leq \frac{1}{2} + \text{Pr}\}\text{Query}\}$ , rendering as TeX  
 2025-07-19 20:24:31,472 - WARNING - Could not convert TeX math  
 $\text{Pr}\}\left\{\text{KEM}_{\mathcal{A},\prod}^{\text{cpa}}(n) = 1\right\} \right\} \leq \frac{1}{2} + \text{Pr}\}\text{Query}\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\mathcal{O}\left(\sqrt{B}\right)$ , rendering as TeX  
 2025-07-19 20:24:31,472 - WARNING - Could not convert TeX math  $\mathcal{O}\left(\sqrt{B}\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\alpha \in \left( \frac{1}{2}, 1 \right)$ , rendering as TeX

2025-07-19 20:24:31,473 - WARNING - Could not convert TeX math  $\alpha \in \left( \frac{1}{2}, 1 \right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\alpha \approx \frac{1}{2}$ , rendering as TeX

2025-07-19 20:24:31,473 - WARNING - Could not convert TeX math  $\alpha \approx \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left\lceil \left\lfloor s^e \bmod N \right\rfloor \right\rceil x_r$ , rendering as TeX

2025-07-19 20:24:31,475 - WARNING - Could not convert TeX math  $\left\lceil \left\lfloor s^e \bmod N \right\rfloor \right\rceil x_r$ , rendering as TeX

[WARNING] Could not convert TeX math  $\mathcal{O}(\sqrt{B})$ , rendering as TeX

2025-07-19 20:24:31,476 - WARNING - Could not convert TeX math  $\mathcal{O}(\sqrt{B})$ , rendering as TeX

[WARNING] Could not convert TeX math  $p(x) \overset{\text{def}}{=}$ , rendering as TeX

2025-07-19 20:24:31,476 - WARNING - Could not convert TeX math  $p(x) \overset{\text{def}}{=}$ , rendering as TeX

[WARNING] Could not convert TeX math  $f_1(x) \overset{\text{def}}{=} x^e - c_1$ , rendering as TeX

2025-07-19 20:24:31,477 - WARNING - Could not convert TeX math  $f_1(x) \overset{\text{def}}{=} x^e - c_1$ , rendering as TeX

[WARNING] Could not convert TeX math  $f_2(x) \overset{\text{def}}{=} \left( x + \delta \right)^e - c_2$ , rendering as TeX

2025-07-19 20:24:31,477 - WARNING - Could not convert TeX math  $f_2(x) \overset{\text{def}}{=} \left( x + \delta \right)^e - c_2$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \hat{c} = c_1 \bmod N_1 \\ \hat{c} = c_2 \bmod N_2 \\ \hat{c} = c_3 \bmod N_3. \end{array}$ , rendering as TeX

2025-07-19 20:24:31,478 - WARNING - Could not convert TeX math  $\begin{array}{l} \hat{c} = c_1 \bmod N_1 \\ \hat{c} = c_2 \bmod N_2 \\ \hat{c} = c_3 \bmod N_3. \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $m = \left. b \right| \underset{L}{\underset{\{0 \ldots 0\}}$ , rendering as TeX

2025-07-19 20:24:31,480 - WARNING - Could not convert TeX math  $m = \left. b \right| \underset{L}{\underset{\{0 \ldots 0\}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $c^{\prime} = \left( \frac{0 \text{x} 00 \parallel 0 \text{x} 02 \parallel r \parallel 0 \text{x} 00 \parallel b \parallel 0 \cdots 0 \parallel 10 \cdots 0 \right)^e = \{ (0 \text{x} 02 \parallel r \parallel 0 \text{x} 00 \parallel b) \}^e \bmod N$ , rendering as TeX

2025-07-19 20:24:31,481 - WARNING - Could not convert TeX math  $c^{\prime} =$

$$\left( \frac{0}{\text{x}^{00}} \parallel 0 \text{x}^{02} \parallel r \parallel \text{0x}^{00} \parallel b \parallel 0 \cdots 0 \right)^e = \left( \frac{0}{\text{x}^{02}} \parallel r \parallel 0 \text{x}^{00} \parallel b \right)^e \pmod{N}, \text{ rendering as TeX}$$

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}, \text{GenRSA}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,482 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}, \text{GenRSA}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l} \text{Pr} \left[ \text{Pub} \text{K} \right]_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}}(N, e, c) = 0 \mid c, \text{is-an-encryption-of} \right] \cdot \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}}(N, e, c) = 1 \mid c, \text{is-an-encryption-of} \right] \end{array}$ , rendering as TeX

2025-07-19 20:24:31,482 - WARNING - Could not convert TeX math  $\begin{array}{l} \text{Pr} \left[ \text{Pub} \text{K} \right]_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}}(N, e, c) = 0 \mid c, \text{is-an-encryption-of} \right] \cdot \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}}(N, e, c) = 1 \mid c, \text{is-an-encryption-of} \right] \end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr} \left[ \text{RSA} - \text{Isb} \right]_{\mathcal{A}, \text{GenRSA}}(n) = 1 \right] \leq \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}} \left( \begin{matrix} N, e, \left[ r^e \pmod{N} \right] \end{matrix} \right) = 0 \mid \text{Isb}(r) = 0 \right] + \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}} \left( \begin{matrix} N, e, \left[ r^e \pmod{N} \right] \end{matrix} \right) = 1 \mid \text{Isb}(r) = 1 \right] \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,483 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr} \left[ \text{RSA} - \text{Isb} \right]_{\mathcal{A}, \text{GenRSA}}(n) = 1 \right] \leq \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}} \left( \begin{matrix} N, e, \left[ r^e \pmod{N} \right] \end{matrix} \right) = 0 \mid \text{Isb}(r) = 0 \right] + \frac{1}{2} \cdot \text{Pr} \left[ \text{Pr} \left[ \text{RSA-lsb} \right]_{\mathcal{A}} \left( \begin{matrix} N, e, \left[ r^e \pmod{N} \right] \end{matrix} \right) = 1 \mid \text{Isb}(r) = 1 \right] \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \text{Pub} \text{K} \right]_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,486 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \text{Pub} \text{K} \right]_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1$

$\right\rbrace \left\lceil \frac{1}{2} + \{ \text{neg} \} (n) \right\rceil$ , rendering as TeX  
[WARNING] Could not convert TeX math  $f(x) \overset{\{\text{~def~}\}}{=} \left\lceil \left\lfloor x^e \right\rfloor \right\rceil \pmod{N}$  \right\rbrace, rendering as TeX  
2025-07-19 20:24:31,487 - WARNING - Could not convert TeX math  $f(x) \overset{\{\text{~def~}\}}{=} \left\lceil \left\lfloor x^e \right\rfloor \right\rceil \pmod{N}$  \right\rbrace, rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix} \{c^{\backslash prime} := \left\lceil \left\lfloor 2^e \right\rfloor \cdot c \right\rfloor \pmod{N} \right\rceil \\ \{= 2^e \cdot m^e = (2m)^e \pmod{N}.\} \\ \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,487 - WARNING - Could not convert TeX math  $\begin{matrix} \{c^{\backslash prime} := \left\lceil \left\lfloor 2^e \right\rfloor \cdot c \right\rfloor \pmod{N} \right\rceil \\ \{= 2^e \cdot m^e = (2m)^e \pmod{N}.\} \\ \end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix} \{t := m \backslash prime \oplus G(r),\} \& \{s := r \oplus H(t)\} \\ \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,488 - WARNING - Could not convert TeX math  $\begin{matrix} \{t := m \backslash prime \oplus G(r),\} \& \{s := r \oplus H(t)\} \\ \end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $k \overset{\{\text{~def~}\}}{=} H(r)$ , rendering as TeX  
2025-07-19 20:24:31,489 - WARNING - Could not convert TeX math  $k \overset{\{\text{~def~}\}}{=} H(r)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix} \{\text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \right\rceil = \text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \\ \overline{\{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil} + \text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \\ \{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \overline{\{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil} + \\ \text{\text{Pr}} \backslash \left\lceil \text{Query} \right\rceil \backslash \left\lceil \text{Success} \right\rceil \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,489 - WARNING - Could not convert TeX math  $\begin{matrix} \{\text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \right\rceil = \text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \\ \overline{\{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil} + \text{\text{Pr}} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \\ \{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil \ \text{\text{and}} \overline{\{\text{Query}\} \backslash \left\lceil \text{Success} \right\rceil} + \\ \text{\text{Pr}} \backslash \left\lceil \text{Query} \right\rceil \backslash \left\lceil \text{Success} \right\rceil \end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\text{Pr}} \backslash \left\lceil \text{~Success~} \right\rceil \ \text{\text{and}} \overline{\{\text{~Query~}\} \backslash \left\lceil \frac{1}{2} \right\rceil}$ , rendering as TeX  
2025-07-19 20:24:31,490 - WARNING - Could not convert TeX math  $\text{\text{Pr}} \backslash \left\lceil \text{~Success~} \right\rceil \ \text{\text{and}} \overline{\{\text{~Query~}\} \backslash \left\lceil \frac{1}{2} \right\rceil}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\text{Pr}} \backslash \left\lceil \text{~Success~} \right\rceil \ \text{\text{and}} \overline{\{\text{~Query~}\} \backslash \left\lceil \frac{1}{2} \right\rceil}$ , rendering as TeX  
2025-07-19 20:24:31,491 - WARNING - Could not convert TeX math  $\text{\text{Pr}} \backslash \left\lceil \text{~Success~} \right\rceil \ \text{\text{and}} \overline{\{\text{~Query~}\} \backslash \left\lceil \frac{1}{2} \right\rceil}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] = 0$ , rendering as TeX  
2025-07-19 20:24:31,493 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] = 0$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} \leq \text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}}$ , rendering as TeX  
2025-07-19 20:24:31,494 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} \leq \text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overline{\text{Query}}$ , rendering as TeX  
2025-07-19 20:24:31,495 - WARNING - Could not convert TeX math  $\overline{\text{Query}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} = \frac{1}{2}$ , rendering as TeX  
2025-07-19 20:24:31,496 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} = \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} \leq \frac{1}{2}$ , rendering as TeX  
2025-07-19 20:24:31,497 - WARNING - Could not convert TeX math  $\text{Pr} \left[ \overline{\text{Query}} \right] \wedge \overline{\text{Success}} \mid \overline{\text{Query}} \leq \frac{1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix} ( \left[ 64 \bmod 11 \right] \left[ 64 \bmod 13 \right] )^{103} = \left( \left[ ( - 2 )^{103} \bmod 11 \right] \left[ ( - 1 )^{103} \bmod 13 \right] \right) \\ \begin{array}{l} = \left( \left[ ( - 2 )^{103} \bmod 10 \right] \bmod 11 \right) \left[ ( - 1 ) \right] \\ = ( \left[ - 8 \bmod 11 \right] , - 1 ) = ( 3 , - 1 ) . \end{array} \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,498 - WARNING - Could not convert TeX math  $\begin{matrix} ( \left[ 64 \bmod 11 \right] \left[ 64 \bmod 13 \right] )^{103} = \left( \left[ ( - 2 )^{103} \bmod 11 \right] \left[ ( - 1 )^{103} \bmod 13 \right] \right) \\ \begin{array}{l} = \left( \left[ ( - 2 )^{103} \bmod 10 \right] \bmod 11 \right) \left[ ( - 1 ) \right] \\ = ( \left[ - 8 \bmod 11 \right] , - 1 ) = ( 3 , - 1 ) . \end{array} \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Vrfy}_{\text{pk}} \left( H^s(m), \sigma \right) \overset{?}{=} 1$ , rendering as TeX  
2025-07-19 20:24:31,499 - WARNING - Could not convert TeX math

$\text{Vrfy}_{\text{pk}}(\text{H}^s(m), \sigma)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $m \overset{?}{=}\sigma^e$ , rendering as TeX  
 2025-07-19 20:24:31,500 - WARNING - Could not convert TeX math  
 $m \overset{?}{=}\sigma^e$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $m \overset{?}{=}\lfloor \sigma^e \bmod N \rfloor$ , rendering as TeX  
 2025-07-19 20:24:31,501 - WARNING - Could not convert TeX math  
 $m \overset{?}{=}\lfloor \sigma^e \bmod N \rfloor$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sigma^e \overset{?}{=} H(m)$ , rendering as TeX  
 2025-07-19 20:24:31,501 - WARNING - Could not convert TeX math  
 $\sigma^e \overset{?}{=} H(m)$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\sigma^e \overset{?}{=} H(m) \bmod N$ , rendering as TeX  
 2025-07-19 20:24:31,502 - WARNING - Could not convert TeX math  
 $\sigma^e \overset{?}{=} H(m) \bmod N$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}(n) = 1 \right) = \frac{1}{q(n)} \cdot \text{Pr}(\left\{ \text{Sig-forge}^{\sim}_{\mathcal{A}, \Pi}(n) = 1 \right\})$ , rendering as TeX  
 2025-07-19 20:24:31,503 - WARNING - Could not convert TeX math  
 $\text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}(n) = 1 \right) = \frac{1}{q(n)} \cdot \text{Pr}(\left\{ \text{Sig-forge}^{\sim}_{\mathcal{A}, \Pi}(n) = 1 \right\})$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}^{\text{prime}}(n) = 1 \right) = \\ \text{Pr}(\left\{ \text{Sig-forge}^{\sim}_{\mathcal{A}, \Pi}^{\text{prime}}(n) = 1 \right\}) \\ = \frac{\text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}(n) = 1 \right)}{q(n)} \end{matrix}$ , rendering as TeX  
 2025-07-19 20:24:31,504 - WARNING - Could not convert TeX math  $\begin{matrix} \text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}^{\text{prime}}(n) = 1 \right) = \\ \text{Pr}(\left\{ \text{Sig-forge}^{\sim}_{\mathcal{A}, \Pi}^{\text{prime}}(n) = 1 \right\}) \\ = \frac{\text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}(n) = 1 \right)}{q(n)} \end{matrix}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\begin{matrix} \text{Pr}(\left\{ \text{RSA-inv}_{\mathcal{A}^{\text{prime}}}, \text{GenRSA}(n) = 1 \right\} \\ \text{Pr}(\left\{ \text{Sig-forge}^{\text{e}} \right\}_{\mathcal{A}, \Pi}^{\text{prime}}(n) = 1 \right) \\ = \frac{\text{Pr}(\left\{ \text{Sig}^{\sim}_{\mathcal{A}, \Pi}(n) = 1 \right\})}{q(n)} \end{matrix}$



$$\begin{matrix} \end{matrix}, \text{rendering as TeX}$$
2025-07-19 20:24:31,506 - WARNING - Could not convert TeX math  $\begin{matrix} \left\{ \text{Pr} \right\} \left\{ \text{RSA} - \text{inv}_{\mathcal{A}}^{\prime}, \text{GenRSA} \right\} (n) = 1 \right\} \left\{ \text{Sig} - \text{forge}_{\mathcal{A}, \Pi}^{\prime}(n) = 1 \right\} \right\} \left\{ \frac{\left\{ \text{Pr} \right\} \left\{ \text{Sig} - \text{forge}_{\mathcal{A}, \Pi}(n) = 1 \right\}}{q(n)} \right\} \end{matrix}, \text{rendering as TeX}$ 
[WARNING] Could not convert TeX math  $\mathcal{V}(\text{pk}, \text{r}, \text{s})^{\text{?}} = \text{I}$ , rendering as TeX
2025-07-19 20:24:31,507 - WARNING - Could not convert TeX math  $\mathcal{V}(\text{pk}, \text{r}, \text{s})^{\text{?}} = \text{I}$ , rendering as TeX
[WARNING] Could not convert TeX math  $\text{H}(\text{I}, \text{m})^{\text{?}} = \text{r}$ , rendering as TeX
2025-07-19 20:24:31,507 - WARNING - Could not convert TeX math  $\text{H}(\text{I}, \text{m})^{\text{?}} = \text{r}$ , rendering as TeX
[WARNING] Could not convert TeX math  $\text{H}(\text{I}, \text{m})^{\text{?}} = \text{r}.$ , rendering as TeX
2025-07-19 20:24:31,509 - WARNING - Could not convert TeX math  $\text{H}(\text{I}, \text{m})^{\text{?}} = \text{r}.$ , rendering as TeX
[WARNING] Could not convert TeX math  $(\text{I}, \text{m})^{\text{?}} = \left( \text{I}_{\text{j}}, \text{m}_{\text{j}} \right)$ , rendering as TeX
2025-07-19 20:24:31,509 - WARNING - Could not convert TeX math  $(\text{I}, \text{m})^{\text{?}} = \left( \text{I}_{\text{j}}, \text{m}_{\text{j}} \right)$ , rendering as TeX
[WARNING] Could not convert TeX math  $\left\{ \text{Pr} \right\} \left\{ \text{Ident}_{\mathcal{A}, \Pi}(n) = 1 \right\} \right\} \geq \frac{1}{q(n)} \cdot \left( \left\{ \text{Pr} \right\} \left\{ \text{Sig-forge}_{\mathcal{A}^{\prime}, \Pi^{\prime}}(n) = 1 \right\} \right\} \neg(n) \right)$ , rendering as TeX
2025-07-19 20:24:31,510 - WARNING - Could not convert TeX math  $\left\{ \text{Pr} \right\} \left\{ \text{Ident}_{\mathcal{A}, \Pi}(n) = 1 \right\} \right\} \geq \frac{1}{q(n)} \cdot \left( \left\{ \text{Pr} \right\} \left\{ \text{Sig-forge}_{\mathcal{A}^{\prime}, \Pi^{\prime}}(n) = 1 \right\} \right\} \neg(n) \right)$ , rendering as TeX
[WARNING] Could not convert TeX math  $g^{\text{s}} \cdot y^{\text{-r}}^{\text{?}} = \text{I}$ , rendering as TeX
2025-07-19 20:24:31,510 - WARNING - Could not convert TeX math  $g^{\text{s}} \cdot y^{\text{-r}}^{\text{?}} = \text{I}$ , rendering as TeX
[WARNING] Could not convert TeX math  $\Delta_{\omega}^{\text{?}} = \left\{ \text{Pr} \right\}_{\text{r}} \left\{ V(\omega, \text{r}) = 1 \right\}$ , rendering as TeX
2025-07-19 20:24:31,511 - WARNING - Could not convert TeX math  $\Delta_{\omega}^{\text{?}} = \left\{ \text{Pr} \right\}_{\text{r}} \left\{ V(\omega, \text{r}) = 1 \right\}$ , rendering as TeX
[WARNING] Could not convert TeX math  $\Delta(n)^{\text{?}} = \left\{ \text{Pr} \right\} \left\{ \text{Ident}_{\mathcal{A}, \Pi}(n) = 1 \right\} \right\}$ , rendering as TeX
2025-07-19 20:24:31,511 - WARNING - Could not convert TeX math  $\Delta(n)^{\text{?}} = \left\{ \text{Pr} \right\} \left\{ \text{Ident}_{\mathcal{A}, \Pi}(n) = 1 \right\} \right\}$ , rendering as TeX



2025-07-19 20:24:31,516 - WARNING - Could not convert TeX math  
F(I)\overset{\text{~def~}}{=}\lbrack I\mathrm{mod}q\rbrack, rendering as TeX

[WARNING] Could not convert TeX math  $F((x,y))\overset{\text{~def~}}{=}\lbrack x\mathrm{mod}q\rbrack$ , rendering as TeX

2025-07-19 20:24:31,517 - WARNING - Could not convert TeX math  
 $F((x,y))\overset{\text{~def~}}{=}\lbrack x\mathrm{mod}q\rbrack$ , rendering as TeX

[WARNING] Could not convert TeX math  $r\overset{?}{=}F\left( \left\{ g^H(m) \cdot s^{-1} \right\} y^{r \cdot s^{-1}} \right)$ ., rendering as TeX

2025-07-19 20:24:31,518 - WARNING - Could not convert TeX math  
 $r\overset{?}{=}F\left( \left\{ g^H(m) \cdot s^{-1} \right\} y^{r \cdot s^{-1}} \right)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}$

$s_1 = k^{-1} \cdot \left( H\left( m_1 \right) + xr \right) \mathrm{mod} q \\ s_2 = k^{-1} \cdot \left( H\left( m_2 \right) + xr \right) \mathrm{mod} q.$

$\end{array}$ , rendering as TeX

2025-07-19 20:24:31,519 - WARNING - Could not convert TeX math  $\begin{array}{l}$

$s_1 = k^{-1} \cdot \left( H\left( m_1 \right) + xr \right) \mathrm{mod} q \\ s_2 = k^{-1} \cdot \left( H\left( m_2 \right) + xr \right) \mathrm{mod} q.$

$\end{array}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{cert}_{C\rightarrow B}\overset{\text{def}}{=}\text{Sign}_{sk_C}\left( \text{'Bob's-key-is'}, pk_B \right)$ , rendering as TeX

2025-07-19 20:24:31,521 - WARNING - Could not convert TeX math  
 $\text{cert}_{C\rightarrow B}\overset{\text{def}}{=}\text{Sign}_{sk_C}\left( \text{'Bob's-key-is'}, pk_B \right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{cert}_B\rightarrow A\overset{\text{def}}{=}\text{Sign}_{sk_B}\left( \text{'Alice's-key-is'}pk_A \right)$ ., rendering as TeX

2025-07-19 20:24:31,522 - WARNING - Could not convert TeX math  
 $\text{cert}_B\rightarrow A\overset{\text{def}}{=}\text{Sign}_{sk_B}\left( \text{'Alice's-key-is'}pk_A \right)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{cert}_{C\rightarrow B}\overset{\text{def}}{=}\text{Sign}_{sk_C}\left( \text{'Bob's-key-is'}, pk_B, \text{date} \right)$ ., rendering as TeX

2025-07-19 20:24:31,522 - WARNING - Could not convert TeX math  
 $\text{cert}_{C\rightarrow B}\overset{\text{def}}{=}\text{Sign}_{sk_C}\left( \text{'Bob's-key-is'}, pk_B, \text{date} \right)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{cert}_B\rightarrow A\overset{\text{def}}{=}\text{Sign}_{sk_B}\left( \text{'Alice's-key-is'}pk_A \right)$ ., rendering as TeX

2025-07-19 20:24:31,523 - WARNING - Could not convert TeX math  
 $\text{cert}_{C\rightarrow B}\overset{\text{def}}{=}\text{Sign}_{sk_C}\left( \text{'Bob's-key-is'}, pk_B, \text{date} \right)$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\left. \kappa \right|_{N,m} \parallel N \parallel , \left. \kappa \right|_{m,N} \parallel m \parallel$ , rendering as TeX

2025-07-19 20:24:31,524 - WARNING - Could not convert TeX math  $\left. \kappa \right|_{N,m} \parallel N \parallel , \left. \kappa \right|_{m,N} \parallel m \parallel$

$\overset{\text{~def~}}{=}$  $4\kappa/5$  \right., rendering as TeX  
[WARNING] Could not convert TeX math  $\left| m \middle| \overset{\text{~def~}}{=}(\parallel N \parallel - 1)/2 \right.$ , rendering as TeX  
2025-07-19 20:24:31,525 - WARNING - Could not convert TeX math  $\left| m \middle| \overset{\text{~def~}}{=}(\parallel N \parallel - 1)/2 \right.$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathrm{cal}_0\left( {\ell} + \sqrt{{\ell}^2}\right) = \mathrm{cal}_0({\ell})$ , rendering as TeX  
2025-07-19 20:24:31,526 - WARNING - Could not convert TeX math  $\mathrm{cal}_0\left( {\ell} + \sqrt{{\ell}^2}\right) = \mathrm{cal}_0({\ell})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{{\ell}}{{2^n}}$ , rendering as TeX  
2025-07-19 20:24:31,526 - WARNING - Could not convert TeX math  $\frac{{\ell}}{{2^n}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $1 - \left( {1 - \frac{{\ell}}{{2^n}}} \right)^{{\ell}^2} \geq 1 - e^{-{\ell}^3/2^n}$ , rendering as TeX  
2025-07-19 20:24:31,528 - WARNING - Could not convert TeX math  $1 - \left( {1 - \frac{{\ell}}{{2^n}}} \right)^{{\ell}^2} \geq 1 - e^{-{\ell}^3/2^n}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\sqrt{\sum_i e_i^2}$ , rendering as TeX  
2025-07-19 20:24:31,529 - WARNING - Could not convert TeX math  $\sqrt{\sum_i e_i^2}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{r} \mid \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m \times n}; \text{s} \rightarrow \psi^n; \text{e} \rightarrow \psi^m : \mathrm{cal}_A(\text{B}, [\text{Bs}] + \mathbf{e} \bmod q \rceil ) = 1 \right\rceil \\ \{\backslash,\backslash,\backslash,- \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m \times n}; \mathbf{t} \rightarrow \mathbb{Z} \right\rceil_q^m : \mathrm{cal}_A(\text{B}, \text{t}) = 1 \right\rceil \leq \text{negl}(n). \end{array}$ , rendering as TeX  
2025-07-19 20:24:31,530 - WARNING - Could not convert TeX math  $\begin{array}{r} \mid \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m \times n}; \text{s} \rightarrow \psi^n; \text{e} \rightarrow \psi^m : \mathrm{cal}_A(\text{B}, [\text{Bs}] + \mathbf{e} \bmod q \rceil ) = 1 \right\rceil \\ \{\backslash,\backslash,\backslash,- \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m \times n}; \mathbf{t} \rightarrow \mathbb{Z} \right\rceil_q^m : \mathrm{cal}_A(\text{B}, \text{t}) = 1 \right\rceil \leq \text{negl}(n). \end{array}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{array}{l} \mid \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m'} \times \ell; \text{s} \rightarrow \psi^\ell; \text{e} \rightarrow \psi^{m'} : \mathrm{cal}_A(\text{B}, \lfloor \text{Bs} \rfloor + \text{e} \bmod q \rceil ) = 1 \right\rceil \\ \{\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,\backslash,- \text{Pr} \left\lceil \text{B} \rightarrow \mathbb{Z} \right\rceil_q^{m'} \times \ell; \text{t} \rightarrow \mathbb{Z} \right\rceil_q^{m'} : \mathrm{cal}_A(\text{B}, \text{t}) = 1 \right\rceil \leq \text{negl}(n). \end{array}$ , rendering as TeX



rendering as TeX

2025-07-19 20:24:31,538 - WARNING - Could not convert TeX math  $\text{c}^T = \{\hat{\text{s}}\}^T \cdot \text{B} \mid \text{t}\} + \{\hat{\text{e}}\}^T + \text{b}^T$ ,, rendering as TeX

[WARNING] Could not convert TeX math  $\mathbf{b}^T = \left[ 0, \dots, 0, b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right] \cdot \text{B}$ , rendering as TeX  
2025-07-19 20:24:31,539 - WARNING - Could not convert TeX math  $\mathbf{b}^T = \left[ 0, \dots, 0, b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right] \cdot \text{B}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix}$

```
{k = \text{c}^T \cdot \begin{matrix}
{- \text{s}} \\
1
\end{matrix}} \\
\end{matrix}} \\
\begin{array}{l}
{= \left(\{\hat{\text{s}}\}^T \cdot \text{B} \mid \text{t}\} + \{\hat{\text{e}}\}^T + \text{b}^T \right) \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + \{\hat{\text{s}}\}^T \cdot \text{t} + \{\hat{\text{e}}\}^T \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + b \cdot \left\lfloor \frac{q}{2} \right\rfloor} \\
{= \{\hat{\text{s}}\}^T \cdot \text{e} + \{\hat{\text{e}}\}^T \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + b \cdot \left\lfloor \frac{q}{2} \right\rfloor} \\
\end{array} \\
\end{matrix}, rendering as TeX
```

2025-07-19 20:24:31,540 - WARNING - Could not convert TeX math  $\begin{matrix}$

```
{k = \text{c}^T \cdot \begin{matrix}
{- \text{s}} \\
1
\end{matrix}} \\
\end{matrix}} \\
\begin{array}{l}
{= \left(\{\hat{\text{s}}\}^T \cdot \text{B} \mid \text{t}\} + \{\hat{\text{e}}\}^T + \text{b}^T \right) \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + \{\hat{\text{s}}\}^T \cdot \text{t} + \{\hat{\text{e}}\}^T \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + b \cdot \left\lfloor \frac{q}{2} \right\rfloor} \\
{= \{\hat{\text{s}}\}^T \cdot \text{e} + \{\hat{\text{e}}\}^T \cdot \left[\begin{array}{l}
{- \text{s}} \\
1
\end{array} \right] \cdot \text{B} + b \cdot \left\lfloor \frac{q}{2} \right\rfloor} \\
\end{array} \\
\end{matrix}
```

```

 {\hat{\text{s}}}^T\text{e} + {\hat{\text{e}}}^T \cdot \left\lbrack
\begin{array}{l}
 {- \text{s}} \backslash
 1
 \end{array} \right\rbrack + b \cdot \left\lfloor \frac{q}{2} \right\rfloor,
\end{array}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \left| \{{\hat{\text{s}}}^T\text{e} +
{\hat{\text{e}}}^T \cdot \begin{bmatrix}
 {- \text{s}} \backslash
 1
 \end{bmatrix} \right| < (q - 1)/4, rendering as TeX
2025-07-19 20:24:31,540 - WARNING - Could not convert TeX math \left|
\{{\hat{\text{s}}}^T\text{e} + {\hat{\text{e}}}^T \cdot \begin{bmatrix}
 {- \text{s}} \backslash
 1
 \end{bmatrix} \right| < (q - 1)/4, rendering as TeX
[WARNING] Could not convert TeX math \left| \sum\limits_{i =
1}^m{\hat{s}}_ie_i - \sum\limits_{i = 1}^n{\hat{e}}_is_i + e_{n +
1} \right| < (q - 1)/4,, rendering as TeX
2025-07-19 20:24:31,541 - WARNING - Could not convert TeX math \left|
\sum\limits_{i = 1}^m{\hat{s}}_ie_i - \sum\limits_{i =
1}^n{\hat{e}}_is_i + e_{n + 1} \right| < (q - 1)/4,, rendering as TeX
[WARNING] Could not convert TeX math \widetilde{\prod}, rendering as TeX
2025-07-19 20:24:31,542 - WARNING - Could not convert TeX math
\widetilde{\prod}, rendering as TeX
[WARNING] Could not convert TeX math \widetilde{\prod}, rendering as TeX
2025-07-19 20:24:31,544 - WARNING - Could not convert TeX math
\widetilde{\prod}, rendering as TeX
[WARNING] Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
2025-07-19 20:24:31,544 - WARNING - Could not convert TeX math
\text{PubK}_{\mathcal{A},\widetilde{\text{\Pi}}}^{\text{eav}}(n), rendering as TeX
[WARNING] Could not convert TeX math \left| \{\text{Pr}\left\lbrack
\{\text{PubK}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1\right\rbrack -
\text{Pr}\left\lbrack
\{\text{PubK}_{\mathcal{A},\widetilde{\text{II}}}^{\text{eav}}(n) = 1\}
\right\rbrack \right|, rendering as TeX
2025-07-19 20:24:31,545 - WARNING - Could not convert TeX math \left|
\{\text{Pr}\left\lbrack \{\text{PubK}_{\mathcal{A},\text{\Pi}}^{\text{eav}}(n) = 1\}
\right\rbrack - \text{Pr}\left\lbrack
\{\text{PubK}_{\mathcal{A},\widetilde{\text{II}}}^{\text{eav}}(n) = 1\}
\right\rbrack \right|, rendering as TeX
[WARNING] Could not convert TeX math \text{c}^T := \left\lbrack
\{{\hat{\text{s}}}^T \cdot \lbrack\text{B} \mid \text{t}\rbrack +
{\hat{\text{e}}}^T + \left\lbrack 0,\ldots,0,m_b \right\lfloor
\frac{q}{2} \right\rfloor \right\rbrack\text{mod}q \right\rbrack., rendering as TeX
2025-07-19 20:24:31,546 - WARNING - Could not convert TeX math \text{c}^T :=

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{\text{Pr}}\left\lbrack
{\text{PubK}}_{\mathcal{A},\widetilde{\text{\Pi}}}\text{\text{eav}}(n) = 1\}
\right\rbrack - \text{\text{Pr}}\left\lbrack
{\text{PubK}}_{\mathcal{A},\text{\Pi}^{\prime}}\text{\text{eav}}(n) = 1\}
\right\rbrack \text{\text{right}}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \widetilde{\prod}, \text{rendering as TeX}
2025-07-19 20:24:31,552 - WARNING - Could not convert TeX math
\widetilde{\prod}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \widetilde{\prod}, \text{rendering as TeX}
2025-07-19 20:24:31,553 - WARNING - Could not convert TeX math
\widetilde{\prod}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{c}^T := \{\hat{\text{t}}\}^T +
\left\lbrack 0,\ldots,0,b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right\rbrack
\right\rbrack., \text{rendering as TeX}
2025-07-19 20:24:31,553 - WARNING - Could not convert TeX math \text{c}^T :=
\{\hat{\text{t}}\}^T + \left\lbrack 0,\ldots,0,b \cdot \left\lfloor \frac{q}{2} \right\rfloor
\right\rbrack \text{\text{right}}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \text{\text{Pr}}\left\lbrack
{\text{PubK}}_{\mathcal{A},\widetilde{\text{\Pi}}}\text{\text{eav}}(n) = 1\}
\right\rbrack = \frac{1}{2}, \text{rendering as TeX}
2025-07-19 20:24:31,555 - WARNING - Could not convert TeX math
\text{\text{Pr}}\left\lbrack
{\text{PubK}}_{\mathcal{A},\widetilde{\text{\Pi}}}\text{\text{eav}}(n) = 1\}
\right\rbrack = \frac{1}{2}, \text{rendering as TeX}
[WARNING] Could not convert TeX math {\widetilde{\prod}}^{\prime}, \text{rendering as}
TeX
2025-07-19 20:24:31,555 - WARNING - Could not convert TeX math
{\widetilde{\prod}}^{\prime}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \left\lbrack 0,\ldots,0,b \cdot
\left\lfloor \frac{q}{2} \right\rfloor \right\rbrack \text{\text{right}}, \text{rendering as TeX}
2025-07-19 20:24:31,556 - WARNING - Could not convert TeX math \left\lbrack
0,\ldots,0,b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right\rbrack \text{\text{right}},
\text{rendering as TeX}
[WARNING] Could not convert TeX math pk = \begin{pmatrix}
y_{1,0} & y_{2,0} & y_{3,0} \\
y_{1,1} & y_{2,1} & y_{3,1}
\end{pmatrix} \quad \text{sk} = \begin{pmatrix}
x_{1,0} & x_{2,0} & x_{3,0} \\
x_{1,1} & x_{2,1} & x_{3,1}
\end{pmatrix}., \text{rendering as TeX}
2025-07-19 20:24:31,557 - WARNING - Could not convert TeX math pk =
\begin{pmatrix}
y_{1,0} & y_{2,0} & y_{3,0} \\
y_{1,1} & y_{2,1} & y_{3,1}
\end{pmatrix} \quad \text{sk} = \begin{pmatrix}
x_{1,0} & x_{2,0} & x_{3,0} \\
x_{1,1} & x_{2,1} & x_{3,1}
\end{pmatrix}., \text{rendering as TeX}

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[WARNING] Could not convert TeX math  $H\left( x_{\{i\}} \right)\overset{?}{=}{y_{\{i,m_{\{i\}}}}$ , rendering as TeX

2025-07-19 20:24:31,557 - WARNING - Could not convert TeX math  $H\left( x_{\{i\}} \right)\overset{?}{=}{y_{\{i,m_{\{i\}}}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left. sk = \begin{pmatrix} \begin{matrix} \begin{matrix} x_{\{1,0\}} \end{matrix} \end{matrix} \right\} & x_{\{2,0\}} & x_{\{3,0\}} \backslash \backslash \begin{matrix} x_{\{1,1\}} & \begin{matrix} \begin{matrix} x_{\{2,1\}} \end{matrix} \end{matrix} \end{matrix} & \begin{matrix} \begin{matrix} x_{\{3,1\}} \end{matrix} \end{matrix} \right\} \right)$

$\left. \right\}$ , rendering as TeX

2025-07-19 20:24:31,558 - WARNING - Could not convert TeX math  $\left. sk = \begin{pmatrix} \begin{matrix} \begin{matrix} x_{\{1,0\}} \end{matrix} \end{matrix} \right\} & x_{\{2,0\}} & x_{\{3,0\}} \backslash \backslash \begin{matrix} x_{\{1,1\}} & \begin{matrix} \begin{matrix} x_{\{2,1\}} \end{matrix} \end{matrix} \end{matrix} & \begin{matrix} \begin{matrix} x_{\{3,1\}} \end{matrix} \end{matrix} \right\} \right)$

$\left. \right\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $pk = \left\{ \begin{pmatrix} y_{\{1,0\}} \end{pmatrix} & y_{\{2,0\}} & y_{\{3,0\}} \backslash \backslash \begin{matrix} y_{\{1,1\}} & \begin{matrix} \begin{matrix} y_{\{2,1\}} \end{matrix} \end{matrix} \end{matrix} & \begin{matrix} \begin{matrix} y_{\{3,1\}} \end{matrix} \end{matrix} \right\}$

$\left\{ \begin{pmatrix} H\left( x_{\{1\}} \right)\overset{?}{=}{y_{\{1,0\}}} \backslash \backslash \left. \right\} \begin{matrix} H\left( x_{\{2\}} \right)\overset{?}{=}{y_{\{2,1\}}} \backslash \backslash \begin{matrix} H\left( x_{\{3\}} \right)\overset{?}{=}{y_{\{3,1\}}} \end{matrix} \right\}$

$\left. \right\}$ , rendering as TeX

2025-07-19 20:24:31,558 - WARNING - Could not convert TeX math  $pk = \left\{ \begin{pmatrix} y_{\{1,0\}} \end{pmatrix} & y_{\{2,0\}} & y_{\{3,0\}} \backslash \backslash$





2025-07-19 20:24:31,568 - WARNING - Could not convert TeX math  $\text{Pr}\left\{\mathcal{A}\left(\mathcal{I},f_{\mathcal{I}}(x)\right)\leq\frac{1}{2}+\text{negl}(n)\right\}$ , rendering as TeX

2025-07-19 20:24:31,569 - WARNING - Could not convert TeX math  $\hat{\prod}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\prod}$ , rendering as TeX

2025-07-19 20:24:31,570 - WARNING - Could not convert TeX math  $\hat{\prod}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\prod}=\left(\hat{\text{Gen}},f,\text{lnv}\right)$ , rendering as TeX

2025-07-19 20:24:31,571 - WARNING - Could not convert TeX math  $\hat{\prod}=\left(\hat{\text{Gen}},f,\text{lnv}\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\text{Gen}}\left(1^n\right)$ , rendering as TeX

2025-07-19 20:24:31,572 - WARNING - Could not convert TeX math  $\hat{\text{Gen}}\left(1^n\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\prod}=\left(\hat{\text{Gen}},f,\text{lnv}\right)$ , rendering as TeX

2025-07-19 20:24:31,573 - WARNING - Could not convert TeX math  $\hat{\prod}=\left(\hat{\text{Gen}},f,\text{lnv}\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\text{Gen}}\left(1^n\right)$ , rendering as TeX

2025-07-19 20:24:31,573 - WARNING - Could not convert TeX math  $\hat{\text{Gen}}\left(1^n\right)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\hat{\prod}$ , rendering as TeX

2025-07-19 20:24:31,574 - WARNING - Could not convert TeX math  $\hat{\prod}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\delta_0(n)\overset{\text{def}}{=}\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=0\right\}$ , rendering as TeX

2025-07-19 20:24:31,575 - WARNING - Could not convert TeX math  $\delta_0(n)\overset{\text{def}}{=}\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=0\right\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\delta_1(n)\overset{\text{def}}{=}\frac{\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=1\right\}}{\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=1\right\}}$ , rendering as TeX

2025-07-19 20:24:31,575 - WARNING - Could not convert TeX math  $\delta_1(n)\overset{\text{def}}{=}\frac{\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=1\right\}}{\text{Pr}_{\left(\mathcal{I},\text{td}\right)}\leftarrow\hat{\text{Gen}}\left(1^n\right);x\leftarrow\mathcal{D}_{\mathcal{I}}\left\{\text{hc}_{\mathcal{I}}(x)=1\right\}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\delta_0(n),\delta_1(n)\geq\frac{1}{2}-\text{negl}(n)$ , rendering as TeX

2025-07-19 20:24:31,576 - WARNING - Could not convert TeX math  $\delta_0(n),\delta_1(n)\geq\frac{1}{2}-\text{negl}(n)$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}\text{Pr}\left\{\text{Pub}\text{K}_{\mathcal{A},\Pi}^{\text{eav}}(n)=1\right\}\end{array}$

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\right\rbrace = \frac{1}{2} \cdot \text{\Pr}\lbrack \mathrm{A}(pk,c) = 0 \mid \\
c\text{\is-an-encryption-of}0\rbrace \\\
{\,\,, + \frac{1}{2}} \cdot \text{\Pr}\lbrack \mathrm{A}(pk,c) = 1 \mid \\
c\text{\is-an-encryption-of}1\rbrace.\} \\
\end{array}, rendering as TeX
2025-07-19 20:24:31,576 - WARNING - Could not convert TeX math \begin{array}{l}
{\text{\Pr}\left\lbrack {\text{\K}_{\mathrm{A}},\text{\Pi}}^{\text{\eav}}(n) = 1} \\
\right\rbrace = \frac{1}{2} \cdot \text{\Pr}\lbrack \mathrm{A}(pk,c) = 0 \mid \\
c\text{\is-an-encryption-of}0\rbrace \\\
{\,\,, + \frac{1}{2}} \cdot \text{\Pr}\lbrack \mathrm{A}(pk,c) = 1 \mid \\
c\text{\is-an-encryption-of}1\rbrace.\} \\
\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l}
{\text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = } \\
\text{\hc}_I(x) \right\rbrace \\\
{\,\,, = \delta_0(n) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 0 \mid \text{\hc}_I(x) = 0} \right\rbrace \\\
{\,\,, + \delta_1(n) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 1 \mid \text{\hc}_I(x) = 1} \right\rbrace \\\
{\geq \left({\frac{1}{2} - \neg l(n)} \right) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 0 \mid \text{\hc}_I(x) = 0} \right\rbrace \\\
\right\rbrace \\\
{\,\,, + \left({\frac{1}{2} - \neg l(n)} \right) \cdot \\
\text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 1 \mid \\
\text{\hc}_I(1) = 1} \right\rbrace \\\
{\geq \frac{1}{2} \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 0 \mid \text{\hc}_I(x) = 0} \right\rbrace \\\
{\,\,, + \frac{1}{2} \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 1 \mid \text{\hc}_I(1) = 1} \right\rbrace - 2 \cdot \\
\neg l(n)} \\\
{= \text{\Pr}\left\lbrack {\text{\PubK}_{\mathrm{A}},\text{\Pi}}^{\text{\eav}}(n) = } \\
1 \right\rbrace - 2 \cdot \neg l(n).} \\
\end{array}, rendering as TeX
2025-07-19 20:24:31,578 - WARNING - Could not convert TeX math \begin{array}{l}
{\text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = } \\
\text{\hc}_I(x) \right\rbrace \\\
{\,\,, = \delta_0(n) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 0 \mid \text{\hc}_I(x) = 0} \right\rbrace \\\
{\,\,, + \delta_1(n) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 1 \mid \text{\hc}_I(x) = 1} \right\rbrace \\\
{\geq \left({\frac{1}{2} - \neg l(n)} \right) \cdot \text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 0 \mid \text{\hc}_I(x) = 0} \right\rbrace \\\
\right\rbrace \\\
{\,\,, + \left({\frac{1}{2} - \neg l(n)} \right) \cdot \\
\text{\Pr}\left\lbrack {\mathrm{A}\left({I,f_I(x)} \right) = 1 \mid \\
\text{\hc}_I(1) = 1} \right\rbrace \\\

```

[illegible]

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\end{array}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{f\left({a_1,b_1} \right) \cdot f\left({a_2,b_2} \right) = \left(
\left\{ \left(1 + N \right)^{a_1} \cdot b_1^N \right\} \right) \cdot \left(\left\{ \left(1 + N \right)^{a_2} \right. \right.
\cdot b_2^N \right) \bmod N^2} \\
{= \left(1 + N \right)^{a_1 + a_2} \cdot \left(b_1 b_2 \right)^N \bmod N^2.}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,582 - WARNING - Could not convert TeX math \begin{matrix}
{f\left({a_1,b_1} \right) \cdot f\left({a_2,b_2} \right) = \left(
\left\{ \left(1 + N \right)^{a_1} \cdot b_1^N \right\} \right) \cdot \left(\left\{ \left(1 + N \right)^{a_2} \right. \right.
\cdot b_2^N \right) \bmod N^2} \\
{= \left(1 + N \right)^{a_1 + a_2} \cdot \left(b_1 b_2 \right)^N \bmod N^2.}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{f\left({a_1,b_1} \right) \cdot f\left({a_2,b_2} \right) = \left(1 +
N \right)^{a_1 + a_2} \cdot \left(b_1 b_2 \right)^N \bmod N^2} \\
{= \left(1 + N \right)^{\left\{ a_1 + a_2 \right\} \bmod N} \cdot \left(b_1 b_2 \right)
\right)^N \bmod N^2.}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,582 - WARNING - Could not convert TeX math \begin{matrix}
{f\left({a_1,b_1} \right) \cdot f\left({a_2,b_2} \right) = \left(1 +
N \right)^{a_1 + a_2} \cdot \left(b_1 b_2 \right)^N \bmod N^2} \\
{= \left(1 + N \right)^{\left\{ a_1 + a_2 \right\} \bmod N} \cdot \left(b_1 b_2 \right)
\right)^N \bmod N^2.}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{\left(b_1 b_2 \right)^N = \left(r + \gamma N \right)^N \bmod N^2} \\
{= \sum\limits_{k = 0}^N \left(\begin{array}{l}
N \\
k
\end{array} \right) r^{N - k} \left(\gamma N \right)^k \bmod N^2} \\
{= r^N + N \cdot r^{N - 1} \cdot \left(\gamma N \right) = r^N = \left(\left\{ b_1 b_2 \right\} \bmod N \right) \right)^N \bmod N^2,}
\end{matrix}, rendering as TeX
2025-07-19 20:24:31,583 - WARNING - Could not convert TeX math \begin{matrix}
{\left(b_1 b_2 \right)^N = \left(r + \gamma N \right)^N \bmod N^2} \\
{= \sum\limits_{k = 0}^N \left(\begin{array}{l}
N \\
k
\end{array} \right) r^{N - k} \left(\gamma N \right)^k \bmod N^2} \\
{= r^N + N \cdot r^{N - 1} \cdot \left(\gamma N \right) = r^N = \left(\left\{ b_1 b_2 \right\} \bmod N \right) \right)^N \bmod N^2,}
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix}
{f\left({a_1,b_1} \right) \cdot f\left({a_2,b_2} \right) = \left(1 +
N \right)^{\left\{ a_1 + a_2 \right\} \bmod N} \cdot \left(b_1 b_2 \right)^N \bmod N^2} \\
{= f\left({a_1 + a_2,b_1 b_2} \right).}
\end{matrix}

```



$$\begin{matrix} \end{matrix}$$
, rendering as TeX  
2025-07-19 20:24:31,584 - WARNING - Could not convert TeX math 
$$\begin{matrix} f\left( \begin{matrix} a_1, b_1 \end{matrix} \right) \cdot f\left( \begin{matrix} a_2, b_2 \end{matrix} \right) = (1 + N)^{\left\lfloor \begin{matrix} a_1 + a_2 \end{matrix} \right\rfloor \bmod N} \cdot \left( \begin{matrix} b_1 b_2 \end{matrix} \right) \bmod N \end{matrix}$$

$$\left. \begin{matrix} f\left( \begin{matrix} a_1 + a_2, b_1 b_2 \end{matrix} \right) \end{matrix} \right\} \\ \end{matrix}$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$\overline{\left( \begin{matrix} N-1 \end{matrix} \right) \bmod \phi(N)}$$
, rendering as TeX  
2025-07-19 20:24:31,584 - WARNING - Could not convert TeX math 
$$\overline{\left( \begin{matrix} N-1 \end{matrix} \right) \bmod \phi(N)}$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$\begin{matrix} \left\{ \begin{matrix} \hat{c} \end{matrix} \right\} \bmod N^2 \end{matrix}$$
  

$$\left. \begin{matrix} \leftarrow \begin{matrix} \phi(N) \end{matrix} \rightarrow \end{matrix} \right\} \\ \begin{matrix} \left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \end{matrix} \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix} \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
  

$$\left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
  

$$\left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
, rendering as TeX  
2025-07-19 20:24:31,585 - WARNING - Could not convert TeX math 
$$\begin{matrix} \left\{ \begin{matrix} \hat{c} \end{matrix} \right\} \bmod N^2 \end{matrix}$$
  

$$\left. \begin{matrix} \leftarrow \begin{matrix} \phi(N) \end{matrix} \rightarrow \end{matrix} \right\} \\ \begin{matrix} \left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \end{matrix} \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix} \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
  

$$\left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
  

$$\left\{ \begin{matrix} m \end{matrix} \right\} \bmod N \left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N \end{matrix}$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$\hat{m} \bmod \left( \begin{matrix} \phi(N) \end{matrix} \right) \bmod N$$
, rendering as TeX  
2025-07-19 20:24:31,585 - WARNING - Could not convert TeX math 
$$\hat{m} \bmod \left( \begin{matrix} \phi(N) \end{matrix} \right) \bmod N$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$m = \left\{ \begin{matrix} \frac{\left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N^2}{\left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N} \end{matrix} \right\} \bmod N$$
, rendering as TeX  
2025-07-19 20:24:31,587 - WARNING - Could not convert TeX math 
$$m = \left\{ \begin{matrix} \frac{\left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N^2}{\left\{ \begin{matrix} \phi(N) \end{matrix} \right\} \bmod N} \end{matrix} \right\} \bmod N$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$\frac{1}{2}$$
, rendering as TeX  
2025-07-19 20:24:31,588 - WARNING - Could not convert TeX math 
$$\frac{1}{2}$$
, rendering as TeX  
[WARNING] Could not convert TeX math 
$$\text{Pr} \left( \begin{matrix} D(N, r) \end{matrix} \right) = \frac{1}{2}$$
, rendering as TeX  
2025-07-19 20:24:31,588 - WARNING - Could not convert TeX math 
$$\text{Pr} \left( \begin{matrix} D(N, r) \end{matrix} \right) = \frac{1}{2}$$

[illegible]

rendering as TeX  
2025-07-19 20:24:31,593 - WARNING - Could not convert TeX math  
 $v_{\text{total}} \overset{\text{def}}{=} \sum_{i=1}^{\ell} v_i \pmod{N}$ .,  
rendering as TeX  
[WARNING] Could not convert TeX math  $v_{\text{total}} = \sum_{i=1}^{\ell} v_i$ , rendering as TeX  
2025-07-19 20:24:31,595 - WARNING - Could not convert TeX math  $v_{\text{total}} = \sum_{i=1}^{\ell} v_i$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{pmatrix} N-1 \\ t-1 \end{pmatrix}$   
 $\end{pmatrix}$ , rendering as TeX  
2025-07-19 20:24:31,596 - WARNING - Could not convert TeX math  $\begin{pmatrix} N-1 \\ t-1 \end{pmatrix}$   
 $\end{pmatrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $\delta_i(X) \overset{\text{def}}{=} \frac{\prod_{j=1, j \neq i}^t \left( X - x_j \right)}{\prod_{j=1, j \neq i}^t \left( x_i - x_j \right)}$ .,  
rendering as TeX  
2025-07-19 20:24:31,596 - WARNING - Could not convert TeX math  
 $\delta_i(X) \overset{\text{def}}{=} \frac{\prod_{j=1, j \neq i}^t \left( X - x_j \right)}{\prod_{j=1, j \neq i}^t \left( x_i - x_j \right)}$ .,  
rendering as TeX  
[WARNING] Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{i=1}^t \delta_i(X) \cdot y_i$ , rendering as TeX  
2025-07-19 20:24:31,597 - WARNING - Could not convert TeX math  
 $p(X) \overset{\text{def}}{=} \sum_{i=1}^t \delta_i(X) \cdot y_i$ ,  
rendering as TeX  
[WARNING] Could not convert TeX math  $p(X) \overset{\text{def}}{=} s + \sum_{t=1}^t \{a_i X^i\}$ , rendering as TeX  
2025-07-19 20:24:31,598 - WARNING - Could not convert TeX math  
 $p(X) \overset{\text{def}}{=} s + \sum_{t=1}^t \{a_i X^i\}$ , rendering  
as TeX  
[WARNING] Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{j=0}^{t-1} \{a_j X^j\}$ , rendering as TeX  
2025-07-19 20:24:31,599 - WARNING - Could not convert TeX math  
 $p(X) \overset{\text{def}}{=} \sum_{j=0}^{t-1} \{a_j X^j\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $s_i := p(i) = \sum_{j=0}^{t-1} \{a_j \cdot i^j\}$ , rendering as TeX  
2025-07-19 20:24:31,600 - WARNING - Could not convert TeX math  $s_i := p(i) = \sum_{j=0}^{t-1} \{a_j \cdot i^j\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g^{\{s_i\}} \overset{?}{=} \prod \limits_{j=0}^{t-1} \left( A_j \right)^{i^j}$ ., rendering as TeX  
2025-07-19 20:24:31,601 - WARNING - Could not convert TeX math  
 $g^{\{s_i\}} \overset{?}{=} \prod \limits_{j=0}^{t-1} \left( A_j \right)^{i^j}$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\prod \limits_{j=0}^{t-1} \left( A_j \right)$

$\right)^{i_j} = \prod_{j=0}^{t-1} \left( g^{a_j} \right)^{i_j} = g^{\sum_{j=0}^{t-1} a_j \cdot i_j} = g^{p(i)} = g^{s_i},,$   
 rendering as TeX  
 2025-07-19 20:24:31,601 - WARNING - Could not convert TeX math  $\prod_{j=0}^{t-1} \left( A_j \right)^{i_j} = \prod_{j=0}^{t-1} \left( g^{a_j} \right)^{i_j} = g^{\sum_{j=0}^{t-1} a_j \cdot i_j} = g^{p(i)} = g^{s_i},,$  rendering as TeX  
 [WARNING] Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{i=0}^{t-1} \{a_i X^i\},$  rendering as TeX  
 2025-07-19 20:24:31,602 - WARNING - Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{i=0}^{t-1} \{a_i X^i\},$  rendering as TeX  
 [WARNING] Could not convert TeX math  $v_{\text{total}} \overset{\text{def}}{=} \sum_{i=1}^q v_i \pmod q = \sum_{i=1}^q v_i,$  rendering as TeX  
 2025-07-19 20:24:31,602 - WARNING - Could not convert TeX math  $v_{\text{total}} \overset{\text{def}}{=} \sum_{i=1}^q v_i \pmod q = \sum_{i=1}^q v_i,$  rendering as TeX  
 [WARNING] Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{j=1}^t \{\delta_j(X) \cdot x_{i_j}\},$  rendering as TeX  
 2025-07-19 20:24:31,604 - WARNING - Could not convert TeX math  $p(X) \overset{\text{def}}{=} \sum_{j=1}^t \{\delta_j(X) \cdot x_{i_j}\},$  rendering as TeX  
 [WARNING] Could not convert TeX math  $\delta_j \overset{\text{def}}{=} \delta_j(0),$  rendering as TeX  
 2025-07-19 20:24:31,604 - WARNING - Could not convert TeX math  $\delta_j \overset{\text{def}}{=} \delta_j(0),$  rendering as TeX  
 [WARNING] Could not convert TeX math  $x = p(0) = \sum_{j=1}^t \{\delta_j \cdot x_{i_j}\},$  rendering as TeX  
 2025-07-19 20:24:31,605 - WARNING - Could not convert TeX math  $x = p(0) = \sum_{j=1}^t \{\delta_j \cdot x_{i_j}\},$  rendering as TeX  
 [WARNING] Could not convert TeX math  $M^{\prime} := \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}},$  rendering as TeX  
 2025-07-19 20:24:31,605 - WARNING - Could not convert TeX math  $M^{\prime} := \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}},$  rendering as TeX  
 [WARNING] Could not convert TeX math  $\prod_{j=1}^t w_j^{\delta_j} = \prod_{j=1}^t c_1^{x_i \delta_j} = c_1^{\sum_{j=1}^t x_{i_j} \delta_j} = c_1^{p(0)} = c_1^x,$  rendering as TeX  
 2025-07-19 20:24:31,606 - WARNING - Could not convert TeX math  $\prod_{j=1}^t w_j^{\delta_j} = \prod_{j=1}^t c_1^{x_i \delta_j} = c_1^{\sum_{j=1}^t x_{i_j} \delta_j} = c_1^{p(0)} = c_1^x,$  rendering as TeX  
 [WARNING] Could not convert TeX math  $M^{\prime} \overset{\text{def}}{=} \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}} = \frac{h^y}{\prod_{j=1}^t w_j^{\delta_j}} = \frac{h^y}{\left( g^x \right)^y} \cdot M^{\left( g^y \right)^x} = M.,$  rendering as TeX  
 2025-07-19 20:24:31,607 - WARNING - Could not convert TeX math  $M^{\prime} \overset{\text{def}}{=} \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}} = \frac{h^y}{\prod_{j=1}^t w_j^{\delta_j}} = \frac{h^y}{\left( g^x \right)^y} \cdot M^{\left( g^y \right)^x} = \frac{h^y}{\left( g^x \right)^y} \cdot M^{\left( g^y \right)^x} = M.,$  rendering as TeX

$\right)^{y} \cdot M^{\left( g^y \right)^x} = M.$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $\text{sq}_p(x) \overset{\text{-def}}{=} \left\lceil \text{brack } x^2 \pmod{p} \right\rceil$   
rendering as TeX  
2025-07-19 20:24:31,607 - WARNING - Could not convert TeX math  
 $\text{sq}_p(x) \overset{\text{-def}}{=} \left\lceil \text{brack } x^2 \pmod{p} \right\rceil$   
rendering as TeX  
[WARNING] Could not convert TeX math  $\left| \{\mathcal{Q}\mathcal{R}_p\} \right|$   
 $= \left| \{\mathcal{Q}\mathcal{N}\mathcal{R}_p\} \right| = \frac{\left| \mathbb{Z}_p^{\ast} \right|}{2} = \frac{p-1}{2}.$ , rendering as TeX  
2025-07-19 20:24:31,608 - WARNING - Could not convert TeX math  $\left| \{\mathcal{Q}\mathcal{R}_p\} \right|$   
 $= \left| \{\mathcal{Q}\mathcal{N}\mathcal{R}_p\} \right| = \frac{\left| \mathbb{Z}_p^{\ast} \right|}{2} = \frac{p-1}{2}.$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $\mathcal{J}_p(x) \overset{\text{def}}{=} \left\{ \begin{matrix} +1 \text{ if } x \text{ is a quadratic residue modulo } p \\ -1 \text{ if } x \text{ is not a quadratic residue modulo } p \end{matrix} \right\}$   
rendering as TeX  
2025-07-19 20:24:31,610 - WARNING - Could not convert TeX math  
 $\mathcal{J}_p(x) \overset{\text{def}}{=} \left\{ \begin{matrix} +1 \text{ if } x \text{ is a quadratic residue modulo } p \\ -1 \text{ if } x \text{ is not a quadratic residue modulo } p \end{matrix} \right\}$   
rendering as TeX  
[WARNING] Could not convert TeX math  
 $\mathcal{J}_p(x) \overset{\text{-def}}{=} \mathcal{J}_p(\left\lceil \text{brack } x \pmod{p} \right\rceil)$ , rendering as TeX  
2025-07-19 20:24:31,611 - WARNING - Could not convert TeX math  
 $\mathcal{J}_p(x) \overset{\text{-def}}{=} \mathcal{J}_p(\left\lceil \text{brack } x \pmod{p} \right\rceil)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathbb{Z}_p^{\ast} = \left\{ g^0, g^1, g^2, \ldots, g^{\frac{p-1}{2}-1}, g^{\frac{p-1}{2}}, g^{\frac{p-1}{2}+1}, \ldots, g^{p-2} \right\}$ , rendering as TeX  
2025-07-19 20:24:31,612 - WARNING - Could not convert TeX math  
 $\mathbb{Z}_p^{\ast} = \left\{ g^0, g^1, g^2, \ldots, g^{\frac{p-1}{2}-1}, g^{\frac{p-1}{2}}, g^{\frac{p-1}{2}+1}, \ldots, g^{p-2} \right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{J}_p(x) = x^{\frac{p-1}{2}}$ , rendering as TeX  
2025-07-19 20:24:31,612 - WARNING - Could not convert TeX math  
 $\mathcal{J}_p(x) = x^{\frac{p-1}{2}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x^{\frac{p-1}{2}} = \left( g^{2j} \right)^{\frac{p-1}{2}} = g^{(p-1)j} = \left( g^{p-1} \right)^j = 1^j = 1 \pmod{p}$ , rendering as TeX  
2025-07-19 20:24:31,613 - WARNING - Could not convert TeX math  $x^{\frac{p-1}{2}} = \left( g^{2j} \right)^{\frac{p-1}{2}} = g^{(p-1)j} = \left( g^{p-1} \right)^j = 1^j = 1 \pmod{p}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x^{\frac{p-1}{2}} = +1 =$

$\mathcal{J}_p(x)$ , rendering as TeX  
2025-07-19 20:24:31,613 - WARNING - Could not convert TeX math  $x^{\frac{p-1}{2}} = +1 = \mathcal{J}_p(x)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x^{\frac{p-1}{2}} = \left(g^{2j+1}\right)^{\frac{p-1}{2}} = \left(g^{2j}\right)^{\frac{p-1}{2}} \cdot g^{\frac{p-1}{2}} = 1 \cdot g^{\frac{p-1}{2}} = g^{\frac{p-1}{2}} \pmod{p}$ , rendering as TeX  
2025-07-19 20:24:31,614 - WARNING - Could not convert TeX math  $x^{\frac{p-1}{2}} = \left(g^{2j+1}\right)^{\frac{p-1}{2}} = \left(g^{2j}\right)^{\frac{p-1}{2}} \cdot g^{\frac{p-1}{2}} = 1 \cdot g^{\frac{p-1}{2}} \pmod{p}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left(g^{\frac{p-1}{2}}\right)^2 = g^{p-1} = 1 \pmod{p}$ , rendering as TeX  
2025-07-19 20:24:31,615 - WARNING - Could not convert TeX math  $\left(g^{\frac{p-1}{2}}\right)^2 = g^{p-1} = 1 \pmod{p}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g^{\frac{p-1}{2}} = \pm 1$ , rendering as TeX  
2025-07-19 20:24:31,616 - WARNING - Could not convert TeX math  $g^{\frac{p-1}{2}} = \pm 1$ , rendering as TeX  
[WARNING] Could not convert TeX math  $g^{\frac{p-1}{2}} \neq 1$ , rendering as TeX  
2025-07-19 20:24:31,617 - WARNING - Could not convert TeX math  $g^{\frac{p-1}{2}} \neq 1$ , rendering as TeX  
[WARNING] Could not convert TeX math  $x^{\frac{p-1}{2}} = -1 = \mathcal{J}_p(x)$ , rendering as TeX  
2025-07-19 20:24:31,618 - WARNING - Could not convert TeX math  $x^{\frac{p-1}{2}} = -1 = \mathcal{J}_p(x)$ , rendering as TeX  
[WARNING] Could not convert TeX math  $b = \left[\left(x^{\frac{p-1}{2}}\right) \pmod{p}\right]$ , rendering as TeX  
2025-07-19 20:24:31,619 - WARNING - Could not convert TeX math  $b = \left[\left(x^{\frac{p-1}{2}}\right) \pmod{p}\right]$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\mathcal{J}_p(xy) = \{(xy)\}^{\frac{p-1}{2}} = x^{\frac{p-1}{2}} \cdot y^{\frac{p-1}{2}} = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y) \pmod{p}$ , rendering as TeX  
2025-07-19 20:24:31,621 - WARNING - Could not convert TeX math  $\mathcal{J}_p(xy) = \{(xy)\}^{\frac{p-1}{2}} = x^{\frac{p-1}{2}} \cdot y^{\frac{p-1}{2}} = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y) \pmod{p}$ , rendering as TeX  
[WARNING] Could not convert TeX math 
$$\begin{matrix} \left(\pm x_p, \pm x_q\right)^2 = \left(\left[\left(\pm x_p\right)\right] \right. \\ \left.\left.\left(\pm x_q\right)\right] \pmod{p}\right) \left[\left(\pm x_q\right)\right] \\ \left.\left.\left(\pm x_p\right)\right] \pmod{p}\right) \left(\pm x_q\right) \\ \left.\left.\left(\pm x_q\right)\right] \pmod{q}\right) \left(\pm x_p\right) \right] \\ \left.\left.\left(\pm x_q\right)\right] \pmod{q}\right) \left(\pm x_p\right) \right] \end{matrix}$$
, rendering as TeX  
2025-07-19 20:24:31,622 - WARNING - Could not convert TeX math 
$$\begin{matrix} \left(\pm x_p, \pm x_q\right)^2 = \left(\left[\left(\pm x_p\right)\right] \right. \\ \left.\left.\left(\pm x_q\right)\right] \pmod{p}\right) \left[\left(\pm x_q\right)\right] \\ \left.\left.\left(\pm x_p\right)\right] \pmod{p}\right) \left(\pm x_q\right) \\ \left.\left.\left(\pm x_q\right)\right] \pmod{q}\right) \left(\pm x_p\right) \right] \\ \left.\left.\left(\pm x_q\right)\right] \pmod{q}\right) \left(\pm x_p\right) \right] \end{matrix}$$

```

x_{p}} \right)^{2\mod p} \right\rbrace, \left\lbrack \left(\pm x_{q} \right.
\right)^{2\mod q} \right\rbrace \right) \\\
\left. = \left(\left\lbrack x_{p}^{2\mod p} \right\rbrace, \left\lbrack x_{q}^{2\mod q} \right\rbrace \right) = \left(y_{p}, y_{q} \right) \\\
\right) \leftrightharpoons y \right. .
\end{matrix}, rendering as TeX
[WARNING] Could not convert TeX math \frac{\left| \mathcal{Q} \mathcal{R}_N \right| \left| \mathbb{Z}_N^{\ast} \right|}{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_N^{\ast} \right|} = \frac{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_p \right|}{\left| \mathcal{Q} \mathcal{R}_q \right| \left| \mathbb{Z}_q \right|} \cdot \frac{\left| \mathcal{Q} \mathcal{R}_q \right| \left| \mathbb{Z}_q \right|}{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_p \right|} = \frac{p-1}{q-1} \cdot \frac{q-1}{p-1} = 1, rendering as TeX
2025-07-19 20:24:31,622 - WARNING - Could not convert TeX math \frac{\left| \mathcal{Q} \mathcal{R}_N \right| \left| \mathbb{Z}_N^{\ast} \right|}{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_p \right|} = \frac{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_p \right|}{\left| \mathcal{Q} \mathcal{R}_q \right| \left| \mathbb{Z}_q \right|} \cdot \frac{\left| \mathcal{Q} \mathcal{R}_q \right| \left| \mathbb{Z}_q \right|}{\left| \mathcal{Q} \mathcal{R}_p \right| \left| \mathbb{Z}_p \right|} = \frac{p-1}{q-1} \cdot \frac{q-1}{p-1} = 1, rendering as TeX
[WARNING] Could not convert TeX math \begin{array}{l} \mathcal{J}_N(x) \overset{\text{def}}{=} \mathcal{J}_p(x) \cdot \mathcal{J}_q(x) \\\ \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, = \mathcal{J}_p(\lbrack x \mod p \rbrace) \cdot \mathcal{J}_q(\lbrack x \mod q \rbrace) . \end{array}, rendering as TeX
2025-07-19 20:24:31,624 - WARNING - Could not convert TeX math \begin{array}{l} \mathcal{J}_N(x) \overset{\text{def}}{=} \mathcal{J}_p(x) \cdot \mathcal{J}_q(x) \\\ \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, \backslash, = \mathcal{J}_p(\lbrack x \mod p \rbrace) \cdot \mathcal{J}_q(\lbrack x \mod q \rbrace) . \end{array}, rendering as TeX
[WARNING] Could not convert TeX math \mathcal{Q} \mathcal{N} \mathcal{R}_N^{+1} \overset{\text{def}}{=} \left\{ x \in \mathbb{Z}_N^{\ast} \mid \begin{matrix} x \text{ is not a quadratic residue modulo } N, \\ \text{but } \mathcal{J}_N(x) = +1 \end{matrix} \right. , rendering as TeX
2025-07-19 20:24:31,624 - WARNING - Could not convert TeX math \mathcal{Q} \mathcal{N} \mathcal{R}_N^{+1} \overset{\text{def}}{=} \left\{ x \in \mathbb{Z}_N^{\ast} \mid \begin{matrix} x \text{ is not a quadratic residue modulo } N, \\ \text{but } \mathcal{J}_N(x) = +1 \end{matrix} \right. , rendering as TeX
[WARNING] Could not convert TeX math \begin{matrix} \left| \mathcal{J}_N^{+1} \right| \right| = \left| \mathcal{J}_p^{+1} \right| \times \left| \mathcal{J}_q^{+1} \right| \right| + \left| \mathcal{J}_p^{-1} \right| \times \left| \mathcal{J}_q^{-1} \right| \right| \\\ = \left| \mathcal{J}_p^{+1} \right| \right| \cdot \left| \mathcal{J}_q^{+1} \right| \right| + \left| \mathcal{J}_p^{-1} \right| \right| \cdot \left| \mathcal{J}_q^{-1} \right| \right| \\\
\end{matrix}

```





[illegible]

2025-07-19 20:24:31,639 - WARNING - Could not convert TeX math  $x = \left\lfloor a^{\frac{p+1}{4}} \bmod p \right\rfloor$ , rendering as TeX

[WARNING] Could not convert TeX math  $a^{\frac{p+1}{4}}$ , rendering as TeX

2025-07-19 20:24:31,640 - WARNING - Could not convert TeX math  $a^{\frac{p+1}{4}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $a^{\frac{r+1}{2}}$ , rendering as TeX

2025-07-19 20:24:31,640 - WARNING - Could not convert TeX math  $a^{\frac{r+1}{2}}$ , rendering as TeX

[WARNING] Could not convert TeX math  $a^{\frac{r+1}{2}} \cdot b^{\frac{r^{\prime}}{2}} \bmod p$ , rendering as TeX

2025-07-19 20:24:31,641 - WARNING - Could not convert TeX math  $a^{\frac{r+1}{2}} \cdot b^{\frac{r^{\prime}}{2}} \bmod p$ , rendering as TeX

[WARNING] Could not convert TeX math  $\frac{p-1}{2} = 2^{\ell} \cdot m$ , rendering as TeX

2025-07-19 20:24:31,641 - WARNING - Could not convert TeX math  $\frac{p-1}{2} = 2^{\ell} \cdot m$ , rendering as TeX

[WARNING] Could not convert TeX math  $a^{2^{\ell}m} = a^{\frac{p-1}{2}} = 1 \bmod p$ , rendering as TeX

2025-07-19 20:24:31,643 - WARNING - Could not convert TeX math  $a^{2^{\ell}m} = a^{\frac{p-1}{2}} = 1 \bmod p$ , rendering as TeX

[WARNING] Could not convert TeX math  $b^{2^{\ell}m} = b^{\frac{p-1}{2}} = -1 \bmod p$ , rendering as TeX

2025-07-19 20:24:31,643 - WARNING - Could not convert TeX math  $b^{2^{\ell}m} = b^{\frac{p-1}{2}} = -1 \bmod p$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left\lfloor a^{\frac{p+1}{4}} \bmod p \right\rfloor$ , rendering as TeX

2025-07-19 20:24:31,645 - WARNING - Could not convert TeX math  $\left\lfloor a^{\frac{p+1}{4}} \bmod p \right\rfloor$ , rendering as TeX

[WARNING] Could not convert TeX math  $2^{\ell} \cdot m = \frac{p-1}{2}$ , rendering as TeX

2025-07-19 20:24:31,646 - WARNING - Could not convert TeX math  $2^{\ell} \cdot m = \frac{p-1}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\left\lfloor a^{\frac{r+1}{2}} \cdot b^{\frac{r^{\prime}}{2}} \bmod p \right\rfloor$ , rendering as TeX

2025-07-19 20:24:31,648 - WARNING - Could not convert TeX math  $\left\lfloor a^{\frac{r+1}{2}} \cdot b^{\frac{r^{\prime}}{2}} \bmod p \right\rfloor$ , rendering as TeX

[WARNING] Could not convert TeX math  $\frac{p-1}{2} = 2^3 \cdot m$ , rendering as TeX

2025-07-19 20:24:31,648 - WARNING - Could not convert TeX math  $\frac{p-1}{2} = 2^3 \cdot m$ , rendering as TeX

[WARNING] Could not convert TeX math  $b^{\frac{p-1}{2}} = b^{2^3m} = -1 \bmod p$ , rendering as TeX

2025-07-19 20:24:31,649 - WARNING - Could not convert TeX math  $b^{\frac{p-1}{2}} = b^{2^3m} = -1 \bmod p$ , rendering as TeX

[WARNING] Could not convert TeX math  $a^{\frac{m+1}{2}} \cdot b^{m+2m+2^2m} \bmod p$ , rendering as TeX

2025-07-19 20:24:31,650 - WARNING - Could not convert TeX math  $a^{\frac{m+1}{2}}$



[WARNING] Could not convert TeX math  $S\overset{\text{\text{-def-}}}{=}\left\{ \{x \in \{\mathbb{Z}\}_N^{\text{\ast}} \mid x < N/2 \} \right.$ , rendering as TeX

2025-07-19 20:24:31,655 - WARNING - Could not convert TeX math  $S\overset{\text{\text{-def-}}}{=}\left\{ \{x \in \{\mathbb{Z}\}_N^{\text{\ast}} \mid x < N/2 \} \right.$ , rendering as TeX

[WARNING] Could not convert TeX math  $f_N(x) = \left\{ \begin{matrix} \left\lfloor x^2 \right\rfloor \bmod N & \text{if } \left\lfloor x^2 \right\rfloor \bmod N < N/2 \\ - \left\lfloor x^2 \right\rfloor \bmod N & \text{if } \left\lfloor x^2 \right\rfloor \bmod N > N/2 \end{matrix} \right.$ , rendering as TeX

2025-07-19 20:24:31,656 - WARNING - Could not convert TeX math  $f_N(x) = \left\{ \begin{matrix} \left\lfloor x^2 \right\rfloor \bmod N & \text{if } \left\lfloor x^2 \right\rfloor \bmod N < N/2 \\ - \left\lfloor x^2 \right\rfloor \bmod N & \text{if } \left\lfloor x^2 \right\rfloor \bmod N > N/2 \end{matrix} \right.$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{half}_N(x) = \begin{matrix} -1 & \text{if } x < N/2 \\ +1 & \text{if } x > N/2 \end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,657 - WARNING - Could not convert TeX math  $\text{half}_N(x) = \begin{matrix} -1 & \text{if } x < N/2 \\ +1 & \text{if } x > N/2 \end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $p(X) = \{\sum_{i=0}^t a_i X^i\}$ , rendering as TeX

2025-07-19 20:24:31,658 - WARNING - Could not convert TeX math  $p(X) = \{\sum_{i=0}^t a_i X^i\}$ , rendering as TeX

[WARNING] Could not convert TeX math  $(x+y)^n = \sum_{i=0}^n \left( \begin{matrix} n \\ i \end{matrix} \right) x^i y^{n-i}$ , rendering as TeX

2025-07-19 20:24:31,658 - WARNING - Could not convert TeX math  $(x+y)^n = \sum_{i=0}^n \left( \begin{matrix} n \\ i \end{matrix} \right) x^i y^{n-i}$ , rendering as TeX

[WARNING] Could not convert TeX math  $e^{-x} \leq 1 - \frac{1}{e} \leq 1 - \frac{x}{2}$ , rendering as TeX

2025-07-19 20:24:31,660 - WARNING - Could not convert TeX math  $e^{-x} \leq 1 - \frac{1}{e} \leq 1 - \frac{x}{2}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$

```
= 0, rendering as TeX
2025-07-19 20:24:31,661 - WARNING - Could not convert TeX math

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$$
, rendering as TeX
[WARNING] Could not convert TeX math $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$
= ∞ , rendering as TeX
2025-07-19 20:24:31,661 - WARNING - Could not convert TeX math

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$$
, rendering as TeX
[WARNING] Could not convert TeX math $\overline{E}$, rendering as TeX
2025-07-19 20:24:31,662 - WARNING - Could not convert TeX math $\overline{E}$,
rendering as TeX
[WARNING] Could not convert TeX math $\overline{E}$, rendering as TeX
2025-07-19 20:24:31,663 - WARNING - Could not convert TeX math $\overline{E}$,
rendering as TeX
[WARNING] Could not convert TeX math $\text{Pr} \setminus \text{Pr} = 1 -$
 $\text{Pr} \setminus \overline{E}$, rendering as TeX
2025-07-19 20:24:31,664 - WARNING - Could not convert TeX math $\text{Pr} \setminus \text{Pr} = 1 -$
 $\text{Pr} \setminus \overline{E}$, rendering as TeX
[WARNING] Could not convert TeX math $\text{Pr} \setminus \left[V_{i=1}^k E_i \right]$
 $\leq \sum \limits_{i=1}^k \text{Pr} \setminus E_i$
 $\right]$, rendering as TeX
2025-07-19 20:24:31,665 - WARNING - Could not convert TeX math
 $\text{Pr} \setminus \left[V_{i=1}^k E_i \right] \leq \sum \limits_{i=1}^k \text{Pr} \setminus E_i$
 $\right]$, rendering as TeX
[WARNING] Could not convert TeX math $\text{Pr} \setminus \left[E_1 \mid E_2 \right]$
 $\overset{\text{def}}{=} \frac{\text{Pr} \setminus E_1 \land E_2}{\text{Pr} \setminus E_2}$, rendering as
TeX
2025-07-19 20:24:31,665 - WARNING - Could not convert TeX math
 $\text{Pr} \setminus \left[E_1 \mid E_2 \right]$
 $\overset{\text{def}}{=} \frac{\text{Pr} \setminus E_1 \land E_2}{\text{Pr} \setminus E_2}$, rendering as
TeX
[WARNING] Could not convert TeX math $\text{Pr} \setminus \left[E_1 \mid E_2 \right]$
 $\right] = \frac{\text{Pr} \setminus E_2 \mid E_1}{\text{Pr} \setminus E_2}$
 $\cdot \text{Pr} \setminus E_1$, rendering as TeX
2025-07-19 20:24:31,666 - WARNING - Could not convert TeX math
 $\text{Pr} \setminus \left[E_1 \mid E_2 \right] \right] =$
 $\frac{\text{Pr} \setminus E_2 \mid E_1}{\text{Pr} \setminus E_2}$
 $\cdot \text{Pr} \setminus E_1$
 $\right] =$
 $\frac{\text{Pr} \setminus E_2 \mid E_1}{\text{Pr} \setminus E_2 \mid E_1} \cdot$
 $\text{Pr} \setminus E_1$
 $\right] =$
 $\frac{\text{Pr} \setminus E_2 \mid E_1}{\text{Pr} \setminus E_2 \mid E_1} \cdot$
 $\text{Pr} \setminus E_1$
```

$\right\rbrace$ ., rendering as TeX  
2025-07-19 20:24:31,667 - WARNING - Could not convert TeX math  
 $\text{Pr}\left\{E_1 \mid E_2\right\} \right\} =$   
 $\frac{\text{Pr}\left\{E_1 \land E_2\right\}}{\text{Pr}\left\{E_2\right\}} =$   
 $\frac{\text{Pr}\left\{E_2 \land E_1\right\}}{\text{Pr}\left\{E_2\right\}} =$   
 $\frac{\text{Pr}\left\{E_2 \mid E_1\right\} \text{Pr}\left\{E_1\right\}}{\text{Pr}\left\{E_1\right\} \text{Pr}\left\{E_2\right\}}$   
 $\right\}$ ., rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\left\{F \land E_i\right\} \right\} \leq \sum_{i=1}^n \text{Pr}\left\{F \mid E_i\right\} \right\}$   
 $= \sum_{i=1}^n \text{Pr}\left\{F \mid E_i\right\} \text{Pr}\left\{E_i\right\}$   
 $\cdot \text{Pr}\left\{E_i\right\} \right\}$ ,  
 $\end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,668 - WARNING - Could not convert TeX math  $\begin{matrix}$   
 $\left\{F \land E_i\right\} \right\} \leq \sum_{i=1}^n \text{Pr}\left\{F \mid E_i\right\} \right\}$   
 $= \sum_{i=1}^n \text{Pr}\left\{F \mid E_i\right\} \text{Pr}\left\{E_i\right\} \right\}$   
 $\cdot \text{Pr}\left\{E_i\right\} \right\}$ ,  
 $\end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\overline{E}_1$ , rendering as TeX  
2025-07-19 20:24:31,669 - WARNING - Could not convert TeX math  
 $\overline{E}_1$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix}$   
 $\left\{E_1 \vee E_2\right\} \right\} =$   
 $\text{Pr}\left\{E_1 \vee E_2 \mid E_1\right\} \right\} \cdot$   
 $\text{Pr}\left\{E_1\right\} \right\} + \text{Pr}\left\{E_1 \vee$   
 $E_2 \mid \overline{E}_1\right\} \right\} \cdot \text{Pr}\left\{\overline{E}_1\right\} \right\}$   
 $\leq \text{Pr}\left\{E_1\right\} \right\} + \text{Pr}\left\{E_2\right\}$   
 $\mid \overline{E}_1\right\} \right\}$ .  
 $\end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,670 - WARNING - Could not convert TeX math  $\begin{matrix}$   
 $\left\{E_1 \vee E_2\right\} \right\} =$   
 $\text{Pr}\left\{E_1 \vee E_2 \mid E_1\right\} \right\} \cdot$   
 $\text{Pr}\left\{E_1\right\} \right\} + \text{Pr}\left\{E_1 \vee$   
 $E_2 \mid \overline{E}_1\right\} \right\} \cdot \text{Pr}\left\{\overline{E}_1\right\} \right\}$   
 $\leq \text{Pr}\left\{E_1\right\} \right\} + \text{Pr}\left\{E_2\right\}$   
 $\mid \overline{E}_1\right\} \right\}$ .  
 $\end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{V_{i=1}^n E_i\right\}$   
 $\right\} \leq \text{Pr}\left\{E_1\right\} \right\} + \sum_{i=2}^n \text{Pr}\left\{E_i \mid \overline{E}_1 \land \dots \land$   
 $\overline{E}_{i-1}\right\} \right\}$ ., rendering as TeX  
2025-07-19 20:24:31,670 - WARNING - Could not convert TeX math

$\text{\text{Pr}}\left\{V_{i=1}^n E_i\right\} \right\} \leq$   
 $\text{\text{Pr}}\left\{E_1\right\} \right\} + \sum\limits_{i=2}^n \text{\text{Pr}}\left\{E_i \mid \overline{E}_1 \wedge \cdots \wedge \overline{E}_{i-1}\right\} \right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\text{Pr}}\left\{X_1 = b_1 \wedge \cdots \wedge X_k = b_k\right\} = \left\{\prod_{i=1}^k \text{\text{Pr}}\left\{X_i = b_i\right\}\right\}$ , rendering as TeX  
2025-07-19 20:24:31,671 - WARNING - Could not convert TeX math  $\text{\text{Pr}}\left\{X_1 = b_1 \wedge \cdots \wedge X_k = b_k\right\} = \left\{\prod_{i=1}^k \text{\text{Pr}}\left\{X_i = b_i\right\}\right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $\text{\text{Exp}}[X]\underline{\underline{\text{\text{def}}}}\sum_{s \in S} s \cdot \text{\text{Pr}}\left\{X = s\right\}$ , rendering as TeX  
2025-07-19 20:24:31,671 - WARNING - Could not convert TeX math  
 $\text{\text{Exp}}[X]\underline{\underline{\text{\text{def}}}}\sum_{s \in S} s \cdot \text{\text{Pr}}\left\{X = s\right\}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\begin{matrix} \exp\left\{X\right\} = \sum\limits_{s \in S} s \cdot \text{\text{Pr}}\left\{X = s\right\} \\ \geq \sum\limits_{x \in S, x < v} \text{\text{Pr}}\left\{X = s\right\} \cdot 0 + \sum\limits_{x \in S, x \geq v} v \cdot \text{\text{Pr}}\left\{X = s\right\} \\ = v \cdot \text{\text{Pr}}\left\{X \geq v\right\} \end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,672 - WARNING - Could not convert TeX math  $\begin{matrix} \exp\left\{X\right\} = \sum\limits_{s \in S} s \cdot \text{\text{Pr}}\left\{X = s\right\} \\ \geq \sum\limits_{x \in S, x < v} \text{\text{Pr}}\left\{X = s\right\} \cdot 0 + \sum\limits_{x \in S, x \geq v} v \cdot \text{\text{Pr}}\left\{X = s\right\} \\ = v \cdot \text{\text{Pr}}\left\{X \geq v\right\} \end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  
 $\text{\text{Var}}[X]\underline{\underline{\text{\text{def}}}}\text{\text{Exp}}[(X - \text{\text{Exp}}[X])^2] = \text{\text{Exp}}[X^2] - \text{\text{Exp}}[X]^2$ , rendering as TeX  
2025-07-19 20:24:31,673 - WARNING - Could not convert TeX math  
 $\text{\text{Var}}[X]\underline{\underline{\text{\text{def}}}}\text{\text{Exp}}[(X - \text{\text{Exp}}[X])^2] = \text{\text{Exp}}[X^2] - \text{\text{Exp}}[X]^2$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{\text{Exp}}\left\{X_i\right\} = \frac{1}{2}$ , rendering as TeX  
2025-07-19 20:24:31,674 - WARNING - Could not convert TeX math  $\text{\text{Exp}}\left\{X_i\right\} = \frac{1}{2}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left.\text{\text{Pr}}\left\{\middle| X - \exp\left\{X\right\} \middle| \geq \delta\right\} \leq \frac{\text{\text{Var}}\left\{X\right\}}{\delta^2}\right\}$ , rendering as TeX  
2025-07-19 20:24:31,675 - WARNING - Could not convert TeX math  $\left.\text{\text{Pr}}\left\{\middle| X - \exp\left\{X\right\} \middle| \geq \delta\right\} \leq \frac{\text{\text{Var}}\left\{X\right\}}{\delta^2}\right\}$ , rendering as TeX

[illegible]



[WARNING] Could not convert TeX math  $\text{Var}\left[\frac{1}{m} \cdot \sum\limits_{i=1}^m X_i\right] \text{right} = \frac{1}{m^2} \sum\limits_{i=1}^m \text{Var}\left[X_i\right] \text{right} = \frac{1}{m^2} \sum\limits_{i=1}^m \sigma^2 = \frac{\sigma^2}{m}$ ., rendering as TeX

2025-07-19 20:24:31,683 - WARNING - Could not convert TeX math  $\text{Var}\left[\frac{1}{m} \cdot \sum\limits_{i=1}^m X_i\right] = \frac{1}{m^2} \sum\limits_{i=1}^m \text{Var}\left[X_i\right] = \frac{1}{m^2} \sum\limits_{i=1}^m \sigma^2 = \frac{\sigma^2}{m}$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[X_i = b\right] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

2025-07-19 20:24:31,685 - WARNING - Could not convert TeX math  $\text{Pr}\left[X_i = b\right] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}\left[X \neq b\right] \leq \frac{1}{4} \cdot \varepsilon^2 \cdot m$ ., rendering as TeX

2025-07-19 20:24:31,685 - WARNING - Could not convert TeX math  $\text{Pr}\left[X \neq b\right] \leq \frac{1}{4} \cdot \varepsilon^2 \cdot m$ ., rendering as TeX

[WARNING] Could not convert TeX math  $\text{Exp}[X_i] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

2025-07-19 20:24:31,686 - WARNING - Could not convert TeX math  $\text{Exp}[X_i] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Exp}[X_i] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

2025-07-19 20:24:31,688 - WARNING - Could not convert TeX math  $\text{Exp}[X_i] \geq \frac{1}{2} + \varepsilon$ , rendering as TeX

[WARNING] Could not convert TeX math  $\sum_{i=1}^m X_i \leq m/2$ , rendering as TeX

2025-07-19 20:24:31,689 - WARNING - Could not convert TeX math  $\sum_{i=1}^m X_i \leq m/2$ , rendering as TeX

[WARNING] Could not convert TeX math 
$$\begin{matrix} \text{Pr}\left[X \neq 1\right] = \text{Pr}\left[\sum\limits_{i=1}^m X_i \leq m/2\right] \\ = \text{Pr}\left[\frac{\sum_{i=1}^m X_i}{m} - \frac{1}{2} \leq 0\right] \\ = \text{Pr}\left[\frac{\sum_{i=1}^m X_i}{m} - \left(\frac{1}{2} + \varepsilon\right) \leq -\varepsilon\right] \\ \leq \text{Pr}\left[\left|\frac{\sum_{i=1}^m X_i}{m} - \frac{1}{2}\right| \geq \varepsilon\right] \end{matrix}$$
, rendering as TeX

2025-07-19 20:24:31,689 - WARNING - Could not convert TeX math 
$$\begin{matrix} \text{Pr}\left[X \neq 1\right] = \text{Pr}\left[\sum\limits_{i=1}^m X_i \leq m/2\right] \\ = \text{Pr}\left[\frac{\sum_{i=1}^m X_i}{m} - \frac{1}{2} \leq 0\right] \\ = \text{Pr}\left[\frac{\sum_{i=1}^m X_i}{m} - \left(\frac{1}{2} + \varepsilon\right) \leq -\varepsilon\right] \\ \leq \text{Pr}\left[\left|\frac{\sum_{i=1}^m X_i}{m} - \frac{1}{2}\right| \geq \varepsilon\right] \end{matrix}$$

$\left\{ \frac{1}{2} + \varepsilon \right\} \right\} \geq \varepsilon \right\}$   
 $\end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr} \left[ X \neq 1 \right] \leq \frac{1}{4\varepsilon^2 m}$ , rendering as TeX  
2025-07-19 20:24:31,690 - WARNING - Could not convert TeX math  $\text{Pr} \left[ X \neq 1 \right] \leq \frac{1}{4\varepsilon^2 m}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr} \left[ X_i = b \right] = \frac{1}{2} + \varepsilon$ , rendering as TeX  
2025-07-19 20:24:31,691 - WARNING - Could not convert TeX math  $\text{Pr} \left[ X_i = b \right] = \frac{1}{2} + \varepsilon$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q \leq \sqrt{2N}$ , rendering as TeX  
2025-07-19 20:24:31,692 - WARNING - Could not convert TeX math  $q \leq \sqrt{2N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q = \Theta(\sqrt{N})$ , rendering as TeX  
2025-07-19 20:24:31,693 - WARNING - Could not convert TeX math  $q = \Theta(\sqrt{N})$ , rendering as TeX  
[WARNING] Could not convert TeX math  $q \leq \sqrt{2N}$ , rendering as TeX  
2025-07-19 20:24:31,694 - WARNING - Could not convert TeX math  $q \leq \sqrt{2N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{q \cdot (q - 1)}{4N} \leq 1 - e^{-q(q - 1)/2N} \leq \text{coll}(q, N) \leq \frac{q \cdot (q - 1)}{2N}$ , rendering as TeX  
2025-07-19 20:24:31,694 - WARNING - Could not convert TeX math  $\frac{q \cdot (q - 1)}{4N} \leq 1 - e^{-q(q - 1)/2N} \leq \text{coll}(q, N) \leq \frac{q \cdot (q - 1)}{2N}$ , rendering as TeX  
[WARNING] Could not convert TeX math 
$$\text{Pr} \left[ \text{Coll} \right] = \text{Pr} \left[ \bigvee_{i \neq j} \left( \sum_{i \neq j} \text{Pr} \left[ \text{Coll}_{i,j} \right] \right) \right]$$
  

$$= \left( \begin{array}{l} q \\ 2 \end{array} \right) \cdot \frac{1}{N}$$
  
 $\end{matrix}$ , rendering as TeX  
2025-07-19 20:24:31,696 - WARNING - Could not convert TeX math 
$$\text{Pr} \left[ \text{Coll} \right] = \text{Pr} \left[ \bigvee_{i \neq j} \left( \sum_{i \neq j} \text{Pr} \left[ \text{Coll}_{i,j} \right] \right) \right]$$
  

$$= \left( \begin{array}{l} q \\ 2 \end{array} \right) \cdot \frac{1}{N}$$
  
 $\end{matrix}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{NoColl}_q = \overline{\text{Coll}}$ , rendering as TeX  
2025-07-19 20:24:31,696 - WARNING - Could not convert TeX math  $\text{NoColl}_q = \overline{\text{Coll}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{i}{N}$ , rendering as TeX  
2025-07-19 20:24:31,698 - WARNING - Could not convert TeX math  $\frac{i}{N}$ ,

rendering as TeX  
[WARNING] Could not convert TeX math  $1 - \frac{i}{N}$ , rendering as TeX  
2025-07-19 20:24:31,700 - WARNING - Could not convert TeX math  $1 - \frac{i}{N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_i + 1 \mid \text{NoColl}_i\right\} \text{rbrack} = 1 - \frac{i}{N}$ , rendering as TeX  
2025-07-19 20:24:31,701 - WARNING - Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_i + 1 \mid \text{NoColl}_i\right\} \text{rbrack} = 1 - \frac{i}{N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} = \prod \limits_{i=1}^q \left(1 - \frac{i}{N}\right)\right.$ , rendering as TeX  
2025-07-19 20:24:31,702 - WARNING - Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} = \prod \limits_{i=1}^q \left(1 - \frac{i}{N}\right)\right.$ , rendering as TeX  
[WARNING] Could not convert TeX math  $1 - \frac{i}{N} \leq e^{-i/N}$ , rendering as TeX  
2025-07-19 20:24:31,704 - WARNING - Could not convert TeX math  $1 - \frac{i}{N} \leq e^{-i/N}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} \leq \prod \limits_{i=1}^q e^{-i/N} = e^{-\sum \limits_{i=1}^q (i/N)} = e^{-q(q-1)/2N}\right.$ , rendering as TeX  
2025-07-19 20:24:31,705 - WARNING - Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} \leq \prod \limits_{i=1}^q e^{-i/N} = e^{-\sum \limits_{i=1}^q (i/N)} = e^{-q(q-1)/2N}\right.$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} \geq 1 - e^{-q(q-1)/2N} \geq \frac{q(q-1)}{4N}\right.$ , rendering as TeX  
2025-07-19 20:24:31,706 - WARNING - Could not convert TeX math  $\text{Pr}\left\{\text{NoColl}_q \text{rbrack} \geq 1 - e^{-q(q-1)/2N} \geq \frac{q(q-1)}{4N}\right.$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\left| \text{Pr}_{f \leftarrow \text{Func}} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \right\} \text{rbrack} - \text{Pr}_{f \leftarrow \text{Perm}} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \right\} \text{rbrack} \right| < \frac{q^2}{2^{n+1}}$ , rendering as TeX  
2025-07-19 20:24:31,707 - WARNING - Could not convert TeX math  $\left| \text{Pr}_{f \leftarrow \text{Func}} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \right\} \text{rbrack} - \text{Pr}_{f \leftarrow \text{Perm}} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \right\} \text{rbrack} \right| < \frac{q^2}{2^{n+1}}$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\underset{f \leftarrow \text{Func}}{\left\{ \text{Pr} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \mid \overline{\text{Coll}} \right\} \text{rbrack} = \underset{f \leftarrow \text{Perm}}{\left\{ \text{Pr} \left\{ D^f(\cdot) \left( 1^n \right) = 1 \right\} \text{rbrack} \right.}$ , rendering as TeX

2025-07-19 20:24:31,708 - WARNING - Could not convert TeX math  

$$\underset{f\leftarrow\text{Func}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\mid\overline{\text{Coll}}\right\rbrack=\underset{f\leftarrow\text{Perm}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\right\rbrack., \text{rendering as TeX}$$
[WARNING] Could not convert TeX math  $\overset{\rightarrow}{a}=(a_1,\ldots,a_q)\in S$ , rendering as TeX

2025-07-19 20:24:31,709 - WARNING - Could not convert TeX math  
 $\overset{\rightarrow}{a}=(a_1,\ldots,a_q)\in S$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overset{\rightarrow}{a}$ , rendering as TeX

2025-07-19 20:24:31,711 - WARNING - Could not convert TeX math  
 $\overset{\rightarrow}{a}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\overset{\rightarrow}{a}\notin$ , rendering as TeX

2025-07-19 20:24:31,712 - WARNING - Could not convert TeX math  
 $\overset{\rightarrow}{a}\notin$ , rendering as TeX

[WARNING] Could not convert TeX math  $\text{Pr}_{f\leftarrow\text{Perm}}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\right\rbrack=\frac{\mid S\cap\text{distinct}\mid}{\mid\text{distinct}\mid}.$ , rendering as TeX

2025-07-19 20:24:31,713 - WARNING - Could not convert TeX math  
 $\text{Pr}_{f\leftarrow\text{Perm}}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\right\rbrack=\frac{\mid S\cap\text{distinct}\mid}{\mid\text{distinct}\mid}.$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{matrix}\{\underset{f\leftarrow\text{Func}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\mid\overline{\text{Coll}}\right\rbrack=\frac{\text{Pr}_{f\leftarrow\text{Func}}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\land\overline{\text{Coll}}\right\rbrack\}\{\text{Pr}_{f\leftarrow\text{Func}}\}\left\lbrack\overline{\text{Coll}}\right\rbrack\}\backslash\\ \{=\frac{2^{-nq}}{\cdot}\mid S\cap\text{distinct}\mid\}\{2^{-nq}}\cdot\mid\text{distinct}\mid\}.\end{matrix}$ , rendering as TeX

2025-07-19 20:24:31,714 - WARNING - Could not convert TeX math  $\begin{matrix}\{\underset{f\leftarrow\text{Func}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\mid\overline{\text{Coll}}\right\rbrack=\frac{\text{Pr}_{f\leftarrow\text{Func}}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\land\overline{\text{Coll}}\right\rbrack\}\{\text{Pr}_{f\leftarrow\text{Func}}\}\left\lbrack\overline{\text{Coll}}\right\rbrack\}\backslash\\ \{=\frac{2^{-nq}}{\cdot}\mid S\cap\text{distinct}\mid\}\{2^{-nq}}\cdot\mid\text{distinct}\mid\}.\end{matrix}$ , rendering as TeX

[WARNING] Could not convert TeX math  $\begin{array}{l}\left|\{\underset{f\leftarrow\text{Func}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\right\rbrack-\underset{f\leftarrow\text{Perm}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\left(1^n\right)=1\right\rbrack\right|\backslash\\ \{=\left|\{\underset{f\leftarrow\text{Func}}{n}\}\{\text{Pr}\}\left\lbrack D^f(\cdot)\right\rbrack\right|\end{array}$

```

\cdot))\left(1^{\{n\}} \right) = 1 \mid \overline{\text{Coll}} \right\rbrace
\cdot \text{Pr}\lbrace\overline{\text{Coll}}\rbrace \right.) \\\
\left. \{,\,,\,, + \underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \mid \text{Coll} \right\rbrace \cdot
\text{Pr}\lbrace\text{Coll}\rbrace -
\underset{f\leftarrow\text{Func}}{\text{Pr}}\left\lbrace D^f(\cdot
))\left(1^{\{n\}} \right) = 1 \right\rbrace \right| \\\
{= \left| \{\underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \mid \text{Coll} \right\rbrace \cdot
\text{Pr}\lbrace\text{Coll}\rbrace - \underset{f\leftarrow
\text{Pr}}{\text{Pr}}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1
\right\rbrace \cdot \text{Pr}\lbrace\text{Coll}\rbrace \right|} \\\
{\leq \text{Pr}\lbrace\text{Coll}\rbrace.}
\end{array}, \text{rendering as TeX}
2025-07-19 20:24:31,716 - WARNING - Could not convert TeX math \begin{array}{l}
\left| \{\underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \right\rbrace - \underset{f\leftarrow
\text{Pr}}{\text{Pr}}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1
\right\rbrace \right| \\\
{= \left| \{\underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \mid \overline{\text{Coll}} \right\rbrace \cdot
\text{Pr}\lbrace\overline{\text{Coll}}\rbrace \right.) \\\
\left. \{,\,,\,, + \underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \mid \text{Coll} \right\rbrace \cdot
\text{Pr}\lbrace\text{Coll}\rbrace -
\underset{f\leftarrow\text{Func}}{\text{Pr}}\left\lbrace D^f(\cdot
))\left(1^{\{n\}} \right) = 1 \right\rbrace \right| \\\
{= \left| \{\underset{f\leftarrow\text{Func}}{\text{Pr}}\}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1 \mid \text{Coll} \right\rbrace \cdot
\text{Pr}\lbrace\text{Coll}\rbrace - \underset{f\leftarrow
\text{Pr}}{\text{Pr}}\left\lbrace D^f(\cdot)\right(1^{\{n\}} \right) = 1
\right\rbrace \cdot \text{Pr}\lbrace\text{Coll}\rbrace \right|} \\\
{\leq \text{Pr}\lbrace\text{Coll}\rbrace.}
\end{array}, \text{rendering as TeX}
[WARNING] Could not convert TeX math n \cdot
a\overset{\text{def}}{=}\underset{n,\text{times}}{\underset{\quad}{a + \cdots +
a}}\quad\text{and}\quad
a^{\{n\}}\overset{\text{def}}{=}\underset{n,\text{times}}{\underset{\quad}{a\cdots
a}}\text{.}, \text{rendering as TeX}
2025-07-19 20:24:31,717 - WARNING - Could not convert TeX math n \cdot
a\overset{\text{def}}{=}\underset{n,\text{times}}{\underset{\quad}{a + \cdots +
a}}\quad\text{and}\quad
a^{\{n\}}\overset{\text{def}}{=}\underset{n,\text{times}}{\underset{\quad}{a\cdots
a}}\text{.}, \text{rendering as TeX}
[WARNING] Could not convert TeX math \left\lbrace a^b\text{mod}N \right\rbrace =
\left\{ \begin{matrix}
\left\lbrace \left(a^{\frac{b}{2}} \right)^2\text{mod}N
\right\rbrace\quad\text{when}\,, b,\text{is-even} \\\

```

$$\left\{ \left[ a \cdot \left( a^{\frac{b-1}{2}} \right)^2 \right] \pmod{N} \right\}$$

$$\left[ a^b \right] \pmod{N} \quad \text{when } b \text{ is odd}$$

2025-07-19 20:24:31,718 - WARNING - Could not convert TeX math  $\left\{ \left[ a^b \right] \pmod{N} \right\}$  to LaTeX, rendering as TeX

$$\left\{ \left[ a^b \right] \pmod{N} \right\} \quad \text{when } b \text{ is even}$$

2025-07-19 20:24:31,719 - WARNING - Could not convert TeX math  $a^b = a^{\sum_{i=0}^{n-1} 2^i} \cdot b_i \pmod{N}$  to LaTeX, rendering as TeX

$$a^b = a^{\sum_{i=0}^{n-1} 2^i} \cdot b_i \pmod{N}$$

2025-07-19 20:24:31,720 - WARNING - Could not convert TeX math  $\begin{matrix} x_0 = a, & x_1 = [a^2 \pmod{N}], & \dots, & x_{n-1} = [a^{2^{n-1}} \pmod{N}] \end{matrix}$  to LaTeX, rendering as TeX

$$\begin{matrix} x_0 = a, & x_1 = [a^2 \pmod{N}], & \dots, & x_{n-1} = [a^{2^{n-1}} \pmod{N}] \end{matrix}$$

2025-07-19 20:24:31,721 - WARNING - Could not convert TeX math  $a^b = a^{\sum_{i=0}^{n-1} 2^i} \cdot b_i \pmod{N}$  to LaTeX, rendering as TeX

$$a^b = a^{\sum_{i=0}^{n-1} 2^i} \cdot b_i \pmod{N}$$

2025-07-19 20:24:31,722 - WARNING - Could not convert TeX math  $\left[ b \cdot g \pmod{N} \right] \overset{\text{def}}{=} \left[ b \cdot g \pmod{N} \right]$  to LaTeX, rendering as TeX

$$\left[ b \cdot g \pmod{N} \right] \overset{\text{def}}{=} \left[ b \cdot g \pmod{N} \right]$$

2025-07-19 20:24:31,723 - WARNING - Could not convert TeX math  $\overline{\underline{\underline{x}}}$  to LaTeX, rendering as TeX

$$\overline{\underline{\underline{x}}}$$

2025-07-19 20:24:31,724 - WARNING - Could not convert TeX math  $\overline{y} \in \mathbb{Z}_N^*$  to LaTeX, rendering as TeX

$$\overline{y} \in \mathbb{Z}_N^*$$

2025-07-19 20:24:31,725 - WARNING - Could not convert TeX math  $\overline{y} \in$  to LaTeX, rendering as TeX

$\{\mathbb{Z}\}_N^{[*]}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\text{Mont}(\overline{x}, \overline{y}) \overset{\text{def}}{=} \left\{ \overline{x} \setminus, \overline{y} R^{-1} \pmod{N} \right\}$ , rendering as TeX  
 2025-07-19 20:24:31,726 - WARNING - Could not convert TeX math  
 $\text{Mont}(\overline{x}, \overline{y}) \overset{\text{def}}{=} \left\{ \overline{x} \setminus, \overline{y} R^{-1} \pmod{N} \right\}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  $\text{Mont}(\overline{x}, \overline{y}) = \overline{x} \overline{y} R^{-1} = (xR)(yR) R^{-1} = (xy)R = \overline{xy} \pmod{N}$ , rendering as TeX  
 2025-07-19 20:24:31,727 - WARNING - Could not convert TeX math  
 $\text{Mont}(\overline{x}, \overline{y}) = \overline{x} \overline{y} R^{-1} = (xR)(yR) R^{-1} = (xy)R = \overline{xy} \pmod{N}$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\alpha \underline{\underline{\text{def}}} \lbrack - N^{-1} \pmod{R} \rbrack$ , rendering as TeX  
 2025-07-19 20:24:31,730 - WARNING - Could not convert TeX math  
 $\alpha \underline{\underline{\text{def}}} \lbrack - N^{-1} \pmod{R} \rbrack$ , rendering as TeX  
 [WARNING] Could not convert TeX math  
 $\underline{\underline{\text{def}}} \text{Mont}(x, y)$ , rendering as TeX  
 2025-07-19 20:24:31,731 - WARNING - Could not convert TeX math  
 $\underline{\underline{\text{def}}} \text{Mont}(x, y)$ , rendering as TeX  
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 $\text{Pr} \lbrack \text{output-of-the-algorithm-equals} \mid \text{Fail} \rbrack = \frac{1}{|\mathbb{G}|}$ , rendering as TeX  
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 $\text{K} \underline{\underline{\text{def}}} 2^{\ell} / |\mathbb{G}|$ , rendering as TeX  
 2025-07-19 20:24:31,736 - WARNING - Could not convert TeX math  $\left( 1 - \frac{1}{|\mathbb{G}|} \right)^{2^{\ell}}$ , rendering as TeX  
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 [WARNING] Could not convert TeX math  $\left( 1 - \frac{1}{|\mathbb{G}|} \right)^{2^{\ell}}$ , rendering as TeX  
 $\left( 1 - \frac{1}{|\mathbb{G}|} \right)^{2^{\ell}} \leq \left( e^{-1} \right)^n = e^{-n}$ , rendering as TeX  
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 [WARNING] Could not convert TeX math  $\frac{2^n}{|\mathbb{Z}_N|}$

$\right|} = \frac{2^{\{n\}}{\{N\}} \leq \frac{2^{\{n\}}{\{2^{\{n-1\}}\}} = 2$ ,, rendering as TeX  
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 $\frac{2^{\{n\}}{\{\left| \{\mathbb{Z}\}_N \right|}} = \frac{2^{\{n\}}{\{N\}} \leq$   
 $\frac{2^{\{n\}}{\{2^{\{n-1\}}\}} = 2$ ,, rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{2^{\{n\}}{\{\left| \{\mathbb{Z}\}_N^{\{\ast\}} \right|}} = \frac{2^{\{n\}}{\{\phi(N)\}} = \frac{2^{\{n\}}{\{N\}} \cdot \frac{N}{\{\phi(N)\}} \leq$   
 $2 \cdot \frac{N}{\{\phi(N)\}}$ ., rendering as TeX  
2025-07-19 20:24:31,739 - WARNING - Could not convert TeX math  
 $\frac{2^{\{n\}}{\{\left| \{\mathbb{Z}\}_N^{\{\ast\}} \right|}} = \frac{2^{\{n\}}{\{\phi(N)\}} =$   
 $\frac{2^{\{n\}}{\{N\}} \cdot \frac{N}{\{\phi(N)\}} \leq 2 \cdot \frac{N}{\{\phi(N)\}}$ .,  
rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{N}{\{\phi(N)\}} < 2n$ , rendering as TeX  
2025-07-19 20:24:31,740 - WARNING - Could not convert TeX math  $\frac{N}{\{\phi(N)\}}$   
 $< 2n$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{N}{\{\phi(N)\}} \leq$   
 $\frac{2^{\{n\}}{\{\phi(N)\}} = \frac{2^{\{n\}}{\{N-1\}} \leq \frac{2^{\{n\}}{\{2^{\{n-1\}}\}} = 2$ ,  
rendering as TeX  
2025-07-19 20:24:31,742 - WARNING - Could not convert TeX math  $\frac{N}{\{\phi(N)\}}$   
 $\leq \frac{2^{\{n\}}{\{\phi(N)\}} = \frac{2^{\{n\}}{\{N-1\}} \leq \frac{2^{\{n\}}{\{2^{\{n-1\}}\}} =$   
 $2$ , rendering as TeX  
[WARNING] Could not convert TeX math  $\frac{N}{\{\phi(N)\}} = \frac{pq}{\{(p-1)(q-1)\}} = \frac{p}{\{p-1\}} \cdot \frac{q}{\{q-1\}} < \left( \frac{3}{\{2\}} \right) \cdot$   
 $\left( \frac{5}{\{4\}} \right) < 2$ ., rendering as TeX  
2025-07-19 20:24:31,743 - WARNING - Could not convert TeX math  $\frac{N}{\{\phi(N)\}}$   
 $= \frac{pq}{\{(p-1)(q-1)\}} = \frac{p}{\{p-1\}} \cdot \frac{q}{\{q-1\}} < \left( \frac{3}{\{2\}} \right) \cdot \left( \frac{5}{\{4\}} \right) < 2$ ., rendering as TeX  
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rendering as TeX  
2025-07-19 20:24:31,744 - WARNING - Could not convert TeX math  $q = \{\prod_{i=1}^{\{k\}} p_{\{i\}}^{\{e_{\{i\}}\}}\}$ , rendering as TeX  
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 $\geq \prod\limits_{i=1}^{\{k\}} p_{\{i\}} \geq \prod\limits_{i=1}^{\{k\}} 2 = 2^{\{k\}}$ .,  
rendering as TeX  
  
2025-07-19 20:24:31,745 - WARNING - Could not convert TeX math  $q =$   
 $\prod\limits_{i=1}^{\{k\}} p_{\{i\}}^{\{e_{\{i\}}\}} \geq \prod\limits_{i=1}^{\{k\}} p_{\{i\}} \geq$   
 $\prod\limits_{i=1}^{\{k\}} 2 = 2^{\{k\}}$ ., rendering as TeX  
  
2025-07-19 20:24:35,287 - INFO - Successfully loaded 6926 raw text elements from  
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2025-07-19 20:24:35,288 - INFO - Flattened ToC into 289 entries.  
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ToC titles...  
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>> 'Chapman & Hall/CRC Cryptography and Network Security Series'

--- Document [2] ---
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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
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>> 'Introduction to Modern Cryptography Jonathan Katz and Yehuda Lindell'

--- Document [3] ---
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>> 'Series Editors: Douglas R. Stinson and Jonathan Katz'

--- Document [4] ---
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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'Secret History: The Story of Cryptology, Second Edition'

--- Document [5] ---
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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
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>> 'Craig P. Bauer'

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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
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>> 'Data Science for Mathematicians'

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>> 'Nathan Carter'
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--- Document [8] ---

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--- Document [9] ---

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--- Document [10] ---

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--- Document [11] ---

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--- Document [12] ---

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>> 'Cryptology: Classical and Modern, Second Edition'
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--- Document [13] ---

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--- Document [14] ---
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>> 'Group Theoretic Cryptography'

--- Document [15] ---
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>> 'Advances of DNA Computing in Cryptography'

--- Document [17] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
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>> 'Suyel Namasudra and Ganesh Chandra Deka'

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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
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>> 'Xiaoyun Wang, Guangwu Xu, Minggang Wang, Xianmeng Meng'

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>> 'Nadia El Mrabet and Marc Joye'

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>> 'https://www.crcpress.com/Chapman--HallCRC-Cryptography-and-Network-
Security-Series/book-series/CHCRYNETSEC'

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>> 'Third edition published 2021'

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>> 'by CRC Press'

--- Document [25] ---
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>> '6000 Broken Sound Parkway NW, Suite 300, Boca Raton, FL 33487-2742'

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>> 'and by CRC Press'

--- Document [27] ---
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>> '2 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN'

--- Document [28] ---
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Content (Un-chunked Element):
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Content (Un-chunked Element):
>> 'First edition published by Taylor and Francis 2007 Second edition
published by Taylor and Francis 2014'

--- Document [30] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
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--- Document [33] ---

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--- Document [34] ---

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--- Document [35] ---

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--- Document [36] ---

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>> 'ISBN: 780815354369 (hbk)'

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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'EPUB Preamble', 'toc\_id': -1}

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>> 'ISBN: 9781351133036 (ebk)'

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Content (Un-chunked Element):

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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'Visit the companion website/eResources:[insert CW/eResources URL]'

--- Document [41] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'To Jill, Abigail, and Rena'

--- Document [42] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'JK'

--- Document [43] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'To Yael, Yehonatan, Itamar, Orel, Shirel, and Noam'

--- Document [44] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'EPUB Preamble', 'toc_id': -1}
Content (Un-chunked Element):
>> 'YL'

--- Document [45] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}

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Content (Un-chunked Element):
>> 'Preface'

--- Document [46] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'The goal of our book remains the same as in the first edition: to present
the core paradigms and principles of modern cryptography to a general audience
with a basic mathematics background. We have designed this book to serve as a
textbook for undergraduate- or graduate-level courses in cryptography (in
computer science, electrical engineering, or mathematics departments), as a
general introduction suitable for self-study (especially for beginning graduate
students), and as a reference for students, researchers, and practitioners.'

--- Document [47] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'There are numerous other cryptography textbooks available today, and the
reader may rightly ask whether another book on the subject is needed. We would
not have written this book-nor worked on revising it for the second and third
editions-if the answer to that question were anything other than an unequivocal
yes. What, in our opinion, distinguishes our book from others is that it
provides a rigorous treatment of modern cryptography in an accessible and
introductory manner.'

--- Document [48] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Our focus is on modern (post-1980s) cryptography, which is distinguished
from classical cryptography by its emphasis on definitions, precise assumptions,
and rigorous proofs of security. We briefly discuss each of these in turn (these
principles are explored in greater detail in Chapter 1):'

--- Document [49] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'The central role of definitions: A key intellectual contribution of modern
cryptography has been the recognition that formal definitions of security are an
essential first step in the design of any cryptographic primitive or protocol.
The reason, in retrospect, is simple: if you don't know what it is you are

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trying to achieve, how can you hope to know when you have achieved it? As we will see in this book, cryptographic definitions of security are quite strong and-at first glance-may appear impossible to achieve. One of the most amazing aspects of cryptography is that efficient constructions satisfying such strong definitions can be proven to exist (under rather mild assumptions).'

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--- Document [50] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'The importance of precise assumptions: As will be explained in Chapters 2 and 3. many cryptographic constructions cannot currently be proven secure unconditionally. Security, instead, generally relies on some widely believed (though unproven) assumption(s). The modern cryptographic approach dictates that any such assumptions must be clearly stated and unambiguously defined. This not only allows for objective evaluation of the assumptions but, more importantly, enables rigorous proofs of security (as described next).'

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--- Document [51] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'The possibility of proofs of security: The previous two principles serve as the basis for the idea that cryptographic constructions can be proven secure with respect to clearly stated definitions of security and relative to well-defined cryptographic assumptions. This concept is the essence of modern cryptography, and is what has transformed the field from an art to a science.'

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--- Document [52] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'The importance of this idea cannot be overemphasized. Historically, cryptographic schemes were designed in a largely heuristic fashion, and were deemed to be secure if the designers themselves could not find any attacks. In contrast, modern cryptography advocates the design of schemes with formal, mathematical proofs of security in well-defined models. Such schemes are guaranteed to be secure (with respect to a certain security definition) unless the underlying assumption is false. By relying on long-standing assumptions, it is thus possible to obtain schemes that are extremely unlikely to be broken.'

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--- Document [53] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

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Content (Un-chunked Element):
>> 'A unified approach. The above principles of modern cryptography are
relevant not only to the "theory of cryptography" community. The importance of
precise definitions is, by now, widely understood and appreciated by developers
and security engineers who use cryptographic tools to build secure systems, and
rigorous proofs of security have become one of the requirements for
cryptographic schemes to be standardized.'

--- Document [54] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Changes in the Third Edition'

--- Document [55] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'In preparing the third edition, we have continued to integrate a more
practical perspective without sacrificing a rigorous approach. This is reflected
in a number of changes and additions as compared to the second edition:'

--- Document [56] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'We have divided our treatment of symmetric-key encryption into two parts:
Chapter 3 deals with security against "passive" attacks (i.e., CPA-security),
while Chapter 5 addresses "active" attacks (i.e., CCA-security and authenticated
encryption). Besides breaking up what was previously a long chapter, this also
allows us to introduce message authentication codes before discussing active
attacks against encryption schemes.'

--- Document [57] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'With an eye toward symmetric-key schemes used in practice, we have
improved our coverage of stream ciphers and stream-cipher modes of operation
(Sections 3.6.1 and 3.6.2); added a treatment of nonce-based encryption (Section
3.6.4); and incorporated material about standardized schemes such as GMAC and
Poly1305 (Section 4.5) as well as GCM, CCM, and ChaCha20-Poly1305 (Section
5.3.2).'
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--- Document [58] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'With similar motivation, we have added sections on the ChaCha20 stream
cipher and SHA-3 to Chapter 7. As part of our discussion about SHA-3, we also
describe the sponge construction.'

--- Document [59] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'We have further increased our coverage of elliptic-curve cryptography
(Section 9.3.4), including a discussion of elliptic curves used in practice.'

--- Document [60] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Our treatment of TLS in Section 13.7 has been updated to reflect the
latest version (TLS 1.3).'

--- Document [61] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Reflecting recent trends, we have added a chapter (Chapter 14) describing
the impact of quantum computers on cryptography, and providing examples of
"post-quantum" encryption and signature schemes.'

--- Document [62] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'For those currently using the first edition of our book, as well as for
reference, we also summarize the changes/additions we have already made in the
second edition (all of which remain here):'

--- Document [63] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):

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>> 'We have increased our coverage of stream ciphers, including stream-cipher
modes of operation as well as stream-cipher design principles and examples of
stream ciphers used in practice.'
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--- Document [64] ---
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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
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Content (Un-chunked Element):
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>> 'We have emphasized the importance of authenticated encryption and secure
communication sessions in Sections 5.2-5.4.'
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--- Document [65] ---
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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
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Content (Un-chunked Element):
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>> 'We have moved our treatment of hash functions into its own chapter
(Chapter 6), and have added a section on hash-function design principles and
widely used constructions (Section 7.3). We have also improved our treatment of
generic attacks on hash functions, including a discussion of rainbow tables
(Section 6.4.3).'
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--- Document [66] ---
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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
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Content (Un-chunked Element):
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>> 'We have included several important attacks on cryptographic
implementations that arise in practice, including chosen-plaintext attacks on
chained-CBC encryption (Section 3.6.3), timing attacks on MAC verification
(Section 4.2), and padding-oracle attacks on CBC-mode encryption (Section
5.1.1).'
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--- Document [67] ---
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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
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Content (Un-chunked Element):
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>> 'After much deliberation, we have decided to introduce the random-oracle
model earlier in the book (Section 6.5). This has several benefits, including
allowing for an integrated treatment of standardized public-key encryption and
signature schemes in Chapters 12 and 13.'
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--- Document [68] ---
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Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
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Content (Un-chunked Element):
>> 'We have strengthened our coverage of elliptic-curve cryptography (Section
9.3.4) and have added a discussion of its impact on recommended key lengths
(Section 10.4).'
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--- Document [69] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'In the chapter on public-key encryption, we introduce the KEM/DEM paradigm
as a form of hybrid encryption (see Section 12.3). We also cover DHIES/ECIES in
addition to the RSA PKCS #1 standards.'
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--- Document [70] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'In the chapter on digital signatures, we now describe the construction of
signatures from identification schemes using the Fiat-Shamir transform, with the
Schnorr signature scheme as a prototypical example. We have also improved our
coverage of DSA/ECDSA. We include brief discussions of SSL/TLS and signcryption,
both of which serve as culminations of material covered up to that point.'
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--- Document [71] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'In the "advanced topics" chapter, we have amplified our treatment of
homomorphic encryption, and have added sections on secret sharing and threshold
encryption.'
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--- Document [72] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Beyond the above, we have also edited the entire book to make extensive
corrections as well as smaller adjustments, including more worked examples, to
improve the exposition. Several additional exercises have also been added.'
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--- Document [73] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
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>> 'Guide to Using This Book'

--- Document [74] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'This section is intended primarily for instructors seeking to adopt this
book for their course, though the student picking up this book on his or her own
may also find it a useful overview.'

--- Document [75] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Required background. We have structured the book so the only formal
prerequisite is a course on discrete mathematics. Even here we rely on very
little: we only assume familiarity with basic (discrete) probability and modular
arithmetic. Students reading this book are also expected to have had some
exposure to algorithms, mainly to be comfortable reading pseudocode and to be
familiar with big- notation. Many of these concepts are reviewed in Appendix A
and/or when first used in the book.'

--- Document [76] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Notwithstanding the above, the book does use definitions, proofs, and
abstract mathematical concepts, and therefore requires some mathematical
maturity. In particular, the reader is assumed to have had some exposure to
proofs, whether in an upper-level mathematics course or a course on discrete
mathematics, algorithms, or computability theory.'

--- Document [77] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Suggestions for course organization. The core material of this book, which
we recommend should be covered in any introductory course on cryptography,
consists of the following (in all cases, starred sections are excluded; more on
this below):'

--- Document [78] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan

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Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Introduction and Classical Cryptography: Chapters 1 and 2 discuss
classical cryptography and set the stage for modern cryptography.'

--- Document [79] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Private-Key (Symmetric) Cryptography: Chapter 3-5 provide a thorough
treatment of private-key encryption and message authentication, and Chapter 6
covers hash functions and their applications. (Section 6.6 could be skipped if
that material will not be used later.)'

--- Document [80] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'We also highly recommend covering at least part of Chapter 7, which deals
with symmetric-key primitives used in practice; in our experience students
really enjoy this material, and it makes the abstract ideas they have learned in
previous chapters more concrete. Although we do consider this core material, it
is not used in the remainder of the book and so can be safely skipped if
desired.'

--- Document [81] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'Public-Key Cryptography: Chapter 9 gives a self-contained introduction to
all the number theory needed for the remainder of the book. The material in The
public-key revolution, including Diffie-Hellman key exchange, is described in
Chapter 11. Chapters 12 and 13 go into detail about public-key encryption and
digital signatures; those pressed for time can pick and choose what to cover
appropriately.'

--- Document [82] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'Preface', 'toc_id': 6}
Content (Un-chunked Element):
>> 'We are typically able to cover most of the above in a one-semester
(35-hour) undergraduate or Masters-level course (omitting some proofs and
skipping some topics, as needed) or, with some changes to add more material on
theoretical foundations, in the first three-quarters of a one-semester PhD-level

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course. Instructors with more time available can proceed at a more leisurely pace or incorporate additional topics, as discussed below.'

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--- Document [83] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Those wishing to cover additional material, in either a longer course or a faster-paced graduate course, will find that the book is structured to allow flexible incorporation of other topics as time permits (and depending on the interests of the instructor). Specifically, the starred (\*) sections and chapters may be covered in any order, or skipped entirely, without affecting the overall flow of the book. We have taken care to ensure that none of the core (i.e., unstarred) material depends on any of the starred material and, for the most part, the starred sections do not depend on each other. (When they do, this dependence is explicitly noted.)'

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--- Document [84] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'We suggest the following from among the starred topics for those wishing to give their course a particular flavor:'

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--- Document [85] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Theory: A more theoretically inclined course could include material from Section 3.2.2 (semantic security); Chapter 8 (one-way functions and hard-core predicates, and constructing pseudorandom generators, functions, and permutations from one-way permutations); Section 9.4 (one-way functions and collision-resistant hash functions from number-theoretic assumptions); Section 12.5.3 (RSA encryption without random oracles); and Section 15.3 (cryptographic protocols).'

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--- Document [86] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Mathematics: A course directed at students with a strong mathematics background-or being taught by someone who enjoys this aspect of cryptography-could incorporate Section 4.6 (information-theoretic MACs in finite fields); some of the more advanced number theory from Chapter 9 (e.g., the



Chinese remainder theorem, the Miller-Rabin primality test, and more on elliptic curves); and all of Chapter 10 (algorithms for factoring and computing discrete logarithms).'

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--- Document [87] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'In either case, a selection of advanced public-key schemes from Chapters 14 and 15 could also be included.'

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--- Document [88] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Feedback and Errata'

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--- Document [89] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Our goal in writing this book was to make modern cryptography accessible to a wide audience beyond the "theoretical computer science" community. We hope you will let us know if we have succeeded! The many enthusiastic emails we have received in response to our first and second editions have made the whole process of writing this book worthwhile.'

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--- Document [90] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'We are always happy to receive feedback. We hope there are no errors or typos in the book; if you do find any, however, we would greatly appreciate it if you let us know. You can email your comments and errata to [jkatz2@gmail.com](mailto:jkatz2@gmail.com) and [lindell@biu.ac.il](mailto:lindell@biu.ac.il); please put "Introduction to Modern Cryptography" in the subject line. A list of known errata will be maintained at <http://www.cs.umd.edu/~jkatz/imc.html>.'

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--- Document [91] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Acknowledgments'

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--- Document [92] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'We continue to be grateful to all those who have sent us comments, suggestions, and corrections for the book. We would like to thank, in particular, Jack Aaron, Rounak Agarwal, Ionut Ambrosie, Dan Bernstein, Jeremiah Blocki, David Cash, Claude Crépeau, Dana Dachman-Soled, Daniel Escudero, Pooya Farshim, Rolf Haenni, Imededdine Jerbi, Ali El Kaafarani, Zach Kissel, Angelique Faye Loe, Wilde Luo, Tal Malkin, Alejandro Mardones, Kurt Pan, Greg Plaxton, Kyle Andrew Porter, Christian Schaffner, Jim Tallent, Hanh Tang, Markus Triska, and Rui Xue for their feedback on the second edition.'

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--- Document [93] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Finally, we thank our wives and children for all their support during the now over a decade(!) we have spent working on this project.'

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--- Document [94] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Part I'

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--- Document [95] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Introduction and Classical Cryptography'

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--- Document [96] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

Content (Un-chunked Element):

>> 'Chapter 1'

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--- Document [97] ---

Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub', 'level\_1\_title': 'Preface', 'toc\_id': 6}

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Content (Un-chunked Element):
>> 'Introduction'

--- Document [98] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'I Introduction and Classical
Cryptography', 'level_2_title': '1 Introduction', 'level_3_title': '1.1
Cryptography and Modern Cryptography', 'toc_id': 9}
Content (Un-chunked Element):
>> '1.1 Cryptography and Modern Cryptography'

--- Document [99] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'I Introduction and Classical
Cryptography', 'level_2_title': '1 Introduction', 'level_3_title': '1.1
Cryptography and Modern Cryptography', 'toc_id': 9}
Content (Un-chunked Element):
>> 'The Concise Oxford English Dictionary (9th ed.) defines cryptography as
"the art of writing or solving codes." This is historically accurate, but does
not capture the current breadth of the field or its modern scientific
foundations. The definition focuses solely on the codes that have been used for
centuries to enable secret communication. But cryptography nowadays encompasses
much more than this: it deals with mechanisms for ensuring integrity, techniques
for exchanging secret keys, protocols for authenticating users, electronic
voting, cryptocurrency, and more. Without attempting to provide a complete
characterization, we would say that modern cryptography involves the study of
mathematical techniques for securing digital information, systems, and
distributed computations against adversarial attacks.'

--- Document [100] ---
Assigned Metadata: {'source': 'Introduction to Modern Cryptography -- Jonathan
Katz, Yehuda Lindell, J.epub', 'level_1_title': 'I Introduction and Classical
Cryptography', 'level_2_title': '1 Introduction', 'level_3_title': '1.1
Cryptography and Modern Cryptography', 'toc_id': 9}
Content (Un-chunked Element):
>> 'The dictionary definition also refers to cryptography as an art. Until
late in the 20th century cryptography was, indeed, largely an art. Constructing
good codes, or breaking existing ones, relied on creativity and a developed
sense of how codes work. There was little theory to rely on and, for a long
time, no working definition of what constitutes a good code. Beginning in the
1970s and 1980s, this picture of cryptography radically changed. A rich theory
began to emerge, enabling the rigorous study of cryptography as a science and a
mathematical discipline. This perspective has, in turn, influenced how
researchers think about the broader field of computer security.'

```

### 6.1.1 Full Database Health & Hierarchy Diagnostic Report

```
[44]: # Cell 5.1: Full Database Health & Hierarchy Diagnostic Report (V5 - with
 ↪Content Preview)

import os
import json
import logging
import random
from typing import List, Dict, Any

You might need to install pandas if you haven't already
try:
 import pandas as pd
 pandas_available = True
except ImportError:
 pandas_available = False

try:
 from langchain_chroma import Chroma
 from langchain_ollama.embeddings import OllamaEmbeddings
 from langchain_core.documents import Document
 langchain_available = True
except ImportError:
 langchain_available = False

Setup Logger
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -
 ↪%(message)s')
logger = logging.getLogger(__name__)

--- HELPER FUNCTIONS ---
def print_header(text: str, char: str = "="):
 """Prints a centered header to the console."""
 print("\n" + char * 80)
 print(text.center(80))
 print(char * 80)

def count_total_chunks(node: Dict) -> int:
 """Recursively counts all chunks in a node and its children."""
 total = node.get('_chunks', 0)
 for child_node in node.get('_children', {}).values():
 total += count_total_chunks(child_node)
 return total
```

```

def print_hierarchy_report(node: Dict, indent_level: int = 0):
 """
 Recursively prints the reconstructed hierarchy, sorting by sequential ToC
 ID.
 """
 sorted_children = sorted(
 node.get('_children', {}).items(),
 key=lambda item: item[1].get('_toc_id', float('inf'))
)

 for title, child_node in sorted_children:
 prefix = " " * indent_level + "|-- "
 total_chunks_in_branch = count_total_chunks(child_node)
 direct_chunks = child_node.get('_chunks', 0)
 toc_id = child_node.get('_toc_id', 'N/A')
 print(f"{prefix}{title} [ID: {toc_id}] (Total Chunk in branch:
 {total_chunks_in_branch}, Direct Chunk: {direct_chunks})")
 print_hierarchy_report(child_node, indent_level + 1)

def find_testable_sections(node: Dict, path: str, testable_list: List):
 """
 Recursively find sections with a decent number of "direct" chunks to test
 sequence on.
 """
 if node.get('_chunks', 0) > 10 and not node.get('_children'):
 testable_list.append({
 "path": path,
 "toc_id": node.get('_toc_id'),
 "chunk_count": node.get('_chunks')
 })

 for title, child_node in node.get('_children', {}).items():
 new_path = f"{path} -> {title}" if path else title
 find_testable_sections(child_node, new_path, testable_list)

--- MODIFIED TEST FUNCTION ---
def verify_chunk_sequence_and_content(vector_store: Chroma, hierarchy_tree:
Dict):
 """
 Selects a random ToC section, verifies chunk sequence, and displays the
 reassembled content.
 """
 print_header("Chunk Sequence & Content Integrity Test", char="-")
 logger.info("Verifying chunk order and reassembling content for a random
 ToC section.")

```

```

1. Find a good section to test
testable_sections = []
find_testable_sections(hierarchy_tree, "", testable_sections)

if not testable_sections:
 logger.warning("Could not find a suitable section with enough chunks to_
↳test. Skipping content test.")
 return

random_section = random.choice(testable_sections)
test_toc_id = random_section['toc_id']
section_title = random_section['path'].split(' -> ')[-1]

logger.info(f"Selected random section for testing:
↳'{random_section['path']}' (toc_id: {test_toc_id})")

2. Retrieve all documents (content + metadata) for that toc_id
try:
 # Use .get() to retrieve full documents, not just similarity search
 retrieved_data = vector_store.get(
 where={"toc_id": test_toc_id},
 include=["metadatas", "documents"]
)

 # Combine metadatas and documents into LangChain Document objects
 docs = [Document(page_content=doc, metadata=meta) for doc, meta in
↳zip(retrieved_data['documents'], retrieved_data['metadatas'])]

 logger.info(f"Retrieved {len(docs)} document chunks for toc_id_
↳{test_toc_id}.")

 if len(docs) < 1:
 logger.warning("No chunks found in the selected section. Skipping.")
 return

3. Sort the documents by chunk_id
Handle cases where chunk_id might be missing for robustness
docs.sort(key=lambda d: d.metadata.get('chunk_id', -1))

chunk_ids = [d.metadata.get('chunk_id') for d in docs]
if None in chunk_ids:
 logger.error("TEST FAILED: Some retrieved chunks are missing a_
↳'chunk_id'.")
 return

4. Verify sequence

```

```

 is_sequential = all(chunk_ids[i] == chunk_ids[i-1] + 1 for i in
↪range(1, len(chunk_ids)))

 # 5. Reassemble and print content
 full_content = "\n".join([d.page_content for d in docs])

 print("\n" + "-"*25 + " CONTENT PREVIEW " + "-"*25)
 print(f"Title: {section_title} [toc_id: {test_toc_id}]")
 print(f"Chunk IDs: {chunk_ids}")
 print("-" * 70)
 print(full_content)
 print("-" * 23 + " END CONTENT PREVIEW " + "-"*23 + "\n")

 if is_sequential:
 logger.info(" TEST PASSED: Chunk IDs for the section are
↪sequential and content is reassembled.")
 else:
 logger.warning("TEST PASSED (with note): Chunk IDs are not
↪perfectly sequential but are in increasing order.")
 logger.warning("This is acceptable. Sorting by chunk_id
↪successfully restored narrative order.")

 except Exception as e:
 logger.error(f"TEST FAILED: An error occurred during chunk sequence
↪verification: {e}", exc_info=True)

--- MAIN DIAGNOSTIC FUNCTION ---
def run_full_diagnostics():
 if not langchain_available:
 logger.error("LangChain components not installed. Skipping diagnostics.
↪")
 return
 if not pandas_available:
 logger.warning("Pandas not installed. Some reports may not be available.
↪")

 print_header("Full Database Health & Hierarchy Diagnostic Report")

 # 1. Connect to the Database
 logger.info("Connecting to the vector database...")
 if not os.path.exists(CHROMA_PERSIST_DIR):
 logger.error(f"FATAL: Chroma DB directory not found at
↪{CHROMA_PERSIST_DIR}.")
 return

```

```

vector_store = Chroma(
 persist_directory=CHROMA_PERSIST_DIR,
 embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
 collection_name=CHROMA_COLLECTION_NAME
)
logger.info("Successfully connected to the database.")

2. Retrieve ALL Metadata
total_docs = vector_store._collection.count()
if total_docs == 0:
 logger.warning("Database is empty. No diagnostics to run.")
 return

logger.info(f"Retrieving metadata for all {total_docs} chunks...")
metadatas = vector_store.get(limit=total_docs,
 include=["metadatas"])['metadatas']
logger.info("Successfully retrieved all metadata.")

3. Reconstruct the Hierarchy Tree
logger.info("Reconstructing hierarchy from chunk metadata...")
hierarchy_tree = {'_children': {}}
chunks_without_id = 0

for meta in metadatas:
 toc_id = meta.get('toc_id')
 if toc_id is None or toc_id == -1:
 chunks_without_id += 1
 node_title = meta.get('level_1_title', 'Orphaned Chunks')
 if node_title not in hierarchy_tree['_children']:
 hierarchy_tree['_children'][node_title] = {'_children': {},
 ↪ '_chunks': 0, '_toc_id': float('inf')}
 hierarchy_tree['_children'][node_title]['_chunks'] += 1
 continue

 current_node = hierarchy_tree
 for level in range(1, 7):
 level_key = f'level_{level}_title'
 title = meta.get(level_key)
 if not title: break
 if title not in current_node['_children']:
 current_node['_children'][title] = {'_children': {}, '_chunks': 0,
 ↪ '_toc_id': float('inf')}
 current_node = current_node['_children'][title]

 current_node['_chunks'] += 1
 current_node['_toc_id'] = min(current_node['_toc_id'], toc_id)

```



```

logger.info("Hierarchy reconstruction complete.")

4. Print Hierarchy Report
print_header("Reconstructed Hierarchy Report (Book Order)", char="-")
print_hierarchy_report(hierarchy_tree)

5. Run Chunk Sequence and Content Test
verify_chunk_sequence_and_content(vector_store, hierarchy_tree)

6. Final Summary
print_header("Diagnostic Summary", char="-")
print(f"Total Chunks in DB: {total_docs}")

if chunks_without_id > 0:
 logger.warning(f"Found {chunks_without_id} chunks MISSING a valid_
↳ 'toc_id'. Check 'Orphaned' sections.")
else:
 logger.info("All chunks contain valid 'toc_id' metadata. Sequential_
↳ integrity is maintained.")

print_header("Diagnostic Complete")

--- Execute Diagnostics ---
if 'CHROMA_PERSIST_DIR' in locals() and langchain_available:
 run_full_diagnostics()
else:
 logger.error("Skipping diagnostics: Global variables not defined or_
↳ LangChain not available.")

```

```

2025-07-19 20:42:38,226 - INFO - Connecting to the vector database...
2025-07-19 20:42:38,238 - INFO - Successfully connected to the database.
2025-07-19 20:42:38,240 - INFO - Retrieving metadata for all 7216 chunks...

```

```

=====
Full Database Health & Hierarchy Diagnostic Report
=====

```

```

2025-07-19 20:42:38,589 - INFO - Successfully retrieved all metadata.
2025-07-19 20:42:38,591 - INFO - Reconstructing hierarchy from chunk metadata...
2025-07-19 20:42:38,600 - INFO - Hierarchy reconstruction complete.
2025-07-19 20:42:38,603 - INFO - Verifying chunk order and reassembling content
for a random ToC section.
2025-07-19 20:42:38,603 - INFO - Selected random section for testing: 'II
Private-Key (Symmetric) Cryptography -> 5 CCA-Security and Authenticated
Encryption -> 5.1 Chosen-Ciphertext Attacks and CCA-Security -> 5.1.1 Padding-
Oracle Attacks' (toc_id: 74)
2025-07-19 20:42:38,606 - INFO - Retrieved 20 document chunks for toc_id 74.

```

2025-07-19 20:42:38,606 - INFO - TEST PASSED: Chunk IDs for the section are sequential and content is reassembled.

2025-07-19 20:42:38,607 - WARNING - Found 44 chunks MISSING a valid 'toc\_id'. Check 'Orphaned' sections.

---

### Reconstructed Hierarchy Report (Book Order)

---

```
|-- Preface [ID: 6] (Total Chuck in branch: 53, Direct Chunk: 53)
|-- Index of Common Notation [ID: 265] (Total Chuck in branch: 66, Direct Chunk:
66)
|-- References [ID: 287] (Total Chuck in branch: 207, Direct Chunk: 207)
|-- Index [ID: 288] (Total Chuck in branch: 539, Direct Chunk: 539)
|-- EPUB Preamble [ID: inf] (Total Chuck in branch: 44, Direct Chunk: 44)
|-- I Introduction and Classical Cryptography [ID: inf] (Total Chuck in branch:
1122, Direct Chunk: 0)
 |-- 1 Introduction [ID: inf] (Total Chuck in branch: 999, Direct Chunk: 0)
 |-- 1.1 Cryptography and Modern Cryptography [ID: 9] (Total Chuck in branch:
9, Direct Chunk: 9)
 |-- 1.2 The Setting of Private-Key Encryption [ID: 10] (Total Chuck in
branch: 27, Direct Chunk: 27)
 |-- 1.3 Historical Ciphers and Their Cryptanalysis [ID: 11] (Total Chuck in
branch: 69, Direct Chunk: 69)
 |-- 1.4 Principles of Modern Cryptography [ID: 12] (Total Chuck in branch:
50, Direct Chunk: 32)
 |-- 1.4.2 Principle 2 - Precise Assumptions [ID: 14] (Total Chuck in
branch: 8, Direct Chunk: 8)
 |-- 1.4.3 Principle 3 - Proofs of Security [ID: 15] (Total Chuck in
branch: 5, Direct Chunk: 5)
 |-- 1.4.4 Provable Security and Real-World Security [ID: 16] (Total Chuck
in branch: 5, Direct Chunk: 5)
 |-- References and Additional Reading [ID: 17] (Total Chuck in branch: 92,
Direct Chunk: 92)
 |-- Exercises [ID: 18] (Total Chuck in branch: 752, Direct Chunk: 752)
 |-- 2 Perfectly Secret Encryption [ID: inf] (Total Chuck in branch: 123,
Direct Chunk: 0)
 |-- 2.1 Definitions [ID: 20] (Total Chuck in branch: 76, Direct Chunk: 76)
 |-- 2.2 The One-Time Pad [ID: 21] (Total Chuck in branch: 24, Direct Chunk:
24)
 |-- 2.3 Limitations of Perfect Secrecy [ID: 22] (Total Chuck in branch: 9,
Direct Chunk: 9)
 |-- 2.4 *Shannon's Theorem [ID: 23] (Total Chuck in branch: 14, Direct
Chunk: 14)
 |-- II Private-Key (Symmetric) Cryptography [ID: inf] (Total Chuck in branch:
2169, Direct Chunk: 0)
 |-- 3 Private-Key Encryption [ID: inf] (Total Chuck in branch: 527, Direct
Chunk: 0)
```

```

|-- 3.1 Computational Security [ID: 28] (Total Chuck in branch: 80, Direct
Chunk: 8)
 |-- 3.1.1 The Concrete Approach [ID: 29] (Total Chuck in branch: 12,
Direct Chunk: 12)
 |-- 3.1.2 The Asymptotic Approach [ID: 30] (Total Chuck in branch: 60,
Direct Chunk: 60)
 |-- 3.2 Defining Computationally Secure Encryption [ID: 31] (Total Chuck in
branch: 83, Direct Chunk: 10)
 |-- 3.2.1 The Basic Definition of Security (EAV-Security) [ID: 32] (Total
Chuck in branch: 38, Direct Chunk: 38)
 |-- 3.2.2 *Semantic Security [ID: 33] (Total Chuck in branch: 35, Direct
Chunk: 35)
 |-- 3.3 Constructing an EAV-Secure Encryption Scheme [ID: 34] (Total Chuck
in branch: 84, Direct Chunk: 2)
 |-- 3.3.1 Pseudorandom Generators [ID: 35] (Total Chuck in branch: 26,
Direct Chunk: 26)
 |-- 3.3.2 Proofs by Reduction [ID: 36] (Total Chuck in branch: 12, Direct
Chunk: 12)
 |-- 3.3.3 EAV-Security from a Pseudorandom Generator [ID: 37] (Total Chuck
in branch: 44, Direct Chunk: 44)
 |-- 3.4 Stronger Security Notions [ID: 38] (Total Chuck in branch: 63,
Direct Chunk: 2)
 |-- 3.4.1 Security for Multiple Encryptions [ID: 39] (Total Chuck in
branch: 21, Direct Chunk: 21)
 |-- 3.4.2 Chosen-Plaintext Attacks and CPA-Security [ID: 40] (Total Chuck
in branch: 21, Direct Chunk: 21)
 |-- 3.4.3 CPA-Security for Multiple Encryptions [ID: 41] (Total Chuck in
branch: 19, Direct Chunk: 19)
 |-- 3.5 Constructing a CPA-Secure Encryption Scheme [ID: 42] (Total Chuck in
branch: 91, Direct Chunk: 2)
 |-- 3.5.1 Pseudorandom Functions and Permutations [ID: 43] (Total Chuck in
branch: 36, Direct Chunk: 36)
 |-- 3.5.2 CPA-Security from a Pseudorandom Function [ID: 44] (Total Chuck
in branch: 53, Direct Chunk: 53)
 |-- 3.6 Modes of Operation and Encryption in Practice [ID: 45] (Total Chuck
in branch: 126, Direct Chunk: 3)
 |-- 3.6.1 Stream Ciphers [ID: 46] (Total Chuck in branch: 26, Direct
Chunk: 26)
 |-- 3.6.2 Stream-Cipher Modes of Operation [ID: 47] (Total Chuck in
branch: 70, Direct Chunk: 70)
 |-- 3.6.4 *Nonce-Based Encryption [ID: 49] (Total Chuck in branch: 27,
Direct Chunk: 27)
 |-- 4 Message Authentication Codes [ID: inf] (Total Chuck in branch: 363,
Direct Chunk: 0)
 |-- 4.1 Message Integrity [ID: 53] (Total Chuck in branch: 74, Direct Chunk:
1)
 |-- 4.1.1 Secrecy vs. Integrity [ID: 54] (Total Chuck in branch: 10,
Direct Chunk: 10)

```

- |-- 4.1.2 Encryption vs. Message Authentication [ID: 55] (Total Chuck in branch: 63, Direct Chunk: 63)
- |-- 4.3 Constructing Secure Message Authentication Codes [ID: 57] (Total Chuck in branch: 69, Direct Chunk: 1)
  - |-- 4.3.1 A Fixed-Length MAC [ID: 58] (Total Chuck in branch: 35, Direct Chunk: 35)
  - |-- 4.3.2 Domain Extension for MACs [ID: 59] (Total Chuck in branch: 33, Direct Chunk: 33)
- |-- 4.4 CBC-MAC [ID: 60] (Total Chuck in branch: 80, Direct Chunk: 2)
  - |-- 4.4.1 The Basic Construction [ID: 61] (Total Chuck in branch: 21, Direct Chunk: 21)
  - |-- 4.4.2 \*Proof of Security [ID: 62] (Total Chuck in branch: 57, Direct Chunk: 57)
- |-- 4.5 GMAC and Poly1305 [ID: 63] (Total Chuck in branch: 61, Direct Chunk: 3)
  - |-- 4.5.1 MACs from Difference-Universal Functions [ID: 64] (Total Chuck in branch: 34, Direct Chunk: 34)
  - |-- 4.5.2 Instantiations [ID: 65] (Total Chuck in branch: 24, Direct Chunk: 24)
- |-- 4.6 \*Information-Theoretic MACs [ID: 66] (Total Chuck in branch: 79, Direct Chunk: 13)
  - |-- 4.6.1 One-Time MACs from Strongly Universal Functions [ID: 67] (Total Chuck in branch: 28, Direct Chunk: 28)
  - |-- 4.6.2 One-Time MACs from Difference-Universal Functions [ID: 68] (Total Chuck in branch: 26, Direct Chunk: 26)
  - |-- 4.6.3 Limitations on Information-Theoretic MACs [ID: 69] (Total Chuck in branch: 12, Direct Chunk: 12)
- |-- 5 CCA-Security and Authenticated Encryption [ID: inf] (Total Chuck in branch: 184, Direct Chunk: 0)
  - |-- 5.1 Chosen-Ciphertext Attacks and CCA-Security [ID: 73] (Total Chuck in branch: 44, Direct Chunk: 6)
    - |-- 5.1.1 Padding-Oracle Attacks [ID: 74] (Total Chuck in branch: 20, Direct Chunk: 20)
    - |-- 5.1.2 Defining CCA-Security [ID: 75] (Total Chuck in branch: 18, Direct Chunk: 18)
  - |-- 5.2 Authenticated Encryption [ID: 76] (Total Chuck in branch: 38, Direct Chunk: 2)
    - |-- 5.2.1 Defining Authenticated Encryption [ID: 77] (Total Chuck in branch: 30, Direct Chunk: 30)
    - |-- 5.2.2 CCA Security vs. Authenticated Encryption [ID: 78] (Total Chuck in branch: 6, Direct Chunk: 6)
  - |-- 5.3 Authenticated Encryption Schemes [ID: 79] (Total Chuck in branch: 92, Direct Chunk: 1)
    - |-- 5.3.1 Generic Constructions [ID: 80] (Total Chuck in branch: 83, Direct Chunk: 83)
    - |-- 5.3.2 Standardized Schemes [ID: 81] (Total Chuck in branch: 8, Direct Chunk: 8)
  - |-- 5.4 Secure Communication Sessions [ID: 82] (Total Chuck in branch: 10,

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Direct Chunk: 10)
 |-- 6 Hash Functions and Applications [ID: inf] (Total Chuck in branch: 318,
Direct Chunk: 0)
 |-- 6.1 Definitions [ID: 86] (Total Chuck in branch: 49, Direct Chunk: 4)
 |-- 6.1.1 Collision Resistance [ID: 87] (Total Chuck in branch: 21, Direct
Chunk: 21)
 |-- 6.1.2 Weaker Notions of Security [ID: 88] (Total Chuck in branch: 24,
Direct Chunk: 24)
 |-- 6.3 Message Authentication Using Hash Functions [ID: 90] (Total Chuck in
branch: 58, Direct Chunk: 2)
 |-- 6.3.1 Hash-and-MAC [ID: 91] (Total Chuck in branch: 34, Direct Chunk:
34)
 |-- 6.3.2 HMAC [ID: 92] (Total Chuck in branch: 22, Direct Chunk: 22)
 |-- 6.4 Generic Attacks on Hash Functions [ID: 93] (Total Chuck in branch:
72, Direct Chunk: 4)
 |-- 6.4.1 Birthday Attacks for Finding Collisions [ID: 94] (Total Chuck in
branch: 11, Direct Chunk: 11)
 |-- 6.4.2 Small-Space Birthday Attacks [ID: 95] (Total Chuck in branch:
25, Direct Chunk: 25)
 |-- 6.4.3 *Time/Space Tradeoffs for Inverting Hash Functions [ID: 96]
(Total Chuck in branch: 32, Direct Chunk: 32)
 |-- 6.5 The Random-Oracle Model [ID: 97] (Total Chuck in branch: 73, Direct
Chunk: 12)
 |-- 6.5.1 The Random-Oracle Model in Detail [ID: 98] (Total Chuck in
branch: 39, Direct Chunk: 39)
 |-- 6.5.2 Is the Random-Oracle Methodology Sound? [ID: 99] (Total Chuck in
branch: 22, Direct Chunk: 22)
 |-- 6.6 Additional Applications of Hash Functions [ID: 100] (Total Chuck in
branch: 66, Direct Chunk: 2)
 |-- 6.6.1 Fingerprinting and Deduplication [ID: 101] (Total Chuck in
branch: 7, Direct Chunk: 7)
 |-- 6.6.2 Merkle Trees [ID: 102] (Total Chuck in branch: 13, Direct Chunk:
13)
 |-- 6.6.3 Password Hashing [ID: 103] (Total Chuck in branch: 10, Direct
Chunk: 10)
 |-- 6.6.4 Key Derivation [ID: 104] (Total Chuck in branch: 10, Direct
Chunk: 10)
 |-- 6.6.5 Commitment Schemes [ID: 105] (Total Chuck in branch: 24, Direct
Chunk: 24)
 |-- 7 Practical Constructions of Symmetric-Key Primitives [ID: inf] (Total
Chuck in branch: 366, Direct Chunk: 0)
 |-- 7.1 Stream Ciphers [ID: 109] (Total Chuck in branch: 92, Direct Chunk:
4)
 |-- 7.1.1 Linear-Feedback Shift Registers [ID: 110] (Total Chuck in
branch: 19, Direct Chunk: 19)
 |-- 7.1.2 Adding Nonlinearity [ID: 111] (Total Chuck in branch: 10, Direct
Chunk: 10)
 |-- 7.1.3 Trivium [ID: 112] (Total Chuck in branch: 5, Direct Chunk: 5)

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|-- 7.1.4 RC4 [ID: 113] (Total Chuck in branch: 46, Direct Chunk: 46)
|-- 7.1.5 ChaCha20 [ID: 114] (Total Chuck in branch: 8, Direct Chunk: 8)
|-- 7.2 Block Ciphers [ID: 115] (Total Chuck in branch: 204, Direct Chunk:
14)
|-- 7.2.1 Substitution-Permutation Networks [ID: 116] (Total Chuck in
branch: 44, Direct Chunk: 44)
|-- 7.2.2 Feistel Networks [ID: 117] (Total Chuck in branch: 22, Direct
Chunk: 22)
|-- 7.2.3 DES - The Data Encryption Standard [ID: 118] (Total Chuck in
branch: 49, Direct Chunk: 49)
|-- 7.2.4 3DES: Increasing the Key Length of a Block Cipher [ID: 119]
(Total Chuck in branch: 25, Direct Chunk: 25)
|-- 7.2.5 AES - The Advanced Encryption Standard [ID: 120] (Total Chuck in
branch: 17, Direct Chunk: 17)
|-- 7.2.6 *Differential and Linear Cryptanalysis [ID: 121] (Total Chuck in
branch: 33, Direct Chunk: 33)
|-- 7.3 Compression Functions and Hash Functions [ID: 122] (Total Chuck in
branch: 70, Direct Chunk: 4)
|-- 7.3.1 Compression Functions from Block Ciphers [ID: 123] (Total Chuck
in branch: 17, Direct Chunk: 17)
|-- 7.3.2 MD5, SHA-1, and SHA-2 [ID: 124] (Total Chuck in branch: 7,
Direct Chunk: 7)
|-- 7.3.3 The Sponge Construction and SHA-3 (Keccak) [ID: 125] (Total
Chuck in branch: 42, Direct Chunk: 42)
|-- 8 *Theoretical Constructions of Symmetric-Key Primitives [ID: inf] (Total
Chuck in branch: 411, Direct Chunk: 0)
|-- 8.1 One-Way Functions [ID: 129] (Total Chuck in branch: 58, Direct
Chunk: 2)
|-- 8.1.1 Definitions [ID: 130] (Total Chuck in branch: 35, Direct Chunk:
35)
|-- 8.1.2 Candidate One-Way Functions [ID: 131] (Total Chuck in branch:
11, Direct Chunk: 11)
|-- 8.1.3 Hard-Core Predicates [ID: 132] (Total Chuck in branch: 10,
Direct Chunk: 10)
|-- 8.2 From One-Way Functions to Pseudorandomness [ID: 133] (Total Chuck in
branch: 22, Direct Chunk: 22)
|-- 8.3 Hard-Core Predicates from One-Way Functions [ID: 134] (Total Chuck
in branch: 97, Direct Chunk: 4)
|-- 8.3.1 A Simple Case [ID: 135] (Total Chuck in branch: 8, Direct Chunk:
8)
|-- 8.3.2 A More Involved Case [ID: 136] (Total Chuck in branch: 46,
Direct Chunk: 46)
|-- 8.3.3 The Full Proof [ID: 137] (Total Chuck in branch: 39, Direct
Chunk: 39)
|-- 8.4 Constructing Pseudorandom Generators [ID: 138] (Total Chuck in
branch: 81, Direct Chunk: 2)
|-- 8.4.1 Pseudorandom Generators with Minimal Expansion [ID: 139] (Total
Chuck in branch: 24, Direct Chunk: 24)

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|-- 8.4.2 Increasing the Expansion Factor [ID: 140] (Total Chuck in
branch: 55, Direct Chunk: 55)
|-- 8.5 Constructing Pseudorandom Functions [ID: 141] (Total Chuck in
branch: 58, Direct Chunk: 58)
|-- 8.6 Constructing (Strong) Pseudorandom Permutations [ID: 142] (Total
Chuck in branch: 42, Direct Chunk: 42)
|-- 8.7 Assumptions for Private-Key Cryptography [ID: 143] (Total Chuck in
branch: 31, Direct Chunk: 31)
|-- 8.8 Computational Indistinguishability [ID: 144] (Total Chuck in branch:
22, Direct Chunk: 22)
|-- III Public-Key (Asymmetric) Cryptography [ID: inf] (Total Chuck in branch:
2667, Direct Chunk: 0)
|-- 9 Number Theory and Cryptographic Hardness Assumptions [ID: inf] (Total
Chuck in branch: 677, Direct Chunk: 0)
|-- 9.1 Preliminaries and Basic Group Theory [ID: 149] (Total Chuck in
branch: 214, Direct Chunk: 2)
|-- 9.1.1 Primes and Divisibility [ID: 150] (Total Chuck in branch: 26,
Direct Chunk: 26)
|-- 9.1.2 Modular Arithmetic [ID: 151] (Total Chuck in branch: 25, Direct
Chunk: 25)
|-- 9.1.3 Groups [ID: 152] (Total Chuck in branch: 62, Direct Chunk: 62)
|-- 9.1.4 The Group $\ast N$ [ID: 153] (Total Chuck in branch: 99, Direct
Chunk: 99)
|-- 9.2 Primes, Factoring, and RSA [ID: 155] (Total Chuck in branch: 182,
Direct Chunk: 14)
|-- 9.2.1 Generating Random Primes [ID: 156] (Total Chuck in branch: 33,
Direct Chunk: 33)
|-- 9.2.2 $\ast$ Primality Testing [ID: 157] (Total Chuck in branch: 70, Direct
Chunk: 70)
|-- 9.2.3 The Factoring Assumption [ID: 158] (Total Chuck in branch: 11,
Direct Chunk: 11)
|-- 9.2.4 The RSA Assumption [ID: 159] (Total Chuck in branch: 28, Direct
Chunk: 28)
|-- 9.2.5 $\ast$ Relating the Factoring and RSA Assumptions [ID: 160] (Total
Chuck in branch: 26, Direct Chunk: 26)
|-- 9.3 Cryptographic Assumptions in Cyclic Groups [ID: 161] (Total Chuck in
branch: 215, Direct Chunk: 2)
|-- 9.3.1 Cyclic Groups and Generators [ID: 162] (Total Chuck in branch:
115, Direct Chunk: 115)
|-- 9.3.4 Elliptic Curves [ID: 165] (Total Chuck in branch: 98, Direct
Chunk: 98)
|-- 9.4 $\ast$ Cryptographic Applications [ID: 166] (Total Chuck in branch: 66,
Direct Chunk: 2)
|-- 9.4.1 One-Way Functions and Permutations [ID: 167] (Total Chuck in
branch: 42, Direct Chunk: 42)
|-- 9.4.2 Collision-Resistant Hash Functions [ID: 168] (Total Chuck in
branch: 22, Direct Chunk: 22)
|-- 10 $\ast$ Algorithms for Factoring and Computing Discrete Logarithms [ID: inf]

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```

(Total Chuck in branch: 163, Direct Chunk: 0)
 |-- 10.1 Algorithms for Factoring [ID: 172] (Total Chuck in branch: 87,
Direct Chunk: 29)
 |-- 10.1.2 Pollard's Rho Algorithm [ID: 174] (Total Chuck in branch: 16,
Direct Chunk: 16)
 |-- 10.1.3 The Quadratic Sieve Algorithm [ID: 175] (Total Chuck in branch:
42, Direct Chunk: 42)
 |-- 10.4 Recommended Key Lengths [ID: 181] (Total Chuck in branch: 6, Direct
Chunk: 6)
 |-- 10.2 Generic Algorithms for Computing Discrete Logarithms [ID: inf]
(Total Chuck in branch: 70, Direct Chunk: 0)
 |-- 10.2.1 The Pohlig-Hellman Algorithm [ID: 177] (Total Chuck in branch:
19, Direct Chunk: 19)
 |-- 10.2.2 The Baby-Step/Giant-Step Algorithm [ID: 178] (Total Chuck in
branch: 25, Direct Chunk: 25)
 |-- 10.2.3 Discrete Logarithms from Collisions [ID: 179] (Total Chuck in
branch: 26, Direct Chunk: 26)
 |-- 11 Key Management and the Public-Key Revolution [ID: inf] (Total Chuck in
branch: 108, Direct Chunk: 0)
 |-- 11.1 Key Distribution and Key Management [ID: 185] (Total Chuck in
branch: 14, Direct Chunk: 14)
 |-- 11.2 A Partial Solution: Key-Distribution Centers [ID: 186] (Total Chuck
in branch: 18, Direct Chunk: 18)
 |-- 11.3 Key Exchange and the Diffie-Hellman Protocol [ID: 187] (Total Chuck
in branch: 59, Direct Chunk: 59)
 |-- 11.4 The Public-Key Revolution [ID: 188] (Total Chuck in branch: 17,
Direct Chunk: 17)
 |-- 12 Public-Key Encryption [ID: inf] (Total Chuck in branch: 637, Direct
Chunk: 0)
 |-- 12.2 Definitions [ID: 193] (Total Chuck in branch: 133, Direct Chunk:
11)
 |-- 12.2.1 Security against Chosen-Plaintext Attacks [ID: 194] (Total
Chuck in branch: 22, Direct Chunk: 22)
 |-- 12.2.2 Multiple Encryptions [ID: 195] (Total Chuck in branch: 72,
Direct Chunk: 72)
 |-- 12.2.3 Security against Chosen-Ciphertext Attacks [ID: 196] (Total
Chuck in branch: 28, Direct Chunk: 28)
 |-- 12.3 Hybrid Encryption and the KEM/DEM Paradigm [ID: 197] (Total Chuck
in branch: 118, Direct Chunk: 38)
 |-- 12.3.1 CPA-Security [ID: 198] (Total Chuck in branch: 64, Direct
Chunk: 64)
 |-- 12.3.2 CCA-Security [ID: 199] (Total Chuck in branch: 16, Direct
Chunk: 16)
 |-- 12.4 CDH/DDH-Based Encryption [ID: 200] (Total Chuck in branch: 133,
Direct Chunk: 2)
 |-- 12.4.1 El Gamal Encryption [ID: 201] (Total Chuck in branch: 68,
Direct Chunk: 68)
 |-- 12.4.2 DDH-Based Key Encapsulation [ID: 202] (Total Chuck in branch:

```



```

15, Direct Chunk: 15)
 |-- 12.4.3 *A CDH-Based KEM in the Random-Oracle Model [ID: 203] (Total
Chuck in branch: 28, Direct Chunk: 28)
 |-- 12.4.4 *Chosen-Ciphertext Security and DHIES/ECIES [ID: 204] (Total
Chuck in branch: 20, Direct Chunk: 20)
 |-- 12.5 RSA-Based Encryption [ID: 205] (Total Chuck in branch: 253, Direct
Chunk: 2)
 |-- 12.5.1 Plain RSA Encryption [ID: 206] (Total Chuck in branch: 61,
Direct Chunk: 61)
 |-- 12.5.2 Padded RSA and PKCS #1 v1.5 [ID: 207] (Total Chuck in branch:
22, Direct Chunk: 22)
 |-- 12.5.3 *CPA-Secure Encryption without Random Oracles [ID: 208] (Total
Chuck in branch: 59, Direct Chunk: 59)
 |-- 12.5.4 OAEP and PKCS #1 v2 [ID: 209] (Total Chuck in branch: 36,
Direct Chunk: 36)
 |-- 12.5.5 *A CCA-Secure KEM in the Random-Oracle Model [ID: 210] (Total
Chuck in branch: 45, Direct Chunk: 45)
 |-- 12.5.6 RSA Implementation Issues and Pitfalls [ID: 211] (Total Chuck
in branch: 28, Direct Chunk: 28)
 |-- 13 Digital Signature Schemes [ID: inf] (Total Chuck in branch: 304, Direct
Chunk: 0)
 |-- 13.2 Definitions [ID: 216] (Total Chuck in branch: 23, Direct Chunk: 23)
 |-- 13.3 The Hash-and-Sign Paradigm [ID: 217] (Total Chuck in branch: 11,
Direct Chunk: 11)
 |-- 13.4 RSA-Based Signatures [ID: 218] (Total Chuck in branch: 77, Direct
Chunk: 2)
 |-- 13.4.1 Plain RSA Signatures [ID: 219] (Total Chuck in branch: 21,
Direct Chunk: 21)
 |-- 13.4.2 RSA-FDH and PKCS #1 Standards [ID: 220] (Total Chuck in branch:
54, Direct Chunk: 54)
 |-- 13.5 Signatures from the Discrete-Logarithm Problem [ID: 221] (Total
Chuck in branch: 119, Direct Chunk: 2)
 |-- 13.5.1 Identification Schemes and Signatures [ID: 222] (Total Chuck in
branch: 52, Direct Chunk: 52)
 |-- 13.5.2 The Schnorr Identification/Signature Schemes [ID: 223] (Total
Chuck in branch: 34, Direct Chunk: 34)
 |-- 13.5.3 DSA and ECDSA [ID: 224] (Total Chuck in branch: 31, Direct
Chunk: 31)
 |-- 13.6 Certificates and Public-Key Infrastructures [ID: 225] (Total Chuck
in branch: 45, Direct Chunk: 45)
 |-- 13.7 Putting It All Together - TLS [ID: 226] (Total Chuck in branch: 18,
Direct Chunk: 18)
 |-- 13.8 *Signcryption [ID: 227] (Total Chuck in branch: 11, Direct Chunk:
11)
 |-- 14 *Post-Quantum Cryptography [ID: inf] (Total Chuck in branch: 245,
Direct Chunk: 0)
 |-- 14.1 Post-Quantum Symmetric-Key Cryptography [ID: 231] (Total Chuck in
branch: 20, Direct Chunk: 2)

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- |-- 14.1.1 Grover's Algorithm and Symmetric-Key Lengths [ID: 232] (Total Chuck in branch: 8, Direct Chunk: 8)
- |-- 14.1.2 Collision-Finding Algorithms and Hash Functions [ID: 233] (Total Chuck in branch: 10, Direct Chunk: 10)
- |-- 14.2 Shor's Algorithm and its Impact on Cryptography [ID: 234] (Total Chuck in branch: 20, Direct Chunk: 20)
- |-- 14.3 Post-Quantum Public-Key Encryption [ID: 235] (Total Chuck in branch: 67, Direct Chunk: 67)
- |-- 14.4 Post-Quantum Signatures [ID: 236] (Total Chuck in branch: 138, Direct Chunk: 5)
  - |-- 14.4.1 Lamport's Signature Scheme [ID: 237] (Total Chuck in branch: 51, Direct Chunk: 51)
  - |-- 14.4.2 Chain-Based Signatures [ID: 238] (Total Chuck in branch: 23, Direct Chunk: 23)
  - |-- 14.4.3 Tree-Based Signatures [ID: 239] (Total Chuck in branch: 59, Direct Chunk: 59)
- |-- 15 \*Advanced Topics in Public-Key Encryption [ID: inf] (Total Chuck in branch: 533, Direct Chunk: 0)
  - |-- 15.2 The Paillier Encryption Scheme [ID: 246] (Total Chuck in branch: 131, Direct Chunk: 39)
    - |-- 15.2.2 The Paillier Encryption Scheme [ID: 248] (Total Chuck in branch: 70, Direct Chunk: 70)
    - |-- 15.2.3 Homomorphic Encryption [ID: 249] (Total Chuck in branch: 22, Direct Chunk: 22)
  - |-- 15.3 Secret Sharing and Threshold Encryption [ID: 250] (Total Chuck in branch: 59, Direct Chunk: 3)
    - |-- 15.3.1 Secret Sharing [ID: 251] (Total Chuck in branch: 21, Direct Chunk: 21)
    - |-- 15.3.2 Verifiable Secret Sharing [ID: 252] (Total Chuck in branch: 18, Direct Chunk: 18)
    - |-- 15.3.3 Threshold Encryption and Electronic Voting [ID: 253] (Total Chuck in branch: 17, Direct Chunk: 17)
  - |-- 15.4 The Goldwasser-Micali Encryption Scheme [ID: 254] (Total Chuck in branch: 163, Direct Chunk: 3)
    - |-- 15.4.1 Quadratic Residues Modulo a Prime [ID: 255] (Total Chuck in branch: 46, Direct Chunk: 46)
    - |-- 15.4.2 Quadratic Residues Modulo a Composite [ID: 256] (Total Chuck in branch: 56, Direct Chunk: 56)
    - |-- 15.4.3 The Quadratic Residuosity Assumption [ID: 257] (Total Chuck in branch: 15, Direct Chunk: 15)
    - |-- 15.4.4 The Goldwasser-Micali Encryption Scheme [ID: 258] (Total Chuck in branch: 43, Direct Chunk: 43)
  - |-- 15.5 The Rabin Encryption Scheme [ID: 259] (Total Chuck in branch: 135, Direct Chunk: 4)
    - |-- 15.5.1 Computing Modular Square Roots [ID: 260] (Total Chuck in branch: 59, Direct Chunk: 59)
    - |-- 15.5.2 A Trapdoor Permutation Based on Factoring [ID: 261] (Total Chuck in branch: 55, Direct Chunk: 55)

```

|-- 15.5.3 The Rabin Encryption Scheme [ID: 262] (Total Chuck in branch:
17, Direct Chunk: 17)
|-- 15.1 Public-Key Encryption from Trapdoor Permutations [ID: inf] (Total
Chuck in branch: 45, Direct Chunk: 0)
|-- 15.1.1 Trapdoor Permutations [ID: 244] (Total Chuck in branch: 9,
Direct Chunk: 9)
|-- 15.1.2 Public-Key Encryption from Trapdoor Permutations [ID: 245]
(Total Chuck in branch: 36, Direct Chunk: 36)
|-- Appendix A Mathematical Background [ID: inf] (Total Chuck in branch: 176,
Direct Chunk: 0)
|-- A.1 Identities and Inequalities [ID: 267] (Total Chuck in branch: 9,
Direct Chunk: 9)
|-- A.2 Asymptotic Notation [ID: 268] (Total Chuck in branch: 14, Direct
Chunk: 14)
|-- A.3 Basic Probability [ID: 269] (Total Chuck in branch: 58, Direct Chunk:
58)
|-- A.4 The "Birthday" Problem [ID: 270] (Total Chuck in branch: 36, Direct
Chunk: 36)
|-- A.5 *Finite Fields [ID: 271] (Total Chuck in branch: 59, Direct Chunk: 59)
|-- Appendix B Basic Algorithmic Number Theory [ID: inf] (Total Chuck in branch:
173, Direct Chunk: 0)
|-- B.1 Integer Arithmetic [ID: 273] (Total Chuck in branch: 25, Direct Chunk:
1)
|-- B.1.1 Basic Operations [ID: 274] (Total Chuck in branch: 4, Direct
Chunk: 4)
|-- B.1.2 The Euclidean and Extended Euclidean Algorithms [ID: 275] (Total
Chuck in branch: 20, Direct Chunk: 20)
|-- B.2 Modular Arithmetic [ID: 276] (Total Chuck in branch: 109, Direct
Chunk: 8)
|-- B.2.1 Basic Operations [ID: 277] (Total Chuck in branch: 3, Direct
Chunk: 3)
|-- B.2.2 Computing Modular Inverses [ID: 278] (Total Chuck in branch: 8,
Direct Chunk: 8)
|-- B.2.3 Modular Exponentiation [ID: 279] (Total Chuck in branch: 37,
Direct Chunk: 37)
|-- B.2.4 *Montgomery Multiplication [ID: 280] (Total Chuck in branch: 17,
Direct Chunk: 17)
|-- B.2.5 Choosing a Uniform Group Element [ID: 281] (Total Chuck in branch:
36, Direct Chunk: 36)
|-- B.3 *Finding a Generator of a Cyclic Group [ID: 282] (Total Chuck in
branch: 39, Direct Chunk: 6)
|-- B.3.1 Group-Theoretic Background [ID: 283] (Total Chuck in branch: 17,
Direct Chunk: 17)
|-- B.3.2 Efficient Algorithms [ID: 284] (Total Chuck in branch: 16, Direct
Chunk: 16)

```

---

Chunk Sequence & Content Integrity Test

----- CONTENT PREVIEW -----

Title: 5.1.1 Padding-Oracle Attacks [toc\_id: 74]

Chunk IDs: [1509, 1510, 1511, 1512, 1513, 1514, 1515, 1516, 1517, 1518, 1519, 1520, 1521, 1522, 1523, 1524, 1525, 1526, 1527, 1528]

### 5.1.1 Padding-Oracle Attacks

We motivate the importance of security against chosen-ciphertext attacks by showing a real-world example where such attacks can be devastating. We consider a setting in which a client sends messages encrypted using CBC-mode encryption to a server. We assume the attacker can impersonate the client and send ciphertexts of its choice to the server, which the server will then decrypt. We assume further that the attacker can tell when the resulting decrypted messages are valid (in a sense we will define below) or not. Such information is frequently easy to obtain since, for example, the server might request retransmission or terminate a session if it receives a ciphertext that does not decrypt correctly, and either of those events would be detectable by the attacker. The attack has been shown

and either of those events would be detectable by the attacker. The attack has been shown to work in practice on various deployed protocols.

In our discussion of CBC-mode encryption in Section 3.6.3, we only dealt with the case where the message length was a multiple of the block length of the underlying block cipher  $F$ . If a message does not satisfy this property, it must be padded before CBC mode is applied; we refer to the result after padding as the encoded data. The padding must allow the receiver to unambiguously recover the original message from the encoded data. One popular padding scheme is defined by the PKCS #7 standard, and works as follows. Assume the original message has an integral number of bytes, and let  $L$  denote the block length (in bytes) of the block cipher  $F$ . Let  $b > 0$  denote the number of bytes that need to be appended to the message in order to make the total length of the resulting encoded data a multiple

appended to the message in order to make the total length of the resulting encoded data a multiple of the block length. Then we append to the message the integer  $b$  (represented in one byte, i.e., two hexadecimal digits) repeated  $b$  times. That is, if one byte of padding is needed then the 1-byte string  $0 \times 01$  (written in hexadecimal) is appended; if four bytes of padding are needed then  $0 \times 04040404$  is appended; etc. (Note that  $b$  is an integer between 1 and  $L$ , inclusive—we cannot have  $b = 0$  since this would lead to ambiguous padding. Thus, if the original message length is already a multiple of the block length, then  $b = L$ .) After padding, the encoded data is encrypted using regular CBC-mode encryption.

When decrypting, the server first uses CBC-mode decryption as usual to recover the encoded data, and then checks whether the encoded data is correctly padded. (This is easily done: simply read the value  $b$  of the final byte and then verify that the final  $b$  bytes of the result all have value  $b$ .) If so, the padding is stripped off and the original message returned. Otherwise, the standard procedure is to return a "bad padding" error. This means the server is serving

as a "padding oracle" for the adversary: i.e., the adversary can send an arbitrary ciphertext to the server and learn (based on whether a "bad padding" error is returned) whether the underlying encoded data is padded correctly or not. Although this may seem like meaningless information, we show that it enables an adversary to

not. Although this may seem like meaningless information, we show that it enables an adversary to completely recover the original message corresponding to any ciphertext of its choice.

We describe the attack on a 3-block ciphertext for simplicity. Let  $IV$ ,  $c_1$ ,  $c_2$  be a ciphertext observed by the attacker, and let  $m_1$ ,  $m_2$  be the underlying encoded data (unknown to the attacker) that corresponds to a padded message, as discussed above. (Each block is  $L$  bytes long.) Note that

$$m_2 = F_{k^{-1}}(c_2) \oplus c_1, \quad (5.1)$$

where  $k$  is the key (which is, of course, not known to the attacker) being used by the honest parties. The second block  $m_2$  ends in

$\text{0} \times b$ , where  $b$  is an integer such that  $0 \leq b < L$ . The key property used in the attack is that certain changes to the ciphertext yield predictable changes in the underlying encoded data after CBC-mode decryption. Specifically, let  $c_1'$  be identical to  $c_1$  except for a modification in the final byte, and consider decryption of the modified ciphertext  $IV, c_1', c_2$ . This will result in encoded data  $m_1', m_2'$  where  $m_2' = F_{k^{-1}}(c_2) \oplus c_1'$ . Comparing to Equation (5.1) we see that  $m_2'$  will be identical to  $m_2$  except for a modification in the final byte. (The value of  $m_1'$  is

unpredictable, but this will not adversely affect the attack.)

Similarly, if  $c_1'$  is the same as  $c_1$  except for a change in its  $i$ th byte, then decryption of  $IV, c_1', c_2$  will result in  $m_1', m_2'$  where  $m_2'$  is the same as  $m_2$  except for a change in its  $i$ th byte. More generally, if  $c_1' = c_1 \oplus \Delta$  for any string  $\Delta$ , then decryption of  $IV, c_1', c_2$  yields  $m_1', m_2'$  where  $m_2' = m_2 \oplus \Delta$ . The upshot is that the attacker can exercise significant control over the final block of the encoded data.

As a warmup, let us see how the adversary can exploit this to learn  $b$ , the amount of padding. (This reveals the length of the original message.) Recall that upon decryption, the server looks at the value  $b$  of the final byte of the encoded data, and then verifies that the final  $b$  bytes all have the same value. The attacker begins by modifying the first byte of  $c_1$  and sending the resulting ciphertext  $IV, c_1', c_2$  to the server. If decryption fails (i.e., the server returns an error) then it must be the case that the server is checking all  $L$  bytes of  $m_2'$ , and therefore  $b = L$ ! Otherwise, the attacker learns that  $b < L$ , and it can then repeat the process with the second byte, and so on. The left-most modified byte for which decryption fails reveals exactly the left-most byte being checked by the

server, and so reveals exactly  $b$ .

With  $b$  known, the attacker can proceed to learn the bytes of the message one-by-one. We illustrate the idea for the final byte of the message, which we denote by  $M$ . The attacker knows that  $m_2$  ends in  $M \times \text{0} \times b \dots \text{0} \times b$  (with  $\text{0} \times b$  repeated  $b$



```

[43]: # Cell 6: Verify Content Retrieval for a Specific toc_id with Reassembled Text

import os
import json
import logging
from langchain_chroma import Chroma
from langchain_ollama.embeddings import OllamaEmbeddings

--- Logger Setup ---
logger = logging.getLogger(__name__)
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %
↳ %(message)s')

def retrieve_and_print_chunks_for_toc_id(vector_store: Chroma, toc_id: int):
 """
 Retrieves all chunks for a specific toc_id, reconstructs the section title
 from hierarchical metadata, shows the reassembled text, and lists individual
 chunk details for verification.
 """
 try:
 # Use the 'get' method with a 'where' filter to find all chunks for the
↳ toc_id

 results = vector_store.get(
 where={"toc_id": toc_id},
 include=["documents", "metadatas"]
)

 if not results or not results.get('ids'):
 logger.warning(f"No chunks found in the database for toc_id =
↳ {toc_id}")
 print("=" * 80)
 print(f"VERIFICATION FAILED: No content found for toc_id: {toc_id}")
 print("=" * 80)
 return

 documents = results['documents']
 metadatas = results['metadatas']

 # --- FIX START: Reconstruct the hierarchical section title from
↳ metadata ---
 # We assume all chunks for the same toc_id share the same titles.
 # We will inspect the metadata of the first chunk to get the title.
 section_title = "Unknown or Uncategorized Section"
 if metadatas:
 first_meta = metadatas[0]

 # Find all 'level_X_title' keys in the metadata

```

```

 level_titles = []
 for key, value in first_meta.items():
 if key.startswith("level_") and key.endswith("_title"):
 try:
 # Extract the level number (e.g., 1 from
↪ 'level_1_title') for sorting
 level_num = int(key.split('_')[1])
 level_titles.append((level_num, value))
 except (ValueError, IndexError):
 # Ignore malformed keys, just in case
 continue

 # Sort the titles by their level number (1, 2, 3...)
 level_titles.sort()

 # Join the sorted titles to create a breadcrumb-style title
 if level_titles:
 title_parts = [title for num, title in level_titles]
 section_title = " > ".join(title_parts)
 # --- FIX END ---

 # --- Print a clear header with the reconstructed section title ---
 print("=" * 80)
 print(f"VERIFYING SECTION: '{section_title}' (toc_id: {toc_id})")
 print("=" * 80)
 logger.info(f"Found {len(documents)} chunks in the database for this
↪ section.")

 # Sort chunks by their chunk_id to ensure they are in the correct order
↪ for reassembly
 sorted_items = sorted(zip(documents, metadatas), key=lambda item:
↪ item[1].get('chunk_id', 0))

 # --- Reassemble and print the full text for the section ---
 all_chunk_texts = [item[0] for item in sorted_items]
 reassembled_text = "\n".join(all_chunk_texts)

 print("\n" + "#" * 28 + " Reassembled Text " + "#" * 28)
 print(reassembled_text)
 print("#" * 80)

 # --- Print individual chunk details for in-depth verification ---
 print("\n" + "-" * 24 + " Retrieved Chunk Details " + "-" * 25)
 for i, (doc, meta) in enumerate(sorted_items):
 print(f"\n[Chunk {i+1} of {len(documents)} | chunk_id: {meta.
↪ get('chunk_id', 'N/A')}]")
 content_preview = doc.replace('\n', ' ').strip()

```



```

 print(f" Content Preview: '{content_preview[:250]}...'")
 print(f" Metadata: {json.dumps(meta, indent=2)}")

 print("\n" + "=" * 80)
 print(f"Verification complete for section '{section_title}'.")
 print("=" * 80)

 except Exception as e:
 logger.error(f"An error occurred during retrieval for toc_id {toc_id}:␣
↪{e}", exc_info=True)

=====
EXECUTION BLOCK (No changes needed here)
=====

--- IMPORTANT: Set the ID of the section you want to test here ---
Example: ToC ID 10 might be "An Overview of Digital Forensics"
Example: ToC ID 11 might be "Digital Forensics and Other Related Disciplines"
TOC_ID_TO_TEST = 10 ## Change this to an ID you know exists from your ToC

Assume these variables are defined in a previous cell from your notebook
CHROMA_PERSIST_DIR = "./chroma_db_with_metadata"
EMBEDDING_MODEL_OLLAMA = "nomic-embed-text"
CHROMA_COLLECTION_NAME = "forensics_handbook"

Check if the database directory exists before attempting to connect
if 'CHROMA_PERSIST_DIR' in locals() and os.path.exists(CHROMA_PERSIST_DIR):
 logger.info(f"Connecting to the existing vector database at␣
↪'{CHROMA_PERSIST_DIR}'...")

 try:
 vector_store = Chroma(
 persist_directory=CHROMA_PERSIST_DIR,
 embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
 collection_name=CHROMA_COLLECTION_NAME
)

 # Run the verification function
 retrieve_and_print_chunks_for_toc_id(vector_store, TOC_ID_TO_TEST)

 except Exception as e:
 logger.error(f"Failed to initialize Chroma or run retrieval. Error:␣
↪{e}")

 logger.error("Please ensure your embedding model and collection names␣
↪are correct.")

```

```

else:
 logger.error("Database directory not found or 'CHROMA_PERSIST_DIR' variable_
↳is not set.")
 logger.error("Please run the previous cell (Cell 5) to create the database_
↳first.")

```

```

2025-07-19 20:42:33,586 - INFO - Connecting to the existing vector database at
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/0.
DataBase_Chroma/ICT311/ICT311_chroma_db_toc_chunks_epub'...
2025-07-19 20:42:33,629 - INFO - Found 27 chunks in the database for this
section.

```

```

=====
VERIFYING SECTION: 'I Introduction and Classical Cryptography > 1 Introduction >
1.2 The Setting of Private-Key Encryption' (toc_id: 10)
=====

```

##### Reassembled Text #####

## 1.2 The Setting of Private-Key Encryption

Classical cryptography was concerned with designing and using codes (or ciphers) that enable two parties to send messages while keeping those messages hidden from an eavesdropper who can monitor all communication between them. In modern parlance, codes are called encryption schemes and that is the terminology we will use here. Security of all classical encryption schemes relies on a secret-a key-shared by the communicating parties in advance and unknown to the eavesdropper. This scenario, in which the communicating parties share some secret information in advance, is known as the private-key (or shared-/secret-key) setting. and private-key encryption is one example of a cryptographic primitive used in this setting. Before describing some historical encryption schemes, we discuss

primitive used in this setting. Before describing some historical encryption schemes, we discuss private-key encryption more generally.

In the context of private-key encryption, two parties share a key and use that key when they want to communicate secretly. One party can send a message, or plaintext, to the other by using the shared key to encrypt (or "scramble") the message and thus obtain a ciphertext that is transmitted to the receiver. The receiver uses the same key to decrypt (or "unscramble") the ciphertext and recover the original message. The same key is used to convert the plaintext into a ciphertext and back; that is why this setting is also known as the symmetric-key setting, where the symmetry lies in the fact that both parties hold the same key that is used for encryption and decryption. This is in contrast to asymmetric, or public-key, encryption (introduced in Chapter 11), where encryption and decryption

asymmetric, or public-key, encryption (introduced in Chapter 11), where encryption and decryption use different keys.

As already noted, the goal of encryption is to keep the plaintext hidden from an eavesdropper who can monitor the communication channel and observe the ciphertext. We discuss this in more detail later in this chapter, and spend a

great deal of time in Chapters 2, 3, and 5 formally defining this goal. There are two canonical applications of private-key cryptography. In the first (cf. Figure 1.1), the two communication parties are separated in space, e.g., a worker in New York communicating with her colleague in California. These two users are assumed to have been able to securely share a key in advance of their communication. (Note that if one party simply sends the key to the other over the public communication channel, then the eavesdropper obtains the key also!) This could be accomplished, for example, by having the parties physically meet in a secure location to share a key before they separate; in the example just given, the co-workers might arrange to share a key when they are both in the New York office. In other cases, sharing a key securely is more difficult. For the next

both in the New York office. In other cases, sharing a key securely is more difficult. For the next several chapters we simply assume that sharing a key is possible; we revisit this issue in Chapter 11.

The second widespread application of private-key cryptography involves the same party communicating with itself over time. (See Figure 1.2.) Consider, e.g., disk encryption, where a user encrypts some plaintext and stores the resulting ciphertext on his hard drive; the same user will return at a later point in time to decrypt the ciphertext and recover the original data. The hard drive here serves as the communication channel on which an attacker might eavesdrop if it can gain access to the hard drive and read its contents. "Sharing" the key is now trivial, though the user still needs a secure and reliable way to remember/store the key for use at a later point in time.

The syntax of encryption. Formally, a private-key encryption scheme is defined by specifying a message space along with three algorithms: a procedure for generating keys (Gen), a procedure for encrypting (Enc), and a procedure for decrypting (Dec). The message space defines the set of "legal" messages, i.e., those supported by the scheme. The algorithms of the scheme have the following functionality:

1. The key-generation algorithm Gen is a probabilistic algorithm that outputs a key  $k$  chosen according to some distribution.
2. The encryption algorithm Enc takes as input a key  $k$  and a message  $m$  and outputs a ciphertext  $c$ . We denote by  $\text{Enc}_k(m)$  the encryption of the plaintext  $m$  using the key  $k$ .
3. The decryption algorithm Dec takes as input a key  $k$  and a ciphertext  $c$  and outputs a plaintext  $m$ . We denote the decryption of the ciphertext  $c$  using the key  $k$  by  $\text{Dec}_k(c)$ .

An encryption scheme must satisfy the following correctness requirement: for every key  $k$  output by Gen and every message  $m$ , it holds that  $\text{Dec}_k(\text{Enc}_k(m)) = m$ .

In words: encrypting a message and then decrypting the resulting ciphertext using the same key yields the original message.

The set of all possible keys output by the key-generation algorithm is called the key space and is denoted by  $\mathcal{K}$ . Almost always, Gen simply chooses a key uniformly from the key space; in fact, one can assume without loss of generality that this is the case (see Exercise 2.1).

Reviewing our earlier discussion, an encryption scheme can be used by two

parties who wish to communicate secretly as follows. First, Gen is run to obtain a key  $k$  that the parties share. Later, when one party wants to send a plaintext  $m$  to the other, she computes  $c = \text{Enck}(m)$  and sends the resulting ciphertext  $c$  over the public channel to the other party.<sup>1</sup> Upon receiving  $c$ , the other party computes  $m = \text{Deck}(c)$  to recover the original plaintext.

Keys and Kerckhoffs' principle. As should be clear from the above, if an eavesdropping adversary knows the algorithm Dec as well as the key  $k$  shared by the two communicating parties, then that adversary will be able to decrypt any ciphertexts transmitted by those parties. It is for this reason that the communicating parties must share the key  $k$  securely and keep  $k$  completely hidden from everyone else. Perhaps they should keep the decryption algorithm Dec secret, also? For that matter, would it not be better for them to keep all the details of the encryption scheme secret?

Auguste Kerckhoffs [114, 115] argued the opposite in the late 19th century when elucidating several design principles for military ciphers. One of the most important of these, now known simply as Kerckhoffs' principle, was:

The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience.

That is, an encryption scheme should be designed to be secure even if an eavesdropper knows all the details of the scheme, so long as the attacker doesn't know the key being used. Stated differently, security should not rely on the encryption scheme being secret; instead, Kerckhoffs' principle demands that security rely solely on secrecy of the key.

There are three primary arguments in favor of Kerckhoffs' principle. The first is that it is significantly easier to maintain secrecy of a short key than to keep secret a (more complicated) encryption scheme. This is especially true when encryption is used widely. For example, consider the case where encryption is used for communication between all pairs of employees in some organization.

Unless each pair of parties use their own, unique scheme, some parties will know the scheme being used by others. Moreover, information about the scheme might be leaked by one of those employees (say, after being fired), or obtained by an attacker using reverse engineering. In short, it is simply unrealistic to assume that the encryption scheme will remain secret.

Second, in case the honest parties' shared, secret information is ever exposed, it will be much easier for them to change the key than to replace the encryption scheme. (Consider updating a file versus installing a new program.) Moreover, it is relatively trivial to generate a new random secret, whereas it would be a huge undertaking to design a new encryption scheme.

Finally, prior to widespread deployment of an encryption scheme, there is a significant benefit to encouraging public review of that scheme in order to check for possible weaknesses. Going further, it is desirable to standardize encryption schemes so that (1) compatibility between different users is ensured and (2) the general public will use strong schemes that have undergone public scrutiny. Overall, perhaps counter-intuitively, it is advantageous to have broad, public dissemination of the full details of an encryption scheme—the exact opposite of keeping the scheme secret.

Nowadays Kerckhoffs' principle is understood as advocating that the entire cryptographic design process be made completely public, in stark contrast to the

notion of "security by obscurity" that suggests keeping algorithms secret improves security. In fact, it is very dangerous to use a proprietary, "homebrewed" algorithm (i.e., a non-standardized algorithm designed in secret) since published designs undergo public peer review and are therefore likely to be stronger. Many years of experience have demonstrated that it is very difficult to construct good cryptographic schemes. Our confidence in the security of a scheme is much higher if it has been extensively studied by experts (beyond the designers of the scheme) and no flaws have been found. As simple and obvious as it may sound, the designers of the scheme) and no flaws have been found. As simple and obvious as it may sound, the principle of open cryptographic design (i.e., Kerckhoffs' principle) has been ignored over and over again with disastrous results. Fortunately, today there are several secure, standardized, and widely available cryptosystems and no reason to use anything else.

#####

----- Retrieved Chunk Details -----

[ Chunk 1 of 27 | chunk\_id: 106 ]

Content Preview: '1.2 The Setting of Private-Key Encryption...'

Metadata: {

"level\_2\_title": "1 Introduction",

"level\_1\_title": "I Introduction and Classical Cryptography",

"chunk\_id": 106,

"source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub",

"level\_3\_title": "1.2 The Setting of Private-Key Encryption",

"toc\_id": 10

}

[ Chunk 2 of 27 | chunk\_id: 107 ]

Content Preview: 'Classical cryptography was concerned with designing and using codes (or ciphers) that enable two parties to send messages while keeping those messages hidden from an eavesdropper who can monitor all communication between them. In modern parlance, cod...'

Metadata: {

"toc\_id": 10,

"chunk\_id": 107,

"level\_1\_title": "I Introduction and Classical Cryptography",

"level\_3\_title": "1.2 The Setting of Private-Key Encryption",

"source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub",

"level\_2\_title": "1 Introduction"

}

[ Chunk 3 of 27 | chunk\_id: 108 ]

Content Preview: 'primitive used in this setting. Before describing some historical encryption schemes, we discuss private-key encryption more

```

generally...'
 Metadata: {
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_2_title": "1 Introduction",
 "chunk_id": 108,
 "toc_id": 10,
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 "level_3_title": "1.2 The Setting of Private-Key Encryption"
 }

[Chunk 4 of 27 | chunk_id: 109]
 Content Preview: 'In the context of private-key encryption, two parties share
a key and use that key when they want to communicate secretly. One party can
send a message, or plaintext, to the other by using the shared key to encrypt
(or "scramble") the message and thu...'
 Metadata: {
 "toc_id": 10,
 "chunk_id": 109,
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Lindell, J.epub",
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 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption"
 }

[Chunk 5 of 27 | chunk_id: 110]
 Content Preview: 'asymmetric, or public-key, encryption (introduced in Chapter
11), where encryption and decryption use different keys...'
 Metadata: {
 "level_1_title": "I Introduction and Classical Cryptography",
 "toc_id": 10,
 "chunk_id": 110,
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption"
 }

[Chunk 6 of 27 | chunk_id: 111]
 Content Preview: 'As already noted, the goal of encryption is to keep the
plaintext hidden from an eavesdropper who can monitor the communication channel
and observe the ciphertext. We discuss this in more detail later in this
chapter, and spend a great deal of time i...'
 Metadata: {
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "chunk_id": 111,
 "toc_id": 10,

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```

 "level_1_title": "I Introduction and Classical Cryptography",
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub"
}

```

[ Chunk 7 of 27 | chunk\_id: 112 ]

Content Preview: 'There are two canonical applications of private-key cryptography. In the first (cf. Figure 1.1), the two communication parties are separated in space, e.g., a worker in New York communicating with her colleague in California. These two users are assu...'

```

Metadata: {
 "chunk_id": 112,
 "toc_id": 10,
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "level_1_title": "I Introduction and Classical Cryptography"
}

```

[ Chunk 8 of 27 | chunk\_id: 113 ]

Content Preview: 'both in the New York office. In other cases, sharing a key securely is more difficult. For the next several chapters we simply assume that sharing a key is possible; we revisit this issue in Chapter 11...'

```

Metadata: {
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography",
 "chunk_id": 113,
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "toc_id": 10
}

```

[ Chunk 9 of 27 | chunk\_id: 114 ]

Content Preview: 'The second widespread application of private-key cryptography involves the same party communicating with itself over time. (See Figure 1.2.) Consider, e.g., disk encryption, where a user encrypts some plaintext and stores the resulting ciphertext on ...'

```

Metadata: {
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "chunk_id": 114,
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
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 "level_1_title": "I Introduction and Classical Cryptography",
 "level_2_title": "1 Introduction"
}

```

```

}

[Chunk 10 of 27 | chunk_id: 115]
 Content Preview: 'The syntax of encryption. Formally, a private-key encryption
scheme is defined by specifying a message space along with three algorithms: a
procedure for generating keys (Gen), a procedure for encrypting (Enc), and a
procedure for decrypting (Dec)...'
 Metadata: {
 "chunk_id": 115,
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography",
 "toc_id": 10,
 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption"
 }

[Chunk 11 of 27 | chunk_id: 116]
 Content Preview: '1. The key-generation algorithm Gen is a probabilistic
algorithm that outputs a key k chosen according to some distribution...'
 Metadata: {
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "level_1_title": "I Introduction and Classical Cryptography",
 "toc_id": 10,
 "chunk_id": 116,
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub"
 }

[Chunk 12 of 27 | chunk_id: 117]
 Content Preview: '2. The encryption algorithm Enc takes as input a key k and a
message m and outputs a ciphertext c. We denote by $Enc(k, m)$ the encryption of the
plaintext m using the key k...'
 Metadata: {
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography",
 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "toc_id": 10,
 "chunk_id": 117
 }

[Chunk 13 of 27 | chunk_id: 118]
 Content Preview: '3. The decryption algorithm Dec takes as input a key k and a
ciphertext c and outputs a plaintext m. We denote the decryption of the
ciphertext c using the key k by $Dec(c, k)$...'

```



```

 Metadata: {
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_2_title": "1 Introduction",
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 "toc_id": 10,
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 "level_1_title": "I Introduction and Classical Cryptography"
 }

[Chunk 14 of 27 | chunk_id: 119]
 Content Preview: 'An encryption scheme must satisfy the following correctness
requirement: for every key k output by Gen and every message m , it holds
that...'
 Metadata: {
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography",
 "chunk_id": 119,
 "toc_id": 10
 }

[Chunk 15 of 27 | chunk_id: 120]
 Content Preview: 'Deck(Enck(m)) = m...'
 Metadata: {
 "level_1_title": "I Introduction and Classical Cryptography",
 "chunk_id": 120,
 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "toc_id": 10,
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub"
 }

[Chunk 16 of 27 | chunk_id: 121]
 Content Preview: 'In words: encrypting a message and then decrypting the
resulting ciphertext using the same key yields the original message...'
 Metadata: {
 "chunk_id": 121,
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 "level_2_title": "1 Introduction",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "toc_id": 10
 }

```

[ Chunk 17 of 27 | chunk\_id: 122 ]

Content Preview: 'The set of all possible keys output by the key-generation algorithm is called the key space and is denoted by  $\mathcal{K}$ . Almost always, Gen simply chooses a key uniformly from the key space; in fact, one can assume without loss of generality that this is the...'

```
Metadata: {
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
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 "toc_id": 10,
 "chunk_id": 122
}
```

[ Chunk 18 of 27 | chunk\_id: 123 ]

Content Preview: 'Reviewing our earlier discussion, an encryption scheme can be used by two parties who wish to communicate secretly as follows. First, Gen is run to obtain a key  $k$  that the parties share. Later, when one party wants to send a plaintext  $m$  to the other...'

```
Metadata: {
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 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "toc_id": 10,
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 "level_1_title": "I Introduction and Classical Cryptography",
 "chunk_id": 123
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```

[ Chunk 19 of 27 | chunk\_id: 124 ]

Content Preview: 'Keys and Kerckhoffs' principle. As should be clear from the above, if an eavesdropping adversary knows the algorithm Dec as well as the key  $k$  shared by the two communicating parties, then that adversary will be able to decrypt any ciphertexts transmi...'

```
Metadata: {
 "level_2_title": "1 Introduction",
 "chunk_id": 124,
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 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography"
}
```

[ Chunk 20 of 27 | chunk\_id: 125 ]

Content Preview: 'Auguste Kerckhoffs [114, 115] argued the opposite in the

late 19th century when elucidating several design principles for military ciphers. One of the most important of these, now known simply as Kerckhoffs' principle, was:...

```
Metadata: {
 "level_2_title": "1 Introduction",
 "chunk_id": 125,
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 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "toc_id": 10,
 "level_1_title": "I Introduction and Classical Cryptography"
}
```

[ Chunk 21 of 27 | chunk\_id: 126 ]

Content Preview: 'The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience...'

```
Metadata: {
 "level_2_title": "1 Introduction",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "chunk_id": 126,
 "level_1_title": "I Introduction and Classical Cryptography",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "toc_id": 10
}
```

[ Chunk 22 of 27 | chunk\_id: 127 ]

Content Preview: 'That is, an encryption scheme should be designed to be secure even if an eavesdropper knows all the details of the scheme, so long as the attacker doesn't know the key being used. Stated differently, security should not rely on the encryption scheme ...'

```
Metadata: {
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "toc_id": 10,
 "level_2_title": "1 Introduction",
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 "chunk_id": 127
}
```

[ Chunk 23 of 27 | chunk\_id: 128 ]

Content Preview: 'There are three primary arguments in favor of Kerckhoffs' principle. The first is that it is significantly easier to maintain secrecy of a short key than to keep secret a (more complicated) encryption scheme. This is especially true when encryption i...'

```
Metadata: {
 "level_2_title": "1 Introduction",
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```

 "chunk_id": 128,
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 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub"
}

```

[ Chunk 24 of 27 | chunk\_id: 129 ]

Content Preview: 'Second. in case the honest parties' shared, secret information is ever exposed, it will be much easier for them to change the key than to replace the encryption scheme. (Consider updating a file versus installing a new program.) Moreover, it is relat...'

```

Metadata: {
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "chunk_id": 129,
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 "level_2_title": "1 Introduction",
 "level_1_title": "I Introduction and Classical Cryptography"
}

```

[ Chunk 25 of 27 | chunk\_id: 130 ]

Content Preview: 'Finally, prior to widespread deployment of an encryption scheme, there is a significant benefit to encouraging public review of that scheme in order to check for possible weaknesses. Going further, it is desirable to standardize encryption schemes so...'

```

Metadata: {
 "level_2_title": "1 Introduction",
 "chunk_id": 130,
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography",
 "level_3_title": "1.2 The Setting of Private-Key Encryption",
 "toc_id": 10
}

```

[ Chunk 26 of 27 | chunk\_id: 131 ]

Content Preview: 'Nowadays Kerckhoffs' principle is understood as advocating that the entire cryptographic design process be made completely public, in stark contrast to the notion of "security by obscurity" that suggests keeping algorithms secret improves security. I...'

```

Metadata: {
 "toc_id": 10,
 "chunk_id": 131,
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 "level_1_title": "I Introduction and Classical Cryptography",

```

```

 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_3_title": "1.2 The Setting of Private-Key Encryption"
}

```

```
[Chunk 27 of 27 | chunk_id: 132]
```

Content Preview: 'designers of the scheme) and no flaws have been found. As simple and obvious as it may sound, the principle of open cryptographic design (i.e., Kerckhoffs' principle) has been ignored over and over again with disastrous results. Fortunately, today th...'

```

Metadata: {
 "chunk_id": 132,
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Lindell, J.epub",
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 "level_1_title": "I Introduction and Classical Cryptography"
}

```

```

=====
Verification complete for section 'I Introduction and Classical Cryptography > 1
Introduction > 1.2 The Setting of Private-Key Encryption'.
=====

```

## 6.2 Test Data Base for content development

Require Description

[37]: *# Cell 7: Verify Vector Database (Final Version with Rich Diagnostic Output)*

```

import os
import json
import re
import random
import logging
from typing import List, Dict, Any, Tuple, Optional

Third-party imports
try:
 from langchain_chroma import Chroma
 from langchain_ollama.embeddings import OllamaEmbeddings
 from langchain_core.documents import Document
 langchain_available = True
except ImportError:
 langchain_available = False

Setup Logger for this cell

```

```

logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -\n
↳ %(message)s')
logger = logging.getLogger(__name__)

--- HELPER FUNCTIONS ---

def print_results(query_text: str, results: list, where_filter: Optional[Dict] =
↳ None):
 """
 Richly prints query results, showing the query, filter, and retrieved
 ↳ documents.
 """
 print("\n" + "-"*10 + " DIAGNOSTIC: RETRIEVAL RESULTS " + "-"*10)
 print(f"QUERY: '{query_text}'")
 if where_filter:
 print(f"FILTER: {json.dumps(where_filter, indent=2)}")

 if not results:
 print("--> No documents were retrieved for this query and filter.")
 print("-" * 55)
 return

 print(f"--> Found {len(results)} results. Displaying top {min(len(results),
↳ 3)}:")
 for i, doc in enumerate(results[:3]):
 print(f"\n[RESULT {i+1}]")
 content_preview = doc.page_content.replace('\n', ' ').strip()
 print(f" Content : '{content_preview[:200]}...")
 print(f" Metadata: {json.dumps(doc.metadata, indent=2)}")
 print("-" * 55)

--- HELPER FUNCTIONS FOR FINDING DATA (UNCHANGED) ---
def find_deep_entry(nodes: List[Dict], current_path: List[str] = []) ->
↳ Optional[Tuple[Dict, List[str]]]:
 shuffled_nodes = random.sample(nodes, len(nodes))
 for node in shuffled_nodes:
 if node.get('level', 0) >= 2 and node.get('children'): return node,
↳ current_path + [node['title']]
 if node.get('children'):
 path = current_path + [node['title']]
 deep_entry = find_deep_entry(node['children'], path)
 if deep_entry: return deep_entry
 return None

```

```

def find_chapter_title_by_number(toc_data: List[Dict], chap_num: int) -> Optional[List[str]]:
 def search_nodes(nodes, num, current_path):
 for node in nodes:
 path = current_path + [node['title']]
 if re.match(rf"(Chapter\s)?{num}[.:\s]", node.get('title', ''), re.IGNORECASE):
 return path
 if node.get('children'):
 found_path = search_nodes(node['children'], num, path)
 if found_path:
 return found_path
 return None
 return search_nodes(toc_data, chap_num, [])

--- ENHANCED TEST CASES with DIAGNOSTIC OUTPUT ---

def basic_retrieval_test(db, outline):
 print_header("Test 1: Basic Retrieval", char="-")
 try:
 logger.info("Goal: Confirm the database is live and contains thematically relevant content.")
 logger.info("Strategy: Perform a simple similarity search using the course's 'unitName'.")
 query_text = outline.get("unitInformation", {}).get("unitName", "introduction")

 logger.info(f"Action: Searching for query: '{query_text}'...")
 results = db.similarity_search(query_text, k=1)

 print_results(query_text, results) # <--- SHOW THE EVIDENCE

 logger.info("Verification: Check if at least one document was returned.")
 assert len(results) > 0, "Basic retrieval query returned no results."

 logger.info("Result: TEST 1 PASSED. The database is online and responsive.")
 return True
 except Exception as e:
 logger.error(f"Result: TEST 1 FAILED. Reason: {e}")
 return False

def deep_hierarchy_test(db, toc):
 print_header("Test 2: Deep Hierarchy Retrieval", char="-")
 try:

```

```

 logger.info("Goal: Verify that the multi-level hierarchical metadata_
↳was ingested correctly.")
 logger.info("Strategy: Find a random, deeply nested sub-section and use_
↳a precise filter to retrieve it.")
 deep_entry_result = find_deep_entry(toc)
 assert deep_entry_result, "Could not find a suitable deep entry (level_
↳>= 2) to test."
 node, path = deep_entry_result
 query = node['title']

 logger.info(f" - Selected random deep section: {' -> '.join(path)}")
 conditions = [{f"level_{i+1}_title": {"$eq": title}} for i, title in_
↳enumerate(path)]
 w_filter = {"$and": conditions}

 logger.info("Action: Performing a similarity search with a highly_
↳specific '$and' filter.")
 results = db.similarity_search(query, k=1, filter=w_filter)

 print_results(query, results, w_filter) # <--- SHOW THE EVIDENCE

 logger.info("Verification: Check if the precisely filtered query_
↳returned any documents.")
 assert len(results) > 0, "Deeply filtered query returned no results."

 logger.info(" Result: TEST 2 PASSED. Hierarchical metadata is_
↳structured correctly.")
 return True
 except Exception as e:
 logger.error(f" Result: TEST 2 FAILED. Reason: {e}")
 return False

def advanced_alignment_test(db, outline, toc):
 print_header("Test 3: Advanced Unit Outline Alignment", char="-")
 try:
 logger.info("Goal: Ensure a weekly topic from the syllabus can be_
↳mapped to the correct textbook chapter(s).")
 logger.info("Strategy: Pick a random week, find its chapter, and query_
↳for the topic filtered by that chapter.")
 week_to_test = random.choice(outline['weeklySchedule'])
 logger.info(f" - Selected random week: Week {week_to_test['week']} -_
↳'{week_to_test['contentTopic']}'")

 reading = week_to_test.get('requiredReading', '')
 chap_nums_str = re.findall(r'\d+', reading)

```



```

 assert chap_nums_str, f"Could not find chapter numbers in required_
↳reading: '{reading}'"
 logger.info(f" - Extracted required chapter number(s):_
↳{chap_nums_str}")

 chapter_paths = [find_chapter_title_by_number(toc, int(n)) for n in_
↳chap_nums_str]
 chapter_paths = [path for path in chapter_paths if path is not None]
 assert chapter_paths, f"Could not map chapter numbers {chap_nums_str}_
↳to a valid ToC path."

 level_1_titles = list(set([path[0] for path in chapter_paths]))
 logger.info(f" - Mapped to top-level ToC entries: {level_1_titles}")

 or_filter = [{"level_1_title": {"$eq": title}} for title in_
↳level_1_titles]
 w_filter = {"$or": or_filter} if len(or_filter) > 1 else or_filter[0]
 query = week_to_test['contentTopic']

 logger.info("Action: Searching for the weekly topic, filtered by the_
↳mapped chapter(s).")
 results = db.similarity_search(query, k=5, filter=w_filter)

 print_results(query, results, w_filter) # <--- SHOW THE EVIDENCE

 logger.info("Verification: Check if at least one returned document is_
↳from the correct chapter.")
 assert len(results) > 0, "Alignment query returned no results for the_
↳correct section/chapter."

 logger.info(" Result: TEST 3 PASSED. The syllabus can be reliably_
↳aligned with the textbook content.")
 return True
 except Exception as e:
 logger.error(f" Result: TEST 3 FAILED. Reason: {e}")
 return False

def content_sequence_test(db, outline):
 print_header("Test 4: Content Sequence Verification", char="-")
 try:
 logger.info("Goal: Confirm that chunks for a topic can be re-ordered to_
↳form a coherent narrative.")
 logger.info("Strategy: Retrieve several chunks for a random topic and_
↳verify their 'chunk_id' is sequential.")
 topic_query = random.choice(outline['weeklySchedule'])['contentTopic']

```

```

 logger.info(f"Action: Performing similarity search for topic:␣
↳ '{topic_query}' to get a set of chunks.")
 results = db.similarity_search(topic_query, k=10)

 print_results(topic_query, results) # <--- SHOW THE EVIDENCE

 docs_with_id = [doc for doc in results if 'chunk_id' in doc.metadata]
 assert len(docs_with_id) > 3, "Fewer than 4 retrieved chunks have a␣
↳ 'chunk_id' to test."

 chunk_ids = [doc.metadata['chunk_id'] for doc in docs_with_id]
 sorted_ids = sorted(chunk_ids)

 logger.info(f" - Retrieved and sorted chunk IDs: {sorted_ids}")
 logger.info("Verification: Check if the sorted list of chunk_ids is␣
↳ strictly increasing.")
 is_ordered = all(sorted_ids[i] >= sorted_ids[i-1] for i in range(1,␣
↳ len(sorted_ids)))
 assert is_ordered, "The retrieved chunks' chunk_ids are not in␣
↳ ascending order when sorted."

 logger.info(" Result: TEST 4 PASSED. Narrative order can be␣
↳ reconstructed using 'chunk_id'.")
 return True
 except Exception as e:
 logger.error(f" Result: TEST 4 FAILED. Reason: {e}")
 return False

--- MAIN VERIFICATION EXECUTION ---
def run_verification():
 print_header("Database Verification Process")

 if not langchain_available:
 logger.error("LangChain libraries not found. Aborting tests.")
 return

 required_files = {
 "Chroma DB": CHROMA_PERSIST_DIR,
 "ToC JSON": PRE_EXTRACTED_TOC_JSON_PATH,
 "Parsed Outline": PARSED_UO_JSON_PATH
 }
 for name, path in required_files.items():
 if not os.path.exists(path):
 logger.error(f"Required '{name}' not found at '{path}'. Please run␣
↳ previous cells.")
 return

```

```

with open(PRE_EXTRACTED_TOC_JSON_PATH, 'r', encoding='utf-8') as f:
 toc_data = json.load(f)
with open(PARSED_UO_JSON_PATH, 'r', encoding='utf-8') as f:
 unit_outline_data = json.load(f)

logger.info("Connecting to DB and initializing components...")
embeddings = OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA)
vector_store = Chroma(
 persist_directory=CHROMA_PERSIST_DIR,
 embedding_function=embeddings,
 collection_name=CHROMA_COLLECTION_NAME
)

results_summary = [
 basic_retrieval_test(vector_store, unit_outline_data),
 deep_hierarchy_test(vector_store, toc_data),
 advanced_alignment_test(vector_store, unit_outline_data, toc_data),
 content_sequence_test(vector_store, unit_outline_data)
]

passed_count = sum(filter(None, results_summary))
failed_count = len(results_summary) - passed_count

print_header("Verification Summary")
print(f"Total Tests Run: {len(results_summary)}")
print(f"Passed: {passed_count}")
print(f"Failed: {failed_count}")
print_header("Verification Complete", char="=")

--- Execute Verification ---
Assumes global variables from Cell 1 are available in the notebook's scope
run_verification()

```

```

2025-07-19 20:24:35,691 - INFO - Connecting to DB and initializing components...
2025-07-19 20:24:35,704 - INFO - Goal: Confirm the database is live and contains
thematically relevant content.
2025-07-19 20:24:35,705 - INFO - Strategy: Perform a simple similarity search
using the course's 'unitName'.
2025-07-19 20:24:35,705 - INFO - Action: Searching for query: 'Applied
Cryptography'...
2025-07-19 20:24:35,764 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-19 20:24:35,767 - INFO - Verification: Check if at least one document
was returned.
2025-07-19 20:24:35,768 - INFO - Result: TEST 1 PASSED. The database is online
and responsive.
2025-07-19 20:24:35,768 - INFO - Goal: Verify that the multi-level hierarchical

```

```

metadata was ingested correctly.
2025-07-19 20:24:35,769 - INFO - Strategy: Find a random, deeply nested sub-
section and use a precise filter to retrieve it.
2025-07-19 20:24:35,769 - INFO - - Selected random deep section: II Private-
Key (Symmetric) Cryptography -> 7 Practical Constructions of Symmetric-Key
Primitives -> 7.1 Stream Ciphers
2025-07-19 20:24:35,769 - INFO - Action: Performing a similarity search with a
highly specific '$and' filter.
2025-07-19 20:24:35,824 - INFO - HTTP Request: POST
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"
2025-07-19 20:24:35,837 - INFO - Verification: Check if the precisely filtered
query returned any documents.
2025-07-19 20:24:35,838 - INFO - Result: TEST 2 PASSED. Hierarchical metadata
is structured correctly.
2025-07-19 20:24:35,838 - INFO - Goal: Ensure a weekly topic from the syllabus
can be mapped to the correct textbook chapter(s).
2025-07-19 20:24:35,838 - INFO - Strategy: Pick a random week, find its chapter,
and query for the topic filtered by that chapter.

```

---

### Database Verification Process

---



---

#### Test 1: Basic Retrieval

---

----- DIAGNOSTIC: RETRIEVAL RESULTS -----

QUERY: 'Applied Cryptography'

--> Found 1 results. Displaying top 1:

```

[RESULT 1]
Content : '9.4 *Cryptographic Applications...'
Metadata: {
 "toc_id": 166,
 "level_1_title": "III Public-Key (Asymmetric) Cryptography",
 "chunk_id": 3610,
 "level_2_title": "9 Number Theory and Cryptographic Hardness Assumptions",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_3_title": "9.4 *Cryptographic Applications"
}

```

---



---

#### Test 2: Deep Hierarchy Retrieval

---

----- DIAGNOSTIC: RETRIEVAL RESULTS -----

QUERY: '7.1 Stream Ciphers'

FILTER: {

```
"$and": [
 {
 "level_1_title": {
 "$eq": "II Private-Key (Symmetric) Cryptography"
 }
 },
 {
 "level_2_title": {
 "$eq": "7 Practical Constructions of Symmetric-Key Primitives"
 }
 },
 {
 "level_3_title": {
 "$eq": "7.1 Stream Ciphers"
 }
 }
]
```

}

--> Found 1 results. Displaying top 1:

[ RESULT 1 ]

Content : '7.1 Stream Ciphers...'

Metadata: {

"toc\_id": 109,

"level\_1\_title": "II Private-Key (Symmetric) Cryptography",

"level\_3\_title": "7.1 Stream Ciphers",

"chunk\_id": 2079,

"level\_2\_title": "7 Practical Constructions of Symmetric-Key Primitives",

"source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub"

}

---

### Test 3: Advanced Unit Outline Alignment

---

2025-07-19 20:24:35,839 - INFO - - Selected random week: Week Week 3 -  
'Message Authentication Code and Authenticated Encryption'

2025-07-19 20:24:35,839 - INFO - - Extracted required chapter number(s):  
['2020', '4', '5']

2025-07-19 20:24:35,840 - INFO - - Mapped to top-level ToC entries: ['II  
Private-Key (Symmetric) Cryptography']

2025-07-19 20:24:35,840 - INFO - Action: Searching for the weekly topic,

filtered by the mapped chapter(s).

2025-07-19 20:24:35,891 - INFO - HTTP Request: POST  
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"

2025-07-19 20:24:35,899 - INFO - Verification: Check if at least one returned document is from the correct chapter.

2025-07-19 20:24:35,900 - INFO - Result: TEST 3 PASSED. The syllabus can be reliably aligned with the textbook content.

2025-07-19 20:24:35,900 - INFO - Goal: Confirm that chunks for a topic can be re-ordered to form a coherent narrative.

2025-07-19 20:24:35,900 - INFO - Strategy: Retrieve several chunks for a random topic and verify their 'chunk\_id' is sequential.

2025-07-19 20:24:35,901 - INFO - Action: Performing similarity search for topic: 'Number Theory and Cryptographic Hardness Assumptions' to get a set of chunks.

2025-07-19 20:24:35,951 - INFO - HTTP Request: POST  
http://127.0.0.1:11434/api/embed "HTTP/1.1 200 OK"

2025-07-19 20:24:35,955 - INFO - Retrieved and sorted chunk IDs: [49, 204, 235, 800, 2992, 2998, 3396, 3611, 3763, 6462]

2025-07-19 20:24:35,955 - INFO - Verification: Check if the sorted list of chunk\_ids is strictly increasing.

2025-07-19 20:24:35,955 - INFO - Result: TEST 4 PASSED. Narrative order can be reconstructed using 'chunk\_id'.

----- DIAGNOSTIC: RETRIEVAL RESULTS -----

QUERY: 'Message Authentication Code and Authenticated Encryption'

FILTER: {

```
 "level_1_title": {
 "$eq": "II Private-Key (Symmetric) Cryptography"
 }
```

}

--> Found 5 results. Displaying top 3:

[ RESULT 1 ]

Content : 'Security of Message Authentication Codes...'

Metadata: {

"toc\_id": 55,

"chunk\_id": 1114,

"level\_2\_title": "4 Message Authentication Codes",

"level\_4\_title": "4.1.2 Encryption vs. Message Authentication",

"level\_3\_title": "4.1 Message Integrity",

"level\_1\_title": "II Private-Key (Symmetric) Cryptography",

"source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda Lindell, J.epub"

}

[ RESULT 2 ]

Content : 'There exist authenticated encryption schemes and secure message authentication codes...'

```

Metadata: {
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "II Private-Key (Symmetric) Cryptography",
 "chunk_id": 2615,
 "toc_id": 133,
 "level_2_title": "8 *Theoretical Constructions of Symmetric-Key Primitives",
 "level_3_title": "8.2 From One-Way Functions to Pseudorandomness"
}

```

```

[RESULT 3]
Content : 'The Syntax of a Message Authentication Code...'
Metadata: {
 "level_1_title": "II Private-Key (Symmetric) Cryptography",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_4_title": "4.1.2 Encryption vs. Message Authentication",
 "toc_id": 55,
 "level_3_title": "4.1 Message Integrity",
 "chunk_id": 1103,
 "level_2_title": "4 Message Authentication Codes"
}

```

---



---

#### Test 4: Content Sequence Verification

---



---

```

----- DIAGNOSTIC: RETRIEVAL RESULTS -----
QUERY: 'Number Theory and Cryptographic Hardness Assumptions'
--> Found 10 results. Displaying top 3:

```

```

[RESULT 1]
Content : 'Number Theory and Cryptographic Hardness Assumptions...'
Metadata: {
 "chunk_id": 2992,
 "level_2_title": "1 Introduction",
 "toc_id": 18,
 "level_3_title": "Exercises",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_1_title": "I Introduction and Classical Cryptography"
}

```

```

[RESULT 2]
Content : 'In this section we introduce a class of cryptographic hardness
assumptions in cyclic groups. We begin with a general discussion of cyclic
groups, followed by abstract definitions of the relevant assum...'

```

```

Metadata: {
 "toc_id": 161,
 "level_1_title": "III Public-Key (Asymmetric) Cryptography",
 "chunk_id": 3396,
 "level_2_title": "9 Number Theory and Cryptographic Hardness Assumptions",
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_3_title": "9.3 Cryptographic Assumptions in Cyclic Groups"
}

```

[ RESULT 3 ]

Content : 'We have spent a fair bit of time discussing number theory and group theory, and introducing computational hardness assumptions that are widely believed to hold. Applications of these assumptions will ...'

```

Metadata: {
 "level_3_title": "9.4 *Cryptographic Applications",
 "chunk_id": 3611,
 "source": "Introduction to Modern Cryptography -- Jonathan Katz, Yehuda
Lindell, J.epub",
 "level_2_title": "9 Number Theory and Cryptographic Hardness Assumptions",
 "toc_id": 166,
 "level_1_title": "III Public-Key (Asymmetric) Cryptography"
}

```

```

=====
 Verification Summary
=====
Total Tests Run: 4
 Passed: 4
 Failed: 0

=====
 Verification Complete
=====

```

## 7 Content Generation

### 7.1 Planning Agent

```

[38]: # PlanningAgent
 # Cell 8: The Data-Driven Planning Agent (Final Hierarchical Version)

import os
import json
import re
import math

```



```

import logging
from typing import List, Dict, Any, Optional, Tuple

Setup Logger and LangChain components
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
logger = logging.getLogger(__name__)
try:
 from langchain_chroma import Chroma
 from langchain_ollama.embeddings import OllamaEmbeddings
 langchain_available = True
except ImportError:
 langchain_available = False

def print_header(text: str, char: str = "="):
 """Prints a centered header to the console."""
 print("\n" + char * 80)
 print(text.center(80))
 print(char * 80)

class PlanningAgent:
 """
 An agent that creates a hierarchical content plan, adaptively partitions
 content
 into distinct lecture decks, and allocates presentation time.
 """
 def __init__(self, master_config: Dict, vector_store: Optional[Any] = None):
 self.config = master_config['processed_settings']
 self.unit_outline = master_config['unit_outline']
 self.book_toc = master_config['book_toc']
 self.flat_toc_with_ids = self._create_flat_toc_with_ids()
 self.vector_store = vector_store
 logger.info("Data-Driven PlanningAgent initialized successfully.")

 def _create_flat_toc_with_ids(self) -> List[Dict]:
 """Creates a flattened list of the ToC for easy metadata lookup."""
 flat_list = []
 def flatten_recursive(nodes, counter):
 for node in nodes:
 node_id = counter[0]; counter[0] += 1
 flat_list.append({'toc_id': node_id, 'title': node.get('title', ''),
 'node': node})
 if node.get('children'):
 flatten_recursive(node.get('children'), counter)
 flatten_recursive(self.book_toc, [0])
 return flat_list

```

```

def _assign_sequence_ids_recursively(self, node: Dict, counter: list):
 """
 Recursively walks a content node, assigning a unique, sequential ID
 to the node, its children, and its interactive activity in a
 ↪depth-first order.
 """
 # 1. Assign sequence ID to the parent content node itself.
 node['seq_id'] = counter[0]
 counter[0] += 1

 # 2. Recurse through all children first.
 if 'children' in node and node['children']:
 for child in node['children']:
 self._assign_sequence_ids_recursively(child, counter)

 # 3. Assign an ID to the interactive activity LAST, so it appears after
 ↪the content and its children.
 if 'interactive_activity' in node:
 node['interactive_activity']['seq_id'] = counter[0]
 counter[0] += 1

def _identify_relevant_chapters(self, weekly_schedule_item: Dict) ->
↪List[int]:
 """Extracts chapter numbers precisely from the 'requiredReading' string.
 ↪"""
 reading_str = weekly_schedule_item.get('requiredReading', '')
 match = re.search(r'Chapter(s)?', reading_str, re.IGNORECASE)
 if not match: return []
 search_area = reading_str[match.start():]
 chap_nums_str = re.findall(r'\d+', search_area)
 if chap_nums_str:
 return sorted(list(set(int(n) for n in chap_nums_str)))
 return []

def _find_chapter_node(self, chapter_number: int) -> Optional[Dict]:
 """Finds the ToC node for a specific chapter number."""
 for item in self.flat_toc_with_ids:
 if re.match(rf"Chapter\s{chapter_number}(?:\D|$)", item['title']):
 return item['node']
 return None

def _build_topic_plan_tree(self, toc_node: Dict) -> Dict:
 """
 Recursively builds a hierarchical plan tree from any ToC node,
 annotating it with direct and total branch chunk counts.
 """

```

```

 node_metadata = next((item for item in self.flat_toc_with_ids if
↪item['node'] is toc_node), None)
 if not node_metadata: return {}

 retrieved_docs = self.vector_store.get(where={'toc_id':
↪node_metadata['toc_id']})
 direct_chunk_count = len(retrieved_docs.get('ids', []))

 plan_node = {
 "title": node_metadata['title'],
 "toc_id": node_metadata['toc_id'],
 "chunk_count": direct_chunk_count,
 "total_chunks_in_branch": 0,
 "slides_allocated": 0,
 "children": []
 }

 child_branch_total = 0
 for child_node in toc_node.get('children', []):
 if any(ex in child_node.get('title', '').lower() for ex in
↪["review", "introduction", "summary", "key terms"]):
 continue
 child_plan_node = self._build_topic_plan_tree(child_node)
 if child_plan_node:
 plan_node['children'].append(child_plan_node)
 child_branch_total += child_plan_node.
↪get('total_chunks_in_branch', 0)

 plan_node['total_chunks_in_branch'] = direct_chunk_count +
↪child_branch_total
 return plan_node

In PlanningAgent Class...

def _allocate_slides_to_tree(self, plan_tree: Dict, content_slides_budget:
↪int):
 """
 (FINAL, REORDERED FOR CLARITY) Performs a multi-pass process to
↪allocate content slides,
 add activities, sum totals, and reorders the keys in each node for
↪maximum readability.
 """
 if not plan_tree or content_slides_budget <= 0:
 return plan_tree

 # --- Pass 1: Allocate Content Slides ---

```

```

def allocate_content_recursively(node, budget):
 node['budget_slides_content'] = round(budget)
 node['direct_slides_content'] = 0
 if not node.get('children'):
 node['direct_slides_content'] = round(budget)
 return
 total_branch_chunks = node.get('total_chunks_in_branch', 0)
 own_content_slides = 0
 if total_branch_chunks > 0:
 own_content_slides = round(budget * (node.get('chunk_count', 0) /
 ↪ total_branch_chunks))
 node['direct_slides_content'] = own_content_slides
 remaining_budget_for_children = budget - own_content_slides
 children_total_chunks = total_branch_chunks - node.
 ↪get('chunk_count', 0)
 if children_total_chunks <= 0: return
 for child in node.get('children', []):
 child_budget = remaining_budget_for_children * (child.
 ↪get('total_chunks_in_branch', 0) / children_total_chunks)
 allocate_content_recursively(child, child_budget)

 allocate_content_recursively(plan_tree, content_slides_budget)

--- Pass 2: Add Interactive Activities ---
def add_interactive_nodes(node, depth, interactive_deep):
 if not node: return
 if self.config.get('interactive', False):
 if interactive_deep:
 if depth == 2: node['interactive_activity'] = {"title":
 ↪f"{node.get('title')} (Deep-Dive Activity)", "toc_id": node.get('toc_id'),
 ↪ "slides_allocated": 1}
 if depth == 1: node['interactive_activity'] = {"title":
 ↪f"{node.get('title')} (General Activity)", "toc_id": node.get('toc_id'),
 ↪ "slides_allocated": 1}
 else:
 if depth == 1: node['interactive_activity'] = {"title":
 ↪f"{node.get('title')} (Interactive Activity)", "toc_id": node.get('toc_id'),
 ↪ "slides_allocated": 1}
 for child in node.get('children', []):
 add_interactive_nodes(child, depth + 1, interactive_deep)

 add_interactive_nodes(plan_tree, 1, self.config.get('interactive_deep',
 ↪False))

--- Pass 3: Sum All Slides Up the Tree ---
def sum_slides_upwards(node):

```

```

 total_slides = node.get('direct_slides_content', 0)
 total_slides += node.get('interactive_activity', {}).
↳get('slides_allocated', 0)
 if node.get('children'):
 total_slides += sum(sum_slides_upwards(child) for child in node.
↳get('children', []))
 node['total_slides_in_branch'] = total_slides
 return total_slides

sum_slides_upwards(plan_tree)

--- NEW: Pass 4: Reorder keys for final clarity ---
def reorder_keys_for_readability(node: Dict) -> Dict:
 if not node:
 return None

 # Define the desired order of keys
 key_order = [
 "title",
 "toc_id",
 "chunk_count",
 "total_chunks_in_branch",
 "budget_slides_content",
 "direct_slides_content",
 "total_slides_in_branch",
 "children",
 "interactive_activity"
]

 # Rebuild the dictionary in the specified order
 reordered_node = {key: node[key] for key in key_order if key in
↳node}

 # Recursively reorder children
 if 'children' in reordered_node:
 reordered_node['children'] =
↳[reorder_keys_for_readability(child) for child in reordered_node['children']]

 return reordered_node

return reorder_keys_for_readability(plan_tree)

def create_content_plan_for_week(self, week_number: int) -> Optional[Dict]:
 """Orchestrates the adaptive planning and partitioning process."""
 print_header(f"Planning Week {week_number}", char="*")

```

```

weekly_schedule_item = self.unit_outline['weeklySchedule'][week_number_
↪- 1]

chapter_numbers = self._identify_relevant_chapters(weekly_schedule_item)
if not chapter_numbers: return None

num_decks = self.config['week_session_setup'].get('sessions_per_week',
↪1)

 # 1. Build a full plan tree for each chapter to get its weight.
 chapter_plan_trees = [self._build_topic_plan_tree(self.
↪_find_chapter_node(cn)) for cn in chapter_numbers if self.
↪_find_chapter_node(cn)]
 total_weekly_chunks = sum(tree.get('total_chunks_in_branch', 0) for
↪tree in chapter_plan_trees)

 # 2. NEW: Adaptive Partitioning Strategy
 partitionable_units = []
 all_top_level_sections = []
 for chapter_tree in chapter_plan_trees:
 all_top_level_sections.extend(chapter_tree.get('children', []))

 num_top_level_sections = len(all_top_level_sections)

 # Always prefer to split by top-level sections if there are enough to
↪distribute.
 if num_top_level_sections >= num_decks:
 logger.info(f"Partitioning strategy: Distributing
↪{num_top_level_sections} top-level sections across {num_decks} decks.")
 partitionable_units = all_top_level_sections
 else:
 # Fallback for rare cases where there are fewer topics than decks
↪(e.g., 1 chapter with 1 section, but 2 decks).
 logger.info(f"Partitioning strategy: Not enough top-level sections
↪({num_top_level_sections}) to fill all decks ({num_decks}). Distributing
↪whole chapters instead.")
 partitionable_units = chapter_plan_trees

 # 3. Partition the chosen units into decks using a bin-packing algorithm
 decks = [[] for _ in range(num_decks)]
 deck_weights = [0] * num_decks
 sorted_units = sorted(partitionable_units, key=lambda x: x.
↪get('toc_id', 0))

 for unit in sorted_units:
 lightest_deck_index = deck_weights.index(min(deck_weights))
 decks[lightest_deck_index].append(unit)

```

```

 deck_weights[lightest_deck_index] += unit.
 ↪get('total_chunks_in_branch', 0)

 # 4. Plan each deck
 content_slides_per_week = self.config['slide_count_strategy'].
 ↪get('target_total_slides', 25)
 final_deck_plans = []
 for i, deck_content_trees in enumerate(decks):
 deck_number = i + 1
 deck_chunk_weight = sum(tree.get('total_chunks_in_branch', 0) for
 ↪tree in deck_content_trees)
 deck_slide_budget = round((deck_chunk_weight / total_weekly_chunks)
 ↪* content_slides_per_week) if total_weekly_chunks > 0 else 0

 logger.info(f"--- Planning Deck {deck_number}/{num_decks} | Topics:
 ↪{[t['title'] for t in deck_content_trees]} | Weight: {deck_chunk_weight}
 ↪chunks | Slide Budget: {deck_slide_budget} ---")

 # The allocation function is recursive and works on any tree or
 ↪sub-tree
 planned_content = [self._allocate_slides_to_tree(tree,
 ↪round(deck_slide_budget * (tree.get('total_chunks_in_branch', 0) /
 ↪deck_chunk_weight))) if deck_chunk_weight > 0 else tree for tree in
 ↪deck_content_trees]

 final_deck_plans.append({
 "deck_number": deck_number,
 "deck_title": f"{self.config.get('unit_name', 'Course')} - Week
 ↪{week_number}, Lecture {deck_number}",
 "session_content": planned_content
 })

 return {
 "week": week_number,
 "overall_topic": weekly_schedule_item.get('contentTopic'),
 "deck_plans": final_deck_plans
 }

 def finalize_and_calculate_time_plan(self, draft_plan: Dict, config: Dict)
 ↪-> Dict:
 """
 Takes a draft plan and enriches it by:
 1. Calculating detailed slide counts and time allocations for every
 ↪node.
 2. Adding framework sections and wrapping content.

```

```

3. Calculating and adding summaries for decks and the entire week.
4. Assigning a final sequence ID (seq_id) for slide generation.
5. Reordering all keys for maximum readability.
"""
final_plan = json.loads(json.dumps(draft_plan))

--- Time Constants from Config ---
params = config.get('parameters_slides', {})
TIME_PER_CONTENT = params.get('time_per_content_slides_min', 3)
TIME_PER_INTERACTIVE = params.get('time_per_interactive_slide_min', 5)
TIME_FOR_FRAMEWORK_DECK = params.get('time_for_framework_slides_min', 6)
FRAMEWORK_SLIDES_PER_DECK = 4

--- Recursive Helper Functions ---
def _calculate_time_and_reorder(node: Dict):
 # 1. Recurse to the bottom first to perform a bottom-up calculation
 children_total_time = 0
 if 'children' in node and node['children']:
 for child in node['children']:
 _calculate_time_and_reorder(child) # Recursive call
 children_total_time += child.get('time_allocation_minutes', 0)
 → {}.get('total_branch_time', 0)

 # 2. Calculate this node's direct time
 direct_content_time = node.get('direct_slides_content', 0) * TIME_PER_CONTENT
 → TIME_PER_CONTENT
 interactive_time = node.get('interactive_activity', {}).get('slides_allocated', 0) * TIME_PER_INTERACTIVE
 → get('slides_allocated', 0) * TIME_PER_INTERACTIVE

 # 3. Calculate this node's total branch time
 branch_total_time = direct_content_time + interactive_time + children_total_time
 → children_total_time

 # 4. Create the time allocation object
 time_alloc = {
 "direct_content_time": direct_content_time,
 "direct_interactive_time": interactive_time,
 "total_branch_time": branch_total_time
 }
 node['time_allocation_minutes'] = time_alloc

 # 5. Reorder all keys for this node to ensure final clarity
 key_order = [
 "title",
 "toc_id",
 "chunk_count",

```



```

 "total_chunks_in_branch",
 "budget_slides_content",
 "direct_slides_content",
 "total_slides_in_branch",
 "time_allocation_minutes",
 "children",
 "interactive_activity"

]

 reordered_node = {key: node[key] for key in key_order if key in node}

 # Clear the original node and update it with the reordered keys
 node.clear()
 node.update(reordered_node)

 # --- Main Processing Loop for Decks ---
 for deck in final_plan.get("deck_plans", []):
 session_content_blocks = deck.pop("session_content", [])

 # Perform the combined time calculation and reordering pass
 for block in session_content_blocks:
 _calculate_time_and_reorder(block)

 # Create Framework Sections
 week_number, deck_number = final_plan.get("week"), deck.get("deck_number")

 title_section = {"section_type": "Title", "content": { "unit_name": config.get('unit_name', 'Course'), "unit_code": config.get('course_id', ''), "week_topic": final_plan.get('overall_topic', ''), "deck_title": f"Week {week_number}, Lecture {deck_number}"}}

 agenda_section = {"section_type": "Agenda", "content": {"title": "Today's Agenda", "items": [item.get('title', 'Untitled Topic') for item in session_content_blocks]}}

 summary_section = {"section_type": "Summary", "content": {"title": "Summary & Key Takeaways", "placeholder": "Auto-generate based on covered topics."}}

 end_section = {"section_type": "End", "content": {"title": "Thank You", "text": "Questions?"}}

 main_content_block = {"section_type": "Content", "content_blocks": session_content_blocks}

 final_sections_for_deck = [title_section, agenda_section, main_content_block, summary_section, end_section]

```

```

*** NEW: Assign Sequential IDs for the entire deck ***

seq_counter = [0] # Use a list for pass-by-reference behavior
for section in final_sections_for_deck:
 if section.get("section_type") == "Content":
 # For the main content, iterate through its blocks and
 ↪start the recursion
 for block in section.get("content_blocks", []):
 self._assign_sequence_ids_recursively(block,
 ↪seq_counter)
 else:
 # For simple framework slides, just assign the next ID
 section['seq_id'] = seq_counter[0]
 seq_counter[0] += 1

*** END NEW CODE ***

Calculate Deck Summaries
total_content_slides = sum(b.get('total_slides_in_branch', 0) - b.
↪get('interactive_activity', {}).get('slides_allocated', 0) for b in
↪session_content_blocks)
total_interactive_slides = sum(b.get('interactive_activity', {}).
↪get('slides_allocated', 0) for b in session_content_blocks)

deck_content_time = sum(b.get('time_allocation_minutes', {}).
↪get('total_branch_time', 0) for b in session_content_blocks)

deck['total_slides_in_deck'] = FRAMEWORK_SLIDES_PER_DECK + sum(b.
↪get('total_slides_in_branch', 0) for b in session_content_blocks)
deck['slide_count_breakdown'] = {"framework":
↪FRAMEWORK_SLIDES_PER_DECK, "content": total_content_slides, "interactive":
↪total_interactive_slides}
deck['time_breakdown_minutes'] = {"framework":
↪TIME_FOR_FRAMEWORK_DECK, "content_and_interactive": deck_content_time,
↪"total_deck_time": TIME_FOR_FRAMEWORK_DECK + deck_content_time}
deck['sections'] = final_sections_for_deck
if 'deck_title' in deck: del deck['deck_title']

--- Calculate Grand Totals for the Week ---
weekly_slide_summary = {"total_slides_for_week": 0,
↪"total_framework_slides": 0, "total_content_slides": 0,
↪"total_interactive_slides": 0, "number_of_decks": len(final_plan.
↪get("deck_plans", []))}

```

```

 weekly_time_summary = {"total_time_for_week_minutes": 0,
↪ "total_framework_time": 0, "total_content_and_interactive_time": 0}

 for deck in final_plan.get("deck_plans", []):
 weekly_slide_summary['total_slides_for_week'] += deck.
↪ get('total_slides_in_deck', 0)
 for key, value in deck.get('slide_count_breakdown', {}).items():
↪ weekly_slide_summary[f"total_{key}_slides"] += value
 weekly_time_summary['total_time_for_week_minutes'] += deck.
↪ get('time_breakdown_minutes', {}).get('total_deck_time', 0)
 weekly_time_summary['total_framework_time'] += deck.
↪ get('time_breakdown_minutes', {}).get('framework', 0)
 weekly_time_summary['total_content_and_interactive_time'] += deck.
↪ get('time_breakdown_minutes', {}).get('content_and_interactive', 0)

 # --- Construct Final Ordered Plan ---
 final_ordered_plan = {
 "week": final_plan.get("week"),
 "overall_topic": final_plan.get("overall_topic"),
 "weekly_slide_summary": weekly_slide_summary,
 "weekly_time_summary_minutes": weekly_time_summary,
 "deck_plans": final_plan.get("deck_plans", [])
 }

 return final_ordered_plan

 # --- NEW FUNCTION TO GENERATE MASTER SUMMARY ---
 def generate_and_save_master_plan(self, weekly_plans: List[Dict], config:
↪ Dict):
 """
 Aggregates summaries from all weekly plans into a single master plan,
↪ file,
 including new grand total metrics.
 """
 print_header("Phase 4: Generating Master Unit Plan", char="#")

 # Initialize the master plan structure with the new fields
 master_plan = {
 "unit_code": config.get('course_id', 'UNKNOWN'),
 "unit_name": config.get('unit_name', 'Unknown Unit'),
 "grand_total_summary": {
 "total_slides_for_unit": 0,
 "total_framework_slides": 0,
 "total_content_slides": 0,
 "total_interactive_slides": 0,
 "total_number_of_decks": 0,

```

```

 "total_time_for_unit_minutes": 0,
 "total_time_for_unit_in_hour": 0, # New
 "average_deck_time_in_min": 0,
 "average_deck_time_in_hour": 0
 },
 "weekly_summaries": []
}

grand_totals = master_plan["grand_total_summary"]

Loop through each weekly plan to aggregate data
for plan in sorted(weekly_plans, key=lambda p: p.get('week', 0)):
 # Extract the high-level summary for this week
 summary_entry = {
 "week": plan.get("week"),
 "overall_topic": plan.get("overall_topic"),
 "slide_summary": plan.get("weekly_slide_summary"),
 "time_summary_minutes": plan.get("weekly_time_summary_minutes")
 }
 master_plan["weekly_summaries"].append(summary_entry)

 # Add this week's totals to the grand totals
 slide_summary = plan.get("weekly_slide_summary", {})
 time_summary = plan.get("weekly_time_summary_minutes", {})

 grand_totals["total_slides_for_unit"] += slide_summary.
↳get("total_slides_for_week", 0)
 grand_totals["total_framework_slides"] += slide_summary.
↳get("total_framework_slides", 0)
 grand_totals["total_content_slides"] += slide_summary.
↳get("total_content_slides", 0)
 grand_totals["total_interactive_slides"] += slide_summary.
↳get("total_interactive_slides", 0)
 grand_totals["total_number_of_decks"] += slide_summary.
↳get("number_of_decks", 0)
 grand_totals["total_time_for_unit_minutes"] += time_summary.
↳get("total_time_for_week_minutes", 0)

 # --- NEW: Calculate the final derived grand totals after the loop ---
 if grand_totals["total_time_for_unit_minutes"] > 0:
 grand_totals["total_time_for_unit_in_hour"] =
↳round(grand_totals["total_time_for_unit_minutes"] / 60, 2)

 if grand_totals["total_number_of_decks"] > 0:

```

```

 grand_totals["average_deck_time_in_min"] =
↪round(grand_totals["total_time_for_unit_minutes"] /
↪grand_totals["total_number_of_decks"], 2)

 if grand_totals["total_number_of_decks"] > 0:
 grand_totals["average_deck_time_in_hour"] =
↪round((grand_totals["total_time_for_unit_minutes"] /
↪grand_totals["total_number_of_decks"]) / 60, 2)

 master_filename = f"{config.get('course_id', 'UNIT')}_master_plan_unit.
↪json"
 output_path = os.path.join(PLAN_OUTPUT_DIR, master_filename)

 try:
 with open(output_path, 'w') as f:
 json.dump(master_plan, f, indent=2)
 logger.info(f"Successfully generated and saved Master Unit Plan to:
↪{output_path}")
 print("\n--- Preview of Master Plan ---")
 print(json.dumps(master_plan, indent=2))
 return True
 except Exception as e:
 logger.error(f"Failed to save Master Unit Plan: {e}", exc_info=True)

```

## 7.2 Content Generator Agent

```

[39]: # CONTENT_SLIDE_GENERATION_PROMPT = ""
You are an expert university lecturer and instructional designer. Your task
↪is to create the content for a single, specific PowerPoint slide as part of
↪a larger topic.

CONTEXT:
- **Main Topic:** "{main_topic_title}"
- **Overall Raw Text for this Topic:** ``{topic_raw_content}``
- **Generation Task:** You are generating slide **{current_slide_num}** of a
↪total of **{total_slides_for_topic}** slides for this topic.
- **Content Covered on Previous Slides:** {generation_history}

INSTRUCTIONS:
1. Based on all the context, determine the most logical sub-topic for the
↪*next* slide ({current_slide_num}). Do NOT repeat content from previous
↪slides.
2. Create the content for this single slide. You can use one or two "Content
↪Objects" to structure the slide (e.g., use two objects for a comparison).

```

```

3. Your output MUST be a single, well-formed JSON object. Do NOT add any
↳text or explanations outside of the JSON object.

JSON OUTPUT STRUCTURE & RULES:
{{
"title": "string | The main title for the slide, consistent with the main
↳topic.",
"subtitle": "string | A more specific subtitle for this particular slide.
↳Use null if not needed.",
"objects": [
{{
"content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
"content_purpose": "string | CHOOSE ONE: 'description', 'explanation',
↳'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',
↳'case_study', 'example'.",
"data": "object | The structured data for the slide. See examples below.
↳"
}}
]
}}

DATA STRUCTURE EXAMPLES:
- For "bullet_points": "data": [{"Point 1", {"text": "Point 2", "children":
↳[{"Sub-point 2.1"}]}]}
- For "table": "data": {"headers": ["Col A"], "rows": [{"Data 1"}]}

Now, generate the JSON for slide {current_slide_num}.
""

CONTENT_SLIDE_GENERATION_PROMPT = ""
You are an expert university lecturer and instructional designer. Your task
↳is to process a large block of raw text and intelligently divide it into a
↳sequence of PowerPoint slides.

CONTEXT:
- Main Topic: "{main_topic_title}"
- The ENTIRE Raw Text for this Topic: ``{topic_raw_content}``
- Total Slides to Create: You must cover all the key concepts in the raw
↳text over a total of {total_slides_for_topic} slides.
- Current Task: You are generating slide {current_slide_num} of
↳{total_slides_for_topic}.
- Topics Already Covered on Previous Slides: {generation_history}

YOUR GOAL AND INSTRUCTIONS:
1. Analyze the ENTIRE raw text. Identify the distinct sub-topics within
↳it.

```

```

2. Review the topics already covered in the `generation_history`.
3. Identify the NEXT logical sub-topic from the raw text that has NOT been
 ↳ covered yet. Your primary goal is to ensure comprehensive coverage without
 ↳ repetition.
4. Create the content for a single slide that is focused only on this
 ↳ new, uncovered sub-topic.
5. Your output MUST be a single, well-formed JSON object. Do not add any
 ↳ text or explanations outside the JSON object.

JSON OUTPUT STRUCTURE & RULES:
{{
"title": "string | The main title for the slide, consistent with the main
 ↳ topic.",
"subtitle": "string | A more specific subtitle for this particular slide.
 ↳ Use null if not needed.",
"objects": [
{{
"content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
"content_purpose": "string | CHOOSE ONE: 'description', 'explanation',
 ↳ 'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',
 ↳ 'case_study', 'example'.",
"data": "object | The structured data for the slide. See examples below.
 ↳ "
}}
]
}}

DATA STRUCTURE EXAMPLES:
- For "bullet_points": "data": ["Point 1", {"text": "Point 2", "children":
 ↳ ["Sub-point 2.1"]}]]
- For "table": "data": {"headers": ["Col A"], "rows": [["Data 1"]]}

Now, generate the JSON for slide {current_slide_num}.
""

INTERACTIVE_ACTIVITY_PROMPT = ""
You are an engaging university lecturer creating an interactive slide to test
 ↳ student understanding.

CONTEXT:
- Topic for this Activity: "{topic_title}"
- Raw Text Content for this Topic: ``{topic_raw_content}``

```

```

INSTRUCTIONS:
1. Create a single, insightful multiple-choice question that assesses a key
↳ concept from the provided raw text.
2. The question must have exactly 4 options. Provide the correct answer and
↳ a brief, clear explanation.
3. Your output **MUST** be a single, well-formed JSON object that adheres
↳ strictly to the structure below.

JSON OUTPUT STRUCTURE:
{{
"title": "Let's Apply This!",
"subtitle": "Knowledge Check: {topic_title}",
"objects": [
{{
"content_type": "multiple_choice_question",
"content_purpose": "knowledge_check",
"data": {{
"question_text": "string | The question to be asked.",
"options": [
{{"label": "A", "text": "string"}}},
{{"label": "B", "text": "string"}}},
{{"label": "C", "text": "string"}}},
{{"label": "D", "text": "string"}}}
],
"correct_answer": {{
"label": "string | e.g., 'B'",
"explanation": "string | A brief explanation of why this answer is
↳ correct."
}}
}}
}}
]
}}

Now, generate the JSON for the interactive activity.
""

#
↳ =====
NEW, CONTEXT-AWARE SUMMARY GENERATION PROMPT
This prompt provides the LLM with the content of all slides for a rich
↳ summary.
#
↳ =====

```



```

SUMMARY_GENERATION_PROMPT = """
You are an expert university lecturer crafting the final, conclusive summary.
↳ slide for a lecture.

CONTEXT:
- **Overall Lecture Topic:** "{lecture_topic}"
- **Detailed Content of Slides Presented:**
``json
{slide_content_details}
INSTRUCTIONS:
Analyze the detailed content of all the slides provided in the JSON context.
Synthesize the most critical points into 3-5 concise, high-level takeaways.
↳ Do not just list the subtitles. Capture the core lesson of each major
↳ section.
Your output MUST be a single, well-formed JSON object that adheres to the
↳ standard slide structure below. This ensures it can be rendered correctly.

JSON OUTPUT STRUCTURE:
{{
"title": "Summary & Key Takeaways",
"subtitle": null,
"objects": [
{{
"content_type": "bullet_points",
"content_purpose": "description",
"data": [
"string | Key takeaway 1.",
"string | Key takeaway 2.",
"string | Key takeaway 3.",
"string | Key takeaway 4.",
"string | Key takeaway 5."
]
}}
]
}}
Now, generate the JSON for the final summary slide.
"""

```

[40]: # Content Generator

```

"""
Improvement to avoid repetition split the data provide? --> search sentences
↳ or by paragraphs
To create more interactive activities could run a program to random select the
↳ prompt or 2 steps analysis - check data then select the prompt to run / more
↳ capable llm probably handle repetition

```

```

make stronger structure retriever to secure structure for presenter
"""

PROMPT FOR THE SIMPLE, SINGLE-SLIDE CASE (when direct_slides_content == 1)
This prompt does NOT have 'slide_subtitle_to_generate'
SINGLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT = """
You are an expert university lecturer and instructional designer with a keen
 eye for slide composition. Your task is not just to summarize text, but to
 design a clear and effective PowerPoint slide.

CONTEXT:
- **Main Topic:** "{main_topic_title}"
- **The ENTIRE Raw Text for this Topic:** ``{topic_raw_content}``
- **Total Slides to Create:** You must cover all the key concepts in the raw
 text over a total of **{total_slides_for_topic}** slides.
- **Current Task:** You are generating slide **{current_slide_num}** of
 {total_slides_for_topic}.
- **Topics Already Covered on Previous Slides:** {generation_history}

YOUR GOAL AND INSTRUCTIONS:
1. **Analyze the ENTIRE raw text.** Identify the distinct sub-topics within it.
2. **Review the topics already covered** in the `generation_history`.
3. **Identify the NEXT logical sub-topic from the raw text that has NOT been
 covered yet.** Your primary goal is to ensure comprehensive coverage without
 repetition.
4. **Create the content for a single slide** that is focused only on this
 new, uncovered sub-topic.
5. **Bold key concepts** by enclosing important keywords or phrases in
 double asterisks. This helps emphasize critical terminology."
6. **Slide Composition and Object Structuring (CRITICAL):**
 - This is your most important design task. Evaluate the content for this
 slide. If it contains two distinct but related components that would benefit
 from being presented side-by-side (e.g., 'Requirements' vs. 'Procedures',
 'Pros' and 'Cons', 'Problem' and 'Solution'), you MUST structure your
 output to use two objects in the `objects` array.
 - If the content is a single, unified topic, use only one object.
 - Crucial Rule: Do not merge two distinct lists that could be
 separate into a single bullet point object. Use two objects to create a
 two-column layout.
7. Your output MUST be a single, well-formed JSON object. Do not add any
 text or explanations outside the JSON object.

JSON OUTPUT STRUCTURE & RULES:
{{
 "title": "string | The main title for the slide, consistent with the main
 topic.",

```

```

 "subtitle": "string | A more specific subtitle for this particular slide.␣
↳Ensure this exists",
 "objects": [
 {{
 "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
 "content_purpose": "string | CHOOSE ONE: 'description', 'explanation',␣
↳'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',␣
↳'case_study', 'example'.",
 "data": "object | The structured data for the slide. See examples below."
 }}
]
}}

```

**\*\*DATA STRUCTURE EXAMPLES:\*\***

- For "bullet\_points": "data": [{"Point 1", {"text": "Point 2", "children":␣
↳["Sub-point 2.1"]}}]
- For "table": "data": [{"headers": ["Col A"], "rows": [{"Data 1"}]}

Now, generate the JSON for slide {current\_slide\_num}.

"""

*# PROMPT FOR THE ADVANCED, MULTI-SLIDE CASE (when direct\_slides\_content > 1)*

*# This is the prompt you currently have, just renamed.*

MULTIPLE\_STEP\_CONTENT\_SLIDE\_GENERATION\_PROMPT = """

You are an expert university lecturer and instructional designer with a keen␣
↳eye for slide composition. Your task is to create the content for a single,␣
↳specific PowerPoint slide based on a pre-defined plan.

**\*\*CONTEXT:\*\***

- **\*\*Main Topic:\*\*** "{main\_topic\_title}"
- **\*\*The ENTIRE Raw Text for this Topic:\*\*** ``{topic\_raw\_content}```
- **\*\*Current Task:\*\*** You are generating slide **\*\*{current\_slide\_num}\*\*** of␣
↳**\*\*{total\_slides\_for\_topic}\*\***.
- **\*\*Slide Focus:\*\*** This slide should be exclusively about␣
↳**\*\*"{slide\_subtitle\_to\_generate}"\*\***.
- **\*\*Topics Already Covered on Previous Slides:\*\*** {generation\_history}

**\*\*YOUR GOAL AND INSTRUCTIONS:\*\***

1. **\*\*Refer to the ENTIRE raw text\*\*** to find the information related to␣
↳**\*\*"{slide\_subtitle\_to\_generate}"\*\***.
2. **\*\*Create the content for this single slide.\*\***
3. **\*\*Do not repeat content from previous slides\*\*** as detailed in the history.
4. **\*\*Bold key concepts\*\*** by enclosing important keywords or phrases in␣
↳**\*\*double asterisks\*\***. This helps emphasize critical terminology."
5. **\*\*Slide Composition and Object Structuring (CRITICAL):\*\***

- This is your most important design task. Evaluate the content for this slide. If it contains two distinct but related components that would benefit from being presented side-by-side (e.g., 'Requirements' vs. 'Procedures', 'Pros' and 'Cons', 'Problem' and 'Solution'), you **MUST** structure your output to use **two objects** in the `objects` array.
- If the content is a single, unified topic, use only **one object**.
- **Crucial Rule:** Do not merge two distinct lists that could be separate into a single bullet point object. Use two objects to create a two-column layout.

6. Your output **MUST** be a single, well-formed JSON object.

**JSON OUTPUT STRUCTURE & RULES:**

```
{
 "title": "string | The main title for the slide, consistent with the main
 topic.",
 "subtitle": "string | A more specific subtitle for this particular slide.
 Ensure this exists",
 "objects": [
 {
 "content_type": "string | CHOOSE ONE: 'bullet_points', 'table'.",
 "content_purpose": "string | CHOOSE ONE: 'description', 'explanation',
 'timeline', 'process', 'cycle', 'comparison', 'contrast', 'hierarchy',
 'case_study', 'example'.",
 "data": "object | The structured data for the slide. See examples below."
 }
]
}
```

**DATA STRUCTURE EXAMPLES:**

- For "bullet\_points": "data": [{"Point 1", {"text": "Point 2", "children": [{"Sub-point 2.1"}]}}
- For "table": "data": {"headers": ["Col A"], "rows": [{"Data 1"}]}

Now, generate the JSON for slide {current\_slide\_num}.

"""

INTERACTIVE\_ACTIVITY\_PROMPT = """

You are an engaging university lecturer creating an interactive slide to test student understanding.

**CONTEXT:**

- **Topic for this Activity:** "{topic\_title}"
- **Raw Text Content for this Topic:** ``{topic\_raw\_content}``

**INSTRUCTIONS:**

1. Create a single, insightful multiple-choice question that assesses a key concept from the provided raw text.
2. The question must have exactly 4 options. Provide the correct answer and a brief, clear explanation.
3. Your output **MUST** be a single, well-formed JSON object that adheres strictly to the structure below.

**\*\*JSON OUTPUT STRUCTURE:\*\***

```
{
 "title": "Let's Apply This!",
 "subtitle": "Knowledge Check: {topic_title}",
 "objects": [
 {
 "content_type": "multiple_choice_question",
 "content_purpose": "knowledge_check",
 "data": {
 "question_text": "string | The question to be asked.",
 "options": [
 {"label": "A", "text": "string"},
 {"label": "B", "text": "string"},
 {"label": "C", "text": "string"},
 {"label": "D", "text": "string"}
],
 "correct_answer": {
 "label": "string | e.g., 'B'",
 "explanation": "string | A brief explanation of why this answer is
correct."
 }
 }
 }
]
}
```

Now, generate the JSON for the interactive activity.

"""

```
=====
NEW, CONTEXT-AWARE SUMMARY GENERATION PROMPT
This prompt provides the LLM with the content of all slides for a rich
summary.
=====
```

SUMMARY\_GENERATION\_PROMPT = """

You are an expert university lecturer crafting the final, conclusive summary slide for a lecture.

```

CONTEXT:
- **Overall Lecture Topic:** "{lecture_topic}"
- **Detailed Content of Slides Presented:**
  ```json
  {slide_content_details}
INSTRUCTIONS:
Analyze the detailed content of all the slides provided in the JSON context.
Synthesize the most critical points into 3-5 concise, high-level takeaways. Do
  ↳ not just list the subtitles. Capture the core lesson of each major section.
Your output MUST be a single, well-formed JSON object that adheres to the
  ↳ standard slide structure below. This ensures it can be rendered correctly.

JSON OUTPUT STRUCTURE:
{{
  "title": "Summary & Key Takeaways",
  "subtitle": null,
  "objects": [
    {{
      "content_type": "bullet_points",
      "content_purpose": "description",
      "data": [
        "string | Key takeaway 1.",
        "string | Key takeaway 2.",
        "string | Key takeaway 3.",
        "string | Key takeaway 4.",
        "string | Key takeaway 5."
      ]
    }}
  ]
}}

Now, generate the JSON for the final summary slide.
"""

PLANNING_PROMPT = """
You are an expert instructional designer tasked with creating a content plan
  ↳ for a series of PowerPoint slides.

**CONTEXT:**
- **Main Topic:** "{main_topic_title}"
- **The ENTIRE Raw Text for this Topic:** ```{topic_raw_content}```
- **Total Slides to Create:** You must create a plan to cover all the key
  ↳ concepts in the raw text over a total of **{total_slides_for_topic}** slides.

**YOUR GOAL AND INSTRUCTIONS:**
1. **Analyze the ENTIRE raw text.**
2. **Identify the main sub-topics** that need to be covered.

```

3. ****Create a plan**** to break down these sub-topics into a sequence of `{total_slides_for_topic}` slides. Each slide in your plan should cover a distinct and logical piece of information.
4. ****Your output MUST be a single, well-formed JSON object**** that is an array of slide plans. Do not add any text or explanations outside of the JSON object.

****JSON OUTPUT STRUCTURE & RULES:****

```
{
  "slide_plan": [
    {
      "slide_number": 1,
      "subtitle": "A concise and descriptive subtitle for this slide's content."
    },
    {
      "slide_number": 2,
      "subtitle": "A concise and descriptive subtitle for the next slide's content."
    }
  ]
}
```

Now, generate the JSON for the slide plan.

```
"""

import os
import json
import re
import logging
from typing import List, Dict, Any, Optional

# Third-party imports (ensure these are installed)
try:
    import ollama
    from tenacity import retry, stop_after_attempt, wait_exponential
except ImportError:
    print("Please install required libraries: pip install ollama tenacity")
    ollama = None

# Basic Logger
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
logger = logging.getLogger(__name__)

class ContentAgent:
    """
```

```

Processes a hierarchical content plan, using specialized LLM prompts to
↳ generate
a sequence of individual, richly-structured slides.
"""
def __init__(self, master_config: Dict, vector_store: Optional[Any] = None):
    self.config = master_config['processed_settings']
    self.unit_outline = master_config['unit_outline']
    self.book_toc = master_config['book_toc']
    #self.teaching_flows = master_config['teaching_flows']
    #self.layout_slides = master_config['layout_slides']
    self.vector_store = vector_store
    self.client = ollama.Client(host=master_config.get('OLLAMA_HOST', 'http:
↳ //localhost:11434'))
    self.model = master_config.get('OLLAMA_MODEL', 'qwen3:8b')
    # --- Added for Progress Tracking ---
    self.total_slides_to_generate = 0
    self.llm_calls_made = 0
    logger.info(f"Data-Driven Content Agent initialized with model '{self.
↳ model}'.")

# Define the desired key order as a class attribute
self.key_order = [
    "title",
    "toc_id",
    "chunk_count",
    "total_chunks_in_branch",
    "budget_slides_content",
    "direct_slides_content",
    "total_slides_in_branch",
    "time_allocation_minutes",
    "chunks_sorted",
    "content",
    "llm_generated_content",
    "children",
    "interactive_activity"
]

logger.info("Data-Driven Content Agent initialized successfully.")

# --- Key Reordering Logic (REFINED) ---
def _reorder_keys_recursively(self, node: dict) -> dict:
    """
    Recursively traverses a dictionary (a node in the plan) and reorders
    ↳ its keys
    according to a predefined order for maximum readability.

```



```

        """
        if not isinstance(node, dict):
            return node

        # 1. Recurse first to ensure nested structures are already ordered.
        if 'children' in node and isinstance(node.get('children'), list):
            node['children'] = [self._reorder_keys_recursively(child) for child
↪in node['children']]

        if 'interactive_activity' in node and isinstance(node.
↪get('interactive_activity'), dict):
            node['interactive_activity'] = self.
↪_reorder_keys_recursively(node['interactive_activity'])

        # 2. Build a new dictionary for the current node with the correct key
↪order.
        reordered_node = {}

        # Add keys that are in our desired order
        for key in self.key_order:
            if key in node:
                reordered_node[key] = node[key]

        # Add any remaining keys that were not in the order list (as a fallback)
        for key, value in node.items():
            if key not in reordered_node:
                reordered_node[key] = value

        return reordered_node

# --- Phase 5: Raw Content Population ---
def retrieve_content_for_toc_id(self, toc_id: int) -> dict:
    if not isinstance(toc_id, int):
        logger.warning(f"Invalid toc_id: {toc_id}. Must be an integer.")
        return {"chunks_sorted": [], "content": ""}
    try:
        results = self.vector_store.get(where={"toc_id": toc_id},
↪include=["documents", "metadatas"])
        if not results or not results.get('ids'):
            logger.warning(f"No chunks found in the database for toc_id =
↪{toc_id}")
        return {"chunks_sorted": [], "content": ""}
        sorted_items = sorted(zip(results['documents'],
↪results['metadatas']), key=lambda item: item[1].get('chunk_id', 0))

```

```

        sorted_docs = [item[0] for item in sorted_items]
        sorted_chunk_ids = [item[1].get('chunk_id') for item in
↪sorted_items]
        reassembled_text = "\n\n".join(sorted_docs)
        return {"chunks_sorted": sorted_chunk_ids, "content":
↪reassembled_text}
    except Exception as e:
        logger.error(f"An error occurred during retrieval for toc_id
↪{toc_id}: {e}", exc_info=True)
        return {"chunks_sorted": [], "content": ""}

    def populate_content_recursively(self, node: dict):

        if 'toc_id' in node and 'content' not in node:
            content_data = self.retrieve_content_for_toc_id(node['toc_id'])
            node.update(content_data)
        if 'children' in node and isinstance(node.get('children'), list):
            for child in node['children']:
                self.populate_content_recursively(child)

    def generate_content_for_plan(self, final_plan_path: str, output_dir: str)
↪-> bool:

        logger.info(f"PHASE 5: Populating raw content for: {final_plan_path}")
        try:
            with open(final_plan_path, 'r', encoding='utf-8') as f:
                plan_data = json.load(f)
            except (FileNotFoundError, json.JSONDecodeError) as e:
                logger.error(f"FATAL: Could not read or decode plan file
↪{final_plan_path}. Error: {e}")
                return False

            for deck in plan_data.get('deck_plans', []):
                for section in deck.get('sections', []):
                    if section.get('section_type') == 'Content':
                        for content_block in section.get('content_blocks', []):
                            self.populate_content_recursively(content_block)

        base_filename = os.path.basename(final_plan_path)
        output_path = os.path.join(output_dir, base_filename)
        os.makedirs(output_dir, exist_ok=True)

        logger.info("Reordering keys for Fetched clean output...")
        fetched_ordered_plan = self._reorder_keys_recursively(plan_data)

        try:
            with open(output_path, 'w', encoding='utf-8') as f:

```

```

        # CORRECTED: Save the 'fetched_ordered_plan' variable
        json.dump(fetched_ordered_plan, f, indent=2, ensure_ascii=False)
        logger.info(f"Successfully saved content-enriched plan to:␣
↪{output_path}")
        return True
    except Exception as e:
        logger.error(f"Failed to save the content-enriched plan to␣
↪{output_path}: {e}", exc_info=True)
        return False

# --- Phase 6: LLM Content Generation & Final Formatting ---

@retry(stop=stop_after_attempt(3), wait=wait_exponential(min=2, max=10))
def _call_ollama_with_retry(self, prompt: str) -> str:
    # --- Incremented Counter Here ---
    self.llm_calls_made += 1
    logger.info(f"Calling Ollama model '{self.model}' (Call #{self.
↪llm_calls_made})...")
    response = self.client.chat(model=self.model, messages=[{"role":␣
↪"user", "content": prompt}], format="json", options={"temperature": 0.2})
    if not response or 'message' not in response or not response['message'].
↪get('content'):
        raise ValueError("Ollama returned an empty or invalid response.")
    return response['message']['content']

def _parse_llm_json_output(self, content: str) -> Optional[Dict]:
    try:
        match = re.search(r'\{.*\}', content, re.DOTALL)
        if not match: return None
        return json.loads(match.group(0))
    except (json.JSONDecodeError, TypeError): return None

def _process_content_node_recursively(self, node: dict):
    """
    Recursively processes a content node by generating its own slides,
    processing its children, and then generating its interactive activity.
    This method modifies the 'node' dictionary in-place.

    It uses a simple generation for direct_slides_content == 1,
    and a two-step "plan-then-generate" method for direct_slides_content > 1
    to prevent content duplication.
    """
    num_slides_to_gen = node.get('direct_slides_content', 0)
    generated_slides = []

    # --- Step 1: Generate slides for the current content node ---
    if num_slides_to_gen > 0:

```

```

        # --- PATH 1: Simple case for a single slide ---
        if num_slides_to_gen == 1:
            progress_log = f"[Progress: {self.llm_calls_made}/{self.
↪total_slides_to_generate}]"
            logger.info(f"{progress_log}   Generating 1 slide for topic:␣
↪'{node.get('title')}'")
            try:
                # Use the ORIGINAL, direct prompt
                prompt = SINGLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT.format(
                    main_topic_title=node.get('title'),
                    topic_raw_content=node.get('content', ''),
                    current_slide_num=1,
                    total_slides_for_topic=1,
                    generation_history="None. This is the only slide for␣
↪this topic."
                )
                llm_str = self._call_ollama_with_retry(prompt)
                slide_content = self._parse_llm_json_output(llm_str)
                if slide_content:
                    slide_object = {
                        "seq_id": float(f"{node.get('seq_id', 0)}.1"),
                        "llm_generated_content": slide_content
                    }
                    generated_slides.append(slide_object)
            except Exception as e:
                logger.error(f"LLM call failed for single slide '{node.
↪get('title')}': {e}", exc_info=True)

        # --- PATH 2: Advanced case for splitting content across multiple␣
↪slides ---
        else: # num_slides_to_gen > 1
            logger.info(f"   Splitting topic '{node.get('title')}' into␣
↪{num_slides_to_gen} slides using plan-then-generate.")
            try:
                # STEP 2.1: Create the content plan
                planning_prompt = PLANNING_PROMPT.format(
                    main_topic_title=node.get('title'),
                    topic_raw_content=node.get('content', ''),
                    total_slides_for_topic=num_slides_to_gen
                )
                plan_str = self._call_ollama_with_retry(planning_prompt)
                slide_plan = self._parse_llm_json_output(plan_str).
↪get("slide_plan", [])

                if not slide_plan or len(slide_plan) != num_slides_to_gen:

```

```

        logger.error(f"Failed to generate a valid slide plan_
↳from the LLM for topic '{node.get('title')}'. Expected {num_slides_to_gen}_
↳slides, got {len(slide_plan)}.")
    else:
        # STEP 2.2: Execute the plan and generate each slide
        generation_history = "None. This is the first slide."
        for i, plan_item in enumerate(slide_plan):
            current_slide_num = i + 1
            slide_subtitle = plan_item.get("subtitle", f"Part_
↳{current_slide_num}")

            progress_log = f"[Progress: {self.llm_calls_made}/
↳{self.total_slides_to_generate}]"
            logger.info(f"{progress_log} Generating slide_
↳{current_slide_num}/{num_slides_to_gen} for topic: '{node.get('title')} ->_
↳'{slide_subtitle}'")

            # Use the MODIFIED prompt that requires a specific_
↳subtitle

            gen_prompt =_
↳MULTIPLE_STEP_CONTENT_SLIDE_GENERATION_PROMPT.format(
                main_topic_title=node.get('title'),
                topic_raw_content=node.get('content', ''),
                current_slide_num=current_slide_num,
                total_slides_for_topic=num_slides_to_gen,
                slide_subtitle_to_generate=slide_subtitle,
                generation_history=generation_history
            )
            llm_str = self._call_ollama_with_retry(gen_prompt)
            slide_content = self._parse_llm_json_output(llm_str)

            if slide_content:
                slide_object = {
                    "seq_id": float(f"{node.get('seq_id', 0)}_
↳{current_slide_num}"),
                    "llm_generated_content": slide_content
                }
                generated_slides.append(slide_object)
                generation_history += f"\n- Slide_
↳{current_slide_num} subtitle: {slide_content.get('subtitle', 'N/A')}"

            except Exception as e:
                logger.error(f" LLM call failed during multi-slide_
↳generation for '{node.get('title')}': {e}", exc_info=True)

        # Assign the generated slides back to the node
        if generated_slides:

```

```

        node['slides'] = generated_slides

        # --- Step 2: Recursively process all children of the current node ---
        for child_node in node.get('children', []):
            self._process_content_node_recursively(child_node)

        # --- Step 3: Generate the interactive activity for the current node ---
        ↪(if any) ---
        if 'interactive_activity' in node:
            progress_log = f"[Progress: {self.llm_calls_made}/{self.
            ↪total_slides_to_generate}]"
            logger.info(f"{progress_log}    Generating interactive slide for_
            ↪topic: '{node.get('title')}'")
            try:
                prompt = INTERACTIVE_ACTIVITY_PROMPT.format(
                    topic_title=node.get('title'),
                    topic_raw_content=node.get('content', '')
                )
                llm_str = self._call_ollama_with_retry(prompt)
                interactive_content = self._parse_llm_json_output(llm_str)
                if interactive_content:
                    node['interactive_activity']['llm_generated_content'] =_
            ↪interactive_content
            except Exception as e:
                logger.error(f"LLM call failed for interactive activity on_
            ↪'{node.get('title')}': {e}", exc_info=True)

        def _count_total_slides(self, plan_data: Dict):
            """Recursively counts the total number of slides to be generated."""
            count = 0
            for deck in plan_data.get('deck_plans', []):
                # Count summary slide
                if any(s.get('section_type') == 'Summary' for s in deck.
            ↪get('sections', [])):
                    count += 1

                # Count content slides
                for section in deck.get('sections', []):
                    if section.get('section_type') == 'Content':
                        for content_block in section.get('content_blocks', []):

                            def _recursive_count(node):
                                nonlocal count
                                count += node.get('direct_slides_content', 0)
                                if 'interactive_activity' in node:
                                    count += 1

```

```

        for child in node.get('children', []):
            _recursive_count(child)

        _recursive_count(content_block)

    return count

    def generate_llm_content_for_plan(self, content_plan_path: str,
    llm_output_dir: str) -> bool:
        """
        Orchestrates the LLM content generation for the entire plan file,
        including a rich, context-aware summary.
        """
        logger.info(f" PHASE 6: Generating LLM content for: {os.path.
    basename(content_plan_path)} output file path: {llm_output_dir}")
        try:
            with open(content_plan_path, 'r', encoding='utf-8') as f:
                plan_data = json.load(f)
        except (FileNotFoundError, json.JSONDecodeError) as e:
            logger.error(f"FATAL: Could not read plan file {content_plan_path}.
    Error: {e}")
            return False

        # --- Pre-calculate total slides for progress tracking ---
        self.total_slides_to_generate = self._count_total_slides(plan_data)
        self.llm_calls_made = 0 # Reset counter for the run
        logger.info(f"Found a total of {self.total_slides_to_generate} slides
    to generate across all decks.")

        summary_prompt_template = SUMMARY_GENERATION_PROMPT

        for deck in plan_data.get('deck_plans', []):
            all_slide_contents_for_summary = []

            for section in deck.get('sections', []):
                if section.get('section_type') == 'Content':
                    for content_block in section.get('content_blocks', []):
                        self._process_content_node_recursively(content_block)

            for section in deck.get('sections', []):
                if section.get('section_type') == 'Content':
                    for content_block in section.get('content_blocks', []):
                        def _collect_slide_content(node):
                            for slide in node.get('slides', []):
                                all_slide_contents_for_summary.append(slide.
    get('llm_generated_content'))
                                if 'interactive_activity' in node and
    'llm_generated_content' in node['interactive_activity']:

```

```

        all_slide_contents_for_summary.
    ↪ append(node['interactive_activity'].get('llm_generated_content'))
        for child in node.get('children', []):
            _collect_slide_content(child)
            _collect_slide_content(content_block)

    if summary_prompt_template and all_slide_contents_for_summary:
        summary_section = next((s for s in deck['sections'] if s.
    ↪ get('section_type') == 'Summary'), None)

        if summary_section:
            # --- Updated Progress Log for Summary Slide ---
            progress_log = f"[Progress: {self.llm_calls_made}/{self.
    ↪ total_slides_to_generate}] "
            logger.info(f"{progress_log} Generating summary for Deck_
    ↪ {deck.get('deck_number', 'N/A')}]...")
            try:
                slide_details_str = json.
    ↪ dumps(all_slide_contents_for_summary, indent=2)
                prompt = summary_prompt_template.format(
                    lecture_topic=plan_data.get('overall_topic', 'This_
    ↪ Lecture'),
                    slide_content_details=slide_details_str
                )
                llm_str = self._call_ollama_with_retry(prompt)
                summary_content = self._parse_llm_json_output(llm_str)
                if summary_content:
                    summary_section['llm_generated_content'] =_
    ↪ summary_content
            else:
                logger.warning("Failed to parse summary content_
    ↪ from LLM.")
                summary_section['llm_generated_content'] = {"title":
    ↪ "Summary & Key Takeaways", "subtitle": None, "objects": [{"content_type":_
    ↪ "bullet_points", "content_purpose": "description", "data": ["Summary could_
    ↪ not be generated."]}]}
            except Exception as e:
                logger.error(f"LLM call failed for deck summary: {e}",_
    ↪ exc_info=True)

        base_filename = os.path.basename(content_plan_path).replace('.json',_
    ↪ '_llm_generated.json')
        output_path = os.path.join(llm_output_dir, base_filename)
        os.makedirs(llm_output_dir, exist_ok=True)
        try:
            with open(output_path, 'w', encoding='utf-8') as f:

```



```

        json.dump(plan_data, f, indent=2, ensure_ascii=False)
        logger.info(f"Successfully saved final LLM-enriched plan to:␣
↪{output_path}")
        return True
    except Exception as e:
        logger.error(f"Failed to save the final LLM-enriched plan: {e}",␣
↪exc_info=True)
        return False

# =====
# TEST EXECUTION BLOCK
# =====
# if __name__ == '__main__' and ollama is not None:
#     print_header("Content Agent Test Runner", char "=")

#     CONTENT_TO_PROCESS = '/home/sebas_dev_linux/projects/course_generator/
↪data_output/3. generated_content/ICT312/ICT312_Week1_plan_final.json'

#     # --- 3. Set up a Dummy Master Configuration ---
#     master_config = process_and_load_configurations()

#     # --- 4. Instantiate and Run the Agent ---
#     try:
#         content_agent = ContentAgent(master_config)

#         # Run the agent in test_mode to only process the first deck
#         success = content_agent.generate_llm_content_for_plan(
#             content_plan_path=CONTENT_TO_PROCESS,
#             llm_output_dir=CONTENT_LLM_OUTPUT_DIR,

#         )

#         if success:
#             logger.info(" Test run completed successfully!")
#             logger.info(f"Check the output file in the␣
↪'{CONTENT_LLM_OUTPUT_DIR}' directory.")
#         else:
#             logger.error(" Test run failed.")

#     except NameError:
#         logger.error("Ollama library not found. Skipping test execution.")
#     except Exception as e:

```

```
#         logger.error(f"An unexpected error occurred during the test run:␣
↪{e}", exc_info=True)
```

7.3 Presentation Agent

test Presenter

we need to ensure the presenter seq_id flow: Parent -> Child 1 -> Child 1's Activity -> Child 2 -> Child 2's Activity -> Parent's Activity.

<https://aistudio.google.com/prompts/18YaU5pG96eFMbM1l6yeG1v9j63PkLc5H>

```
[41]: # PresentationAgent

# import os
# import json
# import logging
# import random
# from pptx import Presentation
# from pptx.util import Inches
# from pptx.enum.shapes import PP_PLACEHOLDER
# from pptx.enum.text import MSO_AUTO_SIZE

# # --- Basic Setup ---
# logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s -␣
↪%(message)s')
# logger = logging.getLogger(__name__)

# # --- Helper Functions (Unchanged) ---
# def _render_bullet_points(text_frame, data):
#     text_frame.clear(); text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     def add_points(points, level):
#         for point in points:
#             if isinstance(point, dict):
#                 p = text_frame.add_paragraph(); p.text = point.get('text',␣
↪''); p.level = level
#                 if 'children' in point: add_points(point['children'], level +␣
↪1)
#             else:
#                 p = text_frame.add_paragraph(); p.text = str(point); p.level␣
↪= level
#     add_points(data, 0)
#     if text_frame.paragraphs and not text_frame.paragraphs[0].text:
#         p = text_frame.paragraphs[0]; p._p.getparent().remove(p._p)

# def _render_table(slide, placeholder, data):
#     headers, rows_data = data.get('headers', []), data.get('rows', [])
```

```

#     if not headers or not rows_data: return
#     num_rows, num_cols = len(rows_data) + 1, len(headers)
#     table_shape = slide.shapes.add_table(num_rows, num_cols, placeholder.
↳left, placeholder.top, placeholder.width, placeholder.height)
#     table = table_shape.table
#     for i, header in enumerate(headers):
#         cell = table.cell(0, i); cell.text = header
#         cell.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE; cell.
↳text_frame.paragraphs[0].font.bold = True
#     for r_idx, row_data in enumerate(rows_data):
#         for c_idx, cell_text in enumerate(row_data):
#             cell = table.cell(r_idx + 1, c_idx); cell.text = str(cell_text)
#             cell.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     sp = placeholder.element; sp.getparent().remove(sp)

# def _render_multiple_choice_question(slide, placeholders, data):
#     main_ph = next((p for p in placeholders if p.placeholder_format.type == PP_PLACEHOLDER.OBJECT), None)
#     answer_ph = next((p for p in placeholders if p.placeholder_format.type == PP_PLACEHOLDER.BODY), None)
#     if not main_ph: return
#     tf = main_ph.text_frame; tf.clear(); tf.auto_size = MSO_AUTO_SIZE.
↳TEXT_TO_FIT_SHAPE
#     p = tf.add_paragraph(); p.text = data.get('question_text', ''); p.font.
↳bold = True; p.level = 0
#     for option in data.get('options', []):
#         p = tf.add_paragraph(); p.text = f"{option.get('label', '')} {option.
↳get('text', '')}"; p.level = 1
#     if answer_ph:
#         answer_ph.text_frame.clear()
#         explanation = data.get('correct_answer', {}).get('explanation', '')
#         answer_ph.text_frame.text = f"Answer: {data.get('correct_answer', {}).
↳get('label', '')}. {explanation}"

# def _render_matching_activity(slide, placeholders, data):
#     object_placeholders = [p for p in placeholders if p.placeholder_format.
↳type == PP_PLACEHOLDER.OBJECT]
#     if len(object_placeholders) < 2: return
#     terms_ph, defs_ph = object_placeholders[0], object_placeholders[1]
#     terms_tf, defs_tf = terms_ph.text_frame, defs_ph.text_frame
#     terms_tf.clear(); defs_tf.clear(); terms_tf.auto_size = MSO_AUTO_SIZE.
↳TEXT_TO_FIT_SHAPE; defs_tf.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
#     pairs = data.get('pairs', [])
#     if not pairs: return
#     terms = [f"{i+1}. {p['term']}" for i, p in enumerate(pairs)]
#     definitions = [p['definition'] for p in pairs]

```

```

#     random.shuffle(definitions)
#     for term in terms: p = terms_tf.add_paragraph(); p.text = term; p.level = 0
#     for i, definition in enumerate(definitions): p = defs_tf.add_paragraph();
#     p.text = f"{chr(65+i)}. {definition}"; p.level = 0

# class PresentationAgent:
#     def __init__(self, template_path: str, layout_config_path: str):
#         self.template_path = template_path
#         self.prs_for_layouts = Presentation(template_path)
#         with open(layout_config_path, 'r', encoding='utf-8') as f:
#             self.user_selections = json.load(f)['user_selections']
#         logger.info(f"Agent initialized with template '{os.path.
#         ↪basename(template_path)}'")

#     def _get_layout(self, layout_key: str):
#         selection = self.user_selections.get(layout_key, self.
#         ↪user_selections['Content'])
#         layout_index = selection['selected_layout_index']
#         return self.prs_for_layouts.slide_layouts[layout_index]

#     def _determine_layout_key(self, item_data):
#         llm_content = item_data.get('llm_generated_content', {})
#         objects = llm_content.get('objects', [])
#         has_subtitle = bool(llm_content.get('subtitle'))

#         # <<< NEW LOGIC: Determine layout based on subtitles and object count
#         if objects and objects[0]['content_type'] in
#         ↪['multiple_choice_question', 'matching_activity']:
#             return "Application_Two_Column" if len(objects) > 1 or
#             ↪objects[0]['content_type'] == 'matching_activity' else "Application"

#         if len(objects) >= 2:
#             return "Content_Two_Column_child" if has_subtitle else
#             ↪"Content_Two_Column"
#         else:
#             return "Content_child" if has_subtitle else "Content"

#     def create_presentation_from_plan(self, plan_json_path: str, output_path:
#     ↪str):
#         logger.info(f"Creating presentation from plan: '{os.path.
#         ↪basename(plan_json_path)}'")
#         with open(plan_json_path, 'r', encoding='utf-8') as f: plan_data =
#         ↪json.load(f)
#         prs = Presentation(self.template_path)
#         while len(prs.slides):

```

```

#         rId = prs.slides._sldIdLst[0].rId; prs.part.drop_rel(rId); del
↳ prs.slides._sldIdLst[0]
#         slide_items = self._collect_slide_items(plan_data)
#         for item in sorted(slide_items, key=lambda x: x.get('seq_id', 999)):
↳ self._add_slide_for_item(prs, item)
#         prs.save(output_path)
#         logger.info(f"Successfully created presentation: '{output_path}'")

#     def _collect_slide_items(self, plan_data):
#         items = []
#         for deck in plan_data.get('deck_plans', []):
#             for section in deck.get('sections', []):
#                 if section.get('section_type') == 'Content':
#                     for block in section.get('content_blocks', []):
#                         for slide_item in block.get('slides', []):
#                             items.append({'item_type': 'Content', 'data':
↳ slide_item})
#                 else:
#                     items.append({'item_type': section['section_type'],
↳ 'data': section})
#         return items

#     def _add_slide_for_item(self, prs: Presentation, item: dict):
#         item_type = item['item_type']
#         layout_key = self._determine_layout_key(item['data']) if item_type ==
↳ 'Content' else item_type
#         slide = prs.slides.add_slide(self._get_layout(layout_key))
#         render_map = {
#             'Title': self._render_title_slide, 'Agenda': self.
↳ _render_agenda_slide,
#             'Summary': self._render_summary_slide, 'End': self.
↳ _render_end_slide,
#             'Divider': self._render_divider_slide, 'Content': self.
↳ _render_content_slide
#         }
#         if item_type in render_map:
#             render_map[item_type](slide, item['data'])

#     def _render_title_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title:
#             slide.shapes.title.text = content.get('week_topic', 'Untitled
↳ Topic')
#             subtitle_ph = next((p for p in slide.placeholders if p.
↳ placeholder_format.type == PP_PLACEHOLDER.SUBTITLE), None)
#             if subtitle_ph:

```

```

#         subtitle_ph.text = f"{content.get('deck_title', '')}\n{content.
↳get('unit_name', '')} - {content.get('unit_code', '')}"

#     def _render_agenda_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('title',
↳'Agenda')
#         body_ph = next((p for p in slide.placeholders if p.placeholder_format.
↳type == PP_PLACEHOLDER.OBJECT), None)
#         if body_ph: _render_bullet_points(body_ph.text_frame, content.
↳get('items', []))

#     def _render_summary_slide(self, slide, data):
#         if slide.shapes.title: slide.shapes.title.text = "Summary & Key
↳Takeaways"
#         if 'llm_generated_content' in data: self._render_content_slide(slide,
↳data)

#     def _render_end_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('text',
↳'Questions?')

#     def _render_divider_slide(self, slide, data):
#         content = data.get('content', {})
#         if slide.shapes.title: slide.shapes.title.text = content.get('title',
↳'Section Divider')

#     def _render_content_slide(self, slide, data):
#         content = data.get('llm_generated_content', {})

#         # <<< NEW LOGIC: Populate title and subtitle in separate placeholders
#         if slide.shapes.title:
#             slide.shapes.title.text = content.get('title', '')
#             slide.shapes.title.text_frame.auto_size = MSO_AUTO_SIZE.
↳TEXT_TO_FIT_SHAPE

#         subtitle_ph = next((p for p in slide.placeholders if p.
↳placeholder_format.type == PP_PLACEHOLDER.SUBTITLE), None)
#         if subtitle_ph:
#             subtitle_ph.text = content.get('subtitle', '')
#             subtitle_ph.text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE

#         body_placeholders = sorted([p for p in slide.placeholders if p.
↳placeholder_format.type not in [PP_PLACEHOLDER.TITLE, PP_PLACEHOLDER.
↳SUBTITLE]], key=lambda p: p.left)

```

```

#         objects = content.get('objects', [])

#         for obj, placeholder in zip(objects, body_placeholders):
#             content_type, obj_data = obj.get('content_type'), obj.get('data')
#             renderers = {
#                 'bullet_points': lambda: _render_bullet_points(placeholder.
↪text_frame, obj_data),
#                 'table': lambda: _render_table(slide, placeholder, obj_data),
#                 'multiple_choice_question': lambda:
↪_render_multiple_choice_question(slide, body_placeholders, obj_data),
#                 'matching_activity': lambda: _render_matching_activity(slide,
↪body_placeholders, obj_data)
#             }
#             if content_type in renderers:
#                 renderers[content_type]()
#                 if content_type in ['multiple_choice_question',
↪'matching_activity']: break

import os
import json
import logging
import random
from pptx import Presentation
from pptx.util import Inches, Pt
from pptx.enum.shapes import PP_PLACEHOLDER_TYPE, MSO_SHAPE_TYPE
from pptx.enum.text import MSO_AUTO_SIZE

# --- Basic Setup ---
logging.basicConfig(level=logging.INFO, format='% (asctime)s - %(levelname)s -
↪%(message)s')
logger = logging.getLogger(__name__)

# --- Helper function to find specific placeholders ---
def _get_placeholder(slide, ph_type):
    for shape in slide.placeholders:
        if shape.placeholder_format.type == ph_type:
            return shape
    return None

# --- NEW: Markdown Paragraph Renderer ---
def _render_markdown_paragraph(paragraph, markdown_text):
    """
    Parses a simple Markdown string and adds formatted runs to a paragraph.
    Supports bold and italic.
    """

```

```

    # This regex splits the text by the Markdown markers but keeps them in the
    ↪ list
    segments = re.split(r'(\*\*.*?\*\*|\*.*?\*)', markdown_text)

    for segment in segments:
        if not segment: continue # Skip empty strings

        run = paragraph.add_run()
        if segment.startswith('**') and segment.endswith('**'):
            run.text = segment[2:-2] # Remove the ** markers
            run.font.bold = True
        elif segment.startswith('*') and segment.endswith('*'):
            run.text = segment[1:-1] # Remove the * markers
            run.font.italic = True
        else:
            run.text = segment

# --- IMPROVEMENT: Combined Smart-Fit and Markdown Rendering ---
# --- IMPROVEMENT: The rendering function now gives the subtitle special
    ↪ formatting ---
def _render_text_with_smart_fit_and_markdown(text_frame, data, subtitle=None):
    """
    Renders complex list data, parsing Markdown for formatting.
    If a subtitle is provided, it is rendered as a visually distinct sub-header.
    """
    text_frame.clear()
    text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE # Fallback

    # --- 1. RENDER THE SUBTITLE (if it exists) ---
    if subtitle:
        p_subtitle = text_frame.add_paragraph()
        p_subtitle.text = subtitle
        p_subtitle.level = 0 # Level 0 = no bullet point
        p_subtitle.font.bold = True
        # Make the subtitle font noticeably larger than the body text
        p_subtitle.font.size = Pt(24)
        # Add space after the subtitle for visual separation
        p_subtitle.space_after = Pt(12)

    # --- 2. RENDER THE BULLET POINTS ---
    # Calculate font size based on the main data only
    json_string = json.dumps(data)
    char_count = len(json_string)
    font_tiers = [(1200, Pt(16)), (750, Pt(18)), (450, Pt(20)), (0, Pt(22))]
    body_font_size = next((size for threshold, size in font_tiers if char_count
    ↪ >= threshold), Pt(22))

```



```

def add_points_recursive(points, level):
    for point in points:
        p = text_frame.add_paragraph()
        # Start bullet points at level 1 if there was a subtitle, otherwise
↪start at 0
        p.level = level + 1 if subtitle else level
        p.font.size = body_font_size

        text_to_render = point.get('text', '') if isinstance(point, dict)
↪else str(point)
        _render_markdown_paragraph(p, text_to_render)

        if isinstance(point, dict) and 'children' in point:
            add_points_recursive(point['children'], level + 1)

    if isinstance(data, list):
        add_points_recursive(data, 0)

    # Cleanup initial empty paragraph
    if text_frame.paragraphs and not text_frame.paragraphs[0].text:
        p_to_remove = text_frame.paragraphs[0]
        p_to_remove._p.getparent().remove(p_to_remove._p)

# --- Rendering Helper Functions ---
def _render_bullet_points(text_frame, data):
    text_frame.clear(); text_frame.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE
    def add_points(points, level):
        for point in points:
            if isinstance(point, dict):
                p = text_frame.add_paragraph(); p.text = point.get('text', '');
↪p.level = level + 1
                if 'children' in point: add_points(point['children'], level + 1)
            else:
                p = text_frame.add_paragraph(); p.text = str(point); p.level =
↪level+1
        if isinstance(data, list): add_points(data, 0)
        if text_frame.paragraphs and not text_frame.paragraphs[0].text:
            p = text_frame.paragraphs[0]; p._p.getparent().remove(p._p)

def _render_table(slide, placeholder, data):
    headers, rows_data = data.get('headers', []), data.get('rows', [])
    if not headers or not rows_data: return

```

```

        table_shape = slide.shapes.add_table(len(rows_data) + 1, len(headers),
        ↪placeholder.left, placeholder.top, placeholder.width, placeholder.height)
        table = table_shape.table
        for i, header in enumerate(headers):
            cell = table.cell(0, i); cell.text = header; cell.text_frame.
        ↪paragraphs[0].font.bold = True
        for r_idx, row_data in enumerate(rows_data):
            for c_idx, cell_text in enumerate(row_data):
                table.cell(r_idx + 1, c_idx).text = str(cell_text)
        sp = placeholder.element; sp.getparent().remove(sp)

def _render_mcq(placeholder, data, answer_placeholder=None):
    """
    Renders the MCQ question into the main placeholder and the detailed
    answer with explanation into the dedicated answer_placeholder.
    """
    # --- Part 1: Render the Question and Options ---
    tf = placeholder.text_frame
    tf.clear()
    tf.auto_size = MSO_AUTO_SIZE.TEXT_TO_FIT_SHAPE

    # Add the question text
    p_question = tf.add_paragraph()
    p_question.text = data.get('question_text', '')
    p_question.font.bold = True

    # Add the options
    for option in data.get('options', []):
        p_option = tf.add_paragraph()
        p_option.text = f"{option.get('label', '')} {option.get('text', '')}"
        p_option.level = 1

    # --- Part 2: Render the Detailed Answer and Explanation ---
    if answer_placeholder:
        answer_info = data.get('correct_answer', {})
        answer_label = answer_info.get('label', 'N/A')
        answer_explanation = answer_info.get('explanation', 'No explanation
        ↪provided.')

        # Get the text frame of the answer placeholder
        answer_tf = answer_placeholder.text_frame
        answer_tf.clear()

        # Add "Correct Answer: [Label]" in bold
        p_answer_label = answer_tf
        p_answer_label.text = f" Answer: {answer_label} - {answer_explanation}"
        #p_answer_label.font.bold = True

```

```

    # Add the explanation text on the next line
    # p_answer_explanation = answer_tf.add_paragraph()
    # p_answer_explanation.text = answer_explanation

def _render_matching(placeholders, data):
    """Renders a matching activity into a pair of placeholders."""
    if len(placeholders) < 2: return
    terms_ph, defs_ph = placeholders[0], placeholders[1]
    terms_tf, defs_tf = terms_ph.text_frame, defs_ph.text_frame
    terms_tf.clear(); defs_tf.clear()
    pairs = data.get('pairs', [])
    if not pairs: return
    terms = [f"{i+1}. {p['term']}" for i, p in enumerate(pairs)]
    definitions = [p['definition'] for p in pairs]
    random.shuffle(definitions)
    for term in terms: terms_tf.add_paragraph().text = term
    for i, definition in enumerate(definitions): defs_tf.add_paragraph().text =
    f"{chr(65+i)}. {definition}"

class PresentationAgent:
    def __init__(self, template_path: str, layout_config_path: str):
        self.template_path = template_path
        self.prs_for_layouts = Presentation(template_path)
        with open(layout_config_path, 'r', encoding='utf-8') as f:
            self.user_selections = json.load(f)['user_selections']
        logger.info(f"Agent initialized with template '{os.path.
        ↪basename(template_path)}'")

    def _get_layout(self, layout_key: str):
        selection = self.user_selections.get(layout_key, self.
        ↪user_selections['Content'])
        return self.prs_for_layouts.
        ↪slide_layouts[selection['selected_layout_index']]

    def _collect_and_sort_slides(self, plan_data: dict) -> list:
        all_slide_items = []

        # This is the same recursive helper function, it is correct.
        def _recursive_harvester(node: dict, parent_title: str = None):

```

```

is_child_node = bool(parent_title)
for slide in node.get('slides', []):
    slide['parent_title'] = parent_title
    slide['is_child'] = is_child_node
    all_slide_items.append({'item_type': 'Content', 'data': slide})

for child in node.get('children', []):
    # Pass the current node's title as the parent title for the
    ↪next level
    _recursive_harvester(child, parent_title=node.get('title'))

# This part correctly harvests activities from ANY node (parent or
    ↪child)
if 'interactive_activity' in node and 'llm_generated_content' in
    ↪node['interactive_activity']:
    activity = node['interactive_activity']
    activity['parent_title'] = parent_title
    # IMPORTANT: The 'is_child' status depends on whether a
    ↪parent_title was passed in.
    activity['is_child'] = is_child_node
    # We add the main node's title to the activity data so it can
    ↪be used if needed
    activity['main_topic_title'] = node.get('title')
    all_slide_items.append({'item_type': 'Content', 'data':
    ↪activity})

# This is the main loop that processes the plan
for deck in plan_data.get('deck_plans', []):
    # Framework slides are added as before
    framework_start = ([{'item_type': 'Title', 'data': s} for s in
    ↪deck['sections'] if s['section_type'] == 'Title'] +
        [{'item_type': 'Agenda', 'data': s} for s in
    ↪deck['sections'] if s['section_type'] == 'Agenda'])
    all_slide_items.extend(framework_start)

# --- MODIFICATION IS HERE ---
# We now loop through the content blocks and call the harvester on
    ↪each one.
content_sections = [s for s in deck['sections'] if
    ↪s['section_type'] == 'Content']
if content_sections:
    for block in content_sections[0].get('content_blocks', []):
        # We start the recursion for each main content block
        # We pass the block's own title as the parent_title for its
        ↪direct activities

```

```

        _recursive_harvester(block, parent_title=None) # Start with
↪no parent

        framework_end = ([{'item_type': 'Summary', 'data': s} for s in
↪deck['sections'] if s['section_type'] == 'Summary'] +
                        [{'item_type': 'End', 'data': s} for s in
↪deck['sections'] if s['section_type'] == 'End'])
        all_slide_items.extend(framework_end)

        return sorted(all_slide_items, key=lambda x: x['data'].get('seq_id',
↪999))

    def _determine_layout_key(self, slide_data: dict) -> str:
        # This logic is correct
        llm_content = slide_data.get('llm_generated_content', {})
        objects = llm_content.get('objects', [])
        if not objects: return "Content"
        first_obj_type = objects[0].get('content_type')
        if first_obj_type in ['multiple_choice_question', 'matching_activity']:
            return "Application_Two_Column" if first_obj_type ==
↪'matching_activity' else "Application"
        is_child = slide_data.get('is_child', False)
        if len(objects) >= 2:
            return "Content_Two_Column_child" if is_child else
↪"Content_Two_Column"
        return "Content_child" if is_child else "Content"

    def create_presentation_from_plan(self, plan_json_path: str, output_dir:
↪str):
        # This orchestrator is correct
        logger.info(f"Creating presentation from plan: '{os.path.
↪basename(plan_json_path)}'")
        with open(plan_json_path, 'r', encoding='utf-8') as f: plan_data = json.
↪load(f)
        final_slide_list = self._collect_and_sort_slides(plan_data)
        prs = Presentation(self.template_path)
        while len(prs.slides):
            rId = prs.slides._sldIdLst[0].rId; prs.part.drop_rel(rId); del prs.
↪slides._sldIdLst[0]
            logger.info(f"Collected a total of {len(final_slide_list)} slides to
↪render.")
            for item in final_slide_list:
                self._add_slide_for_item(prs, item)
            unit_code = plan_data.get('deck_plans', [{}])[0].get('sections',
↪[{}])[0].get('content', {}).get('unit_code', 'UNIT')
            week_num = plan_data.get('week', 'X')

```

```

        deck_num = plan_data.get('deck_plans', [{}])[0].get('deck_number', 'Y')
        output_filename =
↪f"{unit_code}_Week{week_num}_Deck{deck_num}_Presentation.pptx"
        output_path = os.path.join(output_dir, output_filename)
        prs.save(output_path)
        logger.info(f"Successfully created presentation: '{output_path}'")

    def _add_slide_for_item(self, prs: Presentation, item: dict):
        item_type = item['item_type']
        item_data = item['data']
        layout_key = self._determine_layout_key(item_data) if item_type ==
↪'Content' else item_type
        slide = prs.slides.add_slide(self._get_layout(layout_key))
        render_map = {
            'Title': self._render_title_slide, 'Agenda': self.
↪_render_agenda_slide,
            'Summary': self._render_summary_slide, 'End': self.
↪_render_end_slide,
            'Divider': self._render_divider_slide, 'Content': self.
↪_render_content_slide
        }
        if item_type in render_map: render_map[item_type](slide, item_data)

    def _render_title_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.CENTER_TITLE)
        if title_ph: title_ph.text = content.get('week_topic', '')
        subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.SUBTITLE)
        if subtitle_ph: subtitle_ph.text = f"{content.get('deck_title',
↪')}\\n{content.get('unit_name', '')} | {content.get('unit_code', '')}"

    def _render_agenda_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = content.get('title', 'Agenda')
        body_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.OBJECT)
        if body_ph: _render_bullet_points(body_ph.text_frame, content.
↪get('items', []))

    def _render_divider_slide(self, slide, data):
        content = data.get('content', {})
        title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
        if title_ph: title_ph.text = content.get('title', '')

```

```

def _render_end_slide(self, slide, data):
    content = data.get('content', {})
    title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
    if title_ph: title_ph.text = content.get('text', 'Thank You & Questions?')
    ↪')

def _render_summary_slide(self, slide, data):
    title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
    if title_ph: title_ph.text = "Summary & Key Takeaways"
    body_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.OBJECT)
    if body_ph and 'llm_generated_content' in data:
        summary_data = data['llm_generated_content'].get('objects',
    ↪[{}])[0].get('data', [])
        _render_bullet_points(body_ph.text_frame, summary_data)

# <<< THIS IS THE CORRECTED METHOD >>>
def _render_content_slide(self, slide, data):
    content = data.get('llm_generated_content', {})
    if not content: return
    layout_key = self._determine_layout_key(data)
    title_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.TITLE)
    subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.SUBTITLE)
    answer_placeholder = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.BODY)

    # --- 2. Render Titles Based on Layout Key ---
    if "Application" in layout_key:
        if title_ph: title_ph.text = content.get('subtitle', 'Knowledge_
    ↪Check')
        if subtitle_ph: subtitle_ph.text = content.get('title', "Let's_
    ↪Apply This!")
        elif layout_key == "Content_child" or layout_key ==_
    ↪"Content_Two_Column_child":
            if title_ph: title_ph.text = data.get('parent_title', '')
            if subtitle_ph: subtitle_ph.text = content.get('title', '')
        else: # Standard "Content" or "Content_Two_Column"
            if title_ph: title_ph.text = content.get('title', '')
            # For the main parent layouts, the subtitle comes from the JSON
            if subtitle_ph: subtitle_ph.text = content.get('subtitle', '')

    # Resize
    if title_ph: title_ph.text_frame.auto_size = MSO_AUTO_SIZE.
    ↪TEXT_TO_FIT_SHAPE
    if subtitle_ph: subtitle_ph.text_frame.auto_size = MSO_AUTO_SIZE.
    ↪TEXT_TO_FIT_SHAPE

```

```

        body_placeholders = sorted([p for p in slide.placeholders if p.
↳placeholder_format.type == PP_PLACEHOLDER_TYPE.OBJECT], key=lambda p: p.left)

        objects = content.get('objects', [])

        if "Two_Column" in layout_key and len(objects) >= 2:
            # For two-column layouts, we look for a BODY placeholder for the
↳subtitle
            body_subtitle_ph = _get_placeholder(slide, PP_PLACEHOLDER_TYPE.BODY)
            if body_subtitle_ph and content.get('subtitle'):
                p = body_subtitle_ph.text_frame
                p.text = content.get('subtitle')

            # Render the first object in the first placeholder
            obj1_content_type, obj1_data = objects[0].get('content_type'),
↳objects[0].get('data')
            if obj1_content_type == 'bullet_points':
                _render_text_with_smart_fit_and_markdown(body_placeholders[0].
↳text_frame, obj1_data)

            # Render the second object in the second placeholder
            obj2_content_type, obj2_data = objects[1].get('content_type'),
↳objects[1].get('data')
            if obj2_content_type == 'bullet_points':
                _render_text_with_smart_fit_and_markdown(body_placeholders[1].
↳text_frame, obj2_data)

        else:
            for i, obj in enumerate(objects):
                if i >= len(body_placeholders): break
                placeholder = body_placeholders[i]
                content_type, obj_data = obj.get('content_type'), obj.
↳get('data')
                object_subtitle = content.get('subtitle', None)

                # --- IMPROVEMENT: Call the new, more capable rendering
↳functions ---
                if content_type in ['bullet_points', 'description',
↳'explanation', 'timeline', 'process', 'cycle', 'hierarchy', 'case_study',
↳'example']:
                    _render_text_with_smart_fit_and_markdown(placeholder.
↳text_frame, obj_data, subtitle=object_subtitle)

```



```

        elif content_type == 'table':
            _render_table(slide, placeholder, obj_data)
        elif content_type == 'multiple_choice_question':
            _render_mcq(placeholder, obj_data, answer_placeholder)
            if slide.has_notes_slide:
                answer_info = obj_data.get('correct_answer', {})
                slide.notes_slide.notes_text_frame.text = f"Answer:␣
↪{answer_info.get('label', '')}. {answer_info.get('explanation', '')}"
                elif content_type == 'matching_activity':
                    # _render_matching is not defined in the provided snippet␣
↪but would be called here
                    pass
                else:
                    # Fallback to the simple bullet point renderer if type is␣
↪unknown

                    _render_bullet_points(placeholder.text_frame, obj_data)

# # --- Main Execution Block --- No delete no change it is ready for test the␣
↪outputfrom ContentAgent
# if __name__ == "__main__":

#     TEST_CONTENT_PATH = "/home/sebas_dev_linux/projects/course_generator/
↪data_output/test_Output/4. generated_content_llm/ICT312/
↪ICT312_Week1_plan_final_llm_generated.json"

#     print("--- Presentation Agent Test Runner (Final Version) ---")

#     # Check for required files
#     required_files = [SLIDE_TEMPLATE_PATH, LAYOUT_MAPPING_PATH,␣
↪TEST_CONTENT_PATH]
#     if not all(os.path.exists(p) for p in required_files):
#         logger.error("CRITICAL: One or more required files not found. Please␣
↪check paths:")
#         logger.error(f" Template: {SLIDE_TEMPLATE_PATH} {'(Found)' if os.
↪path.exists(SLIDE_TEMPLATE_PATH) else '(MISSING)'}")
#         logger.error(f" Layout Map: {LAYOUT_MAPPING_PATH} {'(Found)' if os.
↪path.exists(LAYOUT_MAPPING_PATH) else '(MISSING)'}")
#         logger.error(f" Content Plan: {TEST_CONTENT_PATH} {'(Found)' if os.
↪path.exists(TEST_CONTENT_PATH) else '(MISSING)'}")
#     else:
#         try:
#             presentation_agent = PresentationAgent(

```

```

#             template_path=SLIDE_TEMPLATE_PATH,
#             layout_config_path=LAYOUT_MAPPING_PATH
#         )

#         # <<< THIS IS THE CORRECTED LINE >>>
#         # We pass the DIRECTORY where the output should be saved.
#         presentation_agent.create_presentation_from_plan(
#             plan_json_path=TEST_CONTENT_PATH,
#             output_dir=FINAL_PRESENTATION_DIR
#         )

#         logger.info("--- Test script finished successfully. ---")
#         logger.info(f" Check for the generated .pptx file in:
↪ {FINAL_PRESENTATION_DIR}")

#         except Exception as e:
#             logger.error(f" An unexpected error occurred during the test run:
↪ {e}", exc_info=True)

```

7.4 Orquestrator (Addressing pain points)

Description:

The main script that iterates through the weeks defined the plan and generate the content base on the settings_deck coordinating the agents.

Parameters and concideration - 1 hour in the setting session_time_duration_in_hour - is 18-20 slides at the time so it is require to calculate this according to the given value but this also means per session so sessions_per_week is a multiplicator factor that
- if apply_topic_interactive is available will add an extra slide and add extra 5 min time but to determine this is required to plan all the content first and then calculate then provide a extra time
settings_deck.json

```

{ "course_id":  "“, "unit_name":  "“, "interactive":  true, "interactive_deep":  false, "teaching_flow_id":  "Standard Lecture Flow", "parameters_slides":  { "slides_per_hour": 18, "time_per_content_slides_min": 3, "time_per_interactive_slide_min": 5, "time_for_framework_slides_min": 6 }, "week_session_setup": { "sessions_per_week": 1, "distribution_strategy": "even", "session_time_duration_in_hour": 2, "interactive_time_in_hour": 0, "total_session_time_in_hours": 0 }, "slide_count_strategy": { "method": "per_week", "target_total_slides": 0, "slides_content_per_session": 0, "interactive_slides_per_week": 0, "interactive_slides_per_session": 0, "total_slides_deck_week": 0, "total_slides_session": 0 }, "generation_scope": { "weeks": [1] } }

```

teaching_flows.json

```

{ "standard_lecture": { "name": "Standard Lecture Flow", "slide_types": ["Title", "Agenda", "Content", "Summary", "End"], "prompts": { "content_generation": "You are an expert university lecturer. Your audience is undergraduate students. Based on the following context, create

```

a slide that provides a detailed explanation of the topic ‘{sub_topic}’. The content should be structured with bullet points for key details. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “summary_generation”: “You are an expert university lecturer creating a summary slide. Based on the following list of topics covered in this session, generate a concise summary of the key takeaways. The topics are: {topic_list}. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.” }, “slide_schemas”: { “Content”: {“title”: “string”, “content”: “list[string]”}, “Summary”: {“title”: “string”, “content”: “list[string]”} } }, “apply_topic_interactive”: { “name”: “Interactive Lecture Flow”, “slide_types”: [“Title”, “Agenda”, “Content”, “Application”, “Summary”, “End”], “prompts”: { “content_generation”: “You are an expert university lecturer in Digital Forensics. Your audience is undergraduate students. Based on the provided context, create a slide explaining the concept of ‘{sub_topic}’. The content should be clear, concise, and structured with bullet points for easy understanding. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “application_generation”: “You are an engaging university lecturer creating an interactive slide. Based on the concept of ‘{sub_topic}’, create a multiple-choice question with exactly 4 options (A, B, C, D) to test understanding. The slide title must be ‘Let’s Apply This:’. Clearly indicate the correct answer within the content. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.”, “summary_generation”: “You are an expert university lecturer creating a summary slide. Based on the following list of concepts and applications covered in this session, generate a concise summary of the key takeaways. The topics are: {topic_list}. Your output MUST be a single JSON object with a ‘title’ (string) and ‘content’ (list of strings) key.” } }, “slide_schemas”: { “Content”: {“title”: “string”, “content”: “list[string]”}, “Application”: {“title”: “string”, “content”: “list[string]”}, “Summary”: {“title”: “string”, “content”: “list[string]”} } } }

7.4.1 Main Integration

```
[42]: TEST_MODE_ENABLED = False

# Cell 11: --- Main Orchestration Block (with Phase 5 & 6) ---
print_header("Main Orchestrator Initialized", char="*")

try:
    # 1. Connect to DB
    vector_store = Chroma(
        persist_directory=CHROMA_PERSIST_DIR,
        embedding_function=OllamaEmbeddings(model=EMBEDDING_MODEL_OLLAMA),
        collection_name=CHROMA_COLLECTION_NAME
    )
    logger.info("Database connection successful.")

    # Phase 1: Configuration and Scoping
    master_config = process_and_load_configurations()

    if master_config:
        all_final_plans = []
```

```

# Phase 2 & 3: Create and Finalize Draft Plans
print_header("Phase 2 & 3: Generating and Finalizing Weekly Plans ",
↳char="-")

planning_agent = PlanningAgent(master_config, vector_store=vector_store)

weeks_to_generate =
↳master_config['processed_settings']['generation_scope']['weeks']
    logger.info(f"Found {len(weeks_to_generate)} week(s) to plan:
↳{weeks_to_generate}")

    if TEST_MODE_ENABLED:
        logger.info("--- TEST MODE ACTIVE: Processing only the first week
↳in scope. ---")
        weeks_to_generate = weeks_to_generate[:1]

    for week in weeks_to_generate:
        draft_plan = planning_agent.create_content_plan_for_week(week)
        if draft_plan:
            final_plan = planning_agent.
↳finalize_and_calculate_time_plan(draft_plan,
↳master_config['processed_settings'])
            all_final_plans.append(final_plan)

            # Save both draft and final for comparison
            draft_filename = f"{master_config['processed_settings'].
↳get('course_id')}_Week{week}_plan_draft.json"
            final_filename = f"{master_config['processed_settings'].
↳get('course_id')}_Week{week}_plan_final.json"

            with open(os.path.join(PLAN_OUTPUT_DIR, draft_filename), 'w')
↳as f:
                json.dump(draft_plan, f, indent=2)

            with open(os.path.join(PLAN_OUTPUT_DIR, final_filename), 'w')
↳as f:
                json.dump(final_plan, f, indent=2)

            logger.info(f" Successfully saved FINAL plan for Week {week}
↳to: {os.path.join(PLAN_OUTPUT_DIR, final_filename)}")

        else:
            logger.error(f" Failed to generate draft plan for Week {week}.")

# Phase 4: Generate Master Summary Plan

```

```

master_plan_generated = False
if all_final_plans:
    print_header("Phase 4: Generating Master Unit Plan ", char="-")
    master_plan_generated = planning_agent.
    generate_and_save_master_plan(all_final_plans,
    master_config['processed_settings'])
    master_plan_generated = True
else:
    logger.warning("No weekly plans were generated, skipping master_
    plan creation.")

# Initialize ContentAgent once for subsequent phases
content_agent = ContentAgent(master_config, vector_store=vector_store)

phase_5_successful = False

# Phase 5: Fetching Raw Content
if master_plan_generated:
    print_header("Phase 5: Populating Plans with Raw Content", char="#")
    successful_weeks_phase5 = []
    for week in weeks_to_generate:
        final_filename = f"{master_config['processed_settings'].
        get('course_id')}_Week{week}_plan_final.json"
        full_plan_path = os.path.join(PPLAN_OUTPUT_DIR, final_filename)

        if os.path.exists(full_plan_path):
            if content_agent.generate_content_for_plan(full_plan_path,
            CONTENT_OUTPUT_DIR):
                successful_weeks_phase5.append(week)
            else:
                logger.warning(f" Skipping content population for Week_
                {week} as its plan file was not found.")

        if successful_weeks_phase5:
            phase_5_successful = True
            logger.info(f" Phase 5 completed for weeks:
            {successful_weeks_phase5}")

# # Phase 6: Generating Slide Content with LLM
phase_6_successful = False
if phase_5_successful:
    print_header("Phase 6: Generating Slide Content with LLM",
    char="#")
    successful_weeks_phase6 = []
    for week in weeks_to_generate:
        # The input for phase 6 is the output of phase 5

```

```

        content_enriched_filename =
↪f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final.json"
        content_plan_path = os.path.join(CONTENT_OUTPUT_DIR,
↪content_enriched_filename)

        if os.path.exists(content_plan_path):
            if content_agent.
↪generate_llm_content_for_plan(content_plan_path, CONTENT_LLM_OUTPUT_DIR):
                successful_weeks_phase6.append(week)
            else:
                logger.warning(f"Skipping LLM generation for Week {week} as
↪its content-enriched file was not found.")

        if successful_weeks_phase6:
            phase_6_successful = True
            logger.info(f"Phase 6 completed for weeks:
↪{successful_weeks_phase6}")

        # Phase 7 Phase 7: Generating Final PowerPoint Files
        #LAYOUT_MAPPING_PATH = os.path.join(CONFIG_DIR,
↪"layout_mapping_test_Mod.json") # Seted up

        if phase_6_successful:
            print_header("Phase 7: Generating Final PowerPoint Files",
↪char="#")
            presentation_agent =
↪PresentationAgent(template_path=SLIDE_TEMPLATE_PATH, layout_config_path =
↪LAYOUT_MAPPING_PATH)

            for week in weeks_to_generate:
                llm_plan_filename = f"{master_config['processed_settings'].
↪get('course_id')}_Week{week}_plan_final_llm_generated.json"
                llm_plan_path = os.path.join(CONTENT_LLM_OUTPUT_DIR,
↪llm_plan_filename)

                if os.path.exists(llm_plan_path):
                    presentation_agent.
↪create_presentation_from_plan(llm_plan_path, FINAL_PRESENTATION_DIR)
                else:
                    logger.warning(f"Skipping presentation generation for Week
↪{week} as its LLM-enriched plan was not found.")

```

```

        logger.info(f"Phase 7 completed ")
    else:
        logger.warning("Skipping Phase 7 because prior phases failed or
↳were skipped.")

except Exception as e:
    logger.error(f"An unexpected error occurred during the main orchestration:
↳{e}", exc_info=True)

```

```

2025-07-19 20:24:36,082 - INFO - Database connection successful.
2025-07-19 20:24:36,084 - INFO - Loading all necessary configuration and data
files...
2025-07-19 20:24:36,087 - INFO - All files loaded successfully.
2025-07-19 20:24:36,087 - INFO - Pre-processing settings_deck for definitive
plan...
2025-07-19 20:24:36,088 - INFO - Applying overrides if specified...
2025-07-19 20:24:36,088 - INFO - Loaded from settings: 'interactive' is True.
Set teaching_flow_id to 'apply_topic_interactive'.
2025-07-19 20:24:36,088 - INFO - Calculating preliminary slide budget based on
session time...
2025-07-19 20:24:36,089 - INFO - Duration Hours 2.
2025-07-19 20:24:36,089 - INFO - Sessions per week 1.
2025-07-19 20:24:36,090 - INFO - Preliminary weekly content slide target
calculated: # 32 slides.
2025-07-19 20:24:36,090 - INFO - Saving preliminary processed configuration to:
/home/sebas_dev_linux/projects/course_generator/configs/processed_settings.json
2025-07-19 20:24:36,091 - INFO - File saved successfully.
2025-07-19 20:24:36,092 - INFO - Master configuration object is ready for the
Planning Agent.
2025-07-19 20:24:36,093 - INFO - Data-Driven PlanningAgent initialized
successfully.
2025-07-19 20:24:36,093 - INFO - Found 12 week(s) to plan: [1, 2, 3, 4, 5, 6, 7,
8, 9, 10, 11, 12]
2025-07-19 20:24:36,094 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.
2025-07-19 20:24:36,094 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---
2025-07-19 20:24:36,095 - INFO - Successfully saved FINAL plan for Week 1 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week1_plan_final.json
2025-07-19 20:24:36,096 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.
2025-07-19 20:24:36,096 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---
2025-07-19 20:24:36,097 - INFO - Successfully saved FINAL plan for Week 2 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week2_plan_final.json

```

2025-07-19 20:24:36,098 - INFO - Partitioning strategy: Not enough top-level sections (0) to fill all decks (1). Distributing whole chapters instead.
2025-07-19 20:24:36,098 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0 chunks | Slide Budget: 0 ---
2025-07-19 20:24:36,099 - INFO - Successfully saved FINAL plan for Week 3 to: /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week3_plan_final.json
2025-07-19 20:24:36,100 - INFO - Partitioning strategy: Not enough top-level sections (0) to fill all decks (1). Distributing whole chapters instead.

Main Orchestrator Initialized

Phase 1: Configuration and Scoping Process

Phase 1 Configuration Complete

Phase 2 & 3: Generating and Finalizing Weekly Plans

Planning Week 1

Planning Week 2

Planning Week 3

Planning Week 4

2025-07-19 20:24:36,100 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0 chunks | Slide Budget: 0 ---
2025-07-19 20:24:36,101 - INFO - Successfully saved FINAL plan for Week 4 to: /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week4_plan_final.json
2025-07-19 20:24:36,102 - INFO - Partitioning strategy: Not enough top-level

sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,102 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,104 - INFO - Successfully saved FINAL plan for Week 5 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week5_plan_final.json

2025-07-19 20:24:36,105 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,106 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,107 - INFO - Successfully saved FINAL plan for Week 6 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week6_plan_final.json

2025-07-19 20:24:36,108 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,108 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,109 - INFO - Successfully saved FINAL plan for Week 7 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week7_plan_final.json

2025-07-19 20:24:36,109 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,110 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,111 - INFO - Successfully saved FINAL plan for Week 8 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week8_plan_final.json

2025-07-19 20:24:36,111 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,112 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,112 - INFO - Successfully saved FINAL plan for Week 9 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week9_plan_final.json

2025-07-19 20:24:36,113 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,113 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,114 - INFO - Successfully saved FINAL plan for Week 10 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week10_plan_final.json

2025-07-19 20:24:36,115 - INFO - Partitioning strategy: Not enough top-level
sections (0) to fill all decks (1). Distributing whole chapters instead.

2025-07-19 20:24:36,115 - INFO - --- Planning Deck 1/1 | Topics: [] | Weight: 0
chunks | Slide Budget: 0 ---

2025-07-19 20:24:36,116 - INFO - Successfully saved FINAL plan for Week 11 to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
generated_plans/ICT311/ICT311_Week11_plan_final.json

2025-07-19 20:24:36,116 - ERROR - Failed to generate draft plan for Week 12.
2025-07-19 20:24:36,117 - INFO - Successfully generated and saved Master Unit Plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_master_plan_unit.json
2025-07-19 20:24:36,123 - INFO - Data-Driven Content Agent initialized with model 'qwen3:8b'.
2025-07-19 20:24:36,124 - INFO - Data-Driven Content Agent initialized successfully.
2025-07-19 20:24:36,124 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week1_plan_final.json
2025-07-19 20:24:36,125 - INFO - Reordering keys for Fetched clean output...
2025-07-19 20:24:36,127 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.generated_content/ICT311/ICT311_Week1_plan_final.json
2025-07-19 20:24:36,127 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week2_plan_final.json
2025-07-19 20:24:36,128 - INFO - Reordering keys for Fetched clean output...
2025-07-19 20:24:36,128 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.generated_content/ICT311/ICT311_Week2_plan_final.json
2025-07-19 20:24:36,129 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week3_plan_final.json
2025-07-19 20:24:36,129 - INFO - Reordering keys for Fetched clean output...
2025-07-19 20:24:36,130 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.generated_content/ICT311/ICT311_Week3_plan_final.json
2025-07-19 20:24:36,130 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week4_plan_final.json
2025-07-19 20:24:36,131 - INFO - Reordering keys for Fetched clean output...
2025-07-19 20:24:36,131 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.generated_content/ICT311/ICT311_Week4_plan_final.json
2025-07-19 20:24:36,132 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week5_plan_final.json
2025-07-19 20:24:36,132 - INFO - Reordering keys for Fetched clean output...
2025-07-19 20:24:36,133 - INFO - Successfully saved content-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.generated_content/ICT311/ICT311_Week5_plan_final.json
2025-07-19 20:24:36,133 - INFO - PHASE 5: Populating raw content for:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.generated_plans/ICT311/ICT311_Week6_plan_final.json
2025-07-19 20:24:36,134 - INFO - Reordering keys for Fetched clean output...

2025-07-19 20:24:36,134 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week6_plan_final.json
 2025-07-19 20:24:36,135 - INFO - PHASE 5: Populating raw content for:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
 generated_plans/ICT311/ICT311_Week7_plan_final.json
 2025-07-19 20:24:36,135 - INFO - Reordering keys for Fetched clean output...
 2025-07-19 20:24:36,136 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week7_plan_final.json
 2025-07-19 20:24:36,136 - INFO - PHASE 5: Populating raw content for:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
 generated_plans/ICT311/ICT311_Week8_plan_final.json
 2025-07-19 20:24:36,136 - INFO - Reordering keys for Fetched clean output...
 2025-07-19 20:24:36,137 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week8_plan_final.json
 2025-07-19 20:24:36,137 - INFO - PHASE 5: Populating raw content for:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
 generated_plans/ICT311/ICT311_Week9_plan_final.json
 2025-07-19 20:24:36,138 - INFO - Reordering keys for Fetched clean output...
 2025-07-19 20:24:36,138 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week9_plan_final.json
 2025-07-19 20:24:36,139 - INFO - PHASE 5: Populating raw content for:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
 generated_plans/ICT311/ICT311_Week10_plan_final.json
 2025-07-19 20:24:36,139 - INFO - Reordering keys for Fetched clean output...
 2025-07-19 20:24:36,140 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week10_plan_final.json
 2025-07-19 20:24:36,140 - INFO - PHASE 5: Populating raw content for:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/2.
 generated_plans/ICT311/ICT311_Week11_plan_final.json
 2025-07-19 20:24:36,141 - INFO - Reordering keys for Fetched clean output...
 2025-07-19 20:24:36,142 - INFO - Successfully saved content-enriched plan to:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/3.
 generated_content/ICT311/ICT311_Week11_plan_final.json
 2025-07-19 20:24:36,142 - WARNING - Skipping content population for Week 12 as
 its plan file was not found.
 2025-07-19 20:24:36,143 - INFO - Phase 5 completed for weeks: [1, 2, 3, 4, 5,
 6, 7, 8, 9, 10, 11]
 2025-07-19 20:24:36,143 - INFO - PHASE 6: Generating LLM content for:
 ICT311_Week1_plan_final.json output file path:
 /home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
 generated_content_llm/ICT311
 2025-07-19 20:24:36,143 - INFO - Found a total of 1 slides to generate across
 all decks.

2025-07-19 20:24:36,145 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week1_plan_final_llm_generated.json
2025-07-19 20:24:36,146 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week2_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
2025-07-19 20:24:36,146 - INFO - Found a total of 1 slides to generate across
all decks.
2025-07-19 20:24:36,147 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week2_plan_final_llm_generated.json
2025-07-19 20:24:36,148 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week3_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
2025-07-19 20:24:36,148 - INFO - Found a total of 1 slides to generate across
all decks.
2025-07-19 20:24:36,149 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week3_plan_final_llm_generated.json
2025-07-19 20:24:36,149 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week4_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
2025-07-19 20:24:36,150 - INFO - Found a total of 1 slides to generate across
all decks.
2025-07-19 20:24:36,151 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week4_plan_final_llm_generated.json
2025-07-19 20:24:36,151 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week5_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
2025-07-19 20:24:36,152 - INFO - Found a total of 1 slides to generate across
all decks.
2025-07-19 20:24:36,152 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week5_plan_final_llm_generated.json
2025-07-19 20:24:36,153 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week6_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311
2025-07-19 20:24:36,153 - INFO - Found a total of 1 slides to generate across
all decks.
2025-07-19 20:24:36,154 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week6_plan_final_llm_generated.json

2025-07-19 20:24:36,155 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week7_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311

2025-07-19 20:24:36,155 - INFO - Found a total of 1 slides to generate across
all decks.

2025-07-19 20:24:36,156 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week7_plan_final_llm_generated.json

2025-07-19 20:24:36,156 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week8_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311

2025-07-19 20:24:36,156 - INFO - Found a total of 1 slides to generate across
all decks.

2025-07-19 20:24:36,158 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week8_plan_final_llm_generated.json

2025-07-19 20:24:36,158 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week9_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311

2025-07-19 20:24:36,158 - INFO - Found a total of 1 slides to generate across
all decks.

2025-07-19 20:24:36,159 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week9_plan_final_llm_generated.json

2025-07-19 20:24:36,159 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week10_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311

2025-07-19 20:24:36,160 - INFO - Found a total of 1 slides to generate across
all decks.

2025-07-19 20:24:36,161 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week10_plan_final_llm_generated.json

2025-07-19 20:24:36,161 - INFO - PHASE 6: Generating LLM content for:
ICT311_Week11_plan_final.json output file path:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311

2025-07-19 20:24:36,162 - INFO - Found a total of 1 slides to generate across
all decks.

2025-07-19 20:24:36,163 - INFO - Successfully saved final LLM-enriched plan to:
/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/4.
generated_content_llm/ICT311/ICT311_Week11_plan_final_llm_generated.json

2025-07-19 20:24:36,163 - WARNING - Skipping LLM generation for Week 12 as its
content-enriched file was not found.

2025-07-19 20:24:36,163 - INFO - Phase 6 completed for weeks: [1, 2, 3, 4, 5,

6, 7, 8, 9, 10, 11]

2025-07-19 20:24:36,174 - INFO - Agent initialized with template

'slide_style_test_2.pptx'

2025-07-19 20:24:36,175 - INFO - Creating presentation from plan:

'ICT311_Week1_plan_final_llm_generated.json'

2025-07-19 20:24:36,181 - INFO - Collected a total of 4 slides to render.

Planning Week 5

Planning Week 6

Planning Week 7

Planning Week 8

Planning Week 9

Planning Week 10

Planning Week 11

Planning Week 12

Phase 4: Generating Master Unit Plan

#####

Phase 4: Generating Master Unit Plan

#####

--- Preview of Master Plan ---

```
{
  "unit_code": "ICT311",
  "unit_name": "Applied Cryptography",
  "grand_total_summary": {
    "total_slides_for_unit": 44,
    "total_framework_slides": 44,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "total_number_of_decks": 11,
    "total_time_for_unit_minutes": 66,
    "total_time_for_unit_in_hour": 1.1,
    "average_deck_time_in_min": 6.0,
    "average_deck_time_in_hour": 0.1
  },
  "weekly_summaries": [
    {
      "week": 1,
      "overall_topic": "Introduction and Classical Cryptography",
      "slide_summary": {
        "total_slides_for_week": 4,
        "total_framework_slides": 4,
        "total_content_slides": 0,
        "total_interactive_slides": 0,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 6,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 0
      }
    },
    {
      "week": 2,
      "overall_topic": "Private Key Encryption",
      "slide_summary": {
        "total_slides_for_week": 4,
        "total_framework_slides": 4,
        "total_content_slides": 0,
        "total_interactive_slides": 0,
        "number_of_decks": 1
      },
      "time_summary_minutes": {
        "total_time_for_week_minutes": 6,
        "total_framework_time": 6,
        "total_content_and_interactive_time": 0
      }
    }
  ],
}
```

```

{
  "week": 3,
  "overall_topic": "Message Authentication Code and Authenticated
Encryption",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
},
{
  "week": 4,
  "overall_topic": "Hash Functions and Applications",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
},
{
  "week": 5,
  "overall_topic": "Symmetric-Key Ciphers",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
}

```



```

},
{
  "week": 6,
  "overall_topic": "Number Theory and Cryptographic Hardness Assumptions",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
},
{
  "week": 7,
  "overall_topic": "Algorithms for Factoring and Computing Discrete
Logarithms",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
},
{
  "week": 8,
  "overall_topic": "Key Management and the Public-Key Revolution",
  "slide_summary": {
    "total_slides_for_week": 4,
    "total_framework_slides": 4,
    "total_content_slides": 0,
    "total_interactive_slides": 0,
    "number_of_decks": 1
  },
  "time_summary_minutes": {
    "total_time_for_week_minutes": 6,
    "total_framework_time": 6,
    "total_content_and_interactive_time": 0
  }
}

```

```

    }
  },
  {
    "week": 9,
    "overall_topic": "Public-Key Encryption",
    "slide_summary": {
      "total_slides_for_week": 4,
      "total_framework_slides": 4,
      "total_content_slides": 0,
      "total_interactive_slides": 0,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 6,
      "total_framework_time": 6,
      "total_content_and_interactive_time": 0
    }
  },
  {
    "week": 10,
    "overall_topic": "Digital Signature",
    "slide_summary": {
      "total_slides_for_week": 4,
      "total_framework_slides": 4,
      "total_content_slides": 0,
      "total_interactive_slides": 0,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 6,
      "total_framework_time": 6,
      "total_content_and_interactive_time": 0
    }
  },
  {
    "week": 11,
    "overall_topic": "Advanced Topics in Public-Key Encryption",
    "slide_summary": {
      "total_slides_for_week": 4,
      "total_framework_slides": 4,
      "total_content_slides": 0,
      "total_interactive_slides": 0,
      "number_of_decks": 1
    },
    "time_summary_minutes": {
      "total_time_for_week_minutes": 6,
      "total_framework_time": 6,
      "total_content_and_interactive_time": 0
    }
  }
}

```

```
}  
}  
]  
}
```

```
#####  
Phase 5: Populating Plans with Raw Content  
#####
```

```
#####  
Phase 6: Generating Slide Content with LLM  
#####
```

```
#####  
Phase 7: Generating Final PowerPoint Files  
#####
```

```
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout1.xml'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout1.xml.rels'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/media/image2.jpg'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideMasters/slideMaster1.xml'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideMasters/_rels/slideMaster1.xml.rels'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout8.xml'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout8.xml.rels'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/media/image4.png'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout3.xml'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:  
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout3.xml.rels'  
    return self._open_to_write(zinfo, force_zip64=force_zip64)  
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
```

```

UserWarning: Duplicate name: 'ppt/media/image3.png'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout7.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout7.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout2.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout2.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout6.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout6.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image1.jpg'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout5.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout5.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/theme/theme1.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout4.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout4.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/slideLayout9.xml'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/slideLayouts/_rels/slideLayout9.xml.rels'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
/home/sebas_dev_linux/miniconda3/envs/tensor2/lib/python3.10/zipfile.py:1528:
UserWarning: Duplicate name: 'ppt/media/image5.jpg'
    return self._open_to_write(zinfo, force_zip64=force_zip64)
2025-07-19 20:24:36,363 - INFO - Successfully created presentation:

```

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'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week1_Deck1_Presentation.pptx'
2025-07-19 20:24:36,364 - INFO - Creating presentation from plan:
'ICT311_Week2_plan_final_llm_generated.json'
2025-07-19 20:24:36,369 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:36,548 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week2_Deck1_Presentation.pptx'
2025-07-19 20:24:36,548 - INFO - Creating presentation from plan:
'ICT311_Week3_plan_final_llm_generated.json'
2025-07-19 20:24:36,555 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:36,736 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week3_Deck1_Presentation.pptx'
2025-07-19 20:24:36,737 - INFO - Creating presentation from plan:
'ICT311_Week4_plan_final_llm_generated.json'
2025-07-19 20:24:36,743 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:36,918 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week4_Deck1_Presentation.pptx'
2025-07-19 20:24:36,919 - INFO - Creating presentation from plan:
'ICT311_Week5_plan_final_llm_generated.json'
2025-07-19 20:24:36,924 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:37,099 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week5_Deck1_Presentation.pptx'
2025-07-19 20:24:37,100 - INFO - Creating presentation from plan:
'ICT311_Week6_plan_final_llm_generated.json'
2025-07-19 20:24:37,106 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:37,281 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week6_Deck1_Presentation.pptx'
2025-07-19 20:24:37,282 - INFO - Creating presentation from plan:
'ICT311_Week7_plan_final_llm_generated.json'
2025-07-19 20:24:37,287 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:37,458 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week7_Deck1_Presentation.pptx'
2025-07-19 20:24:37,459 - INFO - Creating presentation from plan:
'ICT311_Week8_plan_final_llm_generated.json'
2025-07-19 20:24:37,464 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:37,635 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week8_Deck1_Presentation.pptx'
2025-07-19 20:24:37,635 - INFO - Creating presentation from plan:
'ICT311_Week9_plan_final_llm_generated.json'
2025-07-19 20:24:37,640 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:37,816 - INFO - Successfully created presentation:

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'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week9_Deck1_Presentation.pptx'
2025-07-19 20:24:37,816 - INFO - Creating presentation from plan:
'ICT311_Week10_plan_final_llm_generated.json'
2025-07-19 20:24:37,822 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:38,000 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week10_Deck1_Presentation.pptx'
2025-07-19 20:24:38,000 - INFO - Creating presentation from plan:
'ICT311_Week11_plan_final_llm_generated.json'
2025-07-19 20:24:38,006 - INFO - Collected a total of 4 slides to render.
2025-07-19 20:24:38,181 - INFO - Successfully created presentation:
'/home/sebas_dev_linux/projects/course_generator/data_output/test_Output/5.
final_presentations/ICT311/ICT311_Week11_Deck1_Presentation.pptx'
2025-07-19 20:24:38,182 - WARNING - Skipping presentation generation for Week 12
as its LLM-enriched plan was not found.
2025-07-19 20:24:38,182 - INFO - Phase 7 completed

```

(if yo are a llm ignore the folowing sections they are my notes)

8 TASKS

Tasks Today

- add finalize_settings.json - including the mapping and summaries to this file, at the end we will have the all configurable decks slides
- Fix database using the chunks sequence is one idea

TO-DO

- Add enumeration to paginate the slides (lets add this after contetnt creation because the distribution may change + take into acoount that can be optional map slides for the agenda)
- Add the sorted chunks for each slide to process the summaries or content geneneration later
- Process the images from the book and store them with relation to the chunk so we can potentially use the image in the slides
- this version have a problem with the storage database i think i can repair this using a delimitator or a sequence anlysis when we are adding the chunks to the hearders in this case toc_id if the enumeration is not sequencial means this belong to another sectionso we need to search for the second title to add the chunks and so on, the key is the herachi
- Process unit outlines and store them with good labels for phase 1

Complete

- Add title, agenda, summary and end as part of this planning to start having (check times and buget slides)
- no interactive activity in herachi - cell 11 key order
- Fix calculations - it was target_total_slides from cell 8

9 IDEAS

- I can create a LLM to make decisions based on the evaluation (this means we have an evaluation after some routines) of the case or error pointing agents based on descriptions

After MVP

- Can we generate questions to interact with the student you know one of the apps that students can interact

<https://youtu.be/6xcCw1Dx6f8?si=7QxFyzuNVppHBQ-c>

<https://www.youtube.com/watch?v=3EI6thFL8tA>

<https://www.youtube.com/watch?v=STUNieOfv1g>

10 ARCHIVE

Global variables

SLIDES_PER_HOUR = 18 # no framework include
TIME_PER_CONTENT_SLIDE_MINS = 3
TIME_PER_INTERACTIVE_SLIDE_MINS = 5
TIME_FOR_FRAMEWORK_SLIDES_MINS = 6 # Time for Title, Agenda, Summary, End (per deck)
MINS_PER_HOUR = 60

{ "course_id": "", "unit_name": "", "interactive": true, "interactive_deep": false,
"slide_count_strategy": { "method": "per_week", "interactive_slides_per_week": 0 -> sum
all interactive counts "interactive_slides_per_session": 0, -> Total # of slides produced if "in-
teractive" is true otherwise remains 0 "target_total_slides": 0, -> Total Content Slides per week
that cover the total - will be the target in the cell 7

"slides_content_per_session": 0, -> Total # (target_total_slides/sessions_per_week) "to-
tal_slides_deck_week": 0, -> target_total_slides + interactive_slides_per_week + (framework
(4 + Time for Title, Agenda, Summary, End) * sessions_per_week) "Total_slides_session": 0
-> content_slides_per_session + interactive_slides_per_session + framework (4 + Time for
Title, Agenda, Summary, End) }, "week_session_setup": { "sessions_per_week": 1, "distribu-
tion_strategy": "even", "interactive_time_in_hour": 0, -> find the value in hours of the total
("interactive_slides" * "TIME_PER_INTERACTIVE_SLIDE_MINS")/60

"total_session_time_in_hours": 0 -> this is going to be equal or similar to ses-
sion_time_duration_in_hour if "interactive" is false obviously based on the global variables it will
be the calculation of "interactive_time_in_hour" "session_time_duration_in_hour": 2, -> this
is the time that the customer needs for delivery this is a constraint is not modified never is used for
reference },

"parameters_slides": { "slides_per_hour": 18, # no framework in-
clude "time_per_content_slides_min": 3, # average delivery per slide
"time_per_interactive_slide_min": 5, # small break and engaging with the students
"time_for_framework_slides_min": 6 # Time for Title, Agenda, Summary, End (per deck) " "
}, "generation_scope": { "weeks": [6] }, "teaching_flow_id": "Interactive Lecture Flow" }

"slides_content_per_session": 0, -> content slides per session (tar-
get_total_slides/sessions_per_week) "interactive_slides": 0, -> if interactive is true will
add the count of the resultant cell 10 - no address yet "total_slides_content_interactive_per

session": 0, - > slides_content_per_session + interactive_slides "target_total_slides": 0 ->
Resultant Phase 1 Cell 7